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GUYANA'S PETROLEUM TRANSFORMATION

Exploration, the Stabroek agreement, first oil, sovereign wealth, environmental governance and a transferable blueprint for emerging producers.

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64,752 lexical core words and 141 registered sources. The full 120,000–150,000 word target is NOT yet achieved; this edition makes all existing expanded core writing available instead of claiming unfinished research is complete.

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Research integrity and publication status

This integrated development export contains the full ten-Part core manuscript and a dated source register. It replaces the much shorter integrated core of Edition 1.2. Research dossiers and wide technical CSVs remain editable in the companion cloud-worker package rather than being counted as additional core chapters.

Unresolved release gates: sufficient new substantive core research; full source-to-claim verification; precise first-half 2026 NRF cash/accrual bridge; direct legal, environmental and household outcome evidence; external specialist review; and the final graphic, accessibility and editorial QA.

Government-reported H1 2026 petroleum receipts and fund accounting use different timing boundaries. An approximately US\$184 million difference has a likely December 2025 receivables explanation, but the last transaction-level reconciliation remains open.

The ten complete integrated Parts begin on the next page; the accompanying editable CSVs retain the complete technical detail.

PART I | Historical foundations and pre-discovery readiness

Before the Discovery: The Foundations of Guyana's Petroleum Era

Chapter 1. The correct place to begin

A petroleum case study becomes misleading when its history starts with the commercial discovery. The discovery is the point at which petroleum moves from geological possibility toward a bankable development proposition. It is not the point at which a country begins to accumulate law, institutions, technical knowledge, environmental responsibilities, infrastructure constraints or experience with resource governance.

Guyana's modern offshore petroleum story therefore needs two starting dates. One is May 2015, when the Liza discovery changed the commercial outlook for the Stabroek Block. The other lies decades earlier, when successive exploration campaigns, legal reforms, geological surveys and maritime disputes shaped the conditions inherited by the state in 2015. The official petroleum history records exploration activity extending across much of the twentieth century, while the statutory framework for petroleum exploration and production dates to 1986. Environmental law also predates the oil boom. The Environmental Protection Act was enacted in 1996. [S35, S38, S39]

This longer chronology matters for a newcomer country. The central lesson is not that a government should build a large petroleum bureaucracy before finding commercial oil. Such a course risks creating fixed public costs against uncertain resources. The more defensible lesson is narrower. A prospective producer benefits from a scalable legal and technical foundation before the negotiating stakes rise. It benefits from knowing who owns geological information, which agency grants rights, how environmental assessment works, which maritime boundaries remain disputed, who assesses tax and royalty, and how expert advice is procured.

A country that enters a commercial discovery with those questions unresolved faces a compressed learning problem. Geological uncertainty starts to fall while commercial pressure rises. The investor wants to appraise quickly, define a development concept, secure approvals and move toward a final investment decision. Government needs time to test reservoir assumptions, fiscal terms, environmental impacts, infrastructure consequences and long-term liabilities. The negotiation therefore involves both economic value and institutional time.

Guyana's experience is useful because the transition happened quickly enough to expose this time problem. It is also useful because important pre-oil policies existed outside petroleum. The Low Carbon Development Strategy began in 2009. Guyana's forest partnership with Norway also predates commercial offshore oil. These facts make Guyana a more complex case than a simple resource-boom narrative. [S22]

The final case study should therefore resist two stories. The first says oil created every modern institution. The second says pre-existing institutions were fully prepared for the later scale of

deepwater production. The evidence supports neither proposition. The research question is how inherited systems interacted with rapid post-discovery adaptation.

Chapter 2. Economic conditions before commercial oil

A national petroleum strategy is constrained by the economy receiving the resource. Before 2015, Guyana relied on a mix of mining, agriculture, forestry, services and remittance-supported household activity. Gold, bauxite, rice and sugar were important sources of foreign exchange and employment. Mining had already familiarized public institutions with licences, royalties, environmental oversight and commodity price cycles. Yet deepwater oil introduced a different order of capital intensity, technical complexity, marine risk and potential revenue concentration.

The pre-oil economic baseline should be used for comparison rather than nostalgia. It identifies the problems petroleum revenue was expected to help address. Those problems included infrastructure gaps, coastal vulnerability, energy cost and reliability, uneven access to services, limited fiscal space and the need for stronger technical capacity. A credible assessment of post-oil development must later ask whether those constraints improved, shifted or became more expensive under rapid growth.

The baseline also establishes the denominator for institutional absorption. A small economy that receives a sudden increase in foreign exchange and public revenue faces a different challenge from a large diversified economy. Construction demand may rise faster than the supply of engineers, contractors and imported materials. Land prices and rents may respond quickly. Public agencies may lose technical staff to higher-paying private employers. New projects increase future maintenance obligations. These effects do not prove a resource curse. They describe transmission channels that need measurement.

For this reason, the flagship should present several pre-oil baselines rather than one GDP number. The economic baseline needs national output, sector composition, public debt, government revenue, capital spending, external balances, inflation, household conditions and labour-force indicators. The infrastructure baseline needs power generation, outages, roads, bridges, ports, drainage, sea defence, water, hospitals and schools. The institutional baseline needs staffing, budgets and statutory mandates for the agencies later expected to regulate oil.

The historical comparison should preserve data revisions. A national account series revised after 2015 should not be silently mixed with an earlier series using another base year. Household evidence also needs an observation date. A 2018 or 2019 household survey cannot be reported as a 2026 welfare measure merely because it is the latest complete historical dataset in a source. [S61]

The same principle applies to poverty and inequality. Where current representative evidence is absent, the correct statement is that the current outcome is not established by the available survey. Strong GDP growth and large public investment do not logically reveal the distribution of real household consumption. The absence of a current household statistic also does not prove a negative outcome. The paper should keep the gap visible.

This discipline matters for the newcomer blueprint. Before a discovery, a government should freeze a baseline dataset. It should record what electricity costs, how long outages last, where poverty is measured, how much public debt exists, what infrastructure is deficient, what coastal assets are exposed and what technical staff are available. Later claims of transformation then have a documented starting point.

Chapter 3. Petroleum exploration as a long-cycle process

The official exploration history shows that Guyana did not move directly from no petroleum activity to commercial oil in 2015. Exploration stretched across decades. That record included onshore work, offshore seismic acquisition, exploratory wells, changes in investors and periods of reduced activity. [S35]

The important analytical point is uncertainty. An exploration licence grants a right to search under specified conditions. It does not create a commercial reserve. A discovery proves hydrocarbons in a geological structure. It does not automatically establish an economically recoverable field. Appraisal work reduces uncertainty about reservoir size, continuity, pressure, fluid properties and expected recovery. Only after technical and commercial review does a field move toward development.

The distinction has fiscal consequences. Governments are often tempted to discuss future revenue immediately after a discovery. At that stage, several variables remain uncertain. The recoverable volume is not final. The development design is not final. Capital cost is not final. Oil prices over the production life are not known. The timing of first production remains subject to engineering, permitting, procurement and financing. A newcomer country should therefore separate a geological success announcement from the public budget.

Guyana's long exploration history also shows why historical data rights matter. Seismic data, well logs, cores and geological interpretations accumulate across licence generations. If a state does not preserve those records or the legal right to use them after confidentiality periods expire, later negotiators start with less information than private operators. A national data policy is thus a governance instrument.

A robust data policy identifies who owns raw seismic, processed seismic, well logs, samples, test results and interpretation reports. It identifies confidentiality periods and the conditions under which data move into a national repository. It identifies technical formats and metadata standards. It also protects legitimate commercial confidentiality while ensuring the state retains the information needed to regulate future development.

The same logic applies to unsuccessful wells. Dry holes are information. They reduce uncertainty about where petroleum is not present in commercial quantities. A national geological archive therefore has value even if commercial oil is never produced.

This is a practical lesson for small states. Permanent in-house teams need not replicate the technical depth of a global operator. Government still needs enough internal competence to frame questions, preserve data, supervise external advisers and understand the consequences of a technical recommendation. External advisers should add depth without becoming the only holders of institutional memory.

Chapter 4. The 1986 legal foundation

The Petroleum (Exploration and Production) Act 1986 forms part of the legal inheritance that predates Liza. [S38] Its historical importance lies in establishing a statutory framework for exploration and production rights. The final legal analysis needs the exact version of the law that governed each historical decision, together with later amendments, regulations and any successor legislation.

A petroleum act performs several distinct functions. It defines state ownership or control over petroleum. It authorizes the granting of exploration and production rights. It establishes the relationship between licences and agreements. It provides powers for work programmes, information, inspection, conservation and development. It may address assignments, relinquishment, unitization, surrender and termination.

The presence of a statute does not settle whether every operational detail is sufficiently developed. Regulations, licence terms and individual petroleum agreements often carry much of the commercial and technical detail. The state also needs administrative capability to enforce them.

This distinction is useful in comparing legal readiness. A newcomer country should not ask only whether a petroleum act exists. It should ask whether the act and associated regulations define a coherent sequence from acreage award through exploration, discovery, development, production, abandonment and environmental restoration.

The legal review should also test interaction with other law. Petroleum operations involve environmental law, tax, customs, labour, local content, maritime law, public procurement, occupational safety, corporate law and sometimes Indigenous or customary rights. Conflicting or unclear statutes increase administrative discretion and litigation risk.

A practical pre-discovery legal audit should therefore map:

- 1. petroleum title;
- 2. licensing power;
- 3. contract approval;
- 4. environmental assessment;
- 5. tax and royalty authority;
- 6. beneficial ownership;
- 7. audit rights;
- 8. emergency response;
- 9. decommissioning;
- 10. public disclosure;
- 11. judicial review;
- 12. maritime boundaries.

The objective is not to eliminate discretion. Some discretion is unavoidable in complex technical regulation. The objective is to know where discretion sits, who exercises it, what evidence supports a decision and which review mechanism applies.

Chapter 5. Maritime boundaries and petroleum rights

Offshore petroleum policy rests on geography and international law. A state cannot treat a disputed maritime area as if the legal status were irrelevant to investment risk.

The Permanent Court of Arbitration's *Guyana v. Suriname* case record documents an offshore dispute connected to petroleum activity. It records a 2000 incident involving an oil rig and drill ship in a disputed area and Guyana's commencement of arbitration in 2004. [S40] The final legal chapter should rely on the authenticated award for detailed propositions about the boundary and the parties' conduct.

The significance for a petroleum blueprint is broader than the result of one case. Maritime boundary uncertainty affects acreage promotion, seismic acquisition, investor risk, security arrangements and the potential for neighboring-state claims. An operator may hesitate to commit large capital to an unresolved area. A government may face diplomatic pressure if commercial activity starts before a dispute is managed.

A newcomer country therefore needs a maritime boundary register alongside its petroleum block map. The register should show agreed boundaries, pending negotiations, overlapping claims, treaties, provisional arrangements and dispute mechanisms.

Where a boundary is unresolved, options include negotiation, provisional joint arrangements or international adjudication under applicable law. The choice is sovereign and fact specific. The administrative lesson is to make the legal risk visible before a commercial award.

Maritime boundaries also matter after a spill. Oil, response vessels and environmental effects do not remain inside block coordinates. A transboundary preparedness plan should identify notification channels, evidence-sharing, joint response arrangements and compensation issues before an emergency.

Chapter 6. Environmental law before the petroleum boom

The Environmental Protection Act 1996 is another important pre-oil institution. [S39] Its existence shows that environmental impact assessment and pollution control were part of Guyana's legal framework long before large-scale offshore production.

The analytical question is how a general environmental framework adapts to a new industry with deepwater wells, subsea systems, FPSOs, marine transport, flaring, produced water and low-probability high-consequence spill risk.

A general environmental law establishes authority. Project permits translate that authority into operating conditions. Monitoring and enforcement reveal how those conditions function in practice. These stages should be analyzed separately.

The flagship should therefore reconstruct the environmental system at three dates. The first is before Liza. The second is during approval of Liza Phase 1. The third is after several large offshore developments are operating.

The comparison should ask whether permit conditions become more detailed as regulators gain experience. It should ask whether monitoring frequency grows with the number of projects. It should test whether regulator staffing and technical support keep pace with the industry.

This is a better measure of institutional learning than counting amendments alone.

Chapter 7. Low-carbon policy before first oil

Guyana's Low Carbon Development Strategy adds another pre-oil dimension. The strategy began in 2009, before the 2015 commercial discovery and before first oil. [S22]

This chronology is important because later discussions sometimes place petroleum and forest finance into one environmental narrative. The two systems have different origins, accounting rules and environmental effects.

The early LCDS sought to link forest conservation with national development and results-based climate finance. The Norway partnership provided one source of external payments connected to forest outcomes. Later versions of the strategy expanded into jurisdictional carbon markets and other national development priorities.

A petroleum case study should preserve that institutional history. Guyana did not first acquire a forest monitoring system as a reaction to offshore oil. Its forest policy developed under an earlier development and climate framework.

The transferable lesson is not that every producer should sell forest carbon. A small island may have limited forest area and a larger share of national natural capital in reefs, fisheries, mangroves or tourism assets. The stronger lesson concerns natural-capital accounting. A country entering petroleum production benefits from knowing which ecological assets already support livelihoods, public safety and export earnings.

The national balance sheet should therefore treat petroleum as one asset category among several. Extraction increases cash while reducing underground reserves. Public investment may increase produced capital. Forest conservation may protect ecosystem value. Pollution may reduce natural capital. The accounting need not force every asset into a single speculative monetary value. It needs to prevent the analysis from treating petroleum cash as the only form of wealth.

Chapter 8. Coastal geography and development constraints

Guyana's coastal geography shapes the way petroleum revenue interacts with development.

Population, economic activity and public infrastructure are concentrated heavily along a low coastal plain. Flood protection, drainage and sea defence therefore remain central public investment functions. This makes the country different from many petroleum states and

different again from small islands where storm surge, reef degradation and concentrated tourism infrastructure shape risk.

The flagship should establish a coastal baseline before measuring post-oil investment. Key variables include sea-defence condition, drainage capacity, pump infrastructure, historical flood losses, exposed housing, transport corridors and the location of hospitals, schools, utilities and industrial facilities.

This baseline supports a more rigorous evaluation of petroleum-financed infrastructure. A drainage project should not be measured only by contract value. A stronger outcome measure asks whether flood duration falls, whether critical infrastructure remains accessible, whether maintenance is funded and whether the system performs under severe rainfall and tide conditions.

The same approach applies to sea defence. Construction length and expenditure are useful inputs. Outcome assessment needs condition, design standard, maintenance and residual risk.

For a newcomer country, the lesson is to identify national bottlenecks before revenue arrives. Otherwise, the political pressure created by a windfall may produce a long project list without a clear ranking of economic and resilience constraints.

Chapter 9. Institutional capacity as a network

Petroleum governance is often discussed as if one ministry manages the sector. In practice, the state is a network.

The petroleum ministry or regulator handles licences, development plans and operator oversight. The environmental agency handles environmental assessment and permit conditions. The revenue authority handles tax and royalty functions. The finance ministry handles fiscal policy and the budget. The central bank may hold sovereign petroleum assets. The supreme audit institution reviews public accounts. Parliament authorizes spending and scrutinizes reports. Courts review legal disputes. Maritime and emergency agencies handle safety and response.

The quality of the system depends on information moving between these institutions.

A production number recorded by an operator matters to the revenue authority because it affects royalty and entitlement. A cost statement matters to petroleum auditors. A cargo sale matters to the sovereign fund. An environmental incident matters to the EPA and emergency authorities. A public withdrawal matters to parliament and the auditor.

The institutional baseline should therefore map not only organizations but data flows.

For 2014, the map should record which petroleum-specific functions existed, which were located inside broader natural-resource institutions and which depended on external support. For 2019, it should show the systems needed at first oil. For 2026, it should show the scale required by multiple producing FPSOs.

The comparison should avoid an impressionistic score. It should use measurable indicators where records permit. Examples include staffing, budget, audit completion time, permit workload, production reporting frequency, unresolved disputes and publication lag.

Chapter 10. The pre-discovery skills problem

A prospective producer faces a difficult human-capital choice. Deepwater petroleum demands specialized skills that have little domestic market value if no commercial development occurs. Building every capability in-house before discovery is therefore expensive. Waiting until after discovery leaves government under time pressure.

The solution is a layered capacity model.

The first layer is a small permanent public team with strong general competence in law, geology, finance, environmental assessment and public administration. This team owns data, defines questions and supervises advisers.

The second layer is domestic technical institutions, including universities, professional bodies, geological agencies and environmental laboratories. Their capability supports both petroleum and other sectors.

The third layer is external specialist support used for high-complexity tasks such as reservoir simulation, petroleum fiscal modelling, international tax, offshore engineering or catastrophe insurance.

The fourth layer is peer and multilateral support. International institutions and other regulators offer comparative experience, training and technical assistance.

The key governance rule is knowledge transfer. An adviser report that government cannot later reproduce, challenge or update creates dependence rather than capacity.

Contracts for advisers should therefore state what data government receives, what models remain accessible, how staff are trained, what conflicts are prohibited and what intellectual property remains available after the engagement.

Chapter 11. What changed after Liza

The Liza discovery altered the state capacity problem because the probability of large-scale petroleum development rose sharply.

The government then faced parallel workstreams:

- evaluate the discovery and development plans;
- negotiate or administer contractual arrangements;
- review environmental applications;
- prepare production licensing;
- build cost-audit capability;
- develop sovereign revenue management;

- plan local-content policy;
- strengthen public transparency.

The time horizon compressed.

This period is important to the final case because several institutions often described as part of today's petroleum system did not all exist in final form at discovery. Some came before first oil. Others followed first oil. Still others changed again as production expanded.

A rigorous chronology should therefore mark each institution as:

- inherited before 2015;
- created 2015-2019;
- created after first oil;
- later amended.

This avoids retrospective simplification.

Chapter 12. A baseline for evaluating later reform

The value of Part I is that it creates controls for the rest of the publication.

When the paper examines the NRF, Part I identifies what public finance looked like before the windfall.

When the paper examines local content, Part I identifies the skills and supplier base before large-scale petroleum demand.

When the paper examines environmental regulation, Part I identifies the environmental statute and natural-capital policy that already existed.

When the paper examines infrastructure, Part I identifies the earlier gaps and coastal constraints.

When the paper examines institutional capacity, Part I identifies which agencies and mandates predate oil.

The case study therefore avoids attributing every observed change to petroleum. Economic development is influenced by global commodity prices, domestic policy, demographic change, private investment and other sectors.

The final empirical sections should use petroleum timing as one explanatory factor, not as a universal causal claim.

Chapter 13. Transferable lesson for an island or small country

A small state considering petroleum exploration should start with a readiness inventory rather than a sovereign fund law copied from another country.

The readiness inventory should contain:

- maritime boundaries;
- block map;
- petroleum law;
- environmental law;
- tax treatment;
- geological data policy;
- regulator map;
- conflict rules;
- emergency response;
- social and ecological baseline.

If commercial discovery has not occurred, public spending assumptions should exclude future oil revenue.

If a discovery occurs, the inventory becomes a gap assessment. The country identifies what needs to be strengthened before development approval and first production.

An island should add special controls for:

- reefs;
- coastal tourism;
- desalination;
- hurricane disruption;
- single-port dependence;
- limited waste disposal;
- small labour market.

Guyana's forest-finance experience does not translate automatically to an island without comparable forests. The transferable principle is to identify natural capital before extraction changes the economy.

Chapter 14. Counterfactual tests

A strong case study should ask what would weaken its own interpretation.

The proposition that early legal preparation matters would be weakened if a country with no pre-discovery framework consistently achieved equal or stronger revenue, environmental and transparency results without greater transition risk.

The proposition that geological data ownership matters would be weakened if government could reproduce independent technical decisions without retained data.

The proposition that baseline household and environmental information matters would be weakened if credible post-development impact attribution were possible without a starting measure.

The proposition that maritime boundary clarity matters would be weakened if disputed acreage attracted equivalent investment on equivalent terms without legal risk.

These tests keep the analysis falsifiable.

Chapter 15. Part I findings

The documentary record supports several limited findings.

First, Guyana possessed petroleum, environmental and public institutions before the 2015 commercial discovery. [S38, S39]

Second, petroleum exploration stretched across decades rather than beginning with Liza. [S35]

Third, the country entered the oil era with a pre-existing low-carbon and forest-finance strategy. [S22]

Fourth, maritime boundary management formed part of the longer offshore petroleum context. [S40]

Fifth, the commercial discovery created a new scale and speed of institutional demand. The later chapters test the adequacy and adaptation of those systems rather than assuming either success or failure.

For a newcomer state, the strongest lesson from this opening period is chronological discipline. Build what remains useful even if no oil is found. Preserve national data. Clarify legal authority. Measure the environment and economy before the shock. Scale specialist capacity as commercial certainty rises. Do not spend revenue before it exists.

These are administrative design principles. They do not determine a country's political choices about ownership, taxation, extraction pace or the use of future public revenue.

Edition 2.0 deepening: the pre-discovery state was built through repeated cycles

Analytical section 15A. The 1980s petroleum project as institutional prehistory

The modern petroleum story should not begin with the 2015 Liza discovery. It should begin much earlier, when the state was trying to decide whether an uncertain petroleum endowment justified investment in law, data and specialist capacity.

The World Bank's Project Completion Report for the Petroleum Exploration Promotion Project gives unusually useful evidence because it records both the intended design and the shortcomings of an early petroleum-capacity programme. Credit 1208-GUA was approved in 1982. The project sought to attract foreign exploration, examine heavy-oil potential and strengthen the government's ability to administer the petroleum sector. The historical value of this record is not that the project produced oil. It did not. Its importance is that the state had

already begun to assemble the institutional components that later became relevant when commercial discovery finally occurred. [S107]

The project supported a Petroleum Unit, petroleum-potential studies, promotional activity, training and the legal framework needed to negotiate with international companies. The completion report records four exploration contracts negotiated with international firms and describes petroleum legislation being enacted under separate financing. It also records implementation problems, including delays, incomplete components and a capacity-building approach that the Bank considered too weighted toward academic training. [S107]

This makes the project a useful case in institutional sequencing. Capacity created before a discovery can appear underutilized for years. Yet a frontier state that waits for a commercial discovery before building even basic legal and geological capability will enter negotiations at the exact moment when bargaining pressure and information asymmetry become greatest.

The lesson is not that every frontier state should create a large petroleum bureaucracy before hydrocarbons are proven. The lesson is to build scalable capability whose value survives a dry-hole outcome. A small geological data unit, a clear licensing law, a model data-room protocol, trained public lawyers and a documented regulator map can be justified even if exploration never becomes commercial.

Analytical section 15B. Exploration promotion and the problem of credible commitment

A frontier petroleum state faces a strategic contradiction. It wants to attract high-risk capital, but it also wants to preserve the ability to regulate, tax and supervise a discovery if one occurs.

The World Bank completion report shows that this problem was already visible in Guyana's early petroleum programme. The project sought a legal and fiscal structure capable of attracting foreign exploration while preserving a framework for government administration. The report records debate over complex sliding-scale approaches and the need for a structure flexible enough to accommodate different agreement types. [S107]

For a new producer, the deeper issue is credible commitment. Before discovery, the state needs investors to believe that exploration rights, data obligations and development rights are sufficiently clear for capital to be risked. After discovery, citizens need confidence that the same legal system can protect public revenue, environmental obligations and national interests.

These are not opposing objectives if the framework is designed well. A credible petroleum regime should specify what is stable and what is subject to ordinary law. It should define the process for licence renewal, discovery declaration, appraisal, development approval, assignment, unitisation, decommissioning and termination. It should also distinguish fiscal commitments from the state's continuing authority over safety, environment, reporting and emergency response.

The pre-discovery period is therefore the best time to decide which issues belong in statute, which belong in regulation, which belong in the contract and which require case-specific permit conditions.

Analytical section 15C. Data sovereignty is a national asset

Geological and geophysical data can be more durable than any single licence.

The historical petroleum-promotion programme recognized data as an economic and institutional asset. The state sought to synthesize exploration information, use it to promote acreage and retain the ability to support later licensing decisions. [S35, S107]

This principle becomes more important after repeated exploration cycles. A company may leave a basin after drilling one unsuccessful well. The well still produces national information. Seismic processing may be superseded by later technology, but the underlying acquisition remains useful. Core samples, logs, mud reports, pressure data, geochemical results and well-location records can all reduce uncertainty for later explorers and improve government negotiating capacity.

A national petroleum data regime should therefore answer six questions.

First, who owns raw exploration data?

Second, when must contractors submit data to government?

Third, how long may data remain confidential?

Fourth, who is responsible for long-term preservation and metadata quality?

Fifth, under what conditions can the state re-license or commercialize historical data?

Sixth, how are consultants prevented from becoming the only holders of institutional memory?

The state does not need to perform every interpretation itself. It does need to preserve the inputs required for independent interpretation.

Analytical section 15D. Failed exploration is part of the institutional record

A serious case study should account for failed and interrupted exploration rather than tell the story backward from success.

Guyana's record included early wells, periods of inactivity, investor withdrawals, financing constraints and border-related interruptions before Liza. The official exploration history records repeated pauses and restarts across decades. [S35]

This matters because a state can learn from dry holes in at least four ways.

A geological lesson changes basin understanding.

A contractual lesson changes acreage design or work commitments.

A regulatory lesson changes the way licences and reporting obligations are administered.

An institutional lesson changes how much permanent capacity is justified.

A newcomer country should therefore maintain a "failed exploration archive". It should record why each licence ended, which work obligations were completed, what data was produced, whether environmental obligations were closed, and whether any contingent liabilities remain.

That archive prevents a future licensing round from treating the basin as if institutional history began with the current investor.

Analytical section 15E. Maritime boundaries are economic infrastructure

Maritime boundaries are often discussed as questions of sovereignty and international law. For offshore petroleum they are also a form of investment infrastructure.

A block cannot be assessed only by geology. Investors need confidence that the state granting the licence has jurisdiction over the acreage and that drilling can proceed without interruption. Guyana's history includes the 2000 CGX incident with Suriname and the subsequent arbitration process, which became part of the operating context for offshore exploration. [S35, S40]

The transferable lesson is not that all boundary uncertainty must be eliminated before a country conducts any offshore work. Some states explore while boundaries remain contested. The lesson is that boundary risk should be explicit in the petroleum-risk register.

A frontier government should map:

- undisputed acreage;
- overlapping claims;
- areas subject to judicial or arbitral process;
- navigation and fisheries implications;
- coast-guard and maritime-security capability;
- force-majeure implications for contracts;
- insurance implications for operators and contractors.

If acreage is disputed, the economic model should include the possibility of delay, suspension or exclusion.

Analytical section 15F. The institutional minimum viable state

The pre-discovery experience suggests that a new producer does not need every institution at full scale before exploration. It needs a minimum viable petroleum state.

That minimum should include:

- a legally identified licensing authority;
- a geological data custodian;

- public-sector access to petroleum legal expertise;
- a tax authority capable of identifying petroleum-specific risks;
- an environmental authority with offshore jurisdiction;
- a finance ministry able to model contingent future revenue rather than assume it;
- a maritime authority able to coordinate offshore safety and incident response;
- an audit trail for licences, amendments and work obligations.

The institution can be small. What matters is that mandates are clear.

A common failure mode in small states is to compensate for limited staffing by creating committees without assigning ownership. A committee can coordinate work. It cannot substitute for a named institution responsible for a statutory decision.

Analytical section 15G. Capacity should scale with decision risk

Petroleum staffing should not be sized only to production volume.

Before discovery, a handful of specialists may be adequate if the country has limited active acreage.

After a commercial discovery, the state faces a step-change in decision risk. It must review appraisal plans, development economics, field-development concepts, petroleum metering, environmental studies, cost-recovery systems, decommissioning assumptions, marketing arrangements and sovereign-revenue preparation.

This is why discovery should trigger a pre-defined capacity escalation plan.

The plan should identify:

- positions that must become permanent;
- functions that can be outsourced temporarily;
- conflicts that prevent one adviser from serving multiple roles;
- knowledge-transfer requirements;
- records that remain state property;
- the date by which external advisers must be replaced or shadowed by national staff.

The World Bank's earlier finding that academic training alone was insufficient remains relevant. Capacity is not the number of degrees funded. Capacity is the ability to make, document, challenge and repeat a decision under real operating conditions. [S107]

Analytical section 15H. Environmental baselines should precede the investment shock

Environmental baseline work becomes harder after commercial discovery because development pressure accelerates.

The best baseline is created before the location of production infrastructure, support bases, pipelines and tanker routes becomes politically and commercially salient.

For an offshore frontier, baseline ownership should include:

- marine mammals;
- fish and spawning areas;
- seabed conditions;
- water and sediment quality;
- coastal habitats;
- fisheries effort;
- port traffic;
- shoreline sensitivity;
- protected areas;
- disaster-response resources.

The state does not need to measure every variable continuously before a discovery. It should know what data exists, who owns it, whether methods are reproducible and which gaps would become material if development moves forward.

Guyana's pre-existing environmental statute and later project-specific EIA framework make it possible to distinguish law that existed before the petroleum boom from systems created in response to the scale of development. [S39, S48-S56]

Analytical section 15I. Natural capital should be on the same balance sheet as petroleum

A resource-discovery narrative can become distorted if petroleum is treated as the only national asset.

Guyana entered the oil era with forests, biodiversity, freshwater, agriculture, mineral resources and a pre-existing low-carbon strategy. The LCDS and forest-finance architecture therefore belong in the petroleum case study because they demonstrate that the country was already attempting to assign economic value to natural capital before first oil. [S22]

The accounting principle is broader than any specific carbon programme.

A country converts underground petroleum into financial and physical assets. It should simultaneously track whether extraction, infrastructure and associated growth reduce other forms of wealth.

A complete national balance sheet should therefore ask:

- what petroleum stock is being depleted;
- what financial assets are accumulating;
- what infrastructure is being created;

- what human capital is improving;
- what ecosystems are being degraded or protected;
- what future liabilities are being created.

This is the bridge between resource governance and sustainable development.

Analytical section 15J. The discovery shock should trigger a formal readiness review

The moment of discovery is often treated as a celebration or a geological milestone. Administratively, it should trigger a structured readiness review.

Within ninety days of a commercial discovery, a small state should reassess:

- petroleum law and regulations;
- contract administration;
- environmental authority;
- fiscal modelling;
- tax administration;
- sovereign-revenue custody;
- procurement rules;
- beneficial ownership;
- emergency response;
- labour and immigration;
- infrastructure demand;
- public communication.

The output should be a gap register with owners, deadlines and budget.

This is preferable to creating institutions reactively as each new problem appears.

Analytical section 15K. The strongest pre-discovery lesson is optionality

Good pre-discovery institutions preserve national options.

A state with organized data can run a better licensing round.

A state with clear law can choose among fiscal structures with less legal uncertainty.

A state with environmental baselines can compare development options.

A state with trained negotiators can decide when independent advice is necessary.

A state with a maritime-risk map can sequence acreage.

A state with conservative fiscal rules can avoid spending against unproven revenue.

Optionality matters because the state does not know whether the basin will remain marginal, produce one discovery or become a major province.

The best early institutions are therefore those that remain useful across all three outcomes.

Analytical section 15L. Expanded Part I conclusion

The deeper historical record changes the interpretation of Guyana's petroleum rise.

Commercial success arrived quickly after Liza, but the institutional prehistory was long. Petroleum promotion, legislation, geological data management, exploration contracting and public-sector training had been attempted decades earlier. The World Bank project of the 1980s did not solve every institutional problem and did not create a complete petroleum state. It did create a record of how a small frontier country tried to build petroleum capability before commercial certainty. [S107]

That earlier period also reveals a recurring constraint that matters today: institutional capacity is not static. The skills needed to attract an explorer are different from those needed to audit a multibillion-dollar development. The staff required to manage one exploration licence are different from those required to oversee several FPSOs, large petroleum receipts and complex environmental exposure.

For a newcomer, the practical standard is therefore staged readiness.

Before discovery, build the legal map, preserve data and establish environmental and maritime baselines.

After discovery, expand fiscal, legal, engineering and environmental capability before irreversible approvals.

Before first oil, test revenue custody, metering, auditing and emergency response.

After production begins, scale institutions with the number and complexity of projects rather than with the age of the petroleum sector.

Guyana's pre-discovery history is valuable not because it offers a perfect model. It is valuable because it shows that petroleum governance begins long before oil reaches a production vessel.

Edition 2.0 frontier-state deepening: how to govern uncertainty before geology becomes wealth

Analytical section 15M. Exploration policy should price uncertainty honestly

Before a commercial discovery, the state is not allocating proven petroleum wealth.

It is allocating the right to spend capital under uncertainty.

That distinction changes the bargaining problem.

An explorer bears:

- geological risk;
- drilling cost;
- schedule risk;
- capital at risk if no commercial discovery occurs.

The state bears different risks:

- weak terms if a large discovery occurs;
- lost time if acreage is warehoused;
- incomplete data if work obligations are weak;
- environmental exposure;
- institutional distraction.

A licensing system should therefore balance investor risk with state optionality.

The correct question before discovery is not "How much revenue will government receive?"

It is "What value does the state receive in every possible outcome, including a dry hole?"

Analytical section 15N. Work obligations are the frontier-state consideration

In an early exploration basin, a signature bonus can be visible but geological work can be more valuable.

A strong work programme can require:

- seismic acquisition;
- reprocessing;
- geological studies;
- exploration wells;
- minimum expenditure;
- relinquishment.

The benefit to the state survives even if the contractor exits, provided the data are delivered and preserved.

This is why data ownership and work obligations should be modelled as part of licence value.

Analytical section 15O. Acreage should not be allowed to become dormant inventory

An exploration licence can block competing investment if the work programme is weak or extensions become routine.

The state should therefore track:

- award date;
- work commitments;
- milestones;
- expenditure;
- extensions;
- relinquishment;
- reason for delay.

A licence extension may be rational if the contractor has made substantial progress or external conditions changed.

It should still be documented.

This creates a historical record of how efficiently the state converted acreage into information.

Analytical section 15P. Relinquishment is an information-recycling mechanism

Relinquishment is often understood only as returning land or offshore acreage.

Its deeper function is to recycle exploration opportunity.

When acreage returns to the state, the government should already possess:

- seismic;
- wells;
- interpretations required under the licence;
- environmental information;
- final technical report.

The next licensing round can then begin with better information.

This reduces the information advantage of the previous contractor.

Analytical section 15Q. Frontier fiscal terms should be analysed with geological probability

A fiscal regime that looks generous after a large discovery may have been negotiated when discovery probability was low.

That does not automatically make the earlier terms optimal.

It does mean that historical evaluation should include the information set at the negotiation date.

The correct counterfactual is: What terms could the state plausibly have obtained from credible investors under the geological and market conditions then known?

This question is harder than comparing an early contract to terms adopted after billions of barrels had been discovered.

It is also fairer analytically.

Analytical section 15R. Discovery changes the state's reservation value

Once a basin has demonstrated commercial petroleum, later acreage is not the same asset.

The state can reasonably reassess:

- royalty;
- tax;
- cost recovery;
- bonus;
- work commitments;
- relinquishment;
- state participation;
- local content.

This is one reason later model terms should be expected to differ from early frontier terms.

The change itself is not proof that the earlier contract can or should be retroactively rewritten.

It shows that geological information has economic value.

Analytical section 15S. Institutional learning should be priced into later licensing

A second licensing round benefits from more than geology.

The state has learned:

- how long approvals take;
- what data operators request;
- which clauses create audit burden;
- what skills are scarce;
- how environmental review operates.

Later model agreements should therefore reflect both better geology and better government knowledge.

A change log should identify which revisions respond to:

- market conditions;
- contract administration;
- legal reform;

- environmental experience;
- political policy.

Analytical section 15T. The exploration data room is a governance institution

A modern data room should provide investors with controlled access while preserving:

- confidentiality;
- source provenance;
- version;
- user access;
- download log.

The state should know which data were made available to each bidder.

This supports fairness in a competitive round.

It also protects against later claims that bidders were using materially different information sets.

Analytical section 15U. Geological uncertainty should be communicated to the public

Before discovery, resource estimates can be misunderstood as reserves.

After a discovery, early estimates can be misunderstood as recoverable or commercial barrels.

The public communication protocol should distinguish:

- prospect;
- resource;
- contingent resource;
- reserve;
- recoverable estimate;
- production forecast.

The state should avoid building fiscal expectations on speculative resource numbers.

Analytical section 15V. Commerciality is a legal and economic threshold

A discovery can contain petroleum without being commercially developable.

Commerciality depends on:

- reservoir;
- recoverable volume;

- development cost;
- oil/gas price;
- technology;
- infrastructure;
- fiscal terms;
- financing.

The state's technical team should independently understand the commerciality case before a development path becomes irreversible.

Analytical section 15W. The first development creates infrastructure path dependence

An offshore province's first project can influence later choices in:

- shore bases;
- marine routes;
- aviation;
- warehouses;
- training;
- pipeline corridors.

These investments create path dependence.

The first project should therefore be reviewed not only as a standalone development but as a possible foundation for a basin.

This does not justify overbuilding speculative infrastructure.

It justifies considering shared use and future compatibility.

Analytical section 15X. Local-content expectations should begin with a capability inventory

Before imposing targets, government should know the starting economy.

The pre-discovery or early-discovery capability inventory should map:

- marine services;
- fabrication;
- construction;
- food supply;
- legal;
- accounting;
- logistics;

- engineering;
- safety;
- training.

The inventory becomes the baseline against which local-content progress is measured.

Without a baseline, later claims of transformation are difficult to verify.

Analytical section 15Y. Fiscal expectations can destabilize policy before first oil

A commercial discovery can immediately change public expectations even though production cash may be years away.

The state should therefore publish a simple fiscal timeline showing:

- discovery;
- appraisal;
- development approval;
- construction;
- expected first oil;
- expected first revenue;
- uncertainty.

This can reduce pressure to borrow or make permanent commitments against revenue that has not yet arrived.

Analytical section 15Z. Pre-first-oil debt needs a petroleum-risk lens

Debt taken before first oil can create a mismatch.

The country owes debt on a fixed schedule.

Petroleum revenue remains uncertain.

A debt sustainability test should therefore run:

- delayed startup;
- lower price;
- higher development cost if state participation exists;
- slower ramp-up.

The state should know whether ordinary non-petroleum revenue can service the debt if the oil project is delayed.

Analytical section 15AA. Petroleum revenue should not erase ordinary revenue reform

A future oil windfall can weaken incentives to improve tax administration.

That creates long-run risk.

A durable fiscal state should continue to strengthen:

- income tax;
- VAT;
- customs;
- property or other lawful domestic revenue systems;
- compliance.

Petroleum revenue should expand fiscal capacity, not replace the administrative relationship between citizens, firms and the state.

Analytical section 15AB. Environmental institutions should receive pre-first-oil capital

Environmental regulation becomes more expensive after development begins.

The authority may need:

- offshore training;
- marine scientists;
- monitoring equipment;
- legal capacity;
- emergency planning;
- data systems.

These are legitimate pre-first-oil investments.

The state should budget for them before permit workload multiplies.

Analytical section 15AC. Coast guard and maritime administration are petroleum institutions too

Offshore petroleum increases the importance of:

- navigation safety;
- exclusion zones;
- emergency response;
- vessel monitoring;
- pollution reporting;
- search and rescue.

The petroleum institutional map should therefore include maritime agencies rather than treating them as external to the sector.

Analytical section 15AD. First discovery should trigger a whole-of-government risk register

The risk register should include:

- geological;
- commercial;
- fiscal;
- environmental;
- social;
- labour;
- infrastructure;
- legal;
- corruption/conflict;
- maritime;
- macroeconomic.

Each risk should have:

- owner;
- probability range;
- consequence;
- mitigation;
- trigger;
- review date.

This becomes the bridge from exploration policy to national development policy.

Analytical section 15AE. The national narrative should distinguish luck from governance

A major discovery depends partly on geology that the state did not create.

Governance determines what happens around that discovery.

A serious case study should therefore avoid attributing geological abundance to policy skill.

The policy questions begin with:

- how acreage was managed;
- how terms were negotiated;
- how data were retained;

- how approvals were handled;
- how revenue was governed;
- how risks were managed.

This distinction makes the case transferable to countries whose geology is less favourable.

Analytical section 15AF. Pre-discovery conclusion

A newcomer cannot control whether a basin contains a giant field.

It can control whether:

- rights are clearly allocated;
- data are preserved;
- exploration obligations are enforced;
- public institutions learn;
- environmental baselines exist;
- maritime risks are mapped;
- fiscal expectations remain conservative.

Those institutions create value even if no commercial oil is found.

That is the strongest justification for pre-discovery governance.

Edition 2.0 archival case: what Guyana's first petroleum-capacity project actually delivered

A seven-year institutional clock that began before Liza

The original 1993 World Bank completion report supplies a dated counterweight to the idea that Guyana's petroleum administration appeared only after commercial discovery. The Bank's Credit 1208-GUA for the Petroleum Exploration Promotion Project was approved on 16 February 1982. The loan agreement was signed on 24 November 1982, but credit effectiveness came only on 30 March 1983. The project closed retroactively on 16 September 1987, with the final closure decision taken in 1991; the report itself was dated 16 November 1993. These dates describe a very different institutional environment from the post-2015 deepwater development period and should never be conflated with modern project licences. [S107]

The chronology has explanatory value. The Bank's evaluation records initial constraints involving local counterpart funding, project-management staffing and delays in securing consultants, as well as difficulties coordinating external financing. Promotion seminars were eventually held in London and Houston in February 1986. The report attributes four negotiated exploration contracts to the promotion programme, but that does **not** mean that the programme discovered four commercial fields or created four producing operations. [S107]

Separate success in exploration promotion from unfinished components

The report's original objectives were threefold: attract exploration investment, investigate shallow heavy-oil potential and strengthen the government's capacity to administer petroleum operations. The completion record presents evidence of progress in promotion and government institutional development while explicitly stating that the shallow-heavy-oil investigation was not performed. This distinction matters because the absence of a heavy-oil drilling result is not evidence that the drilling found nothing; the test was not completed under the credit. [S107]

The project used consultants to assemble and interpret geological information, prepare marketing material, help negotiate contracts and strengthen state petroleum administration. Government also sought a modern legal framework and training in petroleum operations. The Bank's evaluation credits the Petroleum Unit and contemporary petroleum legislation as institutional products while also finding that the training allocation placed too much weight on longer academic education rather than immediately usable project and technical administration. That is the source's historical assessment, not an independent judgment about the officials involved. [S107]

A quantified implementation gap

The credit was initially SDR 1.7 million, approximately US\$2 million at approval. The World Bank preface states that only 52.1 percent of the credit had been disbursed and that 47.9 percent, recorded as SDR813,862, was cancelled. These percentages measure **credit utilization**, not geological outcomes, total petroleum-sector investment or the Government's own operating expenditure. The report separately explains that the ultimate cancellation amount in its dollar-equivalent presentation was affected by the credit's accounting and timing. A modern case study should preserve the original SDR records instead of multiplying a rounded dollar approval figure by a percentage and presenting the product as an exact historical cancellation. [S107]

This expenditure gap does not erase the contracts negotiated or professional experience acquired. It does, however, demonstrate why a small frontier state's reform programme should be designed to survive delayed effectiveness, staffing constraints, financing interruptions and changes in the commercial environment.

The overlooked counterfactual: institutional value without commercial oil

The economically interesting counterfactual is a country that never reaches first oil. What remains from a petroleum-promotion project if exploratory drilling fails? Geological records, stronger licensing law, trained public officers and a contractual archive may retain value. A large permanent operating company, major support-base investment or debt raised against imagined future production might not. Guyana's earlier World Bank project is particularly useful because its contemporary report was written before the 2015 Liza discovery; its contemporaneous observations therefore do not depend on hindsight about a giant oil province. [S107]

For a new producer, the practical inference is to separate exploratory public goods from scale-contingent commitments. Build the geological and legal files that remain useful under a dry-hole scenario. Preserve every exploration well's data. Establish the public accounting unit before granting complex rights. Delay irreversible public investments that only make economic sense after a development plan is independently validated.

Source integrity and archival limitations

The primary document is World Bank Report No. 12485 (1993), with the funding summary in the preface, institutional findings in the evaluation memorandum and evaluation summary, and detailed chronology in Part I. Its official public PDF text was retrieved for this revision, and the project appears independently in the World Bank Archives Catalogue under record 30370430. The underlying full-text URL remains difficult for GitHub-hosted source monitoring to retrieve, and no authenticated local byte copy with a checksum has yet been archived. The source's quoted or highly specific points must therefore retain their section references, with independent downloadable-PDF preservation still listed as an open evidence task under issue #68. [S107]

This example illustrates the broader discipline required of the book: report an institution's documented accomplishments and limitations together, preserve what was known at the time, and avoid transforming a historic development-bank evaluation into a claim about modern petroleum governance.

PART II | Petroleum contracts and fiscal mechanics

Discovery, Contract Design, Cost Recovery and Revenue Assurance

Chapter 16. Why contract analysis requires historical context

A petroleum contract should be read twice. The first reading asks what the clauses mean. The second asks what information, risks and alternatives existed when the parties agreed to them.

This distinction matters in Guyana because commercial knowledge of the Stabroek Block changed sharply after the 2015 Liza discovery and continued to change as appraisal wells and later discoveries expanded the known resource base. A fiscal arrangement signed under frontier uncertainty and a later model agreement developed after multiple successful discoveries do not arise under the same information set. [S36, S41]

Historical context does not remove accountability. A government remains responsible for understanding the powers it exercises and the agreements it signs. Historical context instead prevents a different error, which is to judge an earlier negotiation solely with geological and commercial information learned years later.

The correct analysis therefore has three layers.

The first layer is the executed legal text. It states the rights and obligations.

The second layer is the information environment at the relevant date. It identifies reserve uncertainty, development cost, oil price, market interest and the state's technical capacity.

The third layer is observed performance. It records development, cost recovery, production, government receipts, audits and disputes.

The flagship uses all three.

Chapter 17. From Liza discovery to the 2016 agreement

The Liza-1 record places the discovery well in May 2015. The official Stabroek Block register records the petroleum agreement date as 27 June 2016 and an effective/licence date of 7 October 2016. [S36, S41]

These dates should not be collapsed.

Between discovery and agreement, the parties were still learning about the field. Appraisal reduced geological uncertainty. Project concepts were being evaluated. Commercial expectations changed as evidence accumulated.

A historical fiscal assessment therefore needs a dated resource-estimate series. It should identify the resource view at:

- discovery;

- first appraisal;
- agreement signature;
- production licence;
- final investment decision;
- first oil.

The same time series should contain oil price conditions and estimated project costs.

Without this chronology, a contract comparison risks hindsight bias.

Chapter 18. What a production-sharing agreement does

A production-sharing agreement allocates petroleum and costs through a contractual formula.

At a simplified level, gross petroleum production is valued. Royalty is calculated according to the agreement. A portion of production or value is allocated to recover eligible costs, subject to the contractual ceiling. The remaining profit petroleum is divided between the state and contractor under the agreed share. Taxes and other obligations then follow the applicable agreement and law.

The simplified arithmetic is useful for teaching.

It is not enough for an audit.

Actual implementation needs:

- measured production;
- crude valuation;
- lifting schedules;
- eligible cost;
- cost allocation;
- currency treatment;
- affiliate transactions;
- tax;
- year-end receivables;
- decommissioning.

The flagship therefore distinguishes a teaching waterfall from the audited account.

Chapter 19. The 2 percent royalty

The public explanation of the 2016 Stabroek fiscal arrangement identifies a 2 percent royalty. [S15]

A royalty is attractive from a revenue-administration perspective because it is calculated before profit and does not depend on whether the contractor has recovered all development cost.

That does not make royalty design simple.

The government still needs:

- verified production;
- verified sales volume;
- a valuation rule;
- due dates;
- currency;
- audit access.

A royalty rate also affects project economics. A higher royalty takes revenue even when costs are high and profit is low. A lower royalty leaves more downside risk with the state in exchange for greater investor cash during cost-intensive periods.

The economic effect should therefore be tested across field quality, capital cost, price and production decline.

A policy argument about a royalty rate should not be reduced to whether the number appears large or small.

Chapter 20. The cost-recovery ceiling

The Stabroek arrangement permits cost recovery subject to a ceiling. Public explanations commonly use 75 percent as the maximum cost-recovery share. [S02, S15]

The word maximum is essential.

If recoverable costs exceed the ceiling in a period, some costs remain unrecovered and move forward according to the contract.

If recoverable costs are below the ceiling, less petroleum is allocated to cost recovery and more remains in the profit pool.

This creates time variation in government receipts even when the formal contract does not change.

The cost bank is therefore a central variable.

Chapter 21. How the cost bank changes government cash

Consider a simplified field at constant production and price.

During early development, large capital costs enter the recoverable pool. If costs exceed the monthly ceiling, the ceiling binds. The residual profit pool is relatively small.

As older development costs are recovered, the actual cost share can fall below the ceiling. The residual profit pool grows and government profit oil rises.

A new sanctioned development can then add a large new set of recoverable costs.

Government cash therefore depends on:

- oil price;
- production;
- recoverable cost;
- timing;
- new investment;
- prior unrecovered balance.

A forecast that assumes a permanently low cost-recovery share after one high-revenue period risks overstating future receipts.

The 2026 government reporting discussed this dynamic when revising expected petroleum deposits and noting that new project development can affect the cost bank. [S07]

Chapter 22. A 100-unit teaching example

Normalize gross petroleum value to 100.

Under a simplified case where cost recovery uses the full 75 percent ceiling:

- 75 units go to cost recovery;
- 25 units remain as profit petroleum;
- a 50 percent government share of profit petroleum equals 12.5 units;
- a 2 percent royalty equals 2 units.

Under the public teaching convention, the combined royalty plus state profit share equals 14.5 units. [S15]

Now reduce cost recovery to 50 units:

- 50 remain as profit petroleum;
- government's half is 25;
- royalty remains 2 under the simplified example.

Combined units become 27.

At 20 units of cost recovery:

- 80 remain as profit petroleum;
- government profit share is 40;
- royalty is 2.

Combined units become 42.

These are teaching cases.

They are not a claim that the state received 14.5, 27 or 42 percent in any specific calendar period.

Chapter 23. Why government take is not one number

Petroleum fiscal analysis often produces a single government-take percentage.

That number depends on assumptions.

A full economic model needs:

- project capex;
- opex;
- production curve;
- oil price;
- royalty;
- cost recovery;
- profit share;
- corporate tax;
- state participation if any;
- bonuses;
- abandonment;
- discount rate.

Government take also has different definitions.

A cash-flow government take is not the same as a share of gross revenue.

A discounted government take is not the same as an undiscounted lifetime share.

A marginal effective tax rate is another concept again.

The flagship should therefore label every ratio with its denominator and model assumptions.

Chapter 24. Cost recovery turns the contract into an audit system

A production-sharing agreement does not regulate revenue only through percentages.

It also regulates which costs are recoverable.

This creates a large administrative task.

The state needs to test:

- whether an expenditure occurred;
- whether the contractor paid it;
- whether it relates to the contract area;
- whether it is permitted;
- whether it is allocated correctly;
- whether it is duplicated;

- whether it is reasonable under the agreement;
- whether affiliate transactions follow the required basis;
- whether timing is correct.

The government-published cost audits provide direct evidence that these questions arise in practice. [S42, S71]

Chapter 25. The 1999-2017 audit as a governance case

The government-published IHS Markit audit covered cost-recovery submissions for 1999-2017. The publication states that the contractor submitted about USD 1.678 billion for review and that the audit identified about USD 214.4 million, plus specified overhead adjustments, as having grounds for dispute. [S71]

Those figures must be described with precise verbs.

An auditor identified or recommended an adjustment.

Government disputed an amount.

A contractor responded.

A sole expert or arbitration process may later determine treatment.

A final accounting credit changes the cost bank.

Cash revenue follows only through the resulting petroleum allocation and sales process.

The word recovered should therefore be reserved for a completed financial outcome.

Chapter 26. The 2018-2020 audit and category-level controls

The 2018-2020 audit offers more detailed examples of cost classification and allocation. [S42]

The audit material contains issues associated with:

- allocation between blocks;
- shared costs;
- office-related construction;
- emergency-response studies;
- pre-effective-date insurance;
- duplicate charges.

The significance is systemic.

Each type of issue suggests a control.

Block-allocation issues require an allocation methodology.

Shared services require cost drivers.

Pre-effective-date costs require contractual date controls.

Duplicate charges require invoice-level testing.

Affiliate charges require transfer-pricing and contractual review.

The paper should use the examples to explain audit design, not to imply that every contractor cost is disputed.

Chapter 27. Effective date as an accounting boundary

The official block register gives 7 October 2016 as an effective/licence date. The 2018-2020 audit also uses the effective date in its treatment of specified costs. [S41, S42]

This shows why contract chronology belongs inside accounting.

A cost incurred before signature may have different treatment from a cost incurred after the effective date.

Some agreements expressly recognize pre-contract costs.

Others limit or condition them.

An audit model must therefore tag every material cost with:

- invoice date;
- service period;
- payment date;
- applicable contractual period.

The final cost-bank dataset should also preserve the audit period in which an adjustment is recognized.

Chapter 28. Affiliate transactions

Large offshore developments rely on specialized global supply chains.

Contractors often procure services from affiliated companies or global framework arrangements.

The issue is not whether affiliate transactions exist.

The issue is whether the agreement's pricing and allocation rules are followed.

Audit controls should identify:

- affiliate;
- service;
- cost basis;
- markup;
- allocation;

- market comparison where required;
- invoice support.

A newcomer country needs access to international tax and petroleum accounting expertise for this work.

Chapter 29. Allocation across multiple blocks and projects

A company operating several blocks or projects may incur costs that benefit more than one area.

Examples include:

- shared office;
- aviation;
- marine support;
- emergency planning;
- technical studies;
- management systems.

The accounting question is how much belongs to each contract area.

An arbitrary allocation can shift recoverable cost into a producing block.

A defensible allocation uses a stated driver such as:

- activity;
- headcount;
- vessel days;
- well count;
- direct usage.

The audit should test the driver and consistency over time.

Chapter 30. Petroleum valuation

Cost oil and profit oil are calculated from physical petroleum and value.

The valuation rule therefore matters.

A fiscal model needs:

- crude grade;
- quality adjustment;
- benchmark;
- sale transaction;
- affiliate/non-affiliate status;
- timing;

- freight treatment.

A state should reconcile contract valuation to actual cargo sales where the agreement permits.

This is distinct from production measurement.

One system answers how many barrels existed.

Another answers what those barrels were worth under the contract.

Chapter 31. Cargo timing and government receipts

Government profit-oil entitlement is often delivered in cargoes.

A cargo is not a monthly production statistic.

Crude produced over several days enters FPSO storage. A tanker lifts a cargo when scheduled. Settlement occurs later.

The 2020 NRF audit record demonstrates the effect. A late-year government lift was recognized as revenue but remained receivable at year end. [S45]

A contract-accounting system should therefore include:

- production period;
- entitlement;
- lift date;
- invoice;
- settlement;
- NRF deposit.

The same cargo should have one persistent identifier across all systems.

Chapter 32. Tax treatment

Petroleum fiscal analysis also needs tax law and contractual tax provisions.

The model should identify:

- taxpayer;
- taxable base;
- rate;
- deductions;
- payment mechanism;
- credits;
- interaction with production-sharing terms.

Tax treatment should not be inferred from royalty and profit oil.

A later model PSA with an explicit corporate tax rate belongs in a different agreement generation. [S14]

Chapter 33. Decommissioning and end-of-life cost

Every offshore project ends.

Wells need abandonment.

Subsea equipment may require removal or safe disposition.

The FPSO leaves.

Environmental restoration and monitoring may continue.

The agreement and production licence should therefore identify:

- who pays;
- when funding begins;
- how security is held;
- what happens if ownership changes;
- what standards apply at abandonment.

A fiscal model that excludes decommissioning overstates project cash available for distribution.

Chapter 34. Later model terms

Guyana later announced model fiscal terms for new offshore licensing that differ materially from the Stabroek arrangement. The official announcement describes a higher royalty, a lower cost-recovery ceiling and a corporate tax for future arrangements. [S14]

The later model shows institutional learning and a changed bargaining environment.

It does not amend Stabroek.

The final comparison should include:

- royalty;
- cost ceiling;
- profit share;
- tax;
- relinquishment;
- work commitments;
- acreage;
- audit;
- local content;

- environmental terms.

Each column should also include the date and information environment.

Chapter 35. Why later terms do not prove earlier terms were available

A later government may offer different terms because:

- geological risk is lower;
- more companies are interested;
- infrastructure exists;
- the regulator has more experience;
- comparable developments reduce technical uncertainty.

The market itself has changed.

A strong historical analysis therefore asks a counterfactual.

What terms would a competitive investor have accepted at the earlier date, given the information then available?

This is an economic question, not a slogan.

It requires contemporaneous frontier-basin agreements, project economics and investor interest.

Chapter 36. Why frontier risk does not eliminate scrutiny

The opposite error also needs attention.

A high-risk frontier does not mean any fiscal term was unavoidable.

Government still has options:

- commission independent advice;
- compare jurisdictions;
- stage commitments;
- use work obligations;
- retain acreage;
- publish contracts after award;
- preserve audit rights.

The final paper should therefore evaluate process quality independently from hindsight.

A process can be strong even if a later price or geological outcome looks unfavorable.

A process can be weak even if good luck produces high revenue.

Chapter 37. Contract transparency

Contract publication allows researchers and citizens to test fiscal claims against the legal text.

Publication also reduces confusion between:

- executed agreements;
- draft model agreements;
- production licences;
- public summaries.

The Petroleum Management Programme publishes petroleum documents, including the Stabroek materials and later model agreements. [S02, S43]

A contract portal should preserve:

- version;
- signature date;
- effective date;
- amendments;
- annexes.

A missing annex can make an apparently transparent contract incomplete for audit.

Chapter 38. Confidentiality and legitimate commercial information

Transparency has limits grounded in law and legitimate commercial protection.

Seismic interpretation, proprietary technology, current bid strategy and some pricing information may require confidentiality for a period.

A strong system defines the exception rather than labeling the entire petroleum relationship confidential.

The public register should state:

- legal basis;
- category;
- duration;
- authority;
- review.

When confidentiality expires, the record should move into the public archive under the relevant law.

Chapter 39. The negotiating team

A newcomer state needs a negotiating team whose composition reflects the contract.

Core disciplines include:

- petroleum law;
- geology/reservoir engineering;
- petroleum economics;
- tax;
- accounting;
- environmental law;
- decommissioning;
- insurance.

The team should have a written mandate.

Advisers should disclose conflicts.

Technical models should remain accessible to the state after the engagement.

This avoids a situation in which a government receives a recommendation without retaining the ability to test it.

Chapter 40. Negotiation decision record

A useful negotiation record is not a transcript.

It is a structured decision log.

For each material clause:

- objective;
- operator position;
- government position;
- alternatives;
- fiscal impact;
- legal risk;
- final outcome.

The log provides institutional memory when officials change.

It also helps later auditors distinguish a negotiated policy choice from an implementation error.

Chapter 41. Contract administration after signature

A signed agreement creates an administrative calendar.

The state must track:

- work programme;
- budget;

- discovery notice;
- appraisal;
- development plan;
- production licence;
- relinquishment;
- audit window;
- royalties;
- tax;
- local content;
- training;
- environmental obligations;
- insurance;
- decommissioning.

A calendar-based contract management system turns clauses into deadlines and evidence.

Chapter 42. Audit capacity as a long-term asset

Petroleum audit should not be treated as an occasional consulting exercise.

The state benefits when each audit improves its permanent ability to:

- understand cost categories;
- query operator systems;
- test allocation;
- reconstruct cargo valuation;
- manage disputes.

External firms remain useful for specialist capacity.

Government still needs an internal team that owns the record.

Chapter 43. Dispute resolution

Large petroleum contracts include mechanisms for technical and legal disputes.

The correct mechanism depends on the issue.

An accounting disagreement may move through audit responses and expert determination.

A legal interpretation dispute may move to arbitration or court, depending on the agreement and law.

The flagship should track disputes by procedural stage.

It should avoid describing an unresolved government claim as a final entitlement.

Chapter 44. Revenue assurance as a system

Revenue assurance links five controls:

- 1. production measurement;
- 2. valuation;
- 3. cost audit;
- 4. fiscal calculation;
- 5. cash reconciliation.

Weakness in any one control affects the final public receipt.

A strong dashboard should therefore show more than money deposited.

It should show whether the underlying physical and contractual calculations reconcile.

Chapter 45. Stress test one: low price

Take the simplified model.

A lower price reduces gross sales.

Royalty falls in proportion to sales.

Profit oil also falls.

If recoverable costs remain high, the state receives a smaller residual profit pool.

The budget impact depends on:

- cash already held in the NRF;
- lawful withdrawal;
- non-oil revenue;
- debt;
- project commitments.

This demonstrates why high-price revenue should not be treated as a permanent baseline.

Chapter 46. Stress test two: new project cost

Suppose a mature field has recovered much of its historical cost and government profit oil rises.

A new development is sanctioned.

New recoverable cost enters the system.

The cost bank increases.

Government profit oil can fall relative to the mature-field period even if production later rises.

This is an important forecasting control.

Chapter 47. Stress test three: production interruption

A production outage reduces physical barrels.

The impact on government cash may arrive later because cargoes and settlements lag production.

A robust fiscal model therefore needs timing rather than annual averages alone.

The new production-shock dataset in the repository combines lower volume, price and cost-recovery assumptions to demonstrate this interaction.

Chapter 48. What another country should copy

The strongest transferable elements are procedural:

- preserve data;
- date every agreement version;
- obtain independent modelling;
- keep an audit calendar;
- reconcile physical and financial records;
- retain dispute status;
- publish executed terms under law;
- maintain institutional memory.

The exact royalty, cost ceiling or profit share is not a template.

Those terms depend on geology, market conditions, state capacity, capital cost and sovereign policy.

Chapter 49. What another country should avoid

A newcomer should avoid:

- budgeting a discovery as proven revenue;
- copying fiscal terms from a mature basin without modelling;
- using a headline government-take number without assumptions;
- treating audit findings as recovered money;
- losing geological data to advisers;
- allowing contract versions to become unclear;
- permitting audit windows to expire through administrative delay.

Chapter 50. Part II findings

The Guyana contract case is valuable because it displays the entire lifecycle of a production-sharing system.

The 2016 Stabroek terms define the baseline. [S02]

The cost audits show the agreement becoming an accounting system. [S42, S71]

The later model terms show a changed fiscal approach under a different information environment. [S14]

The NRF and cargo records show how contract entitlement eventually becomes public cash. [S45, S46]

The case does not yield one political verdict.

It yields a set of technical questions that every new producer should answer before signing, before first oil and throughout production.

Edition 2.0 deepening: from headline fiscal terms to an operating petroleum revenue system

Analytical section 50A. A production-sharing agreement is an operating constitution for the field

A petroleum agreement is often reduced in public discussion to a royalty percentage, a cost-recovery ceiling and a profit split.

Those numbers matter, but they do not describe the whole economic system.

A production-sharing agreement also governs:

- when costs become recoverable;
- which costs are excluded;
- how affiliate charges are tested;
- when audit rights expire;
- how petroleum is valued;
- how production is measured;
- which currency is used;
- how tax obligations are treated;
- how assets are transferred or decommissioned;
- how disputes are escalated;
- how assignments are approved;
- how stabilization and change-of-law provisions interact with later legislation.

The fiscal result is therefore produced by a chain of clauses rather than one percentage.

For audit purposes, every contract analysis should begin with a clause map. The clause map identifies the economic effect of each provision, the institution responsible for administering it, the data needed to test compliance and the time limit for challenge.

The executed 2016 Stabroek agreement remains the controlling baseline for that contract. Later model terms provide evidence of later policy design, not a substitute for the executed agreement. [S02, S112]

Analytical section 50B. Contract economics should be modelled as a waterfall

The most reliable way to explain a production-sharing agreement is to model it as a waterfall.

Start with gross production.

Subtract volumes that do not reach sale because of operational use, loss or other contractually recognized treatment.

Apply the valuation rule to saleable petroleum.

Calculate royalty on the correct base.

Determine the amount of recoverable cost subject to the contractual ceiling.

Calculate the residual profit petroleum.

Apply the profit split.

Add tax treatment.

Then bridge the state's physical entitlement to the timing of actual sale proceeds and cash receipts.

This sequence should be performed for each accounting period.

A single annual average can hide timing effects. A high-cost month, a delayed cargo or a large development expenditure can change the period result even when annual production remains strong.

The teaching value of a waterfall is that it makes every assumption visible.

Analytical section 50C. The cost bank is a balance-sheet item, not a rhetorical category

Cost recovery is frequently discussed as if recovered cost disappears.

In accounting terms, unrecovered cost is better understood as a stock carried forward subject to the agreement.

The opening cost bank increases with eligible current-period cost and may also reflect allowed carry-forward treatment.

It decreases when cost oil is allocated.

The closing cost bank becomes an input into future periods.

This creates an intertemporal link between today's investment and tomorrow's government profit petroleum.

A major new development can therefore increase production capacity while simultaneously increasing the stock of recoverable cost.

The state needs to forecast both.

A petroleum revenue model should report:

- opening recoverable-cost balance;
- current eligible cost;
- adjustments and audit disallowances;
- amount recovered in period;
- closing recoverable-cost balance.

Without this table, a public forecast can make government revenue appear to move mysteriously when the real driver is the cost bank.

Analytical section 50D. Audit rights are perishable assets

An audit right that expires unused has no economic value.

The GRA account of the inaugural cost audit is important because it shows how audit administration actually works. The audit covered pre-contract and exploration/development costs through 31 December 2017. Guyanese officials from several agencies worked with IHS Markit, and the process included draft findings, government review, auditee response and later dispute over the quality and finality of the audit product. [S111]

The administrative lesson is that an audit is not a single event.

It has at least eight stages:

- 1. define the audit population;
- 2. preserve records;
- 3. test invoices and allocation;
- 4. identify exceptions;
- 5. obtain auditee response;
- 6. evaluate supporting evidence;
- 7. settle or dispute findings;
- 8. update the recoverable-cost ledger.

Every stage requires ownership.

A country can have a strong audit clause and still lose value through late mobilisation, incomplete records, weak specialist support or poor dispute tracking.

Analytical section 50E. Audit scope should follow risk rather than invoice volume

Petroleum auditing should not treat every dollar as equally risky.

A risk-based audit plan should prioritize:

- affiliate charges;
- shared-service allocations;
- head-office overhead;
- procurement from related entities;
- financing costs if recoverable;
- inventory and asset transfers;
- pre-contract cost;
- abandonment and decommissioning accruals;
- costs allocated across projects or blocks;
- costs incurred outside the country but charged to the operation;
- abnormal or non-routine events.

The state should maintain a petroleum audit risk register updated after each audit.

Recurring findings should change future audit procedures.

If the same category of cost repeatedly produces disputes, the response should not be another isolated exception. The state should consider clearer guidance, standardized supporting schedules or contract amendments where legally available.

Analytical section 50F. Affiliate transactions require an arm's-length control system

Integrated petroleum groups purchase significant services from affiliated entities.

An affiliate transaction is not automatically improper.

The audit question is whether the charge is permitted, correctly allocated and consistent with the contract's accounting rules.

A robust system should test:

- service description;
- actual service delivery;
- allocation key;
- markup;
- transfer-pricing basis;
- duplication;
- relationship between payer and supplier;

- whether the same cost appears in another block or project.

The state's petroleum audit team and tax authority should coordinate without conflating their legal mandates.

A cost can be deductible for one purpose and non-recoverable for another, depending on the governing rules.

Analytical section 50G. Petroleum valuation deserves its own control framework

Government entitlement is sensitive not only to volume but also to value.

A valuation system should identify:

- crude grade;
- quality adjustment;
- benchmark;
- differential;
- freight;
- timing;
- related-party sales;
- destination;
- sales-agent fee where applicable.

If the government takes physical profit oil and sells its own cargoes, the state also needs a transparent sales protocol.

The sales record should preserve bids, counterparty selection, pricing formula, settlement terms and realized price.

This creates a second audit chain alongside cost recovery.

One chain protects the cost base.

The other protects the value of petroleum sold.

Analytical section 50H. Government take should be reported as a range with assumptions

There is no single government-take number that describes a petroleum agreement under all conditions.

Government take varies with:

- oil price;
- recoverable cost;
- production;

- tax treatment;
- project timing;
- discount rate;
- financing structure;
- decommissioning;
- treatment of bonuses and rentals;
- state participation where applicable.

A defensible fiscal comparison should therefore publish assumptions before the result.

At minimum, a comparative table should use a common oil price path, cost schedule, production profile and discount rate.

The same model should then run both contract systems.

This is especially important when comparing the 2016 Stabroek agreement with the later model PSA. The model agreement includes different fiscal and contractual architecture, including later-generation provisions on financial guarantees, insurance, decommissioning, audit and other matters. [S112]

A difference between the contracts is evidence of policy evolution. It does not establish that both sets of terms were available under the same geological information and market conditions.

Analytical section 50I. Stabilization clauses should be interpreted narrowly and precisely

Stabilization is often discussed in absolute language.

The real legal effect depends on the wording of the clause, the governing law, the type of government action and the available remedy.

A responsible case study should therefore avoid statements such as "the contract freezes all law" unless the text actually supports that conclusion.

The analytical questions are:

- which laws are covered;
- what counts as an adverse change;
- whether environmental and safety measures are treated differently;
- whether the remedy is compensation, renegotiation or another mechanism;
- whether international law limits the clause;
- how the clause interacts with later statutes.

The same discipline applies to tax provisions.

Legal characterization should begin with the executed text.

Analytical section 50J. Decommissioning is a present fiscal issue even when closure is decades away

Decommissioning is often deferred in public debate because production is still expanding.

Financially, it begins much earlier.

The state should understand:

- estimated abandonment cost;
- timing of contributions to any decommissioning fund;
- whether contributions are recoverable;
- asset-transfer obligations;
- residual environmental liability;
- security if a contractor exits or becomes insolvent;
- treatment of unused funds.

A late decommissioning framework creates intergenerational risk.

The public may receive revenue for years and then inherit a liability after production declines.

The later model PSA contains explicit decommissioning architecture that can be compared with the executed Stabroek framework, but the comparison must preserve contract version. [S112]

Analytical section 50K. Negotiation files are national institutional memory

A petroleum agreement records the final bargain.

It does not record all options considered.

For future accountability and learning, the state should retain a negotiation file containing:

- government objectives;
- fiscal model versions;
- adviser reports;
- risk assumptions;
- marked-up drafts;
- legal issues;
- decision memoranda;
- approvals;
- conflict declarations.

Some documents may require lawful confidentiality.

That does not justify institutional amnesia.

A later government should be able to reconstruct why a clause was accepted and what information was available at the time.

This is essential for distinguishing hindsight from genuine process failure.

Analytical section 50L. Contract administration needs a calendar, not just a filing cabinet

The contract should be converted into a compliance calendar.

Every recurring obligation should have:

- responsible institution;
- due date;
- data input;
- review procedure;
- escalation path;
- retention rule.

The calendar should cover:

- work programmes;
- budgets;
- cost statements;
- production reports;
- royalty;
- audit windows;
- training obligations;
- local content;
- insurance;
- guarantees;
- decommissioning updates;
- assignment approvals.

A state that administers the contract only when a dispute arises has already lost leverage.

Analytical section 50M. A petroleum contract should be tested against an adverse case before signature

The fiscal model should not use only a base case.

Before signature, the state should test:

- low oil price;
- high development cost;

- two-year delay;
- dry appraisal;
- rapid cost escalation;
- major outage;
- high decommissioning cost;
- contractor insolvency;
- disputed affiliate charges;
- major environmental event.

The purpose is not to predict these events.

The purpose is to identify which risks remain with the contractor, which move to the state and which become shared.

A contract that looks strong in a single central scenario can behave very differently under stress.

Analytical section 50N. Later model terms should be used as a reform crosswalk

The 2023 draft model PSA is valuable because it shows a later policy architecture in a basin with more geological information and a different institutional setting. It includes explicit articles on financial guarantees, liability and insurance, local content, cost recovery, taxation and royalty, decommissioning, audits, dispute resolution and stabilization. [S112]

A rigorous crosswalk should classify each change into one of five categories:

- 1. fiscal;
- 2. administrative;
- 3. risk allocation;
- 4. transparency and reporting;
- 5. end-of-life responsibility.

The crosswalk should then ask whether the change arose from:

- stronger geological knowledge;
- lessons from contract administration;
- changed market conditions;
- public-policy reform;
- a different licensing strategy.

This produces more insight than a simple "old versus new" percentage table.

Analytical section 50O. Expanded Part II conclusion

The contract chapter should leave the reader with one central idea: petroleum revenue is administered, not merely received.

The agreement creates rights, but government value depends on institutions converting those rights into measured production, validated costs, priced cargoes, settled disputes and reconciled cash.

Guyana's cost-audit history illustrates both the importance and difficulty of this work. The GRA account shows that government agencies had to coordinate with external specialists, challenge audit methodology, provide the contractor a response process and continue working through unresolved issues. [S111]

The later model PSA shows how contract architecture can evolve as a petroleum province matures. [S112]

For a newcomer, the practical blueprint is therefore to build four systems before first oil:

- a contract register;
- a cost-recovery ledger;
- a valuation and cargo-sales ledger;
- an audit and dispute calendar.

The state should then reconcile those systems to the sovereign revenue account.

If that chain cannot be reproduced, headline fiscal terms provide less protection than they appear to offer.

Edition 2.0 clause deepening: reading the Stabroek agreement as an integrated fiscal machine

Analytical section 50P. Article 11 cannot be analysed without Article 13 and Article 15

The cost-recovery formula does not operate on an abstract barrel.

The agreement first needs a value.

Article 13 provides the petroleum-valuation architecture.

Article 15 provides royalty and tax treatment.

Article 11 then determines cost recovery and production sharing. [S02]

The fiscal model should therefore follow the contractual order.

If valuation is wrong, the royalty base and profit calculation can also be wrong.

If cost is wrong, the residual profit petroleum changes.

If tax treatment is described incorrectly, government cash and tax accounting can be confused.

The clause crosswalk now added to Edition 2.0 makes these dependencies explicit.

Analytical section 50Q. Article 11 is an intertemporal mechanism

Article 11 and Annex C should be understood across time, not just within one month.

When eligible recoverable costs exceed the amount that can be recovered from current production, the unrecovered balance carries forward under the agreement's accounting mechanics. [S02]

That means a development sanctioned today can affect the state's share of profit petroleum in later periods.

The fiscal model should therefore display the cost bank as a rolling stock.

A snapshot of one month's government share cannot explain the economics of the block over its life.

Analytical section 50R. Article 13 creates a price-governance problem

Petroleum valuation determines the monetary base on which fiscal flows are calculated.

The audit team should therefore understand not only the contract wording but the actual commercial sale process.

The controls should include:

- benchmark selection;
- quality differential;
- cargo date;
- loading window;
- freight adjustment;
- related-party transaction;
- provisional versus final price.

A state can have correct metering and still lose revenue if valuation is weak.

Analytical section 50S. Article 15 contains several distinct fiscal mechanisms

Public discussion often compresses Article 15 into "2% royalty and taxes paid by government".

The actual analysis should separate at least:

- royalty;
- income/corporation tax architecture;

- pay-on-behalf mechanics;
- statutory exemptions/orders;
- contractor tax filing;
- government accounting of tax certificates.

GRA has separately clarified that the 2% royalty should not be treated as Cost Oil and has distinguished the tax regime from the cost-recovery regime. [S15]

This is an important control because a payment can have different treatment in:

- petroleum cost recovery;
- corporate tax;
- sovereign revenue reporting.

Analytical section 50T. Article 20 and environmental permits solve different problems

Article 20 contains contract-level provisions concerning assets and insurance. [S02]

Environmental permits create project-specific regulatory conditions.

The Oil Pollution Act creates a later statutory layer.

The paper should not use one as a substitute for another.

A complete financial-protection assessment asks:

- What does the petroleum contract require?
- What does the permit require?
- What does statute require?
- What executed insurance or guarantee exists?
- Which obligation applies to which type of loss?

This is the legal architecture behind the environmental financial-assurance discussion in Part V.

Analytical section 50U. Article 23 is only useful if audit systems are operational

A contractual accounting and audit right has to be translated into:

- data access;
- audit plan;
- specialist staff;
- deadline monitoring;
- finding classification;
- dispute management.

The GRA account of the IHS audit shows that the first major audit required interagency participation, external expertise, drafts, responses and disagreements about the process. [S111]

That history should be presented as an operating case.

The lesson is not simply "audit earlier".

It is "build the workflow needed to convert a contractual audit right into a settled accounting adjustment".

Analytical section 50V. Annex C is the economic grammar of cost recovery

Annex C is not an administrative appendix of secondary importance.

It defines how costs are classified, allocated and treated.

The crosswalk should therefore identify:

- exploration cost;
- development cost;
- operating cost;
- service cost;
- general and administrative cost;
- overhead;
- recoverable categories;
- non-recoverable categories;
- ministerial approval categories.

A reviewer cannot evaluate the cost bank without understanding these classifications.

Analytical section 50W. Cost allocation across fields should be modelled explicitly

Annex C allows field-level cost-recovery statements to feed the contractor's total allocation under the agreement's accounting procedure. [S02]

This is the accounting issue often discussed publicly as ring-fencing.

The book should define the mechanism rather than rely on the label.

The fiscal question is whether costs associated with one development can reduce profit petroleum generated by production elsewhere within the contract area under the executed accounting system.

The model should show the result numerically.

This is more informative than describing "ring-fencing" as inherently good or bad.

Analytical section 50X. Article 26 creates a dispute timeline

Arbitration is not just an end-stage legal remedy.

Its existence affects how unresolved audit, stabilization or other contractual disputes are administered.

The dispute register should show:

- issue;
- notice date;
- negotiation period;
- government position;
- contractor position;
- amount at issue;
- accounting treatment while pending;
- final resolution;
- implementation.

This prevents a disputed amount from disappearing from institutional memory.

Analytical section 50Y. Article 32 should be quoted with its remedial structure

The official agreement text shows that Article 32 addresses increases in economic burden, materially adverse effects and remedial action, with a path that can proceed to arbitration if unresolved. [S02]

That is more precise than saying "the contract freezes future law".

The final legal chapter should reproduce a narrowly paraphrased sequence:

- identify change;
- assess material adverse economic effect;
- notify;
- negotiate remedial action;
- use dispute process if unresolved.

Whether a specific later law triggers that sequence is a legal question requiring the actual circumstances.

Analytical section 50Z. Later law creates a version-management obligation

The Petroleum Activities Act 2023 is part of the modern legal framework. [S87]

The executed Stabroek agreement predates it.

Every legal proposition should therefore have a "governing date" field.

The field should identify:

- statute in force;
- contract version;
- licence;
- permit;
- later amendment;
- transitional rule.

This prevents anachronistic analysis.

Analytical section 50AA. Model PSA comparison should be clause by clause

The 2023 draft model PSA has its own articles on financial guarantees, insurance, cost recovery, tax, royalty, local content, decommissioning, audit, dispute resolution and stabilization. [S112]

The comparison should not be "2016 agreement versus 2023 agreement" as if they were signed under the same conditions.

It should be:

- clause;
- 2016 mechanism;
- 2023 model mechanism;
- change in geological information;
- change in policy;
- change in risk allocation;
- implementation consequence.

This reveals where reform focused.

Analytical section 50AB. Contract transparency should include machine-readable metadata

Publishing the PDF is important.

A contract register should also provide structured fields:

- parties;
- block;
- effective date;
- term;

- royalty;
- cost ceiling;
- profit split;
- tax treatment;
- audit period;
- assignment rules;
- stabilization;
- dispute forum.

The machine-readable layer does not replace the text.

It makes comparison and monitoring easier.

Analytical section 50AC. A fiscal model should publish its clause map

Every model assumption should have a contract source.

For example:

- royalty rate -> Article 15.6;
- cost ceiling -> Article 11;
- valuation -> Article 13;
- tax treatment -> Article 15;
- cost classification -> Annex C.

This creates model lineage.

If a clause interpretation changes, the analyst can identify which model cell is affected.

Analytical section 50AD. Clause-level audit conclusion

Edition 2.0 adds `PSA\_CLAUSE\_CROSSWALK.csv` and the companion methodology note.

This materially changes Part II.

The chapter is no longer asking the reader to accept a summary of the contract.

It provides an audit map from mechanism to clause to model consequence.

The remaining legal task is to deepen selected provisions with exact subsection/page pinpoints and independent petroleum-law review.

That external legal review remains a release gate rather than being simulated by the author.

Edition 2.0 fiscal deepening: how contract design becomes cash

Analytical section 50AE. Fiscal modelling should have three views

A petroleum fiscal model should produce three different views.

Project economics asks whether the development generates sufficient contractor return.

Government take asks how much economic rent or project cash flows to government through all fiscal instruments.

Government cash flow asks when cash actually becomes available to the budget or sovereign fund.

These views are related but not interchangeable.

A regime can have a high lifetime government take but low early cash because development cost is being recovered.

A regime can generate large early cash at high oil prices while long-run government take remains sensitive to cost.

The book should state which view is being discussed every time it uses a percentage.

Analytical section 50AF. Nominal and discounted government take answer different questions

An undiscounted lifetime share treats a dollar received twenty years from now like a dollar received today.

A discounted measure reflects timing.

The discount rate is itself an assumption.

A fair comparison should therefore publish both:

- nominal undiscounted government take;
- discounted government take at stated rates.

This avoids using one selected rate to create an impression of precision.

Analytical section 50AG. The fiscal model should show contractor cash too

A government-only model can hide the economic trade-off that made investment possible.

The model should show:

- contractor capital;
- operating cost;
- cost recovery;
- contractor profit petroleum;
- tax treatment;
- net cash.

This allows readers to understand how changes in fiscal terms affect investment return and state revenue together.

Analytical section 50AH. The model should separate sunk and future cost

Once a project is producing, some capital has already been spent.

A decision about a later expansion or contract amendment should distinguish:

- sunk historical cost;
- committed future cost;
- avoidable future cost.

This matters for bargaining.

The contractor's investment decision before FID is different from its economic position after billions have been sunk.

Analytical section 50AI. Oil price should be modelled as a path, not a constant

A single flat price can be useful for illustration.

A full model should also test:

- low early price;
- high early price;
- boom then decline;
- persistent low price.

Timing matters because cost recovery is concentrated in particular years.

Two price paths with the same average can produce different government cash timing.

Analytical section 50AJ. Inflation should be explicit

Development and operating costs can rise with inflation.

If the fiscal model uses nominal oil price but constant real cost, the result is inconsistent.

The model should specify whether each input is:

- nominal;
- real;
- indexed;
- fixed.

This is especially important when comparing projects sanctioned in different years.

Analytical section 50AK. Financing costs require contract-specific treatment

Petroleum projects can be financed through parent funding, affiliate loans, external debt or internal cash.

Whether interest is recoverable or deductible depends on the governing contract and tax rules.

The audit should therefore identify:

- lender;
- rate;
- principal;
- purpose;
- related-party status;
- recoverability;
- tax treatment.

A high interest charge can affect project economics even when production is unchanged.

Analytical section 50AL. Overhead should be audited as a service, not a percentage alone

Head-office and affiliate overhead can be difficult to verify because the service is delivered outside the host country.

The audit should ask:

- what service was performed;
- for which project;
- allocation key;
- whether the cost is already charged elsewhere;
- contract cap or formula.

Annex C should be the controlling accounting reference. [S02]

Analytical section 50AM. Shared cost should have a transparent allocation rule

A contractor may use one service for several fields or blocks.

Shared cost should be allocated using a documented driver such as:

- time;
- headcount;
- equipment use;
- production;

- activity.

The choice of driver can materially affect recoverable cost.

The audit should test whether the driver is consistent and economically reasonable.

Analytical section 50AN. Cost forecasting should show committed versus forecast cost

A cost bank includes actual eligible cost.

Government forecasting also needs future expected cost.

Those are different evidence classes.

The model should separate:

- audited actual;
- unaudited actual;
- committed contract;
- operator forecast;
- government forecast;
- contingency.

This makes revisions easier to explain.

Analytical section 50AO. Development cost overruns should trigger causal review

If a project cost increases materially, the state should identify why.

Possible drivers include:

- scope change;
- inflation;
- supply chain;
- engineering;
- delay;
- exchange rate;
- additional wells.

An overrun is not automatically evidence of improper cost.

It is a signal that the cost-recovery forecast and project economics need to be updated.

Analytical section 50AP. Reserve revisions should feed the fiscal model

A development may recover more or less petroleum than originally forecast.

Reserve or resource revisions affect:

- production life;
- unit cost;
- decommissioning timing;
- government revenue.

The state should therefore maintain its own versioned production and reserves model.

Analytical section 50AQ. Fiscal sensitivity should be presented as a surface

Instead of one low, base and high case, the final book can present a matrix.

For example: rows = oil price columns = recoverable-cost share.

Each cell shows government cash share for the period.

This reveals which variable dominates the result.

It also makes the 14.5% binding-cap example easier to understand as one specific cell rather than a universal government take. [S15]

Analytical section 50AR. The 14.5% example should be taught carefully

GRA explains that when the 75% cost ceiling binds, the government's 12.5% share of the remaining profit petroleum plus the 2% royalty produces a 14.5% gross-revenue share under that simplified condition. [S15]

That number is useful pedagogically.

It should not be described as:

- lifetime government take;
- minimum cash in every month;
- total fiscal burden.

As costs fall, more production becomes profit petroleum and the government share changes.

Tax treatment and other flows also need separate analysis.

Analytical section 50AS. Tax certificates and sovereign cash should be separately reconciled

The pay-on-behalf arrangement creates an accounting relationship between contractor tax, government profit petroleum and tax certification.

The book should present a flowchart showing:

- contractor taxable income;

- GRA assessment;
- ministerial/government pay-on-behalf treatment;
- tax certificate;
- effect on government share/accounting.

The flowchart should be reviewed by a petroleum tax specialist before final publication.

Analytical section 50AT. Ring-fencing analysis should use a two-project numeric example

The final fiscal appendix should model:

- Project A producing with low remaining cost;
- Project B under development with high recoverable cost.

The model then tests whether Block-level aggregation allows Project B costs to reduce profit petroleum generated from Project A production.

The result should be derived from the executed agreement's accounting procedure.

This turns a contested policy phrase into reproducible arithmetic.

Analytical section 50AU. Decommissioning should appear in both fiscal and environmental models

Decommissioning has at least three dimensions:

- recoverable cost;
- secured funding;
- physical environmental obligation.

The fiscal model should show when cost is recognized or funded.

The environmental register should show the actual closure obligation.

The two should reconcile.

Analytical section 50AV. Government negotiation should use a shadow model

Before accepting a contractor fiscal model, the state should maintain an independent shadow model.

It does not have to reproduce every corporate financing detail.

It should be capable of:

- project cash flow;
- cost recovery;
- profit share;

- tax;
- price sensitivity;
- production sensitivity;
- decommissioning.

The state cannot negotiate effectively if it only observes the contractor's result.

Analytical section 50AW. Model governance needs its own audit trail

Fiscal models change.

Every version should record:

- author;
- date;
- data source;
- assumptions changed;
- formula changes;
- reviewer;
- approval.

A spreadsheet without version control can become a hidden policy instrument.

Analytical section 50AX. Fiscal deepening conclusion

The fiscal chapter should make one idea unavoidable:

Contract terms do not generate public wealth automatically.

They generate formulas and rights.

Institutions convert those formulas into cash through measurement, valuation, accounting, audit, dispute resolution and fund custody.

Edition 2.0 should therefore present the Stabroek regime as a complete operating system with explicit data lineage from clause to model to ledger.

Edition 2.0 original case study: the dated 1999–2023 cost-audit pipeline

From a submitted expense to a lawful revenue adjustment

The public dispute over offshore cost recovery offers a rare case in which three audit periods, three different procedural stages and the same overarching contractual accounting architecture can be observed. The relevant units must be kept separate. A contractor-submitted cost is an input to the accounting process. An auditor's exception is a challenge to that input. A government rejection or dispute is an institutional position. A contractual determination is a further step. A cost-bank adjustment changes subsequent entitlement calculations. Actual recovered cash, if any, is a final and different evidentiary category.

A reader who combines these units can produce an apparently large number without demonstrating any fiscal transfer. Edition 2.0 therefore adds ``PETROLEUM_COST_AUDIT_STATUS.csv``, a dated audit-status register that records reported submitted expenses, exceptions, procedural status and what remains unverified. The source distinction matters: the full IHS final PDF was unavailable to the GitHub-hosted source checker, whereas the Ministry of Natural Resources' April 2024 official executive-summary excerpt remains available on the Ministry's own website. The narrower public status update issued in May 2025 is independently usable for its then-current procedural account. [S71, S72]

The inaugural audit, 1999–2017: a disputed cost is not revenue recovered

The official April 2024 publication notice describes an IHS Markit audit of US\$1,677,774,727 in expenses submitted for the 1999–2017 period. The quoted executive summary reports reasonable grounds to dispute approximately US\$214.4 million plus overhead adjustments. The indicated US\$214.4 million alone is approximately 12.78 percent of the reported submission; this calculation is a ratio of **audit exceptions to reported submitted expenses**, not a measure of lost government revenue and not a final contractual disallowance. [S71]

By the Ministry's May 2025 update, the Government reported that it had communicated the dispute to the co-venturers and that the parties had agreed to use the production-sharing agreement's sole-expert procedure. The Ministry reported nominations and a requested operator response by 31 May 2025. These facts establish the Government's stated procedural position at that point. They do not establish what a later expert decided or what amount, if any, was finally removed from the cost bank. [S72]

That distinction also limits the book's language. The accurate formulation is a **government-disputed audit exception**, not “US\$214.4 million recovered”. The full report, supporting schedules and later final disposition remain explicit evidence gaps under issue #69. Where the chapter discusses the scope of the inaugural audit itself, its documentary chain must include the published government excerpt, the GRA's earlier explanation of the audit

process, and any subsequently authenticated full report rather than an inaccessible PDF link alone. [S71, S111]

The 2018–2020 audit: different period, different procedural posture

The same Ministry status release reported that the second audit reviewed approximately US\$7.2 billion in expenses incurred in 2018–2020. Government stated that US\$65.1 million had not been accepted and that the co-venturers had furnished updated submissions still under review. The US\$65.1 million is roughly 0.90 percent of the rounded stated audit population, but this percentage should not be turned into a judgment that the second period had less or more legitimate expense than the first. The project mix, accounting populations, maturity of developments, audit methods, identified issues and procedural status differ between periods. [S42, S72]

For fiscal analysis, the useful unanswered questions are concrete. What portion of the US\$65.1 million was supported after the revised submissions? Which exceptions were reclassified, accepted, disallowed or referred to dispute? Were costs associated with one field attributed to another under the applicable Annex C procedure? Did any adjustment affect government entitlement during a producing month, or did it change only a carry-forward cost balance? Until those records are publicly available, the model should use **open scenarios**, not a claimed final recovery.

The 2021–2023 audit: completion of contractor review is not completion of public process

In the same dated 2025 release, the Ministry stated that the third cost audit covering 2021–2023 had been completed and submitted to Government, with the Guyana Revenue Authority reviewing the initial report. The release did not provide the same kind of settled exception and adjustment schedule for that period. [S72]

This creates a third public category: **submitted report with government review underway**. A complete Edition 2.0 account should record the date of the first report, the dates of public responses, the scope and the resolution of exceptions when those materials become available. Until then, no exception amount should be invented or inferred from either of the previous audits.

Why the apparent percentages cannot be converted directly into a fiscal loss

The public figures are tempting to add together. On the May 2025 account, US\$214.4 million and US\$65.1 million imply roughly US\$279.5 million in reported exceptions or disputed/not-accepted expense **before any unknown overhead adjustment and later resolution**. That arithmetic says nothing directly about immediate cash recovery.

Suppose, for illustration, that a US\$1 million cost were ultimately disallowed. The fiscal effect depends on the accounting period, whether the cost had already been recovered, whether the 75 percent monthly cost ceiling was binding, the closing unrecovered-cost balance, later

production and oil prices. If the cost sat in an unrecovered carry-forward balance, removal might affect future profit petroleum rather than today's cash. If an earlier producing period had been misallocated, the contractual adjustment process and time-value effects would matter. The book's synthetic fiscal model can demonstrate these mechanics without presenting a simulated outcome as an actual audit settlement. [S02, S15, S72]

Institutional learning from the three-period comparison

The three periods show why the Government's audit system requires both continuity and independent technical capacity. The inaugural audit involved a contested interpretation of audit findings and the prospect of a sole expert. The second required government consideration of revised contractor submissions. The third had reached initial government review by the published May 2025 status date. [S72, S111]

An auditable national cost-recovery register should therefore preserve, for every audit period, the original cost submission, audit team and methodology, draft findings, contractor response, reasoned government position, formal dispute notice, expert/arbitral disposition where used, cost-bank entries, and any proven revenue adjustment. The register should explicitly date the last confirmed status. Later silence in the public record is not proof of resolution or abandonment.

This case offers a transferable lesson for a new producer: there is no single “cost-audit completion” event. An effective system separates technical fieldwork, government acceptance, contractual dispute procedure, accounting correction and cash realization, each with a different institution, deadline and evidence requirement. That is how a percentage-based production-sharing agreement becomes a genuinely enforceable public revenue system.

PART III | Development approvals and offshore production

From First Oil to Large-Scale Production

Chapter 51. First oil is a chain, not a single event

The phrase first oil is useful shorthand. It also hides several different milestones.

A development moves through project sanction, engineering, construction, drilling, installation, commissioning and production startup. Once crude begins flowing, it still needs processing, storage, lifting, sale and settlement before public petroleum cash appears in the sovereign account.

For a case study, each milestone should have its own date.

Guyana's Liza Phase 1 began production in December 2019. Liza Phase 2 started production on 11 February 2022. Payara started in November 2023. Yellowtail began production on 8 August 2025. [S81, S82, S83, S84]

This sequence demonstrates the speed at which one offshore development became a multi-FPSO production system.

The sequence also created a regulatory scaling problem.

Every new FPSO increases:

- production measurement;
- cargo scheduling;
- cost submissions;
- environmental monitoring;
- inspections;
- local-content reporting;
- emergency exposure;
- future decommissioning obligations.

The number of barrels is therefore also a measure of administrative workload.

Chapter 52. Liza Phase 1 as the first operating test

The first producing project is the point at which petroleum governance moves from plans to live transactions.

For Liza Phase 1, the state needed to supervise:

- the production licence;
- environmental permit;

- production reporting;
- cargo entitlement;
- royalty;
- profit oil;
- cost recovery;
- cash receipt;
- NRF accounting.

The first project therefore functions as an institutional test bench.

It reveals whether laws written for exploration translate into operating controls.

It also exposes gaps that are hard to identify on paper.

Examples include:

- mismatch between production and sales periods;
- accounting receivables at year end;
- operating flaring;
- timing of cost audits;
- public reporting definitions.

The later permit modification associated with Liza Phase 1 flaring provides one example of regulation responding to observed operations rather than remaining fixed at approval. [S55]

Chapter 53. The production licence

A petroleum agreement establishes the contractual relationship.

A production licence authorizes development and production under law.

The distinction matters.

The production licence identifies:

- area;
- project;
- development plan;
- term;
- reporting;
- operational obligations;
- environmental requirements;
- local-content requirements;
- abandonment.

The official petroleum portal publishes the Liza production licence and later project licences. [S86]

The flagship should build a production-licence register for every development and preserve amendments separately.

A newcomer country benefits from the same discipline. Contract rights and project approval should not be treated as one undifferentiated document.

Chapter 54. Liza Unity and the move to two FPSOs

Liza Phase 2 began production on 11 February 2022. ExxonMobil stated that the Liza Unity FPSO was expected to ramp toward 220,000 barrels per day. [S82]

The startup doubled the complexity of several systems.

Production reports now needed project identification.

Environmental monitoring needed to separate FPSOs.

Cargoes needed entitlement tracking across two sources.

Cost accounting needed project-level detail.

Emergency planning needed to consider concurrent operations.

This is an important governance threshold.

A regulator that works comfortably with one producing asset does not automatically scale to two, four or six.

Chapter 55. Payara and Prosperity

The government announced first oil at the Prosperity FPSO in November 2023. [S83]

By this stage, the production system contained:

- Liza Destiny;
- Liza Unity;
- Prosperity.

The third project added another production licence, environmental permit, cost stream, local-content plan and operating history.

The flagship should use Payara to test institutional standardization.

Questions include:

- Did production data use the same definitions?
- Were environmental reporting formats comparable?
- Were audit procedures consistent?
- Were public project documents easier to access?

- Did lessons from earlier permits appear in later conditions?

This is how regulatory learning is measured.

Chapter 56. Yellowtail and the fourth operating development

Yellowtail began production on 8 August 2025. The operator stated that the ONE GUYANA FPSO increased combined installed capacity above 900,000 barrels per day. [S84]

In November 2025, the operator announced a daily production milestone of 900,000 barrels. [S85]

These two facts should remain separate.

Installed capacity is an engineering measure.

A daily milestone is a point observation.

Neither is the same as annual average production.

The final production chapter should display all three on separate lines.

Chapter 57. Capacity versus realized production

A recurring error in petroleum reporting is to multiply nameplate capacity by 365 and treat the result as actual production.

Realized output is affected by:

- ramp-up;
- maintenance;
- well availability;
- reservoir performance;
- operating constraints;
- unplanned shutdowns.

The official production dashboard extends from December 2019 through 31 July 2026. [S81]

The flagship should calculate:

- monthly barrels;
- monthly daily average;
- annual barrels;
- annual daily average;
- utilization relative to stated capacity.

The utilization ratio should be treated cautiously because nameplate capacity can change after optimization.

Chapter 58. Production optimization

Offshore facilities sometimes operate above original nameplate assumptions after engineering review and optimization.

This does not mean the original capacity figure was false.

It means facility constraints, well performance or operating procedures changed.

The data series should therefore preserve:

- original design;
- approved optimization;
- revised operating capacity;
- realized output.

A newcomer regulator should require technical support for material capacity increases, including environmental and safety review where required.

Chapter 59. Production measurement

Petroleum revenue begins with physical measurement.

A credible measurement system requires:

- calibrated meters;
- sampling;
- temperature and pressure correction;
- volume reconciliation;
- tank measurement;
- cargo measurement.

The government needs independent access to the measurement chain.

A contract formula is only as accurate as the physical data entering it.

The flagship should identify which standards govern custody transfer and which agency verifies metering.

Chapter 60. Production data architecture

The public production database should use persistent identifiers.

Each record should include:

- project;
- FPSO;
- field;
- date;

- oil;
- gas;
- water;
- flaring;
- reinjection;
- downtime.

Production data should link to:

- environmental data;
- cargo data;
- government entitlement.

This makes cross-checking possible.

Chapter 61. Cargo lifting

FPSOs store produced crude until a tanker lifts a cargo.

The cargo date therefore follows production.

A cargo register should include:

- vessel;
- load date;
- barrels;
- crude grade;
- entitlement owner;
- pricing method;
- sale date;
- settlement date.

For government cargoes, the record should link to the NRF cash deposit.

The same cargo identifier should appear in both petroleum and financial records.

Chapter 62. Government entitlement

Under the production-sharing arrangement, government receives a share of profit oil and royalty.

Government entitlement is influenced by:

- total production;
- valuation;
- recoverable cost;
- profit pool;

- contractual share.

Entitlement does not guarantee cash on the same day.

The government may lift a cargo and receive sale proceeds later.

This creates timing differences across production, fiscal and NRF reports.

Chapter 63. The first-year receivable lesson

The 2020 audited NRF statements illustrate timing directly.

Four government profit-oil lifts were recognized for 2020, while the last lift remained a receivable at year end. [S45]

This is a small accounting detail with major analytical importance.

A calendar-year chart using cash deposits would show one number.

An accrual financial statement would show another.

Both can be correct.

The paper should never reconcile them by forcing one definition onto the other.

Chapter 64. Royalty timing

Royalty is calculated from petroleum revenue under the applicable agreement.

Payment timing may differ from cargo sale timing.

The annual audited statements and petroleum reporting should be reconciled to the same transaction register.

Fields should include:

- royalty period;
- amount;
- due date;
- payment date;
- exchange rate;
- fund deposit.

This supports audit and public understanding.

Chapter 65. The 2026 scale

By the first half of 2026, four FPSOs were operating.

The Ministry of Finance reported substantial profit-oil receipts and royalties during the period and a large NRF balance at end-June. [S07]

These figures establish scale.

They do not settle sustainability.

A high-revenue period can reflect:

- strong production;
- oil price;
- lower cost recovery;
- cargo timing.

The fiscal chapter therefore needs the physical production chapter underneath it.

Chapter 66. Regulatory staffing versus asset count

A useful performance indicator is the ratio between regulator capacity and operating complexity.

Possible measures include:

- petroleum engineers per FPSO;
- environmental specialists per active permit;
- auditors per annual contractor cost submission;
- inspection days per project;
- unresolved audit value;
- publication delay.

The objective is not to establish an arbitrary international benchmark.

The objective is to detect whether workload grows faster than institutional resources.

Chapter 67. Production and environmental monitoring

Production expansion changes environmental exposure.

Each project adds:

- emissions;
- flaring risk;
- produced water;
- vessel traffic;
- waste;
- subsea infrastructure.

Monitoring should therefore be both project-specific and cumulative.

A regulator needs to answer two questions.

Is each permit being followed?

What is the basin-wide effect of several projects operating together?

The second question becomes more important as the fleet expands.

Chapter 68. Production and local content

More FPSOs increase procurement and employment demand.

This can widen local opportunities.

It can also strain:

- skilled labour;
- port capacity;
- housing;
- transport.

The local-content chapter should therefore align procurement data with project startup dates.

A rise in supplier spending immediately before first oil may reflect construction and commissioning rather than steady-state operations.

Chapter 69. Production and infrastructure

Offshore operations rely on shore-based systems.

These include:

- port services;
- aviation;
- warehousing;
- waste handling;
- fuel;
- roads;
- telecommunications.

The final case study should distinguish privately financed oil-support infrastructure from public national infrastructure.

This avoids attributing every new facility to public petroleum spending.

Chapter 70. Production and emergency readiness

Each additional offshore asset increases the number of possible simultaneous incidents.

Emergency planning needs to test concurrency.

Examples:

- a medical evacuation during severe weather;

- an operational incident on one FPSO while another is in maintenance;
- spill-response equipment already deployed elsewhere.

The response plan should state which resources are dedicated and which are shared.

Chapter 71. Production data as a transparency tool

Public production data let citizens and analysts verify several other claims.

They support:

- royalty estimation;
- export analysis;
- emissions intensity;
- production forecasts;
- fund receipt expectations.

A public dashboard has greater value when it provides downloadable data rather than only a chart.

The dataset should retain historical revisions.

Chapter 72. Forecasting production

A production forecast should show:

- producing projects;
- sanctioned projects;
- unsanctioned prospects.

Each category has different confidence.

Producing projects use observed decline and operating data.

Sanctioned projects use construction and startup assumptions.

Unsanctioned prospects should remain scenario cases.

A national budget should not treat all three categories as equivalent.

Chapter 73. Startup risk

A project scheduled for first oil in a given quarter may face:

- fabrication delay;
- subsea delay;
- drilling delay;
- weather;
- commissioning issue.

A prudent fiscal forecast should include schedule sensitivity.

The public document should show which production year depends on a new project starting on time.

Chapter 74. Downtime risk

A large FPSO contributes a material share of national output.

An outage therefore affects:

- exports;
- government entitlement;
- royalties;
- GDP;
- foreign exchange.

The model suite now includes lower-volume stress cases to demonstrate the arithmetic effect.

A national fiscal framework should test multi-month interruption, not only daily variation.

Chapter 75. Price and production together

Price and volume are not independent risk controls.

A lower oil price can coincide with production interruption.

The combined effect on gross petroleum value is multiplicative.

Government cash then depends on cost recovery and fiscal terms.

This is why the stress tests use both price and volume changes.

Chapter 76. First-oil readiness for another country

A newcomer should rehearse the first cargo before it happens.

The exercise should assign:

- producer;
- petroleum regulator;
- revenue authority;
- sales agent;
- central bank;
- finance ministry;
- auditor.

A fictional cargo should include one deliberate error.

The system passes only when institutions identify and correct it.

Chapter 77. A production data room for government

The government should maintain a permanent production data room containing:

- licence;
- development plan;
- meter calibration;
- daily production;
- cargo;
- price;
- royalty;
- cost statement;
- entitlement;
- permit monitoring.

Access controls should protect commercial data.

Institutional memory should survive personnel changes.

Chapter 78. A public petroleum ledger

Not every field in the government data room belongs in public.

The public ledger should still allow reconciliation of:

- production;
- government cargoes;
- public petroleum receipts;
- NRF deposits.

It should define every term.

Publication should follow a fixed calendar.

Chapter 79. What rapid scale-up demonstrates

Guyana's production sequence demonstrates that institutional preparation is not a one-time first-oil task.

The state needs to scale with the sector.

A system adequate for one project may become inadequate at four.

Laws need implementation.

Audits need completion.

Environmental monitoring needs expansion.

Data systems need standardization.

Chapter 80. Part III findings

The production record establishes a fast move from first oil in December 2019 to four operating FPSOs by August 2025. [S81-S85]

The main transferable lesson is not the speed itself.

It is the need to separate every physical and financial stage.

Discovery is not production.

Capacity is not output.

Production is not cargo.

Cargo is not cash.

Cash is not budget spending.

A newcomer country that preserves these distinctions gains a traceable petroleum system.

That traceability becomes the foundation for sovereign wealth management, environmental oversight and public accountability in the later parts of this study.

Edition 2.0 deepening: production scale-up as an integrated control system

Analytical section 80A. First oil should be decomposed into a readiness sequence

"First oil" compresses many separate events into one phrase.

Before production begins, a state has already had to approve or monitor:

- appraisal;
- commerciality;
- field-development design;
- production licence;
- environmental authorization;
- FPSO construction and commissioning;
- subsea installation;
- metering;
- lifting procedures;
- customs and import treatment;
- emergency arrangements;

- revenue custody.

The correct readiness test is therefore not whether the FPSO can produce.

It is whether the state can independently account for the first barrel from reservoir to public receipt.

A new producer should conduct a full simulated production month before startup. The exercise should create synthetic daily production, meter data, cargo nominations, lifting statements, benchmark pricing, royalty calculations, cost-recovery entries, government entitlement and a mock NRF deposit.

Every responsible institution should sign off on the same transaction.

Analytical section 80B. FPSO-by-FPSO expansion changes regulatory complexity

Guyana moved from the Liza Destiny to additional FPSOs in a short period. The production dashboard and project records allow each startup to be dated separately. [S81-S85]

The policy significance is not only higher capacity.

Each additional FPSO creates another operating system with:

- production meters;
- maintenance schedules;
- flare and emissions records;
- tanker traffic;
- helicopter and marine support;
- environmental monitoring;
- emergency-response requirements;
- cargo schedules;
- project-specific cost accounts.

A regulator that was adequate for one project may not remain adequate for four or more.

Capacity planning should therefore use project count and control points, not only barrels per day.

Analytical section 80C. Installed capacity, observed output and annual production are different measures

An FPSO's nameplate capacity is an engineering design parameter.

Observed daily production is a point-in-time operating result.

Monthly production is an aggregation of realized output.

Annual production incorporates ramp-up, downtime, maintenance and reservoir performance.

These measures should never be substituted for one another.

The distinction becomes especially important during rapid scale-up. A new vessel may add large installed capacity but contribute only part of a year's production because of startup timing.

Forecasting should therefore be done monthly, project by project.

The public dashboard should display capacity and realized production on separate axes.

Analytical section 80D. Metering is the first line of revenue assurance

Revenue assurance begins before price and cost recovery.

It begins with volume.

The state needs confidence in:

- meter specification;
- calibration;
- proving;
- failure procedures;
- correction factors;
- custody-transfer points;
- data retention.

If a production meter fails, the agreement and licence should define how production is reconstructed.

The regulator should preserve calibration certificates and exception logs.

This is not an engineering detail separate from fiscal governance.

A systematic measurement error becomes a revenue error.

Analytical section 80E. Cargo accounting is a bridge between reservoir and treasury

The public revenue chain should be reproducible at cargo level.

For each cargo, government should be able to connect:

- production period;
- entitlement calculation;
- lifting date;
- volume;
- quality;
- buyer;

- pricing formula;
- realized price;
- settlement date;
- cash receipt;
- NRF deposit.

This table should not expose legitimate commercial secrets unnecessarily.

It should exist inside government.

A summarized public version can then reconcile total government cargoes to reported petroleum receipts.

The advantage of cargo-level accounting is that timing differences become visible. Production in one month may generate a sale in another and a cash receipt later still.

Analytical section 80F. Production forecasting should be probabilistic

A single production forecast hides operational risk.

A more useful framework separates:

- base production;
- planned maintenance;
- startup ramp;
- reservoir decline;
- unplanned downtime;
- new project contribution.

Government revenue forecasting should then run at least three production cases.

The low case should not be an arbitrary percentage reduction. It should represent plausible operational conditions such as delayed startup or extended maintenance.

This produces a more credible fiscal contingency range.

Analytical section 80G. Maintenance is part of sovereign revenue risk

As an offshore province matures, maintenance becomes economically material.

Shutdowns affect:

- production;
- cargo timing;
- cost recovery;
- government revenue;
- service demand;

- safety risk.

The state should therefore distinguish planned from unplanned downtime.

A public production series can show the output effect.

The regulator's internal record should also preserve the reason for material outages where legally reportable.

This matters because repeated operational interruptions can signal a different risk from one scheduled maintenance campaign.

Analytical section 80H. Production data should be governed as public infrastructure

A production dashboard is not merely a communications product.

It is part of national statistical infrastructure.

The series should have:

- documented units;
- project identifiers;
- revision policy;
- observation frequency;
- publication lag;
- metadata;
- archival versions.

If historical values are revised, the state should preserve the earlier release and explain the revision.

This makes the dataset usable for fiscal analysis, academic work and public accountability.

The official Guyana production series through July 2026 provides a foundation for this approach. [S81]

Analytical section 80I. Production and cost data should meet at project level

A national production total is not enough for petroleum economics.

The state should be able to associate each project with:

- cumulative production;
- development cost;
- operating cost;
- recoverable-cost balance;
- government entitlement;

- environmental obligations;
- decommissioning estimate.

This creates a project economics ledger.

The ledger helps distinguish whether lower government profit petroleum results from price, production, cost recovery or project mix.

Analytical section 80J. Rapid scale creates a regulator financing question

A producing state should ask how regulatory capacity is funded.

If the regulator depends entirely on annual general-budget appropriations, staffing may lag sector growth.

Alternative mechanisms can include lawful licence fees, regulatory charges or dedicated appropriations.

The design must preserve independence and public accountability.

The key principle is that the state should not be forced to supervise a multibillion-dollar industry with a staffing model designed for a pre-production frontier.

Analytical section 80K. Production expansion should trigger emergency-response scaling

Every additional offshore project changes the response problem.

Response planning should account for:

- simultaneous operations;
- distance between installations;
- capping-resource availability;
- vessel capacity;
- aviation capacity;
- medical evacuation;
- shoreline sensitivity;
- regional assistance.

The emergency plan should test concurrent incidents rather than assume only one project can fail at a time.

This links Part III directly to the environmental analysis in Part V.

Analytical section 80L. Mature production requires decline planning

A petroleum strategy should not focus only on startup.

Every field eventually declines.

The state should monitor:

- reservoir performance;
- water cut;
- gas handling;
- infill drilling;
- enhanced-recovery proposals;
- operating-cost changes;
- cessation-of-production planning.

A late-stage field may produce fewer barrels while carrying higher operating cost.

That can affect both cost recovery and government cash.

A country should therefore begin decline and decommissioning analysis while production is still growing.

Analytical section 80M. Expanded Part III conclusion

Guyana's rapid production scale-up makes the country especially useful for a newcomer blueprint because institutional stress becomes visible quickly.

The transition from one FPSO to several projects compressed regulatory learning into a few years. [S81-S85]

The central control principle is traceability.

Every barrel should pass through a chain of measurement, entitlement, sale, receipt and public accounting.

Every project should have a corresponding chain of licence, environmental authorization, operating data, cost account and decommissioning estimate.

The production chapter should therefore be read as a data-governance chapter as much as an engineering chapter.

A newcomer that builds these controls before first oil creates the evidence base needed for every later question in the book: fiscal performance, environmental impact, local content, public investment and intergenerational wealth.

Edition 2.0 project deepening: seven development decisions as one evolving system

Analytical section 80N. Liza Phase 1 as the institutional first test

Liza Phase 1 should be treated as more than the first production project.

It was the first time the state had to apply the new offshore petroleum system to a commercial deep-water development.

The project created simultaneous demands for:

- development-plan review;
- production licensing;
- environmental authorization;
- metering;
- cargo accounting;
- royalty administration;
- cost recovery;
- emergency response;
- public revenue custody.

First oil on 20 December 2019 therefore marks the beginning of a new operating regime, not simply a production milestone. [S01, S81, S86]

The project is useful for audit because it establishes the baseline against which later changes can be measured.

For every major rule or institutional improvement introduced later, the book should ask whether the control existed at Liza Phase 1 startup or was added after operating experience.

This produces an institutional learning timeline.

Analytical section 800. Liza Phase 2 as the first scale replication test

Liza Phase 2 reached first oil on 11 February 2022. [S82]

The significance of the second development is that a country no longer has the excuse of first-project novelty.

The regulator should be able to compare:

- permit conditions;
- development cost;
- production capacity;
- project schedule;
- environmental monitoring;
- metering architecture;
- local content;
- startup ramp;
- operating incidents.

A second project creates the first true benchmark.

If a requirement changed between Phase 1 and Phase 2, the state should be able to explain whether the change resulted from new evidence, different project design or policy reform.

Analytical section 80P. Payara as the multi-project transition

Payara's first oil in November 2023 marked the transition from a two-project system to a multi-project basin. [S83]

This changes the regulator's workload non-linearly.

Several projects can now:

- submit cost statements simultaneously;
- schedule maintenance at overlapping times;
- compete for response resources;
- use the same supply base;
- employ the same contractors;
- affect the same fisheries and marine environment.

The state should therefore stop treating projects as isolated administrative files.

It needs portfolio management.

A portfolio dashboard should show the cumulative effect of all approved developments on:

- production;
- cost bank;
- regulator staffing;
- marine traffic;
- emergency resources;
- local labour;
- public revenue.

Analytical section 80Q. Yellowtail as the rapid-capacity step

Yellowtail began production on 8 August 2025, becoming the fourth producing project. [S84]

The later November 2025 operator report that production reached a daily milestone of 900,000 barrels should be treated as a point observation. [S85]

This is a useful example of the distinction between capacity and realized output.

The correct time series should preserve:

- startup date;
- monthly output;
- daily maximum where relevant;

- maintenance periods;
- national total.

A newcomer should resist the temptation to convert a production milestone into an annual revenue assumption without accounting for ramp-up and downtime.

Analytical section 80R. Uaru as the cutoff-date discipline test

Uaru is particularly important for the current edition because it demonstrates why an evidence cutoff matters.

The Errea Wittu FPSO arrived in Guyana in August 2026. Government and contractor sources described commissioning work ahead of first oil and identified a design capacity around 250,000 barrels per day. [S124, S125]

At the 27 September 2026 evidence cutoff used by this publication, the source set reviewed here does not establish an observed first-oil event.

The book should therefore say "commissioning ahead of first oil" rather than convert an expected September or fourth-quarter startup into an observed event.

This may become outdated immediately after publication.

That is acceptable.

A dated publication should prefer a correct cutoff over a premature milestone.

The corrections register can record the eventual startup in a later edition.

Analytical section 80S. Whiptail as an under-construction comparison

Whiptail provides a useful case because the state can observe development while another project is still being commissioned.

The 2026 maritime record confirms continuing offshore development activity, while project reporting identifies the Jaguar FPSO and a planned capacity around 250,000 barrels per day. [S126]

The correct analytical treatment is prospective.

The book can examine:

- approved project design;
- construction;
- subsea installation;
- expected capacity;
- stated startup target.

It should not include future Whiptail barrels in observed production.

This distinction becomes crucial in fiscal forecasting.

Analytical section 80T. Hammerhead as the seventh-development governance test

Hammerhead received development approval and a production licence in September 2025. Government and operator sources describe an estimated US\$6.8 billion project, planned production capacity of approximately 150,000 barrels per day and an anticipated 2029 startup. [S127, S128]

The project's analytical significance is that the state was approving a seventh development roughly a decade after the original Liza discovery.

This raises a cumulative governance question.

How should fiscal terms, environmental controls, financial assurance, local content and data requirements evolve as geological uncertainty declines and institutional experience increases?

The answer should be based on executed legal instruments and project-specific approvals.

It should not assume that later policy preferences automatically alter earlier contractual rights.

Analytical section 80U. Project approval should create a national commitments register

Each approved project creates future commitments for both operator and state.

The register should include:

- approved capital estimate;
- well count;
- FPSO;
- design capacity;
- startup target;
- production licence conditions;
- environmental permit;
- financial assurance;
- local-content plan;
- decommissioning plan;
- expected regulator staffing.

This allows government to see the full forward workload created by cumulative project approvals.

Analytical section 80V. Project cost should be normalized before comparison

Headline project costs cannot be compared without context.

A useful comparison should consider:

- design capacity;
- number of wells;
- water depth;
- subsea complexity;
- inflation;
- supply-chain conditions;
- construction timing;
- storage capacity;
- gas handling;
- local-content requirements.

The purpose is not to determine automatically whether one project is expensive.

It is to identify the drivers of cost.

This also improves cost-recovery forecasting.

Analytical section 80W. Production capacity should be connected to environmental capacity

Every new FPSO adds production capacity.

The environmental system must also add:

- monitoring capacity;
- inspection capacity;
- incident response;
- financial assurance;
- data review.

A country should therefore maintain an "environmental regulatory capacity per active project" indicator.

Possible measures include:

- inspectors per producing project;
- permit reports reviewed on time;
- drills completed;
- independent samples;
- unresolved permit findings.

This does not create a universal staffing ratio.

It makes regulator scaling visible.

Analytical section 80X. Marine traffic should be part of the project register

Offshore development changes tanker, supply-vessel, helicopter and support traffic.

The project register should therefore connect each development to:

- cargo frequency;
- support-vessel requirements;
- aviation demand;
- shore-base demand;
- navigational notices.

This helps quantify cumulative maritime exposure.

Analytical section 80Y. Development plans should include data-delivery obligations

The development approval should specify the data government expects during construction and operation.

At minimum:

- well data;
- reservoir model updates;
- cost forecasts;
- procurement;
- schedule;
- installation progress;
- commissioning;
- production;
- maintenance;
- incidents;
- decommissioning estimate.

A government that receives only final summary reports cannot independently update its own national model.

Analytical section 80Z. Startup readiness should be project-specific

A national first-oil system can be reused.

Each project still needs its own startup certification.

The checklist should confirm:

- permit conditions satisfied;

- metering commissioned;
- emergency plans tested;
- financial assurance valid;
- cargo and valuation protocols ready;
- government entitlement system configured;
- reporting templates active.

This avoids assuming that readiness for Liza automatically proves readiness for Uaru or Whiptail.

Analytical section 80AA. The project portfolio should be stress-tested as a system

A multi-project basin can experience common shocks.

Examples include:

- hurricane or severe weather affecting several facilities;
- helicopter-service constraint;
- port disruption;
- contractor failure;
- common equipment defect;
- cyber incident;
- shortage of specialist personnel.

The regulator should therefore run portfolio-level stress scenarios in addition to project-specific ones.

Analytical section 80AB. The offshore project register is now a core publication dataset

Edition 2.0 adds `OFFSHORE\_PROJECT\_REGISTER.csv` as a machine-readable control.

The register classifies each major project by observed or planned status.

Its most important rule is simple:

A planned startup remains planned until a dated source establishes production.

This rule should govern every chart, narrative and fiscal scenario in the final publication.

Analytical section 80AC. Project-level conclusion

The seven-development sequence shows why Guyana cannot be described only as a story of rapid production growth.

It is also a story of repeated development approvals, each adding another layer of fiscal, environmental, operational and administrative exposure.

For a newcomer, the key lesson is to build a project portfolio system before the second development is approved.

The state should be able to compare projects on common fields, identify cumulative burdens and preserve a dated record of what was forecast versus what actually occurred.

That is how rapid development becomes an auditable institutional learning process rather than a succession of isolated announcements.

PART IV | The Natural Resource Fund and public wealth

The Natural Resource Fund and the Conversion of Petroleum into National Wealth

Chapter 81. The fund begins before the money

A sovereign resource fund is most useful when its rules exist before large recurring resource receipts create pressure for immediate spending.

Guyana's first Natural Resource Fund law was enacted in January 2019, before first offshore production in December 2019. [S37]

This timing is one of the most important documented features in the Guyana case.

The timing does not prove that every later rule was optimal.

The fund framework changed.

The 2019 law was replaced in 2021.

Withdrawal provisions changed again in 2024. [S04, S05]

The best-practice question is therefore not whether Guyana established one perfect rule before first oil.

The better question is whether the country created a formal petroleum-revenue institution before recurring production, published financial records, subjected the fund to audit and later revised the framework through law.

Chapter 82. Why a fund exists

A petroleum fund solves several administrative problems.

It gives resource revenue a defined legal destination.

It separates receipt of petroleum cash from automatic public spending.

It creates a custodian.

It defines investment authority.

It creates reporting and audit obligations.

It supports a visible bridge from resource extraction to the national budget.

A fund does not decide whether money is spent well after transfer.

That is the job of public investment, procurement, service delivery and audit systems.

The distinction is central to this paper.

Chapter 83. The legal objective

A resource-fund law should state its purpose clearly.

Possible objectives include:

- stabilization;
- intergenerational savings;
- budget support;
- macroeconomic management;
- preservation of national wealth.

One fund can combine several objectives.

Combining objectives creates trade-offs.

A highly liquid stabilization portfolio is designed differently from a long-term growth portfolio.

A rule designed to support major development spending differs from one designed to preserve capital indefinitely.

The legal purpose should therefore connect to the investment and withdrawal rules.

Chapter 84. The custodian

Guyana's NRF uses the Bank of Guyana as a central operational institution.

The audited statements and quarterly reports provide the financial record. [S44-S47]

A custodian performs functions different from the political budget authority.

The custodian records:

- deposits;
- investments;
- income;
- expenses;
- transfers;
- closing assets.

The budget authority determines lawful public spending under the national fiscal framework.

Keeping the two roles clear improves accountability.

Chapter 85. Cash versus accrual

The 2020 audited NRF statements show why fund analysis needs accounting precision. [S45]

Petroleum revenue recognized in the financial statements included a year-end receivable.

Cash at year end was therefore lower than the accumulated recognized income.

A researcher who reports the cash balance and a researcher who reports accrual revenue are not necessarily contradicting each other.

They are answering different questions.

A national dashboard should state the accounting basis beside every number.

Chapter 86. The 2020 opening record

The first audited period establishes a reference point.

The 2020 statement records royalties, profit oil, limited investment income and no government withdrawal. [S45]

The account also identifies the receivable tied to a late-year government lift.

This gives the flagship a complete example of petroleum moving through three stages:

- 1. cargo entitlement;
- 2. accounting recognition;
- 3. later cash settlement.

The example belongs in the executive training module because the timing issue appears in many resource states.

Chapter 87. The 2021 record

The 2021 audited statements show a larger fund and still no government withdrawal during the year. [S46]

The statement also carries restated comparative data.

Restatement is not an error to hide.

It is part of the financial reporting process.

The flagship should use the latest audited comparative where a later report formally restates an earlier period.

A public data system should preserve both the original publication and the restatement history.

Chapter 88. The first withdrawals

Withdrawals began after the early accumulation period.

The Bank of Guyana reporting provides a dated series of transfers to the Consolidated Fund. [S47]

The analysis should separate:

- legal maximum;
- parliamentary appropriation;
- actual transfer;
- ultimate public expenditure.

A legal maximum is not an instruction to withdraw the full amount.

An appropriation is not a completed project.

A transfer is not an outcome.

Chapter 89. The consolidated budget boundary

Once NRF money enters the Consolidated Fund, it enters the broader public budget.

At this point, the petroleum origin of the cash is economically important but accounting attribution to a specific project depends on budget structure.

If petroleum revenue is pooled with taxes and borrowing, a claim that one named road was paid for by one specific oil cargo requires a traceable appropriation mechanism.

The flagship should therefore distinguish macro-fiscal financing from project tracing.

Chapter 90. The withdrawal rule

A withdrawal rule transforms fund assets into authorized budget space.

A rule can use:

- recent petroleum revenue;
- fund balance;
- expected long-run return;
- a fixed formula;
- a multi-tier formula;
- a fiscal-balance target.

Every formula creates different sensitivity to oil prices and asset values.

Guyana's legal rule changed over time.

The final crosswalk should show each version and apply it only to the periods in which it governed.

Chapter 91. Rules and discretion

A mechanical formula reduces some forms of discretion.

It does not remove political choice.

Parliament still chooses appropriations within the legal framework.

Government still chooses projects.

A formula also depends on data definitions.

If the rule references revenue, the law must define which revenue and which period.

If it references assets, valuation matters.

A strong rule is therefore both mathematical and legal.

Chapter 92. Stabilization

Petroleum prices move sharply.

Production also moves.

A stabilization objective seeks to prevent the public budget from rising and falling one-for-one with resource receipts.

The fund can support this by accumulating assets during high-revenue periods and providing lawful budget support during weaker periods.

The size of the buffer depends on:

- revenue volatility;
- budget rigidity;
- debt;
- import dependence;
- disaster exposure.

An island exposed to hurricanes may need a different liquidity profile from a continental state.

Chapter 93. Intergenerational equity

Oil left underground has option value.

Oil produced today becomes cash.

If all public cash is consumed, future generations inherit less of the original asset unless current spending creates durable capital.

Intergenerational equity therefore does not necessarily mean saving all petroleum revenue as financial assets.

A country can convert resource wealth into:

- education;
- health;
- infrastructure;

- productive capacity;
- resilience.

The challenge is measuring whether the converted asset lasts.

Chapter 94. Financial saving versus domestic investment

External financial saving has advantages.

It diversifies national wealth away from the domestic economy.

It creates liquid assets.

It reduces pressure on local construction markets.

Domestic investment also has advantages when high-return public projects remove constraints.

The comparison depends on project quality.

A poorly designed road is not automatically better than a global bond portfolio because it is domestic.

A high-return drainage, power or education investment is not automatically worse than financial saving because it is spent.

The case study should compare expected public value, risk and maintenance.

Chapter 95. Absorptive capacity

A government can have the legal authority and cash to invest faster than the economy can efficiently deliver.

Absorptive limits include:

- engineering staff;
- contractor capacity;
- imported materials;
- land;
- procurement;
- project management.

When demand exceeds supply, project costs rise.

This is why the IMF's 2026 mission discussed public investment monitoring and absorptive capacity. [S64]

The statement should remain attributed to IMF staff.

The flagship should then test the proposition with independent data.

Chapter 96. Construction inflation

Rapid public construction can increase:

- cement cost;
- aggregate cost;
- equipment rental;
- wages;
- land prices.

The national project register should record original and final contract cost.

Cost growth needs a cause classification.

Possible causes include:

- design change;
- inflation;
- scope expansion;
- delay;
- unforeseen conditions.

This gives a stronger basis for fiscal analysis than the size of the capital budget alone.

Chapter 97. Project sequencing

A large revenue envelope creates a temptation to launch many projects simultaneously.

Project sequencing asks which constraints should be removed first.

For example, a new industrial facility may deliver less value if power and transport remain unreliable.

A new hospital may struggle if staff housing and training are not available.

The public investment system should therefore identify dependencies.

A project portfolio should be reviewed as a network.

Chapter 98. Public investment appraisal

Before procurement, a major project should have an appraisal.

The appraisal should state:

- problem;
- alternatives;
- capital cost;
- operating cost;

- demand;
- economic benefit;
- environmental effect;
- climate resilience;
- implementation risk.

The strongest option may not be the largest project.

A transparent appraisal allows later comparison between forecast and outcome.

Chapter 99. Procurement

A resource windfall increases procurement volume.

This increases the importance of:

- competition;
- qualification;
- bid evaluation;
- contract publication;
- variation control.

Procurement reform is therefore part of resource governance even when it sits outside petroleum law.

The supreme audit institution and parliamentary oversight become part of the petroleum wealth-conversion chain.

Chapter 100. Completion is not operation

A physical asset produces value only after commissioning and operation.

A hospital building without staff is not full health capacity.

A power plant without dependable fuel or transmission is not reliable electricity.

A road without maintenance loses value.

The project register should therefore have separate fields for:

- physical completion;
- commissioning;
- service start;
- service performance.

Chapter 101. Maintenance as a hidden liability

Capital expenditure creates future recurrent spending.

Roads need repair.

Drainage pumps need fuel and maintenance.

Hospitals need staff, medicines and power.

Power systems need overhaul.

A public balance sheet should estimate lifecycle cost.

This prevents the country from building a portfolio whose future maintenance burden exceeds stable revenue after oil growth slows.

Chapter 102. Human capital

Education and health spending create assets that do not appear in conventional government balance sheets.

A petroleum wealth strategy should still measure them.

Indicators include:

- learning;
- certification;
- labour productivity;
- life expectancy;
- disease burden;
- access to care.

Spending alone is insufficient.

The national wealth framework should connect expenditure to outcome.

Chapter 103. Natural capital

Petroleum is part of natural capital.

So are forests, fisheries, water and ecological systems.

Resource extraction reduces one natural asset.

Pollution can reduce others.

Conservation can preserve them.

A complete wealth framework should therefore report natural assets beside financial and produced capital.

This is particularly relevant in Guyana because forest finance predates the oil boom.

Chapter 104. The non-oil primary balance

The non-oil primary balance measures the fiscal position excluding petroleum revenue and interest.

It helps show how dependent public expenditure has become on resource income.

A large non-oil deficit is not automatically unsustainable.

A developing country may choose temporary resource-financed investment.

The indicator still reveals the scale of the transition.

The IMF suggested consideration of a longer-run guide based on the non-oil primary balance. [S64]

The paper should present that as an external proposal rather than its own political recommendation.

Chapter 105. Public debt

A country with petroleum assets may still borrow.

Debt can finance projects before petroleum cash arrives, smooth project delivery or support capital markets.

Debt also creates fixed future payments.

A sovereign balance sheet should therefore report both fund assets and public debt.

Gross fund size alone can overstate net financial wealth.

Chapter 106. Exchange-rate and import effects

Large public spending often increases imports.

Construction equipment, machinery and specialized services may come from abroad.

This creates foreign-exchange demand.

A petroleum fund invested abroad partially offsets domestic liquidity pressure until money is withdrawn.

The macroeconomic section should link withdrawals to imports, exchange rates and inflation.

Chapter 107. Fund investment policy

A sovereign fund needs a risk policy.

Key dimensions include:

- liquidity;
- currency;

- duration;
- credit risk;
- equities;
- governance;
- benchmark.

A young fund expecting high withdrawals has different liquidity needs from a mature fund with stable fiscal use.

The investment policy should match the withdrawal profile.

Chapter 108. Investment return

Investment return should be separated into:

- interest;
- dividends;
- realized gains;
- unrealized gains.

Market value can fall in a year even if petroleum deposits are strong.

The synthetic stress model includes a negative-return year followed by recovery to illustrate this point.

It is not a forecast.

Chapter 109. Fund expenses

Professional management, custody and administration cost money.

A full return measure should be net of fund expenses.

The audit should identify:

- investment fees;
- custody;
- administration;
- external managers.

The materiality of these costs grows with the fund.

Chapter 110. Transparency

A useful public fund report should provide:

- opening balance;
- deposits;
- investment return;

- withdrawals;
- expenses;
- closing balance.

It should state currency and accounting basis.

Annual audit provides assurance over the financial statements.

Quarterly reporting improves timeliness.

The Bank of Guyana publishes both annual and quarterly material. [S44, S47]

Chapter 111. Audit

An external financial audit answers whether statements are fairly presented under the identified reporting framework.

It does not answer whether withdrawal policy was economically optimal.

It does not answer whether projects delivered value.

The paper should preserve these boundaries.

Different reviewers answer different questions.

Chapter 112. Stress test one: low petroleum revenue

The fund-stress model includes a synthetic case where deposits fall for several years.

The purpose is to test:

- whether withdrawals remain affordable;
- whether liquid assets are sufficient;
- whether the budget needs adjustment.

The case does not predict Guyana's future receipts.

Chapter 113. Stress test two: high spending

A second synthetic case increases withdrawals while deposits remain high.

The fund still grows in the example, but more slowly.

This demonstrates that a rising balance does not reveal the opportunity cost of spending.

Project quality remains the key question.

Chapter 114. Stress test three: market loss

A fund invested in markets can experience negative return.

The synthetic model starts with a market loss and later positive returns.

The case tests whether spending rules force asset sales after a loss.

A liquidity reserve can reduce that risk.

Chapter 115. Stress test four: combined disaster

A small producer should test a natural disaster occurring during weaker petroleum revenue.

Emergency spending rises while revenue falls.

Infrastructure damage may also interrupt production.

This combined stress case belongs in the island-state blueprint.

Chapter 116. Reporting public wealth

A national petroleum dashboard should therefore have four panels.

Panel one shows petroleum production and government entitlement.

Panel two shows the NRF.

Panel three shows the national budget and public investment.

Panel four shows outcomes and national assets.

This structure prevents a large fund balance from becoming a substitute for development evidence.

Chapter 117. What Guyana established early

The strongest factual statement is about timing.

A statutory resource fund existed before first oil. [S37]

The country also established an operational relationship with the central bank before significant production revenue accumulated.

This sequence gives new producers a concrete institutional milestone to examine.

Chapter 118. What changed later

The framework did not remain fixed.

Later legislation changed governance and withdrawal rules.

This shows that pre-first-oil preparation is not the end of policy design.

A resource framework needs lawful revision as evidence changes.

The revision process itself should remain transparent.

Chapter 119. Transferability

A newcomer country should decide first what problem its fund is intended to solve.

It should then align:

- deposit rules;
- investment;
- withdrawal;
- budget integration;
- audit.

It should not copy a withdrawal percentage without modelling:

- reserves;
- revenue;
- debt;
- development gap;
- disaster risk.

Chapter 120. Part IV findings

Guyana's NRF case offers a sequence rather than a fixed formula.

The fund was established before first oil.

Audited financial statements provide a public account.

Withdrawals later moved revenue into the budget.

The law changed over time. [S37, S44-S47]

The blueprint lesson is to build the petroleum-to-budget accounting chain early, preserve independent financial reporting, and evaluate public investment separately from fund accounting.

A country ultimately needs both.

A clean fund without effective public investment leaves development opportunities unrealized.

Large public investment without financial control risks converting a finite resource into weak assets.

The purpose of the system is to convert depleting petroleum into a durable national balance sheet.

Edition 2.0 deepening: the Natural Resource Fund as one account inside a national wealth system

Analytical section 120A. A sovereign resource fund should solve a defined problem

A resource fund can serve several purposes, but those purposes should not be blurred.

It can:

- stabilize the budget;
- save for future generations;
- separate volatile petroleum receipts from annual spending decisions;
- preserve foreign assets;
- improve transparency;
- create a rules-based bridge into the national budget.

These objectives can conflict.

A rule optimized for stabilization may not maximize long-run saving.

A rule designed to build financial assets may constrain near-term development spending.

A state with large infrastructure gaps may decide to spend more than a mature high-income producer.

The design question is therefore not whether a country should "save" or "spend".

The question is how much can be spent productively without creating macroeconomic instability, weak projects or permanent expenditure commitments that outlive volatile petroleum revenue.

Analytical section 120B. The fund should be reconstructed as a stock-flow identity

The cleanest audit of a sovereign fund uses a simple identity:

opening balance plus petroleum inflows plus investment income plus or minus valuation and currency effects minus lawful withdrawals equals closing balance.

Every annual and quarterly report should reconcile to this identity.

The Bank of Guyana reporting architecture makes it possible to reconstruct the NRF across audited annual statements and quarterly or monthly reports. [S44-S47, S88, S109-S110]

The value of the reconstruction is not only arithmetic.

It separates four concepts that are often confused:

- petroleum revenue;

- investment return;
- withdrawal;
- fund balance.

A large withdrawal can occur in a year when petroleum receipts are also large.

A lower closing balance does not automatically mean the fund "lost" money.

The explanation may be lawful transfers to the Consolidated Fund.

Analytical section 120C. Cash and accrual accounting should not be mixed casually

Petroleum cash flows can straddle reporting periods.

A cargo lifted in December can be paid in January.

A quarterly report may recognize an accrual before the cash reaches the bank account.

A government budget may refer to expected receipts for a full year.

These are different accounting boundaries.

The Q4 2025 NRF report explicitly describes accrual accounting and notes petroleum payments due after year-end for some December lifts. [S109]

This is a valuable teaching example.

A publication should never add an accrual figure to a later cash receipt without checking whether they refer to the same transaction.

The NRF reconciliation workbook should therefore include:

- transaction date;
- recognition date;
- cash receipt date;
- accounting basis;
- source report.

Analytical section 120D. The custody chain is part of sovereign risk management

The resource fund is not simply a line in the budget.

It is a custody arrangement.

The state should know:

- where the assets are held;
- who can authorize transfers;

- who reconciles the account;
- who audits the financial statements;
- what investment mandate applies;
- what operational controls prevent unauthorized movement.

Bank of Guyana reporting identifies the central bank's operational role and the investment mandate. [S109]

A new producer should test the entire custody chain before the first material petroleum receipt.

A simulated transfer should verify bank instructions, dual authorization, accounting entries, reconciliation and audit evidence.

Analytical section 120E. Investment policy should match the purpose and horizon of the fund

A fund that may be used heavily for development spending has a different liquidity need from a mature intergenerational fund.

Investment policy should therefore distinguish:

- liquidity reserve;
- near-term withdrawal needs;
- long-duration savings;
- currency exposure;
- market risk;
- operational risk.

A conservative early-stage mandate can be rational when withdrawal needs are uncertain and institutions are still maturing.

Over time, the state can decide whether a diversified portfolio is appropriate.

The decision should be based on explicit risk tolerance, not a desire to imitate another sovereign fund.

Analytical section 120F. Withdrawal rules should be tested against revenue volatility

A withdrawal formula can look prudent under a high-price base case and become procyclical under stress.

The rule should therefore be tested under:

- oil price decline;
- delayed startup;
- major outage;

- cost-recovery increase;
- global recession;
- domestic disaster;
- investment-market loss.

The question is not only whether the formula is legal.

The question is whether the resulting budget path remains sustainable.

A country should know how quickly public spending would need to adjust if petroleum receipts fall.

Analytical section 120G. The non-oil primary balance is the key macro bridge

The fund balance alone does not show the fiscal impulse delivered to the domestic economy.

The non-oil primary balance is useful because it asks how large the fiscal position would be without petroleum revenue and interest.

A rapidly widening non-oil deficit can be sustainable for a period if the state is deliberately converting resource wealth into high-return public assets.

It becomes risky when spending growth exceeds implementation capacity or creates permanent recurrent obligations without future financing.

The indicator should therefore be read with public-investment quality, not in isolation.

Analytical section 120H. Absorptive capacity is measurable

"Absorptive capacity" should not remain a vague warning.

A government can track it through:

- project completion delays;
- procurement lead times;
- construction-cost inflation;
- skilled-labour vacancies;
- imported construction inputs;
- contractor concentration;
- project redesign;
- maintenance backlogs;
- land and housing prices.

If these indicators deteriorate as public capital spending rises, the marginal return on additional spending may be falling.

The response is not necessarily to stop investment.

It may be to sequence projects, expand engineering capacity, improve procurement and prioritize maintenance.

Analytical section 120I. Public investment should be tracked through an asset register

A budget appropriation is not a national asset.

A completed structure is not necessarily an operating public service.

The public-investment chain should therefore record:

- 1. project concept;
- 2. appraisal;
- 3. budget;
- 4. procurement;
- 5. contract;
- 6. construction progress;
- 7. commissioning;
- 8. operating agency;
- 9. maintenance plan;
- 10. service outcome.

For a resource-rich state, this asset register is the other half of the sovereign fund.

The NRF records financial wealth.

The public asset register records what withdrawals became.

Analytical section 120J. Maintenance converts construction into durable wealth

Resource booms create incentives to build visible capital projects.

The less visible question is whether those assets can be operated and maintained.

Every major public project should therefore have a lifecycle-cost estimate before construction begins.

The estimate should include:

- staffing;
- utilities;
- spare parts;
- periodic rehabilitation;
- digital systems;
- insurance;

- replacement cycle.

A hospital without recurrent staffing or a road without maintenance is not equivalent to a durable increase in national wealth.

Analytical section 120K. Human capital belongs in the wealth-conversion ledger

Petroleum revenue can finance education and health, but expenditure is not the same as human-capital formation.

The state should track:

- learning outcomes;
- tertiary completion;
- technical certification;
- health access;
- mortality and morbidity indicators;
- labour-force participation;
- occupational shortages.

The public wealth framework should therefore include financial, physical and human capital.

This makes it possible to compare competing uses of a finite resource.

Analytical section 120L. Natural capital prevents false accounting

A state can increase financial assets while degrading ecosystems.

A comprehensive wealth framework should record major natural-capital changes alongside fiscal assets.

For Guyana this is especially relevant because petroleum development coexists with forest-carbon policy, biodiversity assets, freshwater systems and coastal vulnerability.

The objective is not to reduce every ecological value to one monetary number.

The objective is to prevent the public balance sheet from treating natural assets as if they were free.

Analytical section 120M. A national wealth statement would improve accountability

An annual national wealth statement could present five accounts:

- petroleum depletion;
- NRF financial assets;
- public physical assets;
- human-capital indicators;

- natural-capital indicators.

The statement would not need to claim perfect valuation.

Its value would be conceptual discipline.

Citizens could see whether petroleum extraction is being matched by growth in other durable assets.

This is a stronger intergenerational test than the size of the NRF alone.

Analytical section 120N. The 2025 NRF record shows why reconciliation matters

The Bank of Guyana's Q4 2025 report provides a useful case because it records the year-end market value, quarterly inflows, transfers, investment return and the investment mandate in one report. [S109]

A reader can therefore observe that fund performance is affected simultaneously by new petroleum revenue, investment income and withdrawals.

This is exactly why political or media discussion should not infer fund performance from one number.

The same discipline should be carried into 2026 using the monthly reports and Ministry of Finance mid-year reporting. [S07, S110]

Analytical section 120O. Expanded Part IV conclusion

The Natural Resource Fund should be treated as a control account in a much larger national wealth system.

Its first job is to record and safeguard petroleum receipts under law.

Its second job is to create a disciplined interface with the budget.

Its third job is to preserve evidence about how much of a finite resource has been converted into financial assets.

But the development question begins after the withdrawal.

A state ultimately succeeds only if withdrawals become productive, maintained and socially valuable assets without destabilizing the economy.

For a newcomer, the operating model should therefore link four ledgers:

- petroleum entitlement;
- sovereign fund;
- national budget;
- public asset and outcome register.

The country should be able to trace one petroleum dollar through all four.

That is the standard required for true resource-to-wealth accountability.

Edition 2.0 evidence deepening: the year-by-year NRF record

Analytical section 120P. The NRF should be read as a continuous ledger

Edition 2.0 adds `NRF\_ANNUAL\_LEDGER.csv` because the sovereign-fund story is best understood as a continuous stock-flow account rather than a sequence of press releases.

The table separates:

- opening balance;
- petroleum and other eligible inflows;
- investment return;
- withdrawals;
- closing balance;
- accounting basis;
- reconciliation status.

This immediately reveals whether an apparent change in the fund is driven by petroleum revenue, financial return, withdrawal or accounting timing.

Analytical section 120Q. 2020: the first accounting year

The early NRF record contains an important accounting lesson.

The 2020 Q4 reporting shows both cash and a profit-oil receivable. That means a cargo-related amount could belong to the economic activity of one period while cash settlement occurred in the next. [S45]

The final historical ledger should therefore preserve receivables rather than treating every year-end balance as cash physically received by 31 December.

This is why the 2020 and 2021 rows remain explicitly marked as provisional early-year bridges until the audited annual statements are transcribed line by line into the master dataset.

The choice to label an early row provisional is a strength, not a weakness.

It prevents false precision.

Analytical section 120R. 2021: accumulation before withdrawals

The 2021 year-end fund balance increased materially while the fund had not yet entered the later period of large budget transfers.

This provides a useful baseline for evaluating the transition from accumulation to withdrawal.

The institutional question is not whether withdrawal is inherently better or worse than accumulation.

It is whether the transition occurs through the legal budget process and whether the assets financed by withdrawals can be traced.

Analytical section 120S. 2022: the first full withdrawal cycle

The 2022 record is the first clear example in the ledger of large petroleum inflows occurring alongside material withdrawals.

Official reporting records approximately US\$607.65 million transferred to the Consolidated Fund during the year and a year-end fund value around US\$1.43 billion. [S44, S47]

This is the point at which sovereign-fund governance and public-investment governance become inseparable.

The fund can remain correctly accounted for while the development value of the withdrawal depends on a separate budget and project-delivery system.

Analytical section 120T. 2023: the importance of accounting basis

The 2023 annual reporting states petroleum inflows around US\$1.608 billion, withdrawals around US\$1.002 billion and investment return around US\$86.84 million, with a year-end market value around US\$2.122 billion. [S44]

A separate public oversight source reported a lower closing figure in a different context.

This discrepancy is analytically useful.

A responsible publication should not silently choose the number that best fits the narrative.

It should first determine whether the difference reflects:

- cash versus accrual;
- receivables;
- reporting date;
- market value versus cash;
- transcription.

Edition 2.0 therefore treats the Bank of Guyana fund statement as the controlling financial source and records the alternate figure as a reconciliation note until the basis is fully matched.

Analytical section 120U. 2024: simultaneous high inflow and high withdrawal

The 2024 year illustrates why a closing balance cannot be read without the flow statement.

Official reporting records annual petroleum inflows of approximately US\$2.568 billion and withdrawals of approximately US\$1.586 billion. The year-end market value was approximately US\$3.246 billion. [S44]

Using the official GYD stock-flow figures, the implied annual investment result is approximately US\$141 million, consistent with public reporting on interest income.

The publication labels that annual return as derived from the official identity unless and until the exact annual line is transcribed from the final annual statement.

That disclosure matters because a derived figure should not be represented as a direct quote.

Analytical section 120V. 2025: withdrawals approach annual inflows

The 2025 record is particularly important for long-run fiscal analysis.

Annual reporting records inflows of approximately US\$2.510 billion, investment income around US\$142.13 million and withdrawals around US\$2.463 billion, with a year-end fund value around US\$3.435 billion. [S88, S109]

The key observation is not that the fund balance still rose.

It is that annual withdrawals were close in magnitude to annual petroleum inflows.

A fiscal sustainability analysis should therefore ask:

- Is this level of withdrawal temporary or structural?
- How much recurrent spending does it support?
- What happens if oil price or production falls?
- How much is being converted into productive assets?
- How quickly is the fund expected to grow under stress?

Those are forward-looking policy questions.

The ledger supplies the historical evidence needed to model them.

Analytical section 120W. H1 2026: partial-year data must stay partial

The H1 2026 official mid-year reporting records government petroleum receipts and a June 30 fund balance after first-half withdrawals. [S07]

The book should resist a common analytical shortcut: doubling the first-half number and calling it an annual result.

Second-half production, cargo timing, prices, startup schedules and withdrawals can differ materially.

The row is therefore labelled H1 2026 and remains visibly separate from audited full-year rows.

Analytical section 120X. The withdrawal-to-inflow ratio is a useful but incomplete indicator

One useful annual indicator is:

withdrawals / petroleum inflows.

A rising ratio shows that a larger share of current-year petroleum revenue is being transferred to the budget.

The indicator does not answer whether the withdrawal is prudent.

It must be read with:

- fund stock;
- expected future revenue;
- public debt;
- non-oil primary balance;
- investment quality;
- recurrent spending.

It is a diagnostic, not a policy verdict.

Analytical section 120Y. The fund-to-budget ratio should be paired with capital execution

A second indicator is petroleum-funded budget transfer relative to total central-government expenditure or capital expenditure.

This shows the fiscal importance of the fund.

Again, the ratio is not sufficient.

If capital execution is weak or projects are delayed, a high transfer may increase domestic liquidity faster than productive capacity.

The fiscal dashboard should therefore pair:

- withdrawal;
- capital appropriation;
- capital execution;
- commissioned assets.

Analytical section 120Z. The NRF ledger should reconcile to petroleum entitlement

The fund ledger begins at the point of revenue deposit.

The petroleum revenue chain begins earlier.

Edition 2.0 should ultimately connect:

- government cargo entitlement;
- sale;
- royalty invoice;
- settlement;
- NRF deposit.

If a government cargo is lifted in December and paid in January, the bridge should show that timing.

This creates one continuous chain from physical petroleum to sovereign financial asset.

Analytical section 120AA. The ledger should be reproducible by an outside reviewer

The final dataset should provide:

- source document;
- table/page;
- GYD figure;
- USD figure;
- exchange rate where relevant;
- accounting basis;
- whether direct or derived;
- formula.

An outside reviewer should be able to reconstruct the annual identity without asking the author for a private spreadsheet.

Analytical section 120AB. A fund can grow while fiscal dependence also grows

This is one of the most important conceptual points in resource governance.

A larger year-end fund balance can coexist with:

- higher annual withdrawals;
- a wider non-oil deficit;
- higher recurrent expenditure;
- larger public investment;
- rising dependence on petroleum transfers.

The fund stock and the fiscal flow answer different questions.

A country can be accumulating financial assets and increasing budget dependence at the same time.

Analytical section 120AC. The long-run stress model should start from the reconciled history

Illustrative future scenarios should not begin from an arbitrary opening balance.

They should begin from the most recent verified closing fund balance.

They should then model:

- petroleum receipts;
- investment return;
- withdrawal rule;
- expenditure adjustment;
- production delay;
- price shock.

This makes the scenario model auditable.

The synthetic stress tables already in the repository should therefore be linked to the NRF annual ledger in the final Edition 2.0 methods appendix.

Analytical section 120AD. Evidence conclusion

The year-by-year ledger changes Part IV from an institutional description into a financial history.

It shows exactly why fund governance should be judged through both stock and flow.

The final book should continue this record annually.

A sovereign-fund system becomes more credible when every new year can be added to the same reconciliation without changing definitions.

Case study: how a prior-year petroleum receivable can distort a current-year revenue narrative

The question an auditor must answer

The central test in petroleum wealth accounting is whether the state can follow one underlying economic transaction through production entitlement, cargo lifting, sale, accrual recognition, cash receipt, sovereign-fund custody and budget transfer. The entries occur in different records and sometimes in different calendar years. In a fast-growing oil province, aggregate receipts can become large enough for a single delayed cargo to produce a material discrepancy between otherwise credible official reports. This is not necessarily an accounting error. It is a requirement for a documented timing bridge.

Guyana's 2025 year-end and first-half 2026 Natural Resource Fund reports provide a concrete example. The Bank of Guyana's audited annual accounts establish the position at 31 December 2025. Its December monthly statement separately reports cash-basis holdings. The government's mid-year report describes cash received during January-June 2026, including settlement of some cargoes lifted in the last quarter of 2025. A researcher who treats all these amounts as new 2026 accrual revenue will double-count part of the economic flow. The case is important precisely because the primary sources, taken individually, can all be correct while an improperly assembled chart is not. [S07, S88, S134, S137]

What each primary record measures

The December 2025 monthly statement reports about US\$3,250.41 million of cash and cash equivalents. The audited year-end and subsequent Q1 comparative reports give a larger total fund position of approximately US\$3,434.53 million because a receivable of GYD 38,387,708 thousand was recognized at year-end. At the report's stated GYD 208.5/US\$ conversion, that receivable is about US\$184.11 million. The equality between the audited position and cash plus receivables can be checked against the Bank's original GYD-thousand figures before any currency conversion. [S88, S134, S137]

These are two legitimate measures with different boundaries. A monthly cash statement answers how much had reached the bank by a particular date. The year-end accrual position also includes amounts earned and recognized but not yet received in cash. Neither measure should be called a more current or more accurate version of the other without specifying the question.

The Ministry's first-half 2026 report records US\$1,778.6 million of government profit-oil cash receipts and US\$218.4 million in royalties, or US\$1,997.0 million combined. Its accompanying narrative says the profit-oil receipts include payments for three government lifts made in the final quarter of 2025 as well as receipts from 2026 lifts. The same report describes about US\$66.9 million of first-half investment income, US\$1,020 million of withdrawals and a US\$4,294.2 million fund balance at 30 June. The Ministry's statement is a receipt-period account and should be cited as such. [S07]

Reproduce the apparent discrepancy before interpreting it

Starting with the rounded audited 2025 closing position of US\$3,434.53 million, suppose the full US\$1,997.0 million of reported H1 cash receipts is entered again as 2026 fund inflow. Add US\$66.9 million of return and subtract US\$1,020 million of transfers. The resulting hypothetical closing balance is US\$4,478.43 million. That is US\$184.23 million higher than the reported June closing balance of US\$4,294.2 million.

It would be incorrect to present this US\$184.23 million as money missing from the fund. The starting balance is an accrual-based stock that already contains a prior-period claim, while the supposed current-year inflow includes cash settlements of prior-period transactions. The legitimate reconciliation problem is to determine how much of the US\$1,997 million represents settlement of an asset already recognized in the opening stock.

Rearrange the stock-flow identity without inventing that missing detail. If the rounded values contain no other measurement adjustments, the inflow required to reproduce the closing balance is US\$4,294.2m minus US\$3,434.53m plus US\$1,020m minus US\$66.9m, or US\$1,812.77 million. This is an *implied balancing inflow*, not an independently observed 2026 revenue figure. It is intentionally left separate from the government's US\$1,997m cash-receipt total in the working ledger.

The resulting US\$184.23 million difference is extraordinarily close to the roughly US\$184.11 million receivable shown by the 2025 closing and Q1 2026 comparative records. On rounded figures, the difference after accounting for that receivable is about US\$0.12 million. A candidate explanation is therefore that collection of a prior-year profit-oil receivable in 2026 appears among the cash receipts described by the Ministry but was already present in the audited opening fund position. The Ministry's reference to three late-2025 lifts supports the timing hypothesis. It does not, by itself, identify every transaction in the bank ledger. [S07, S88, S134, S137]

The evidence needed to certify the bridge

A final audit should match the three 2025 cargoes individually. For each, it should establish the production and entitlement period, lift date, sale invoice, valuation formula, receivable recognized at 31 December, cash settlement date, bank deposit posting and reporting-period classification. The aggregate should equal the reduction or realization of year-end receivables in the Q1 and Q2 2026 fund statements, subject to independently documented adjustments.

The review must also compare like currency bases. The Bank reports original GYD-thousand values and a stated conversion rate in its monthly and quarterly materials. A Ministry report may use rounded US-dollar equivalents. A hundred thousand US dollars of apparent discrepancy can arise from rounding at several steps; but the reviewer should show the rounding calculation rather than simply assume it explains the difference.

The independent check should use the exact Q1 and Q2 2026 Bank of Guyana financial summaries, not a government speech or secondary tracker in place of the bank's ledger. The Q1 report records its own opening balance, inflows, return, Consolidated Fund transfer and closing position in GYD thousands. That record can be verified directly against the previous quarter's closing balance and the subsequent quarter's opening balance. The Q2 report is listed in the Bank's public quarterly index; a page-specific extraction of its full original table remains a final-evidence task. [S134, S136]

A prudent reviewer should also check whether the Ministry's reporting language refers strictly to deposits in the quarter, cash collections from cargo sales or revenue recognized during the period. That definition—not a conveniently chosen total—should govern the comparison. If the definition changed between reporting years, both series must be presented separately or restated with a documented method.

Consequences for the fiscal and development chapters

A mislabeled revenue denominator propagates beyond the NRF table. If a current-year withdrawal ratio uses cash receipts while an annual fund accumulation calculation uses accrual income, the two indicators cannot be compared without a bridge. If a forecast assumes all late-year lifts settle before 31 December, its cash balance may overstate available end-year liquidity. A budget stress test based on the larger number could then exaggerate near-term financing space. These risks apply to any first-oil state with relatively few, large cargoes.

The publication should therefore maintain three distinct petroleum series: **economic entitlement by production period, revenue recognized by financial reporting period,** and **cash deposited by settlement period**. Each series has a legitimate use. The first supports petroleum fiscal modeling and government-take analysis. The second supports audited financial statements. The third supports liquidity and budget cash management. The cargo-level reconciliation links them.

This also clarifies how a resource fund should communicate its performance to citizens. A cash balance, total net assets, annual petroleum inflows, investment return and transfers are not interchangeable terms. An increase in audited assets can include receivables; a rise in cash received can include settlement of claims recognized the previous year; a large transfer to the budget may leave a different net asset trajectory. The public account should show the principal reconciliation entries alongside its headline figures.

Release disposition

The Edition 2.0 working ledger flags its first-half 2026 stock-flow row **OPEN**. The 2025 receivable is a strongly evidenced candidate that accounts for nearly all the rounded apparent gap. Until the individual cargo settlements and exact GYD Q2 entries have been matched, neither the underlying revenue allocation nor the small residual should be described as independently reconciled. This narrower conclusion is more informative than either concealing the discrepancy or calling the difference an unexplained loss. It demonstrates the method the book expects other new producers to apply to their own petroleum receipts.

PART V | Environmental governance and spill readiness

Environmental Governance, Offshore Risk and Climate Coexistence

Chapter 121. Environmental governance is a chain

Environmental performance should not be inferred from the existence of an environmental law.

A functioning offshore environmental system contains several distinct steps:

- baseline science;
- impact assessment;
- public participation;
- authorization;
- operating conditions;
- monitoring;
- inspection;
- incident reporting;
- enforcement;
- remediation;
- financial responsibility.

Each step produces its own evidence.

A permit proves that an authority granted approval under specified conditions.

A monitoring report shows what the operator measured.

An inspection shows what the regulator observed.

A laboratory result provides a measurement.

An enforcement notice shows how a deviation was addressed.

A restoration assessment shows whether ecological conditions recovered.

The Guyana case needs all of these layers.

Chapter 122. The legal foundation

Guyana's Environmental Protection Act predates commercial offshore oil. [S39]

This gives the state a general environmental framework before the petroleum boom.

The petroleum-specific question is how that framework is applied to deepwater development.

A general environmental statute needs project-level implementation through:

- EIA requirements;
- permits;
- standards;
- plans;
- audits;
- enforcement.

The final case should therefore move from statute to project.

Chapter 123. Liza Phase 1 environmental sequence

The official permit record gives a clear chronology.

The project application was submitted on 5 July 2016.

The approved EIA and Environmental and Socioeconomic Management Plan are dated 1 June 2017.

The permit was issued on 1 June 2017.

The permit record also references revised oil-spill and wildlife-response plans.

The permit was later modified on 12 May 2021. [S48]

This is important because it shows environmental authorization occurring before first production.

It also shows the permit changing after operating experience.

Chapter 124. Environmental impact assessment

An EIA is a forecast and decision-support document.

It identifies:

- baseline conditions;
- project activities;
- impacts;
- alternatives;
- mitigation;
- residual effects;
- monitoring.

An EIA is strongest when assumptions are explicit.

For an offshore project, important assumptions include:

- currents;

- spill rate;
- spill duration;
- oil weathering;
- drilling waste;
- produced water;
- flaring;
- vessel traffic;
- marine species.

The final paper should compare predicted conditions with observed operations where later data exist.

Chapter 125. Baseline ownership

Environmental baseline data have public value beyond one project.

The government should know:

- who collected the data;
- who paid;
- who owns the raw data;
- whether methods are reproducible;
- whether independent researchers have access.

A baseline locked inside a project consultant's report is less useful for cumulative basin management.

A newcomer country should negotiate data rights before approvals.

Chapter 126. Public participation

Public participation should be measured by more than a notice.

The Yellowtail public record provides an example of an EIA requirement and public submission process. [S52]

The flagship should ask:

- who received notice;
- how long the comment period lasted;
- where documents were available;
- which groups participated;
- what issues were raised;
- what the authority changed;
- what remained unresolved.

This produces a participation audit trail.

Chapter 127. Permit conditions

A permit converts environmental findings into enforceable conditions.

Important offshore conditions include:

- flaring;
- emissions reporting;
- discharge;
- waste;
- monitoring;
- spill plans;
- wildlife response;
- audits;
- reporting deadlines.

The permit should be treated as a versioned document.

Amendments should remain visible.

Chapter 128. Liza Phase 1 permit modification

The EPA's public notice on the Liza Phase 1 modification states that the revised permit added emissions reporting and flaring-related requirements, including specified time constraints and a payment associated with flaring beyond those timelines. [S55]

This is an example of adaptive regulation.

The operating issue did not remain solely inside the original EIA.

The regulator changed permit conditions.

The next question is whether the later operating record shows compliance.

That requires actual flaring data.

Chapter 129. Flaring

Flaring has several categories.

Routine flaring is different from:

- commissioning flaring;
- emergency flaring;
- maintenance flaring;
- equipment-failure flaring.

The environmental impact depends on volume, composition and duration.

The final project table should record:

- gas flared;
- reason;
- start;
- end;
- permit condition;
- emissions;
- regulator response.

A company statement that flaring returned to background level is useful evidence of its position, not a substitute for the regulator's measured record.

Chapter 130. Gas reinjection

Offshore FPSOs often reinject produced gas to support reservoir pressure and reduce routine flaring.

The system therefore connects production optimization and environmental performance.

A compression failure can affect both.

The environmental chapter should explain:

- design;
- redundancy;
- downtime;
- restart;
- flare impact.

This helps readers understand why flaring may occur during commissioning or equipment failure.

Chapter 131. Produced water

Produced water is another operating stream.

The environmental assessment should state:

- expected volume;
- treatment;
- discharge standard;
- monitoring;
- chemical content.

The final case needs project-level observed data before drawing conclusions.

A permit limit is not an observed discharge.

Chapter 132. Drilling waste

Development drilling produces cuttings, fluids and other waste.

The EIA and permit should identify:

- treatment;
- disposal;
- discharge;
- transport;
- hazardous classification.

A basin with many wells accumulates waste across projects.

Cumulative management matters.

Chapter 133. Marine traffic

Offshore petroleum increases:

- supply vessels;
- tankers;
- helicopters;
- support traffic.

These create:

- collision risk;
- noise;
- emissions;
- wildlife interaction.

A cumulative assessment should include marine traffic density rather than analyzing each FPSO as if it operates alone.

Chapter 134. Underwater noise

Seismic acquisition, drilling and vessel operations create underwater sound.

Potential receptors include:

- marine mammals;
- fish;
- turtles.

A strong monitoring framework records:

- source level;

- duration;
- season;
- species observations;
- mitigation.

The final paper should state where direct Guyana-specific data are absent.

Chapter 135. Fisheries

Fisheries matter economically and socially.

A fisheries baseline should include:

- catch;
- landed value;
- employment;
- export;
- subsistence;
- season;
- location.

Without a baseline, compensation after an incident becomes more difficult to evaluate.

An island producer with high tourism and fishery dependence faces especially concentrated marine exposure.

Chapter 136. Biodiversity

Guyana's offshore environment sits beside a country with extensive terrestrial biodiversity and forests.

The petroleum case should not allow the visibility of forests to obscure marine biodiversity.

The environmental asset register should contain both.

The national wealth chapter should avoid subtracting one environmental gain from another loss without a defensible common accounting method.

Chapter 137. Oil spill risk

Oil spill risk is a combination of:

- probability;
- release size;
- duration;
- trajectory;
- receptor vulnerability;

- response.

A worst-case scenario does not state the probability of that event.

It tests whether institutions and resources are sufficient if it occurs.

This distinction should remain explicit.

Chapter 138. Prevention

Spill governance begins before response.

Prevention includes:

- well design;
- barriers;
- blowout preventers;
- maintenance;
- operating procedures;
- training;
- inspection.

A response plan does not substitute for prevention.

The final case should separate prevention requirements from emergency assets.

Chapter 139. Detection and notification

A response begins with detecting an event and notifying the right authorities.

The operational standard should define:

- detection method;
- notification deadline;
- regulator;
- maritime authority;
- emergency command;
- neighboring states.

A tabletop exercise should measure actual notification time.

Chapter 140. Incident command

An offshore spill involves more than the operator.

Government responsibilities can include:

- environmental decisions;
- maritime safety;

- fisheries;
- health;
- coast guard;
- civil defence;
- public communication.

The incident command structure should identify who leads each function.

Ambiguity during an emergency increases delay.

Chapter 141. Capping stack

A subsea loss of well control may require capping equipment.

The Uaru production licence contains spill-response requirements including capping capability.
[S56]

The final review should identify:

- equipment source;
- location;
- mobilization time;
- compatibility;
- transport;
- installation.

A subscription agreement is not the same as equipment located at the well.

Both facts matter.

Chapter 142. Dispersants

Dispersant use involves environmental trade-offs.

A response plan should define:

- approval authority;
- product;
- stock;
- toxicity data;
- conditions for use;
- monitoring.

The presence of dispersant inventory does not establish that use is appropriate in every spill.

Chapter 143. Mechanical recovery

Booms and skimmers work differently under different sea conditions.

The response capability assessment should record:

- equipment;
- capacity;
- sea-state limits;
- storage;
- waste disposal.

Nominal equipment capacity should be distinguished from realistic offshore recovery.

Chapter 144. Shoreline protection

If a spill threatens the coast, priorities need to be established before impact.

Sensitive sites include:

- mangroves;
- fisheries;
- communities;
- industrial intakes.

A shoreline-response atlas should rank protection actions based on ecological and social evidence.

Chapter 145. Transboundary response

Ocean currents do not follow national boundaries.

A serious offshore incident can affect neighboring states.

A transboundary plan needs:

- notification;
- joint surveillance;
- evidence;
- responder access;
- claims;
- diplomacy.

Maritime boundary agreements do not automatically create environmental response arrangements.

Chapter 146. The 2025 oil-pollution law

Parliament records the Oil Pollution Prevention, Preparedness, Response and Responsibility Act 2025. [S53]

The Official Gazette records publication. [S54]

At the research cut-off, the project has not treated every provision as fully operational because commencement and implementing regulations still require verification.

This is a critical legal-writing rule.

Enacted does not necessarily mean commenced.

Commenced does not mean fully implemented.

Chapter 147. Financial responsibility

Spill response depends on money.

Financial responsibility is not one instrument.

It can include:

- operator assets;
- insurance;
- parent guarantee;
- affiliate support;
- response contracts;
- public emergency spending.

Each layer has different triggers and limits.

The paper should not combine them into one headline figure without reading the instruments.

Chapter 148. Legal liability versus security

Legal liability describes what a party owes under law.

Financial security describes funds or guarantees available to satisfy obligations.

An operator might have broad legal liability while a specific guarantee has a stated limit.

The two concepts should remain separate.

This distinction is central to the financial-assurance litigation.

Chapter 149. Court review

A 2026 appellate dispute concerning financial assurance is material to the environmental case.

The final manuscript must use the authenticated judgment before making a legal interpretation.

Government summaries and stakeholder commentary should be attributed.

The case illustrates why environmental governance includes courts alongside regulators.

Chapter 150. Insurance

A useful insurance review asks:

- insured party;
- insurer;
- policy period;
- limit;
- deductible;
- exclusions;
- territory;
- pollution coverage;
- defense cost;
- claims trigger.

An insurance certificate alone provides less information than the full policy.

Where the full instrument is not public, the evidence gap should be stated.

Chapter 151. Parent guarantees

A parent guarantee can add financial strength.

Its value depends on:

- guarantor;
- amount;
- trigger;
- law;
- duration;
- termination;
- enforcement.

A guarantee should be evaluated on its legal terms and guarantor capacity.

Chapter 152. Cleanup cost and ecological damage

A serious spill creates several cost categories:

- source control;
- offshore response;
- shoreline cleanup;
- waste;
- fishery loss;
- tourism loss;

- ecological restoration;
- monitoring;
- legal claims.

A financial assurance assessment should test all material categories.

One cleanup-cost estimate should not be treated as a complete loss estimate.

Chapter 153. Long-term monitoring

Environmental damage may persist after visible cleanup ends.

Monitoring can include:

- sediment;
- water;
- fisheries;
- habitat;
- species;
- health.

The responsible party and funding duration should be established in law or settlement.

Chapter 154. Cumulative impact

The cumulative environmental question becomes more important as the number of projects grows.

Each project may satisfy its permit and still contribute to a larger cumulative load.

Relevant basin-level measures include:

- total flaring;
- total emissions;
- total produced water;
- vessel traffic;
- underwater noise;
- waste;
- incident frequency.

The regulator needs project and basin views.

Chapter 155. Regulatory workload

Environmental oversight has a capacity dimension.

Metrics include:

- permits per specialist;

- inspections;
- audit reviews;
- data volume;
- unresolved notices.

As production expands, staffing and data systems should scale.

A permit system that works for one project should not be assumed adequate for six.

Chapter 156. Independent monitoring

Independent monitoring can strengthen confidence where:

- methods are public;
- laboratories are accredited;
- raw data are accessible;
- conflicts are controlled.

Independence should be demonstrated by governance and method rather than the label of a contractor.

Chapter 157. Environmental data portal

A national portal should publish:

- permit;
- EIA;
- monitoring summary;
- incident;
- enforcement;
- permit amendment.

Sensitive commercial or security information can be protected.

The public should still be able to trace material environmental obligations.

Chapter 158. LCDS as a separate system

Guyana's Low Carbon Development Strategy began before oil. [S22]

The strategy therefore has an institutional history distinct from petroleum.

The later carbon-credit programme should be analyzed under its own accounting.

The petroleum system should not absorb or redefine it.

Chapter 159. Forest-carbon accounting

A forest-credit programme has several stages:

- measured forest outcome;
- verification;
- issuance;
- sale;
- registry transfer;
- cash receipt;
- allocation.

These are separate.

A contract value is not a cash receipt.

A credit issued is not necessarily sold.

A payment allocated is not necessarily a completed project.

The same traceability principles used for oil should apply.

Chapter 160. Community allocation

LCDS materials describe a share of forest-carbon revenues directed to participating Indigenous villages under village-led planning. [S62, S63]

The final social outcome review should trace:

- eligibility;
- allocation;
- receipt;
- plan;
- procurement;
- completion;
- outcome.

Government programme records are one source.

Village and independent evidence are also needed.

Chapter 161. Petroleum emissions

Offshore petroleum has direct operational emissions.

Sources include:

- turbines;
- flaring;
- vessels;
- helicopters;

- shore facilities.

The inventory should state scope and method.

An emissions-intensity measure needs consistent production units.

Chapter 162. Downstream combustion

Most emissions associated with crude occur when fuels produced from oil are later combusted.

Those emissions often occur outside Guyana.

They remain relevant to global climate analysis even if they sit outside the national territorial inventory.

The paper should state the accounting boundary.

Chapter 163. Forest credits do not automatically cancel petroleum emissions

A forest credit represents an environmental claim under a specified methodology.

Petroleum emissions arise under a different accounting system.

The paper should not simply subtract one from the other and call the result national neutrality.

Any net claim needs a recognized accounting basis and complete boundaries.

Chapter 164. Domestic energy transition

Petroleum development also interacts with Guyana's domestic power system.

Gas-to-energy, solar, hydro and efficiency policies affect domestic emissions, cost and reliability.

These policies should be measured on their own outcomes.

Lower domestic power emissions do not change the physical combustion of exported oil.

Chapter 165. Climate resilience

Petroleum wealth can finance adaptation.

Guyana's coastal exposure gives this issue particular importance.

Adaptation investment should be measured by reduced risk.

Examples include:

- flood depth;
- outage duration;
- protected population;
- drainage performance.

Spending volume alone is not resilience.

Chapter 166. Island-state adaptation

For a small island producer, the environmental chapter should put added weight on:

- reefs;
- beach tourism;
- desalination;
- small fisheries;
- hurricanes;
- one or two major ports.

A spill and a storm could occur together.

The response plan should test that combined case.

Chapter 167. Stress test one: blowout during poor weather

Assume a subsea loss of well control during adverse weather.

Ask:

- Is detection immediate?
- Is capping equipment available?
- Can aircraft operate?
- Can vessels mobilize?
- Who approves dispersant?
- Are neighboring states notified?

The scenario tests readiness, not event probability.

Chapter 168. Stress test two: major spill with insurance dispute

Assume physical response begins while an insurer disputes part of coverage.

The state still needs cash for immediate actions.

The financial assurance structure should identify who pays before the dispute is resolved.

This is why liquidity and legal liability are different questions.

Chapter 169. Stress test three: simultaneous incidents

Assume response assets are already deployed to a smaller event when a second event occurs.

The plan should state:

- reserve capacity;

- external support;
- prioritization.

A multi-project basin needs concurrency planning.

Chapter 170. What another country should adopt

The transferable environmental principles are procedural:

- baseline before development;
- public EIA;
- versioned permit;
- independent monitoring;
- incident transparency;
- tested emergency command;
- enforceable financial assurance;
- cumulative assessment.

The exact permit conditions depend on ecosystem and project.

Chapter 171. Part V findings

Guyana's environmental case is not one decision.

It is a sequence from the 1996 statutory framework through project EIAs and permits, later permit modification, production-licence obligations, a dedicated 2025 pollution statute and judicial review. [S39, S48, S53-S56]

The existence of this architecture is a documented fact.

Its operating performance requires project-level evidence.

The blueprint lesson is equally precise.

A newcomer should not wait until a spill to define response authority, and it should not treat a permit as the end of environmental governance.

The environmental system should be designed to learn from operations, publish evidence, enforce conditions and finance the consequences of failure.

Edition 2.0 deepening: environmental governance as a lifecycle assurance system

Analytical section 171A. Environmental governance should begin before project approval

An offshore environmental regime should be designed as a lifecycle system.

The sequence begins with baseline data and strategic assessment.

It continues through project EIA, permit conditions, monitoring, incident reporting, enforcement, financial assurance, emergency response and decommissioning.

Each stage should answer a different question.

The EIA asks what impacts are reasonably foreseeable before approval.

The permit converts selected mitigation and monitoring requirements into enforceable obligations.

Operational monitoring asks what is actually occurring.

Incident reporting captures deviations and failures.

Financial assurance asks whether the operator can finance the consequences of non-compliance or an accident.

Decommissioning asks what remains after production ends.

Treating any one of these as the complete environmental system creates a blind spot.

Analytical section 171B. Baseline quality determines the credibility of later impact claims

A post-development environmental measurement has limited interpretive value if the pre-development condition is unknown.

This is particularly important in marine systems because fish abundance, currents, water quality and species distribution can vary naturally.

The baseline should therefore record:

- methodology;
- sampling locations;
- season;
- equipment;
- detection limits;
- uncertainty;
- raw data ownership;
- laboratory quality controls.

A baseline is not credible merely because it exists in an EIA.

It should be reproducible.

Where monitoring methods change, the record should explain whether new and old series remain comparable.

Analytical section 171C. Strategic assessment is different from project assessment

An EIA for one FPSO cannot answer every cumulative question created by a basin with several producing projects.

Strategic environmental and social assessment asks system-level questions:

- total tanker traffic;
- aggregate flaring and emissions;
- cumulative noise;
- fisheries interaction;
- waste handling;
- shared support bases;
- response capacity across simultaneous operations;
- onshore infrastructure effects.

The Guyana Oil and Gas Capacity Building Project's strategic-assessment terms identify many of these categories. [S94]

The transferable lesson is that project-by-project authorization should be supplemented by basin-wide assessment as development scales.

Analytical section 171D. Permit conditions should be transformed into a compliance matrix

A long environmental permit is difficult to administer if obligations remain embedded in prose.

Each permit should be converted into a compliance matrix containing:

- condition number;
- obligation;
- frequency;
- responsible party;
- evidence required;
- regulator reviewer;
- public disclosure status;
- breach consequence.

This makes compliance auditable.

The matrix should distinguish one-time conditions from recurring conditions.

A permit that requires annual reporting should generate an expected evidence item every year.

If the evidence is missing, the regulator should know immediately.

Analytical section 171E. Financial assurance has multiple layers

Financial assurance should not be reduced to a single headline guarantee.

The Uaru permit provides a useful structure because conditions 14.3 to 14.5 distinguish credible-cost estimation, liability insurance and parent or affiliate guarantee obligations. [S99]

These instruments serve different purposes.

Insurance transfers defined risks to an insurer subject to coverage terms, exclusions and limits.

A parent guarantee depends on the guarantor's obligation and financial capacity.

Other financial-assurance instruments may address specific regulatory obligations.

The audit questions are therefore:

- what loss is covered;
- what amount is available;
- when can it be called;
- who can draw;
- what exclusions apply;
- what happens if the instrument expires;
- what happens if the obligor becomes insolvent.

A stated liability principle is not the same thing as immediately available cash.

Analytical section 171F. The distinction between liability and financial security is fundamental

An operator may have legal responsibility for environmental damage while the state still faces a separate question about how quickly money can be accessed after a major event.

This is why legal liability and financial assurance should be reported separately.

The 2026 appellate litigation illustrates the importance of this distinction. Government and contemporaneous reporting describe the Court of Appeal as overturning the earlier unlimited-guarantee order and treating financial assurance as distinct from unlimited liability. [S101, S105]

The publication should remain narrow until authenticated full reasons are available.

The transferable point does not depend on the outcome of one case.

Every producer should distinguish:

- liability;
- insurance;
- guarantee;

- liquid emergency funding;
- ultimate recovery.

Analytical section 171G. A national spill plan should be built around decision rights

Oil-spill plans often contain equipment lists.

The more important question is who decides.

The plan should identify authority for:

- incident command;
- dispersant approval;
- fishing closure;
- shoreline protection;
- public health advice;
- international notification;
- wildlife response;
- claims administration;
- termination of emergency status.

These decisions should be exercised in drills.

A plan that has never been tested across agencies is not equivalent to operational readiness.

Analytical section 171H. Response equipment should be assessed by deployment time

An inventory can create false confidence.

A capping stack or response vessel has little operational value if it cannot arrive within the scenario window assumed by the response plan.

The state should therefore measure:

- location;
- mobilization time;
- customs and border requirements;
- vessel availability;
- weather limits;
- compatible connectors;
- trained personnel;
- maintenance status.

For a small state, regional agreements may be essential.

The response plan should show which resources are national, which are operator-owned and which depend on external assistance.

Analytical section 171I. Fisheries require both ecological and economic baselines

A fisheries impact programme should not rely only on biological sampling.

It should also document:

- fishing grounds;
- vessel types;
- seasonality;
- landing sites;
- value chain;
- household dependence;
- exclusion zones;
- navigation changes;
- complaints.

This creates the evidence needed to distinguish ecological change from access or economic disruption.

If compensation is contemplated, the baseline should define eligibility before a dispute arises.

Analytical section 171J. Flaring should be treated as an operating-performance indicator

Flaring has climate, air-quality and operational dimensions.

The regulator should distinguish:

- commissioning flaring;
- routine flaring;
- safety flaring;
- equipment failure;
- gas-injection constraints.

A single annual flare total does not explain the cause.

The operating record should therefore include duration, volume, reason and corrective action for material events.

This helps distinguish a temporary startup condition from persistent system performance.

Analytical section 171K. Produced water and drilling waste need a mass-balance approach

Environmental reporting becomes stronger when it moves from narrative statements to mass balance.

For produced water, the record should track:

- volume generated;
- treatment;
- discharge;
- quality;
- monitoring location.

For drilling waste, it should track:

- material type;
- quantity;
- treatment;
- discharge or disposal;
- destination.

This creates an auditable environmental flow account.

Analytical section 171L. Incident transparency should use severity tiers

Not every incident warrants the same reporting protocol.

A national system can define severity tiers based on:

- quantity released;
- toxicity;
- injury;
- operational shutdown;
- protected-area impact;
- shoreline impact;
- cross-border significance.

Higher tiers should trigger faster public reporting and more detailed investigation.

The system should preserve minor incidents as data rather than treating them as irrelevant.

Patterns can matter even when individual events are small.

Analytical section 171M. Independent verification is most valuable where information asymmetry is highest

Operator monitoring will always be a major source of environmental data.

Independence matters most where:

- the impact is difficult to observe;
- the operator controls the measurement system;
- consequences are large;
- public trust is low;
- disputes are likely.

Independent sampling does not require the regulator to duplicate every operator measurement.

It requires a risk-based verification programme.

Analytical section 171N. The 2025 pollution statute should be tracked from enactment to operation

The Official Gazette confirms publication of Act 6 of 2025. [S98]

That is a legal milestone.

Operational readiness requires more.

A complete implementation tracker should identify:

- commencement;
- regulations;
- designated authorities;
- national contingency arrangements;
- claims procedures;
- financial-assurance rules;
- offences and penalties;
- cross-border coordination.

The document should use "published" or "enacted" where those facts are established and reserve "in force" or "operational" for provisions supported by authoritative commencement evidence.

Analytical section 171O. Decommissioning is an environmental project

End-of-life planning should begin before facilities approach closure.

The environmental decommissioning plan should address:

- well abandonment;
- subsea infrastructure;
- residual contamination;
- waste;
- seabed recovery;
- monitoring period;
- liability after surrender.

The financial model should connect decommissioning obligations to secured funding.

A state should not assume that decades of production guarantee the operator or parent company will still have the same financial capacity at closure.

Analytical section 171P. Climate accounting should use explicit boundaries

Guyana's forest-carbon programmes and petroleum production belong in the same national development story, but they use different accounting systems.

Forest-carbon credits should not be presented as automatically cancelling petroleum lifecycle emissions.

The paper should distinguish:

- domestic operational emissions;
- exported-fuel combustion;
- national greenhouse-gas inventory treatment;
- forest-carbon accounting;
- buyer retirement claims;
- corresponding adjustments where applicable.

This avoids double counting and misleading equivalence. [S22, S95]

Analytical section 171Q. Expanded Part V conclusion

A strong environmental regime is not defined by the number of permits issued.

It is defined by the evidence chain from baseline to outcome.

Guyana has statutory, project-level, spill-response, financial-assurance and later legislative components that can be documented. [S39, S48-S56, S98-S100]

The next level of scrutiny is performance evidence.

For each material permit condition, the state should be able to show what was required, what evidence was submitted, what the regulator concluded and what corrective action followed where necessary.

For a newcomer, the blueprint is therefore to build five linked registers:

- baseline register;
- permit compliance matrix;
- incident register;
- response-capability register;
- financial-assurance register.

Together they turn environmental governance from a set of documents into an operating assurance system.

Edition 2.0 project deepening: seven projects, seven assurance files

Analytical section 171R. The environmental chapter should be read project by project

Edition 2.0 adds `OFFSHORE\_ENVIRONMENTAL\_ASSURANCE\_MATRIX.csv` because a basin-level description can hide important differences between project authorizations.

For each development, the matrix separates:

- licence and permit;
- EIA and management documents;
- financial assurance;
- spill response;
- observed performance;
- unresolved evidence.

This prevents evidence from one development from being generalized to another.

Analytical section 171S. Liza Phase 1 is the baseline permit case

The Liza Phase 1 authorization record is especially important because it is the first production project and the source of later financial-assurance litigation.

The official project record shows the permit history and incorporation of environmental and social management, oil-spill response and wildlife-response materials. [S100]

The authenticated 2024 CCJ reasons identify the litigation concerning renewed permit condition 14 and the Environmental Protection Act framework. [S103]

The analytical lesson is not a legal verdict.

It is that permit wording, regulator discretion, insurance, guarantee instruments and liability can become distinct legal questions.

The book should preserve all five layers.

Analytical section 171T. Permit modification should be treated as version control

An environmental permit can change over time.

The regulator and publication should therefore maintain:

- original permit;
- modification date;
- modified text;
- reason;
- conditions affected;
- transition arrangements.

A later permit should not be quoted as if it was the condition in force at an earlier date.

This is the environmental equivalent of contract-version control.

Analytical section 171U. Liza Phase 2 and Payara should be compared with the first-project baseline

The second and third developments create an opportunity to test institutional learning.

A project comparison should ask whether later permits became more specific on:

- flaring;
- monitoring;
- spill response;
- financial assurance;
- reporting;
- wildlife;
- decommissioning.

A stronger later condition may reflect a different project, new evidence, legal evolution or a regulatory lesson.

The comparison should not assume one cause without documentary support.

Analytical section 171V. Yellowtail marks the shift to basin-scale cumulative impact

By the fourth producing development, cumulative effects become harder to treat as incidental.

The environmental question expands from "What is the impact of this FPSO?" to:

- What is the total vessel traffic?
- What is the cumulative produced-water discharge?
- What is aggregate flaring?

- How much response equipment is shared?
- What is the total area affected by exclusion zones?
- How does monitoring distinguish project signal from basin background?

This is why strategic assessment must sit above project EIAs.

Analytical section 171W. Uaru provides the clearest financial-assurance pinpoints

The Uaru permit is useful because it provides explicit financial-assurance architecture.

Condition 14.3 addresses credible-cost estimation.

Condition 14.4 requires liability insurance including environmental liability insurance.

Condition 14.5 addresses parent or affiliate guarantee arrangements and resource availability following default. [S99]

These conditions should be quoted or paraphrased carefully in the final legal annex with page and condition pinpoints.

They establish the obligation structure.

They do not establish:

- the exact current policy limit;
- every exclusion;
- guarantor liquidity;
- actual draw mechanics under a future event.

The book should state that distinction prominently.

Analytical section 171X. Financial assurance should be stress-tested, not merely checked for existence

A document can exist and still be inadequate for a defined scenario.

A regulator should stress-test financial assurance against:

- small operational spill;
- prolonged cleanup;
- fisheries interruption;
- shoreline impact;
- cross-border claim;
- well-control event;
- contractor insolvency.

For each scenario, the state should identify:

- first source of cash;
- insurer response;
- deductible;
- guarantee trigger;
- government emergency liquidity;
- later recovery route.

The purpose is to identify timing gaps.

Analytical section 171Y. Whiptail is a pre-production assurance opportunity

Whiptail was under development at the evidence cutoff. [S126]

That creates an opportunity to test the full assurance system before production.

The regulator can verify:

- baseline completeness;
- permit matrix;
- response plan;
- insurance;
- guarantee;
- commissioning drill;
- monitoring equipment.

The value of a pre-start audit is that deficiencies can be corrected before operating risk begins.

Analytical section 171Z. Hammerhead should be evaluated under the evolved regulatory system

Hammerhead is the seventh approved Stabroek development, with production expected later in the decade. [S127, S128]

A mature regulator should be able to demonstrate how the environmental review for the seventh project incorporates lessons from earlier projects.

The final Edition 2.0 project profile should therefore compare Hammerhead's:

- EIA scope;
- permit conditions;
- financial assurance;
- spill response;
- monitoring;
- decommissioning.

This comparison should be documentary, not rhetorical.

Analytical section 171AA. Compliance evidence should be condition-specific

A statement that "the project is compliant" is too broad.

Compliance should be reported as:

- condition;
- reporting period;
- evidence submitted;
- regulator finding;
- corrective action.

A project can comply with one condition and have an open issue under another.

This structure is both fairer and more auditable.

Analytical section 171AB. Inspection results should be separated from operator self-reporting

Operator reports are necessary because the operator controls much of the operating data.

Regulatory inspection provides a different evidence class.

Independent sampling provides another.

The publication should label the evidence source.

A claim based on operator monitoring should not be described as independent verification unless a separate verifier performed the measurement.

Analytical section 171AC. Environmental incidents should have a common reporting schema

The basin incident register should record:

- date;
- project;
- incident type;
- quantity;
- affected medium;
- immediate response;
- regulator notification;
- investigation;
- corrective action;

- public disclosure.

The schema should include near misses where legally reportable.

Near misses can reveal control weaknesses before harm occurs.

Analytical section 171AD. Flaring should be linked to cause and duration

A flaring total can be misleading.

A high figure caused by temporary commissioning has a different regulatory implication from repeated equipment failure.

The environmental dataset should therefore classify flare events by:

- commissioning;
- maintenance;
- safety;
- equipment failure;
- gas-system constraint;
- other.

This creates a performance trend rather than a single emissions number.

Analytical section 171AE. Marine monitoring should preserve raw data

Environmental interpretation changes over time.

The state should therefore retain raw monitoring data, not only annual summaries.

Raw data allow:

- reanalysis;
- method comparison;
- cumulative trend assessment;
- independent research.

Metadata should identify station, depth, method, laboratory and quality control.

Analytical section 171AF. Fisheries evidence should include effort and income

A decline in fish landings can reflect:

- stock condition;
- weather;
- fuel cost;

- market price;
- effort;
- access restriction.

The fisheries baseline should therefore include vessel activity and economic variables as well as biological data.

This avoids attributing every change to offshore petroleum without causal evidence.

Analytical section 171AG. Spill modelling should be treated as a scenario, not a forecast

Oil-spill trajectory models are useful for planning.

They are sensitive to:

- release location;
- volume;
- current;
- wind;
- weather;
- oil characteristics.

The result is a scenario distribution.

It should not be described as a prediction of where oil "will" travel in a future incident.

Analytical section 171AH. Cross-border response should be pre-negotiated

Offshore spills can cross maritime boundaries or require foreign equipment.

A national plan should therefore pre-negotiate:

- customs clearance;
- aviation;
- vessel entry;
- command liaison;
- data sharing;
- claims cooperation.

The time to resolve those issues is before an incident.

Analytical section 171AI. Environmental assurance conclusion

The project matrix makes the evidence standard concrete.

The state should be able to open one file for each development and answer:

What was approved?

What was required?

What financial protection existed?

What monitoring occurred?

What incidents occurred?

What did the regulator conclude?

What remains unresolved?

That is a much stronger basis for scrutiny than either the claim that permits prove good performance or the opposite claim that development necessarily proves regulatory failure.

Original case study: Uaru's permit as a test of financial assurance, verification and public disclosure

A permit is a control architecture, not a financial asset

One of the most consequential distinctions in offshore environmental policy is between a regulatory obligation and the financial capacity available to discharge that obligation after an accident. Guyana's Uaru environmental permit provides a useful, unusually detailed case for teaching that distinction. It establishes a credible-cost basis for assurance, identifies several insurance requirements, requires parent or affiliate guarantees and provides a separate monitoring and independent-audit framework. The question for a technical reviewer is not whether these clauses sound comprehensive. It is how to verify the existence, continuing validity, scope and responsiveness of the actual instruments and related operating controls. [S99, S141]

Uaru's signed permit reference 2023-023-EEFGL is effective for the period 27 April 2023 through 26 April 2028, subject to the annual-review language in condition 15.7. The period is not evidence that the conditions can never change; the permit itself allows defined review and modification processes. It is also not evidence that production had begun. Government and contractor notices in August 2026 placed the Errea Wittu FPSO in Guyana for commissioning ahead of first oil. A September 2026 case study should assess Uaru's pre-production assurance and readiness without inventing a post-start environmental performance record. [S99, S124, S125]

The most useful starting point is the permit's own separation of liability and financial security. The EPA's authorization announcement describes permit-holder liability for restoration, rehabilitation and compensation related to defined pollution and other covered harm. It also describes insurance, annual financial-capacity reporting and a parent/affiliate guarantee architecture. These are connected obligations but not interchangeable accounts. A tort or statutory claim may exceed the amount immediately available under one policy; an insurance claim may be contested; and a guarantor may need time to supply funds. None of these possibilities proves that a given instrument is invalid or adequate. They explain why a serious

audit must review the instruments individually rather than treat an announced number as immediate, unconditional liquidity. [S141]

The four-column financial-assurance test

Condition 14.3 instructs that assurance be guided by a reasoned estimate of credible costs arising from a permit breach. Its scope explicitly includes response, cleanup, remediation and monitoring. It also recognizes that some unidentifiable or unquantifiable losses and continuing damage may be pursued through civil action. That distinction makes two separate analytical documents necessary: a scenario-based financial-security estimate for identifiable response costs, and a legal exposure map for liabilities that may be larger or harder to estimate. Neither can replace the other. [S99]

Condition 14.4 requires liability insurance, including environmental liability cover, on terms and amounts customary for comparable international petroleum operations. It includes insurer financial-strength criteria and specified categories of loss. An auditor should therefore obtain the actual policy schedule and endorsements, not only the permit. The schedule should identify named insureds, project and period, occurrence and aggregate limits, deductibles, exclusions, environmental and well-control triggers, sublimits, applicable law, notice deadlines and insurer ratings. The annual renewal record should be retained because a certificate issued in one year is not proof of coverage throughout a later year. A high policy limit may also be less accessible than its headline suggests if a disputed exclusion is material. [S99]

Condition 14.5 addresses parent or affiliate guarantees and the provision of adequate resources after notice of default within the time structure stated in the permit. A guarantee audit should obtain the executed document, identify each guarantor and beneficiary, match the guarantee's governing law and triggering conditions to the permit, test the current financial condition of the responsible entities, and verify renewal or continuation across assignments and corporate changes. The permit requirement alone cannot prove the full executed amount or ability to draw cash when needed. The official EPA authorization notice indicated that guarantee negotiations were an important condition of approval; a dated announcement and an executed financial instrument are distinct evidence classes. [S99, S141]

For audit purposes, the four columns in the project file should therefore be **liability, insurance, guarantee** and **public emergency liquidity**. The first states legal responsibility. The second identifies contractually insured events and their limits. The third identifies an additional obligor and the conditions under which its support becomes payable. The fourth establishes what cash or other assets can be mobilized immediately while a dispute or claims adjustment is pending. Public emergency liquidity is not a waiver of the operator's legal liability. It addresses the timing mismatch between prompt response and eventual cost recovery.

A test scenario without invented expected damages

Imagine, solely to test the administrative controls, an offshore event requiring US\$150 million in prompt response and clean-up work during the first month and another US\$400 million of remediation and monitoring over the following two years. Assume that the operator can supply US\$50 million immediately, that an insurer accepts US\$75 million of undisputed first-month costs, and that remaining costs may require additional insurance or guarantee claims. These figures are synthetic stress inputs, **not a credible-loss estimate for Uaru**, not a prediction of a Guyana oil spill, and not evidence of the actual Uaru policy or guarantee amounts.

In the synthetic first month, US\$150 million of urgent costs minus US\$50 million of immediate operator liquidity minus US\$75 million of undisputed insurance proceeds leaves a US\$25 million **timing requirement**. The review should ask which entity advances those funds and under what authority, whether an additional insurance layer is contractually responsive, when a parent guarantee may be called, and how recovered money is allocated after an interim public advance. The later US\$400 million should be stress-tested separately because delayed ecological remediation may operate under different policy periods, monitoring standards and claims evidence. The scenario does not permit the analyst to assume that every unspent dollar of stated insurance is accessible without adjustment.

Now add a second complication: the carrier requires more time to resolve whether one category of environmental monitoring expense falls within the policy. The response cannot stop while the accounting classification is litigated. A robust regime should have a predefined order for action, emergency procurement, evidence preservation, notification and later recovery. That is why an insurance certificate and a response-deployment plan belong in the same assurance file even though they serve different functions.

Audit environmental performance, not merely permit existence

Uaru's permit also contains the independent-audit provisions that can turn paper controls into measured performance. Condition 13.24 requires an environmental management system and at least annual internal audits, with findings addressed as specified. Condition 13.25 separately requires annual independent external compliance audits on the embedded controls, including the operating integrity management system and oil-spill response plan, with the final report submitted to the Agency within the specified thirty-day period after receipt. The permit also reserves regulatory rights concerning independent sampling, modification and other compliance mechanisms. The actual independent-audit reports and corrective-action closure records must be requested or identified before the book can state that a given year's requirements were met. [S99]

An audit-grade record should therefore look like a dated matrix, not a verdict. For condition 13.25, the fields are audit year; auditor identity and independence; scope; original report date; delivery date to the EPA; material findings; operator responses; corrective-action deadlines; closure evidence; and regulator follow-up. For a discharge condition, the fields are permitted parameter, method, station, sample date, result, measurement uncertainty, independent verification and relevant corrective action. An operator-submitted value is informative but should not be called regulator-tested or independently sampled unless such review occurred.

An incident provides a second distinction between initial reporting and final verification. On 24 August 2026, the Ministry of Natural Resources reported that fire/heat detectors activated in the Liza Unity living-quarters laundry room and that a small fire had been extinguished, with personnel safe and regulatory follow-up expected. That is useful evidence that an initial notification was publicly made. It is **not** the final root-cause investigation and should not be used to claim either that safety performance was verified or that the event proves systemic failure. The model reporting record should track the follow-up investigation and any corrective actions, including an explicit "not yet available" status where appropriate. [S138]

Cumulative risk in a multi-FPSO province

Project-by-project authorization is necessary, but the scale of the basin changes the nature of the environmental problem. By the time the fifth FPSO was being commissioned, several earlier developments were producing and further projects were in development. Each project may have its own environmental permit and response plan, while the same coast guard, medical evacuation capacity, shore bases, marine contractors, specialist response equipment and regulatory inspectors may support several fields. A national readiness assessment should therefore examine simultaneous demand and common-mode failure rather than assume every project can use the full emergency resource inventory at the same moment. [S81-S85, S124-S128]

A cumulative assurance review can begin with six questions: how much response capacity is physically in-country; what is shared or contracted abroad; what is the fastest proven mobilization time; what customs or aviation approvals are needed; how many concurrent response operations can be sustained; and how would an offshore incident during a major coastal flood affect command and logistics? The environmental strategic-assessment terms already identify fisheries, marine ecosystems, accidental spills and shore-based activities as interacting risk categories. Permit files should be integrated with such basin-wide analysis rather than substitute for it. [S94]

Publication disclosure and remaining evidence

The final chapter should include a two-level disclosure table. Level one lists the permit obligations that can be read directly from authoritative documents: term, key insurance and guarantee conditions, audit obligations, reporting requirements, and the regulator's powers. Level two identifies the evidence needed to demonstrate performance: executed policies and guarantees, audited company financial statements, annual independent compliance reports, drill evaluations, discharge and marine-monitoring data, incident investigations, and response-resource contracts or deployment tests. Access restrictions may apply to certain documents, but their nonpublication must be described as an evidence limitation rather than proof they do not exist.

The transferable result is a rigorous distinction between **an environmental authorization, a legally enforceable liability, financial resources available after default, operational response capability**, and **observed environmental performance**. The Uaru file illustrates how to build the framework before first oil. The wider Guyana sequence illustrates why the

framework must be revisited as the number of operating projects grows. This is a more useful lesson for a small new producer than claiming that permit issuance by itself establishes comprehensive environmental security.

PART VI | Local content and social distribution

Local Content, Human Capital and the Distribution of Opportunity

Chapter 172. Why local content requires more than a procurement number

A petroleum industry creates demand for goods, services and labour.

A local-content policy attempts to increase the share of that demand supplied by nationals and nationally owned businesses.

The resulting headline number is often gross procurement.

Gross procurement is useful.

It is not the same as domestic value added.

A locally registered company may import a finished product and resell it.

Another local company may fabricate, engineer, test and maintain a product inside the country.

Both transactions can appear in local procurement.

Their contribution to national capability differs.

The flagship therefore separates:

- supplier registration;
- beneficial ownership;
- procurement;
- imported content;
- domestic wages;
- retained profit;
- training;
- capability.

Chapter 173. The 2021 statutory framework

Guyana's Local Content Act was passed at the end of 2021 and published in early 2022. [S57]

This means the formal statutory framework arrived after first oil.

The chronology is important.

The industry had already begun hiring and procurement before the law.

The final chapter should therefore divide local-content history into:

- pre-law industry activity;
- statutory implementation;
- later reporting and target adjustments.

This gives a more accurate view of policy development.

Chapter 174. Definitions

The legal definition of a Guyanese company matters.

A meaningful local-content rule needs to address:

- ownership;
- voting control;
- management;
- beneficial ownership;
- residency;
- workforce.

A paper company with nominal domestic ownership but external control presents a different economic case from a business whose management, assets and profits are genuinely domestic.

The final legal crosswalk should quote the operative definitions from the Act.

Chapter 175. The supplier register

The official register provides a structured route for company certification. [S58]

The register supports:

- identification;
- procurement;
- compliance.

Certification should not be interpreted as an independent measure of economic value.

A verification sample should test:

- ownership;
- employees;
- domestic operations;
- imports;
- related parties.

The purpose is measurement, not suspicion.

Chapter 176. Scheduled targets

A local-content schedule identifies services where the law expects a stated level of Guyanese participation.

A target should reflect actual national capacity.

A target set far above domestic capability can lead to:

- inflated cost;
- nominal compliance;
- subcontracting;
- supply bottlenecks.

A target set too low can miss realistic opportunities.

The policy should therefore be revised using evidence.

Chapter 177. Capacity assessment

Before changing a target, government should know:

- number of qualified suppliers;
- capacity;
- safety record;
- price;
- certification;
- equipment;
- workforce.

The 2025 government reporting described ongoing capacity assessment and wider reporting. [S60]

The final paper should compare target changes with documented supplier capability.

Chapter 178. Domestic value-added accounting

A practical domestic value formula starts with contract value.

Subtract:

- imported goods;
- imported subcontracted services;
- foreign intellectual property;
- external management charges directly tied to delivery.

The residual approximates domestic gross value added.

That residual can be divided into:

- wages;
- domestic intermediate purchases;
- taxes;
- operating surplus.

This produces a clearer picture than procurement alone.

Chapter 179. Customs data

Customs records help estimate imported content.

A supplier reporting USD 10 million in local procurement may have imported most of the underlying equipment.

This does not make the transaction valueless.

The local contribution may include logistics, inventory, financing, installation and maintenance.

The research should quantify the difference.

Chapter 180. Payroll data

Payroll provides evidence of:

- local employment;
- wage;
- occupation.

A local supplier with high domestic payroll generates a different value profile from a reseller with minimal employment.

Payroll analysis should protect personal data.

Aggregate occupational statistics are sufficient for the public paper.

Chapter 181. Retained profit

Domestic ownership matters partly because profit can remain in the national economy.

Retained profit can support:

- reinvestment;
- equipment;
- expansion;
- tax;
- wealth formation.

The research should avoid assuming that every locally owned firm's profit remains inside the country.

Actual company data would be needed.

Chapter 182. Technology and know-how

Local value also includes knowledge.

A training programme that qualifies technicians to work across mining, power and marine sectors creates value beyond one petroleum contract.

The flagship should therefore identify transferable skills.

Examples:

- welding;
- nondestructive testing;
- electrical;
- mechanical maintenance;
- marine safety;
- logistics;
- environmental monitoring;
- project controls.

Chapter 183. Direct employment

Direct employment is the most visible local-content metric.

The 2024 government mid-year report included employment information from reporting petroleum companies and a broader estimate of Guyanese industry employment. [S59]

The paper should preserve the reporting population.

A number from 40 reporting companies should not be presented as a census unless the source says so.

Chapter 184. Occupation matters

Ten jobs do not have equal capability effect.

The workforce dataset should distinguish:

- entry level;
- skilled trade;
- technician;
- engineer;
- supervisor;
- manager;
- professional services.

Progression into higher-responsibility roles is a better capability measure than headcount alone.

Chapter 185. Expatriate succession

A petroleum industry often relies on expatriate specialists in early years.

A local-content strategy can define succession pathways.

A credible pathway states:

- role;
- competency;
- training;
- mentor;
- target date;
- assessment.

A quota without competency development can create safety and performance risk.

Chapter 186. Safety

Employment quantity should never displace occupational safety.

The local-value dashboard should include:

- lost-time incidents;
- training;
- certification;
- offshore medical fitness;
- safety competence.

A workforce programme that expands rapidly needs a matching safety system.

Chapter 187. Wage pressure

Petroleum demand can raise wages for scarce occupations.

Higher wages benefit workers.

They can also increase costs for:

- construction;
- utilities;
- mining;
- public agencies.

The national labour analysis should compare petroleum wages with other sectors and vacancy rates.

This is part of the absorption-capacity question.

Chapter 188. Public-sector capacity

The government competes for some of the same specialists it regulates.

Engineers, accountants, lawyers and environmental professionals can move to private employers.

A resource strategy should therefore include public-sector retention.

Options include:

- career progression;
- training;
- specialist pay structures;
- secondments;
- targeted external expertise.

The policy choice belongs to government.

The analytical requirement is to measure vacancy and turnover.

Chapter 189. Education pipeline

Long-run localization begins before recruitment.

The skills pipeline includes:

- secondary science;
- vocational education;
- university;
- apprenticeship;
- professional certification.

The flagship should map annual graduates against projected occupational demand.

This reduces the temptation to set targets based only on aspiration.

Chapter 190. Training expenditure versus training outcome

Training expenditure is an input.

The outcome is:

- completion;
- certification;
- employment;
- promotion;

- retention;
- wage progression.

The local-content report should publish both.

A scholarship without completion data provides an incomplete picture.

Chapter 191. Supplier finance

Local firms often face working-capital constraints.

Large petroleum contracts may require:

- equipment;
- inventory;
- insurance;
- payroll before payment.

The supplier-development system should examine:

- payment terms;
- collateral;
- banking;
- invoice financing.

Access to a contract does not mean a firm has the balance sheet to deliver it.

Chapter 192. Standards and certification

Petroleum supply chains use demanding standards.

Local companies need:

- quality systems;
- safety certification;
- testing;
- traceability.

Supplier development is stronger when these capabilities also support exports and other sectors.

Chapter 193. Procurement transparency

Local suppliers need visibility into opportunities.

A transparent procurement system should state:

- expected tender;
- qualification;
- schedule;

- contact.

Post-award reporting should state aggregate domestic participation.

Commercially sensitive bid information can remain protected under law.

Chapter 194. Measuring capability after the contract

The best long-run local-content question is what remains after a petroleum contract ends.

Possible durable assets include:

- certified technicians;
- equipment;
- workshops;
- management systems;
- export clients;
- retained earnings.

This measure connects local content to diversification.

Chapter 195. Household welfare

Petroleum employment reaches only part of the population directly.

National development therefore needs household evidence.

The Bureau of Statistics states that the major Household Budget and Living Condition Survey was fielded from late 2018 through late 2019. [S61]

That observation period is close to first oil.

It is not a 2026 household measurement.

The research should make the date visible beside every indicator.

Chapter 196. Why GDP and household welfare differ

GDP records production.

Household welfare depends on:

- income;
- prices;
- taxes;
- transfers;
- public services;
- housing;
- environment.

A rise in petroleum GDP can coincide with a smaller change in median household consumption.

It can also support large public investments that improve welfare later.

Only household and service data resolve the question.

Chapter 197. Real income

Nominal wage growth is not enough.

The household analysis should adjust for:

- food;
- rent;
- transport;
- utilities.

A household whose wage rises 10 percent while housing cost rises 20 percent experiences a different result from the nominal wage statistic.

Chapter 198. Housing

Rapid economic growth can increase demand for:

- rentals;
- construction land;
- mortgages.

Housing policy can expand supply.

The final chapter should use:

- rent;
- homeownership;
- construction;
- household size;
- housing quality.

These measures show whether the housing response keeps pace.

Chapter 199. Health

Petroleum-financed public investment can expand health infrastructure.

The outcome test should include:

- staffing;
- service availability;
- waiting time;

- referral;
- health outcomes.

A completed hospital is an important capital asset.

Its social return depends on operation.

Chapter 200. Education

Education investment should be measured through:

- enrollment;
- attendance;
- completion;
- learning;
- teacher availability;
- technical certification.

A resource windfall creates an opportunity to invest in capabilities that remain after oil.

The evidence should test the outcome.

Chapter 201. Water and sanitation

Access to reliable water and sanitation is a direct development measure.

The chapter should report:

- connection;
- quality;
- continuity;
- regional gaps.

Large national GDP growth does not reveal these household conditions.

Chapter 202. Regional distribution

Guyana's ten administrative regions differ in geography and economic structure.

National averages can hide:

- coastal/hinterland differences;
- urban/rural differences;
- infrastructure access.

The final social dashboard should therefore report regional measures.

Chapter 203. Amerindian and hinterland development

Indigenous communities have distinct governance, land and cultural contexts.

The paper should not reduce their development experience to petroleum transfers.

Relevant systems include:

- land titling;
- public services;
- connectivity;
- renewable energy;
- forest-carbon benefit sharing.

LCDS materials describe a community allocation mechanism and village-led plans. [S62, S63]

Independent community evidence is still required.

Chapter 204. Village plans

A village-led plan should be traced from:

- consultation;
- approval;
- funding;
- procurement;
- completion;
- outcome.

This creates the same accountability structure used for national public investment.

The scale is different.

The principle is the same.

Chapter 205. Consent and participation

A public programme should record more than the number of consultations.

The paper should seek:

- topics raised;
- dissent;
- revisions;
- grievance.

Government programme descriptions should be paired with community and independent sources.

Where positions conflict, the paper should state the disagreement.

Chapter 206. Land rights

Resource development and forest finance interact with land rights.

The final chapter should use legal and community sources to distinguish:

- titled land;
- pending title;
- extension;
- customary claim.

No national development claim should imply settlement of a land dispute that remains open.

Chapter 207. Gender

Local-content analysis should include gender.

Indicators include:

- employment;
- occupation;
- wage;
- training;
- management;
- supplier ownership.

A petroleum industry with high male participation may require deliberate skills pathways for broader access.

The paper should measure rather than assume.

Chapter 208. Youth

Young people experience the resource transition through:

- education;
- jobs;
- housing;
- migration;
- expectations.

A boom can create pressure to leave education early for short-term work.

It can also finance technical education.

Longitudinal data are needed.

Chapter 209. Migration

High wages can attract workers from abroad and encourage return migration.

Migration expands labour supply.

It also increases housing and service demand.

The labour-market chapter should include legal migration data where available.

Chapter 210. Cost of living

A credible social dashboard should track:

- CPI;
- food;
- rent;
- transport;
- construction.

Average CPI can hide pressure in categories that dominate lower-income budgets.

The household analysis should identify expenditure shares.

Chapter 211. Social transfers

Government can use petroleum-supported fiscal space for transfers or subsidies.

The distributional effect depends on:

- eligibility;
- amount;
- coverage;
- inflation;
- duration.

The paper should separate transfer spending from infrastructure investment.

Both affect welfare through different channels.

Chapter 212. Grievance mechanisms

Stakeholders need routes to challenge:

- environmental permits;
- labour practices;
- local-content certification;
- public procurement;
- community programmes.

Each mechanism should state:

- who receives complaints;
- deadline;
- decision;
- appeal;
- publication.

A process that receives complaints but never records outcomes is incomplete.

Chapter 213. Social licence

Social licence is not a legal licence.

It describes public acceptance and trust.

The concept should be treated carefully.

Polling or consultation provides evidence of attitudes.

It does not replace legal rights or environmental standards.

The paper should not infer consent from the absence of protest.

Chapter 214. Measuring distribution

A petroleum transition can be evaluated across four distribution channels:

- 1. labour income;
- 2. business income;
- 3. public services;
- 4. direct transfers.

The final paper should map which groups receive each channel.

This gives a clearer picture than asking whether "the country" benefited in the abstract.

Chapter 215. A local-content adverse case

Assume procurement rises rapidly while most high-value inputs remain imported.

Gross local procurement rises.

Domestic value added rises much less.

The policy response should depend on the cause.

If local manufacturing is uneconomic at national scale, forcing it may increase cost.

If capability exists but firms lack certification or finance, targeted support may have a higher return.

The audit should identify the bottleneck.

Chapter 216. A labour adverse case

Assume petroleum wages attract skilled electricians from the power utility.

Oil localization improves.

Grid maintenance suffers.

This is a national allocation problem.

The workforce strategy should therefore monitor vacancies outside petroleum.

Chapter 217. A household adverse case

Assume nominal wages rise but rent and food costs increase faster for lower-income households.

GDP and employment improve while real household conditions diverge by income group.

A current household survey is required to detect the effect.

Chapter 218. New-producer checklist

Before first oil, a new producer should have:

- household baseline;
- labour baseline;
- supplier register;
- beneficial ownership;
- realistic local targets;
- training pathway;
- grievance process.

After first oil it should update these on a fixed schedule.

A country should not wait a decade for the next household benchmark.

Chapter 219. Part VI findings

Guyana's local-content framework formalized after first oil and now provides a statutory platform for procurement, employment and supplier participation. [S57-S60]

The economic question goes beyond compliance.

The strongest future evidence will show how much domestic value, skill and business capability remains after contracts are delivered.

The social question is broader again.

Petroleum revenue, jobs and public investment need to be connected to current household and regional outcomes.

Where current representative data are absent, the white paper should preserve the gap instead of substituting an old figure.

A newcomer country should build those measurement systems before the resource transition makes the baseline harder to reconstruct.

Edition 2.0 deepening: local content as capability formation, not only compliance

Analytical section 219A. Local content should be measured through retained domestic value

A procurement value can overstate local economic contribution.

A company registered in Guyana may import most of the goods it supplies.

A contract awarded to a local firm may pass a large share of the revenue to foreign manufacturers, expatriate labour or imported equipment.

The stronger metric is retained domestic value.

For each procurement category, government should estimate:

- local wages;
- local ownership;
- imported inputs;
- domestic operating expenditure;
- retained profit;
- local tax contribution;
- training expenditure;
- technology or process capability retained after contract completion.

This produces a domestic-value bridge rather than a gross-spend headline.

The Local Content Secretariat's reporting framework already requires detailed information on expenditure, employment, salary scales, supplier participation, training and capacity development. [S108]

The next analytical step is to convert those reporting fields into value-retention metrics.

Analytical section 219B. Local ownership should be verified economically, not only legally

A company can meet formal registration requirements while exercising limited local economic control.

A credible ownership test should therefore examine:

- beneficial ownership;
- voting rights;
- management control;
- financing;
- profit distribution;
- related-party arrangements;
- operational capability.

This does not imply that joint ventures or foreign capital are undesirable.

It means that "local" should describe the economic relationship being measured.

A transparent beneficial-ownership regime is therefore part of local-content governance.

Analytical section 219C. Category targets should reflect capability pathways

Not every petroleum good or service should have the same localization strategy.

Some categories can be localized quickly because the required capability already exists.

Others require years of training, certification, capital or international partnership.

A rational local-content plan should classify categories into:

- 1. immediately competitive;
- 2. competitive with certification or finance;
- 3. competitive after workforce development;
- 4. unlikely to be efficient at domestic scale;
- 5. strategic capability worth developing despite a longer horizon.

This avoids treating every target as a static percentage.

The policy objective becomes capability progression.

Analytical section 219D. Supplier development should be tracked after the petroleum contract ends

A supplier programme succeeds when firms become more capable, not merely when they win one petroleum contract.

The audit should ask whether participating firms later:

- won larger contracts;
- exported services;
- entered other sectors;

- improved safety performance;
- obtained international certification;
- invested in equipment;
- hired and trained permanent staff.

This creates a longitudinal supplier dataset.

The strongest local-content outcome is a firm that becomes competitive beyond the protected or targeted procurement category.

Analytical section 219E. Workforce localization should be occupation-specific

A national employment percentage can hide skill concentration.

The workforce dataset should distinguish:

- executive and managerial;
- petroleum engineering;
- geoscience;
- subsea;
- marine;
- process operations;
- maintenance;
- safety;
- finance;
- legal;
- logistics;
- administrative support.

A country may have a high national employment share while specialist decision-making remains concentrated among expatriates.

The succession plan should therefore be occupation-specific.

Analytical section 219F. Training should be evaluated by competence, not attendance

Training expenditure and participant count are inputs.

The output is demonstrated competence.

A training programme should identify:

- role;
- required competency;

- training method;
- assessment;
- supervised practice;
- certification;
- date of independent responsibility.

This is especially important for safety-critical and technical roles.

A scholarship can contribute to long-term capability.

It is not the same as operational qualification.

Analytical section 219G. The public sector competes for the same scarce skills

A petroleum boom can improve private wages while weakening regulators, utilities and ministries if scarce specialists leave.

The workforce strategy should therefore track vacancies and turnover in:

- environmental regulation;
- tax administration;
- public engineering;
- power utilities;
- maritime authorities;
- statistics;
- public procurement.

If public institutions cannot retain critical staff, localization inside the private petroleum sector may coexist with weaker state capacity.

Compensation reform, specialist career tracks and targeted secondments may be required.

Analytical section 219H. Migration is part of the labour-market response

A small economy cannot always train workers fast enough to meet a sudden investment boom.

Migration can relieve labour shortages.

It can also increase demand for housing, transport, health care and schools.

The labour-market chapter should therefore track:

- work permits;
- occupation;
- duration;
- housing pressure;

- remittance flows;
- transition to local workers.

The objective is not to treat migration as success or failure.

It is to include it in capacity planning.

Analytical section 219I. Wage growth should be interpreted with prices

Nominal wage growth does not establish improved household welfare.

Real household conditions depend on:

- rent;
- food;
- transport;
- utilities;
- health costs;
- household size;
- employment stability.

A petroleum boom can increase average wages while widening differences between households connected to the new economy and those facing higher prices without equivalent income gains.

This is why current household data are essential.

Analytical section 219J. A rolling household survey system is part of resource governance

A country experiencing rapid structural change should not wait many years between household surveys.

A rolling system can combine:

- annual labour-force data;
- periodic household expenditure modules;
- price surveys;
- administrative social-protection data;
- regional service indicators.

The purpose is not to publish one poverty rate more frequently.

It is to understand how the transition is distributed across income, geography and household type.

The 2018/2019 household survey remains valuable as a baseline, but it should not be described as a current 2026 outcome. [S61]

Analytical section 219K. Regional equity should be measured through access and outcomes

National averages can obscure geographic gaps.

A regional dashboard should include:

- household income or consumption;
- electricity;
- water;
- internet;
- travel time to health care;
- school outcomes;
- housing;
- roads;
- employment;
- business registration.

For hinterland and riverine areas, physical distance can make the same budget allocation produce different service access.

This is especially relevant when national resource revenue expands rapidly.

Analytical section 219L. Indigenous and community benefit systems require transaction-level evidence

Community allocations should be evaluated through a chain similar to public investment.

The chain should record:

- eligibility;
- consultation;
- decision;
- allocation;
- receipt;
- procurement;
- asset or service;
- maintenance;
- grievance;
- outcome.

Official programme documents and critical stakeholder accounts should remain distinguishable. [S62-S63, S93]

The strongest evidence is village-level implementation that can be verified by the community and an independent reviewer.

Analytical section 219M. Grievance systems should generate data

A grievance mechanism should not be a mailbox.

It should produce a dataset containing:

- category;
- location;
- date;
- institution responsible;
- resolution time;
- outcome;
- appeal;
- recurrence.

The data can reveal systemic issues in land access, employment, procurement, environmental impact or public services.

Recurring grievances should trigger policy review.

Analytical section 219N. Gender analysis should move beyond participation rates

A petroleum transition can affect women and men differently through:

- occupational segregation;
- unpaid care;
- migration;
- safety;
- access to technical training;
- entrepreneurship;
- land ownership;
- household price exposure.

The analysis should therefore track not only labour-force participation but also occupational level, pay, training completion, business ownership and regional access.

Analytical section 219O. Local content should have an exit test

A local-content rule should not be judged only by whether the target was met.

It should also ask whether protection or preference remains necessary.

An exit test can assess:

- number of capable suppliers;
- competitive pricing;
- quality;
- export success;
- safety performance;
- financing access.

A mature category may eventually need less preference.

A weak category may need a different intervention.

This keeps local content focused on capability rather than permanent insulation from competition.

Analytical section 219P. Expanded Part VI conclusion

Local content is one of the clearest places where petroleum policy can create durable capability, but only if the measurement moves beyond gross spend.

The reporting architecture already collects many of the fields needed for a more rigorous system. [S108]

The next-generation framework should reconcile procurement, employment, ownership, imports, wages, retained profit, training and supplier progression.

The social side requires equally disciplined evidence.

GDP growth, public expenditure and petroleum employment are not substitutes for current household and regional data.

For a newcomer, the practical blueprint is to build three linked systems before first oil:

- supplier and beneficial-ownership register;
- occupation and skills ledger;
- household and regional outcomes baseline.

These systems make it possible to distinguish temporary boom participation from lasting national capability.

Edition 2.0 evidence deepening: from reported local spend to verified economic retention

Analytical section 219Q. The H1 2026 procurement figure is an input, not the conclusion

The 2026 mid-year reporting provides a useful contemporary observation because it places a number on procurement within the categories prioritized for Guyanese participation. The reported first-half amount was approximately US\$466.2 million. The comparable first-half 2025 figure cited in the same reporting cycle was lower. [S07]

The correct analytical treatment is deliberately narrow.

The figure demonstrates that substantial procurement activity is being routed through categories subject to local-content policy.

It does not, by itself, establish:

- how much of the contract value remained in Guyana;
- how much represented imported equipment or material;
- how much became Guyanese wages;
- how much profit accrued to verified Guyanese beneficial owners;
- how much capability remained after the contract ended.

The final Edition 2.0 tables should therefore place reported procurement beside a second column labelled "retained domestic value: not yet independently reconciled" unless the underlying supplier data permit the calculation.

This is an example of how the book should distinguish a valid government statistic from a broader economic inference that the statistic alone cannot support.

Analytical section 219R. Cumulative procurement needs consistent start dates

Cumulative industry-spend figures are sensitive to the date from which the calculation begins.

One source may measure spending since 2015.

Another may measure spending since first oil in 2019.

Another may measure spending since the Local Content Act came into force.

These totals are not directly comparable.

Every cumulative local-content statistic should therefore include:

- start date;
- end date;
- included companies;

- whether goods and services are gross or net of imports;
- whether wages are included;
- currency conversion method.

A cumulative figure without those fields should be treated as a communications statistic rather than a full economic account.

Analytical section 219S. The supplier registry should be connected to contract outcomes

The number of registered suppliers measures access to the system.

It does not measure whether those suppliers received work.

A stronger registry should connect each supplier to:

- beneficial ownership;
- capability category;
- certifications;
- bids submitted;
- contracts won;
- contract value;
- contract performance;
- training received;
- later non-petroleum or export contracts.

This would allow the state to distinguish registration growth from supplier development.

It would also show whether procurement is concentrated among a small number of firms.

Analytical section 219T. Concentration is a local-content risk

A local-content regime can meet aggregate targets while opportunities remain concentrated.

The analysis should therefore calculate:

- top-five supplier share;
- top-ten supplier share;
- median contract value;
- number of first-time suppliers;
- number of repeat suppliers;
- region of supplier ownership;
- sector of supplier capability.

High concentration is not automatically evidence of failure.

Some petroleum services require scale and certification.

It is evidence that the policy should examine whether barriers to entry are technical, financial or informational.

Analytical section 219U. Procurement categories should be reviewed against cost and quality

Local-content targets can create value when they accelerate learning and market formation.

They can create cost when supply is constrained or firms exist only nominally.

For each reserved or targeted category, government should periodically compare:

- local price;
- international benchmark;
- quality;
- delivery time;
- safety performance;
- number of qualified suppliers.

The objective is to understand whether the category is moving toward competitive domestic capability.

A category that remains structurally expensive after years of support may need a different strategy.

Analytical section 219V. Offshore catering is a useful micro-case

Catering illustrates how a seemingly simple service can create a wider local value chain.

The economic chain can include:

- agriculture;
- food processing;
- cold storage;
- transport;
- hygiene standards;
- offshore logistics;
- workforce training.

A procurement contract therefore has greater developmental value when local firms progressively capture more of the upstream chain rather than only the final service.

The audit should ask what share of food was grown or processed domestically, which certifications were required and whether the supplier later gained access to other industrial clients.

Analytical section 219W. Fabrication illustrates the difference between assembly and capability

Fabrication can range from basic metal work to high-specification components requiring welding procedures, traceability, non-destructive testing and international certification.

A "local fabrication" statistic should therefore identify the technical level of the work.

Capability indicators could include:

- coded welders;
- certified procedures;
- quality-management systems;
- pressure or structural classes;
- client acceptance;
- export qualification.

This makes the local-content discussion relevant to industrial policy.

Analytical section 219X. Finance is a hidden constraint on supplier development

A local company can have technical capability and still fail to win large petroleum contracts because it cannot finance:

- inventory;
- equipment;
- insurance;
- performance bonds;
- long payment cycles.

The policy toolkit should therefore distinguish a capability gap from a financing gap.

Potential responses include:

- receivables finance;
- guarantee schemes;
- supplier-development lending;
- transparent contract pipelines;
- prompt-payment standards.

The risk is that public support socializes weak commercial decisions.

Financial support should therefore be linked to verified contracts and capability.

Analytical section 219Y. Safety performance is part of local value

A supplier that wins work but cannot meet petroleum safety standards is not durable local capacity.

Supplier-development reporting should include:

- safety training;
- incident rates;
- audit findings;
- quality non-conformances;
- corrective actions.

This prevents local content from being treated as a separate objective from operational integrity.

Analytical section 219Z. Local content should be connected to the education system

The workforce forecast should be shared with:

- technical institutes;
- universities;
- apprenticeship programmes;
- scholarship administrators.

This does not mean training institutions should produce workers for only one sector.

It means that labour-demand information should help shape course capacity and technical equipment.

The strongest skills investment creates portable competencies useful in mining, power, construction, manufacturing and maritime services as well as petroleum.

Analytical section 219AA. Skills portability is the post-petroleum test

A petroleum-specific skill can be valuable during production but vulnerable to sector decline.

A portable skill creates a stronger national asset.

The workforce strategy should therefore favour competencies such as:

- electrical systems;
- mechanical maintenance;
- process control;
- welding;
- instrumentation;
- marine operations;

- data analysis;
- safety management;
- project management.

The policy should measure where trained workers move over time.

Analytical section 219AB. Housing pressure belongs in the distribution analysis

A resource boom can raise rents rapidly near employment centres.

For households not receiving petroleum-linked wage gains, higher housing cost can reduce real welfare even while national income rises.

The social dashboard should therefore include:

- rent index;
- house-price index;
- household overcrowding;
- commute time;
- new housing supply.

These indicators help explain why GDP and household experience can diverge.

Analytical section 219AC. Food-price exposure matters for lower-income households

Imported demand, wage growth and logistics constraints can increase food prices.

The distributional effect is larger for households that spend a high share of income on food.

The paper should therefore present inflation by major household category where data permit rather than only the headline consumer-price index.

Analytical section 219AD. Regional investment should be tracked to service outcomes

Resource revenue may finance roads, schools, health facilities or digital infrastructure outside the main economic corridor.

The audit question is not only whether money was allocated.

It is whether access improved.

For each region, the public-investment register should track:

- facility or infrastructure delivered;
- operational status;
- staffing;

- utilization;
- maintenance;
- service outcome.

This is the social equivalent of the petroleum transaction ledger.

Analytical section 219AE. A local-content annual account

The final book should propose an annual local-content account with five statements.

Statement 1: Procurement. Gross local procurement by category and contractor.

Statement 2: Domestic value. Imported inputs, Guyanese wages, local profit, taxes and retained value.

Statement 3: Employment. Guyanese employment by occupation and seniority.

Statement 4: Capability. Training completion, certification, supplier progression and exports.

Statement 5: Distribution. Regional supplier participation, household indicators and community outcomes.

This would convert local content from a target-compliance exercise into a national capability account.

Analytical section 219AF. Evidence conclusion

The existence of detailed reporting requirements is a strength because it means many of the required data fields already exist in the regulatory system. [S108]

The analytical challenge is aggregation and verification.

Edition 2.0 should therefore avoid both extremes.

It should not dismiss official procurement numbers because they are not perfect measures of value added.

It should not present them as proof that the entire amount remained in-country.

The defensible position is to use the reported figure for what it measures and build the additional reconciliation needed to answer the deeper question.

Original case study: from Guyanese procurement to verifiable domestic value added

Why the headline procurement total is not the result

Guyana's local-content regime creates a case study in the distinction between **access to a petroleum supply chain** and **durable value retained by the host economy**. The policy has a substantial and increasingly detailed reporting architecture. The Local Content Secretariat's June 2025 submission guidance specifies categories for procurement, employment, salary scales, training and capability development and expects reporting against approved plans. By June 2026, government reporting described approval of more than forty companies' annual local-content plans. These are important controls because they establish a reporting population and a recurring opportunity to test progress. They are not, on their own, an independent valuation of what the country retains. [S108, S139]

The reported amounts make careful boundary-setting particularly important. Government reporting for 2024 cites approximately US\$743 million spent with Guyanese suppliers, with about US\$612 million attributed to the forty designated areas. A June 2026 government release attributed approximately US\$3.6 billion in cumulative spending with local businesses and engagement of roughly 2,000 suppliers by the end of 2025 to statements made during the annual-plan approval process. Another official report dated September 2026 described approximately 500 registrations in the first half and preliminary first-half procurement of US\$466.2 million in priority sectors. These figures refer to different times, institutional populations and categories. Summing or contrasting them without harmonizing the reporting universe risks creating a false trend. [S139, S140]

Consider the first-half number. Its legitimate meaning is that a substantial volume of goods and services procurement in identified priority categories was reported for the six-month period. It is not an estimate of how much domestic value was created. An imported heavy-equipment item sold through a qualified local distributor may appear in gross procurement even when most of the final contract price covers foreign manufactured content. Conversely, an engineering contract with high Guyanese payroll and local intellectual property may generate far more domestic value per dollar of spending. Both contracts can be genuine local-content participation but they answer different development questions.

The procurement dataset should therefore distinguish at least the following *non-additive* measures: gross awarded or invoiced local contracts, spending within designated categories, cash paid during the reporting period, participating legal entities, independently verified beneficial ownership, wage income earned by Guyanese nationals, imported goods/services, domestic production taxes, operating surplus generated in Guyana, foreign factor income and supplier investment in transferable capacity. A source that gives only the first two measures cannot support claims about all the others.

A worked example with separate accounting boundaries

Suppose a hypothetical Guyanese petroleum-service company earns a US\$100 million contract over one accounting period. Assume US\$56 million of purchased inputs and specialized services are imported from abroad; US\$24 million goes to Guyanese labour; US\$4 million represents qualifying domestic production taxes net of subsidies; and US\$16 million is gross operating surplus generated by the contractor. For simplicity, this synthetic illustration has no other intermediate inputs, inventory timing or capital transfers, and the categories are mutually exclusive.

Under that strictly simplified production-account boundary, **gross domestic value added is US\$44 million**: contract output of US\$100 million less US\$56 million of imported inputs. The same amount can be described from its components—US\$24 million wages, US\$4 million net taxes on production and US\$16 million gross operating surplus. This is an illustration of a value-added bridge, not an actual audit result for Guyana, not an estimate of the multiplier effect and not a claim that all local-content suppliers have this cost structure.

Now introduce an ownership question. Suppose the evidence shows that only US\$10 million of the US\$16 million surplus is attributable to verified domestic beneficial owners and that the remaining US\$6 million represents foreign factor income under the defined accounting convention. Under a simplified **retained domestic income** concept, the same contract may retain US\$38 million in domestic wages, production taxes and domestic-owned surplus, even though its domestic value added was US\$44 million and its headline local procurement was US\$100 million. If the domestic-owned surplus is later reinvested in imported equipment, still another cash measure changes. The term "retained" therefore requires a precise time horizon, ownership test and definition. It should never be presented as identical to GDP value added or assumed from a company registration certificate.

A credible public dashboard should report these three measures side by side rather than select the one with the largest total. Under the synthetic assumptions the report reads: **US\$100m gross contract value; US\$44m domestic value added; US\$38m income retained by domestic workers, government and domestic owners**. The gap between them is the reason the import content and ownership records matter. This worked example should be included as an auditable exhibit with its inputs clearly marked hypothetical and a warning that real-world value-added accounting must also account for local intermediate purchases, depreciation, cross-company transactions and fiscal classifications.

Test capability, not registration alone

Supplier registration is a legitimate access metric. The strong development outcome is whether the supplier can subsequently compete without relying on preferred procurement. The longitudinal supplier register should track each firm from first registration through certification, bidding, award, delivery, audit results, workforce training and later export or non-petroleum contracts. The quality of evidence changes at each step. A business registration establishes legal entry. A signed purchase order establishes a contract. A paid invoice establishes a cash transaction. A quality or safety audit establishes a specific compliance observation. A contract won from an unrelated international buyer provides evidence of external competitiveness. It would be misleading to label all five observations simply "supplier development".

The same logic applies to employment. A headcount of Guyanese workers should be broken down by occupation, seniority, pay and competency. An employer can create substantial employment in accommodation or transport without yet building local subsea design or reservoir-modeling capability; each activity matters, but the skills pathways differ. The Secretariat's existing salary, training and capacity-development reporting fields provide a starting architecture for this more detailed assessment. A national skills ledger should link employers' forecasts to independent assessment, supervised practice and certification rather than counting course attendance as completed technical transfer. [S108]

One useful sectoral micro-case is offshore catering. A catering contract may create a domestic platform for agriculture, food processing, cold storage, logistics, hygiene certification and maritime delivery. Its long-term value depends on whether more of those upstream activities become locally supplied and whether participating firms later win other industrial clients. In contrast, a fabrication contract may require expensive equipment, coded welders, quality-control systems and internationally accepted traceability. An identical spending target can therefore imply very different investment, risk and training requirements. The government should classify capability progression by category rather than expect one localization percentage to capture all industrial learning.

Audit the distributional claim independently

A local-content policy is also a claim about who benefits. Enterprise data cannot substitute for household evidence. Urban workers in petroleum logistics and specialist construction may gain income while households in geographically remote regions face rising food or transport costs without corresponding employment. A genuinely national assessment should use current household and labour data broken down by region, gender, occupation and living cost. It should not transfer welfare estimates from the 2018/2019 household survey into 2026 as if the latter period had been measured. Where current representative microdata are unavailable, that is a limitation to disclose and a data-collection task to assign. [S61]

Ownership and regional access require equally careful treatment. A qualified firm based in Georgetown may serve an offshore operation while drawing workers, profits or imported materials from a range of origins. A national supplier total does not identify which regions receive wages or investment. An integrated system would collect the location of the productive

activity, residence of workers, location of subcontractors and beneficial ownership separately. It would also track grievance mechanisms and whether procurement opportunities are posted in a form accessible to new entrants. None of these measures requires the author to assume that programme success or failure follows automatically from the statute's enactment.

For village or Indigenous benefit schemes financed by separate natural-capital programmes, use a distinct account. Official allocations, village-level receipts, procurement, asset delivery, maintenance and community verification should be independently traceable. Critical perspectives on consultation and safeguard procedures should be attributed and assessed against programme documents and direct community evidence rather than ignored or treated as automatically conclusive. This is the same resource-to-outcome discipline applied to a different rights and institutional context. [S62, S63, S93]

What the next annual local-content account should publish

A full annual account could be organized into five connected but non-overlapping statements. The first gives gross procurement by reporting population and category, with award, invoice and cash-payment periods distinguished. The second shows a consistent domestic-value bridge from gross output to imported and domestic intermediate inputs, value added and factor-income retention, with independent ownership checks. The third shows employment by occupation and pay alongside competency-based training. The fourth traces supplier progression, quality, safety and new non-petroleum customers. The fifth presents the geographical and household indicators needed to test distribution, using the latest representative evidence with explicit dates.

This is also how the book should present the reported US\$466.2 million of first-half 2026 priority-sector procurement. The figure belongs in the gross-activity statement, not in the verified domestic-value statement unless the underlying supplier accounts have been independently reconciled. The government's June approval of more than forty company plans belongs in the compliance-process account, not in a realized-outcomes chart. The operator-attributed supplier and cumulative-spend data belong in an explicitly labelled industry-reporting series. Keeping these boundaries visible allows readers to recognize the substantial scale of participation without overstating how much wealth remained in the country. [S139, S140]

The transferable blueprint for another small producer is therefore a set of linked registers rather than a headline target alone. Build a beneficial-ownership and supplier register; require comparable invoice and employment reports; calculate value added using transparent accounting boundaries; verify training through demonstrated capability; and follow a cohort of firms beyond the first petroleum contract. If the regime is intended to create long-lived domestic industry, its results should ultimately be observable after petroleum procurement growth slows.

PART VII | Economic transformation and diversification

Economic Transformation, Infrastructure, Energy and Diversification

Chapter 220. The development question after the windfall

Petroleum changes the scale of national income and public finance.

It does not automatically answer the development question.

A resource-rich state still needs to decide how to convert temporary extraction income into assets and capabilities that continue producing value after the fastest period of oil growth ends.

The framework used in this study therefore separates:

- petroleum output;
- non-oil output;
- financial saving;
- public capital;
- human capital;
- private productive capacity;
- natural capital.

The objective is national wealth conversion.

Chapter 221. GDP as a production measure

GDP measures economic production within the country.

Guyana's petroleum expansion has produced exceptional headline growth.

IMF staff reported real GDP growth above 19 percent in 2025 after average growth near 40 percent during 2023 and 2024. [S64]

These figures establish the scale of the macroeconomic transition.

They do not show how income is distributed.

They do not show service quality.

They do not show the durability of infrastructure.

The flagship should therefore use GDP as the first line of analysis, not the final line.

Chapter 222. Oil and non-oil GDP

A resource economy needs both total and non-resource measures.

Oil GDP responds to:

- production;
- price-volume accounting;
- new FPSO startup.

Non-oil GDP captures activity in:

- construction;
- agriculture;
- manufacturing;
- services;
- mining outside petroleum.

A strong non-oil growth rate can reflect genuine diversification.

It can also reflect oil-financed construction and services.

Sector detail is needed.

Chapter 223. Construction

Construction is often one of the fastest channels through which a resource boom enters the domestic economy.

Government builds infrastructure.

Private firms build offices, housing and industrial facilities.

Households increase demand.

Construction growth creates jobs.

It also competes for:

- labour;
- cement;
- aggregate;
- equipment;
- land.

The final chapter should track output and cost together.

Chapter 224. Agriculture

A petroleum boom can support agriculture through:

- roads;
- drainage;
- electricity;
- cold storage;
- finance.

It can also challenge agriculture through:

- higher wages;
- imported food;
- land prices.

The diversification analysis should use:

- production;
- yield;
- exports;
- imports;
- farm income.

A policy announcement about food security should be tested against those indicators.

Chapter 225. Manufacturing

Manufacturing benefits when energy and logistics become more reliable.

It struggles when wages and imported input costs rise.

The paper should measure:

- manufacturing value added;
- employment;
- exports;
- electricity cost;
- industrial capacity.

Petroleum-related fabrication should be identified separately from broader manufacturing.

Chapter 226. Mining

Gold and other mining sectors formed part of Guyana's economy before oil.

The petroleum era changes their relative share of GDP.

A declining share does not necessarily mean lower mining output.

The denominator has expanded.

The analysis should therefore report both absolute output and share.

Chapter 227. Services

Oil creates demand for:

- finance;
- legal work;
- accounting;
- transport;
- logistics;
- hospitality;
- engineering.

Some services build exportable capability.

Others depend almost entirely on the oil investment cycle.

The diversification chapter should identify which have clients outside petroleum.

Chapter 228. Public investment as a transmission channel

Petroleum revenue influences the non-oil economy strongly through public investment.

The 2026 Budget Speech records a large programme of capital and social spending. [S65]

A budget is an authorization and policy statement.

The final case needs a project outcome register.

For each major project it should show:

- appropriation;
- award;
- construction;
- completion;
- operation.

Chapter 229. Roads

Road investment can reduce:

- travel time;
- vehicle cost;
- geographic isolation.

A road also creates:

- maintenance;
- land-use change;

- environmental effect.

The project outcome should therefore include:

- traffic;
- time saved;
- condition;
- maintenance cost.

Chapter 230. Bridges

A major bridge can change national transport patterns.

The appraisal should compare:

- existing crossing;
- traffic;
- travel time;
- capacity;
- resilience;
- toll or operating model.

The post-completion review should compare actual use with forecast.

Chapter 231. Ports

Petroleum and broader trade both depend on ports.

Port investment should be analyzed through:

- depth;
- berth;
- cargo throughput;
- turnaround;
- cost.

A petroleum shorebase is not automatically a national public port asset.

Ownership and financing should be recorded.

Chapter 232. Airports

Economic growth increases passenger and cargo traffic.

Airport investment can support:

- tourism;
- business;

- emergency response.

The infrastructure ledger should distinguish capital expansion from operating performance.

Chapter 233. Drainage and sea defence

For Guyana, climate resilience is economic infrastructure.

A flood-protection project should be assessed through:

- protected area;
- pump capacity;
- drainage time;
- failure frequency;
- maintenance.

This allows petroleum-financed resilience to enter the national wealth framework.

Chapter 234. Housing

Housing investment affects:

- supply;
- rent;
- household wealth;
- construction demand.

A national housing programme should be evaluated through occupied units and infrastructure access, not allocated lots alone.

Chapter 235. Hospitals

Health capital creates value only when combined with:

- staff;
- equipment;
- supplies;
- operating budget.

The public-investment register should include those recurrent requirements.

Chapter 236. Schools and universities

Education infrastructure should support a broader human-capital strategy.

The petroleum transition increases demand for:

- engineering;
- technology;

- finance;
- project management.

A building without qualified teachers does not close the skills gap.

Chapter 237. Electricity as a binding constraint

Power cost and reliability affect households and firms.

Guyana's energy strategy includes gas-to-energy and renewable resources. [S67]

The correct performance measures include:

- dependable capacity;
- generation;
- outages;
- losses;
- tariff;
- fuel cost;
- emissions.

Installed megawatts alone are not enough.

Chapter 238. Gas-to-energy

A domestic gas project can change the power system through:

- fuel cost;
- generation efficiency;
- supply diversity.

A full project assessment needs:

- pipeline;
- processing;
- power plant;
- grid;
- financing;
- operating cost;
- gas terms.

The outcome should be tested through observed generation, reliability and tariff.

Chapter 239. Project delay risk

Large integrated energy projects contain several dependencies.

A delay in one component can delay system benefits.

The project register should therefore use:

- original schedule;
- revised schedule;
- commissioning.

A policy target should not be described as an achieved outcome.

Chapter 240. Demand growth

Economic growth increases power demand.

New housing, industry, commercial buildings and data infrastructure add load.

A generation plan must therefore compare new supply with demand.

The paper should not report a new power plant as excess capacity without a demand forecast.

Chapter 241. Energy efficiency

The Guyana Energy Agency's 2026 demand-side management plan targets significant savings by 2030. [S66]

The final assessment should distinguish target from verified saving.

Efficiency programmes should report:

- participants;
- measure;
- cost;
- verified kWh saved.

Demand reduction is a resource.

Chapter 242. Renewable energy

Solar, hydro and other renewable resources diversify the electricity mix.

A national energy comparison should use:

- installed capacity;
- annual generation;
- capacity factor;
- storage;
- grid integration;
- cost.

A renewable project should not be measured only by nameplate MW.

Chapter 243. Reliability

Reliability has direct economic value.

Common measures include:

- outage frequency;
- outage duration;
- unserved energy.

The final paper should establish a pre-oil baseline and post-investment series.

This tests whether capital spending improves service.

Chapter 244. Energy affordability

The household and business impact depends on the final tariff and quality of service.

A tariff reduction achieved through hidden subsidy has a different fiscal effect from one achieved through lower generation cost.

The paper should report:

- cost;
- tariff;
- subsidy.

Chapter 245. Export concentration

Petroleum rapidly changes export composition.

A diversification strategy should therefore monitor concentration.

The Herfindahl index offers one measure.

The paper should also show non-oil exports in absolute terms.

A country can become more petroleum-concentrated even while non-oil exports grow.

Chapter 246. Dutch-disease channels

The classic resource-boom concern operates through several channels.

Higher resource income increases spending.

Demand for non-tradable goods rises.

Wages and prices can increase.

Labour shifts toward resource-linked sectors.

The real exchange rate appreciates through nominal exchange rate movement, domestic prices or both.

Tradable non-resource firms become less competitive.

These channels should be tested, not assumed.

Chapter 247. IMF 2026 assessment

IMF staff in July 2026 stated that available indicators did not show clear signs of overheating or resource-driven competitiveness pressure at that time, while identifying wage and foreign-exchange indicators for continued monitoring. [S64]

The paper should preserve the date.

A 2026 assessment is not a permanent conclusion.

The independent dashboard should update each indicator.

Chapter 248. Wage-based indicators

Wage growth can reflect:

- productivity;
- labour shortage;
- inflation;
- sector demand.

The paper should compare wages with productivity.

A wage increase supported by productivity has different competitiveness implications from one without matching output.

Chapter 249. Real exchange rate

The real exchange rate compares domestic costs with trading partners.

In a relatively stable nominal exchange-rate system, adjustment may appear through domestic inflation and wages.

The paper should use more than one indicator.

Chapter 250. Imports

Large investment raises imports of:

- machinery;
- vehicles;
- materials;
- services.

Import growth is not automatically a weakness.

Capital imports can expand future productive capacity.

The analysis should distinguish capital, intermediate and consumption imports.

Chapter 251. Non-oil exports

The most direct diversification test is whether sectors earn foreign exchange outside petroleum.

The paper should track:

- gold;
- agriculture;
- manufactured goods;
- services;
- tourism.

A stronger export base reduces long-run oil dependence.

Chapter 252. Private investment

Government spending is only one development engine.

The private sector needs:

- reliable power;
- logistics;
- rule clarity;
- finance;
- skills.

The flagship should track non-oil private investment and firm formation.

This provides evidence about whether public infrastructure crowds in private activity.

Chapter 253. Financial sector

Rapid growth affects banks through:

- deposits;
- credit;
- mortgages;
- corporate lending.

A resource boom can increase asset prices and concentration.

The financial stability chapter should monitor:

- credit growth;
- nonperforming loans;
- sector exposure;

- foreign currency.

Chapter 254. Land prices

Infrastructure and income growth can increase land values.

This creates wealth for owners and affordability pressure for buyers.

A land-price series would strengthen the distribution analysis.

Where data are absent, the gap should be noted.

Chapter 255. Public-sector wages

Government competes with private employers.

Public wage policy affects:

- staffing;
- inflation;
- service delivery.

The paper should analyze technical occupations separately from broad wage averages.

Chapter 256. National productivity

The most durable economic test is productivity.

Infrastructure and human capital should allow workers and firms to produce more value per unit of input.

The flagship should calculate productivity cautiously because petroleum output dominates aggregate measures.

Non-oil productivity is more informative for diversification.

Chapter 257. National wealth accounting

The economic transformation should ultimately appear in a balance-sheet view.

Petroleum reserves fall.

Financial assets rise.

Public infrastructure rises.

Human capital may rise.

Natural capital can rise or fall.

Debt and maintenance liabilities change.

The country should track these categories separately.

Chapter 258. Conversion efficiency

A conceptual wealth-conversion ratio can compare the present value of durable assets created with petroleum revenue used.

The paper should not publish a precise ratio without defensible valuation.

It can still use a dashboard.

For example:

- financial assets;
- infrastructure condition;
- educational outcomes;
- health outcomes;
- natural capital.

Chapter 259. A low-price development test

Assume petroleum receipts fall for three years.

Which public investments continue?

Which recurrent programmes depend on oil?

Which projects can be delayed without large economic loss?

A resilient development plan should identify priorities before the shock.

Chapter 260. A capacity test

Assume capital budgets rise 30 percent but contractor capacity does not.

Expected effects include:

- fewer bids;
- higher price;
- delay.

The response options include sequencing, imports, training and procurement changes.

This is the practical meaning of absorptive capacity.

Chapter 261. A maintenance test

Assume oil revenue growth slows after a large infrastructure build.

Can stable non-oil revenue cover:

- maintenance;
- staffing;

- operating cost?

If not, the asset portfolio creates future fiscal pressure.

This test should be part of appraisal.

Chapter 262. New-producer blueprint

A newcomer should establish before first revenue:

- national infrastructure baseline;
- project appraisal;
- procurement control;
- maintenance accounting;
- non-oil GDP series;
- price and wage monitoring;
- export concentration;
- household baseline.

The country should then update them annually.

Chapter 263. Part VII findings

Guyana's growth record demonstrates the scale of a petroleum-led macroeconomic transition.
[S64]

The strongest development case will rest on what the petroleum period leaves behind.

That includes:

- financial assets;
- infrastructure that works;
- educated and healthy people;
- competitive firms;
- resilient natural systems.

Budget size and GDP growth are important evidence.

They are not sufficient evidence of durable national wealth.

The blueprint lesson is to measure the conversion, not only the windfall.

Edition 2.0 deepening: turning a petroleum boom into a second growth engine

Analytical section 263A. The development test is productivity outside petroleum

Non-oil GDP growth is important, but it does not by itself establish diversification.

Construction, retail, transport and professional services can expand because petroleum investment and public spending are increasing.

A stronger diversification test asks whether the economy is building activities that can remain competitive when petroleum investment and revenue growth slow.

The core indicators should therefore include:

- non-oil exports;
- labour productivity;
- firm survival;
- private investment outside petroleum;
- energy intensity;
- logistics cost;
- export-market concentration;
- technological capability.

This distinguishes demand spillovers from structural transformation.

Analytical section 263B. Export diversification should be measured in foreign-exchange terms

An economy can produce more goods and services domestically while remaining dependent on petroleum for foreign exchange.

The diversification dashboard should therefore track:

- petroleum export share;
- gold and other mineral share;
- agricultural exports;
- manufactured exports;
- tourism receipts;
- business-service exports;
- regional trade.

The objective is not to reduce petroleum exports artificially.

It is to create additional sources of foreign exchange before the petroleum cycle matures.

Analytical section 263C. Public infrastructure should reduce economy-wide transaction cost

The development value of a road, bridge, port or airport is not the construction expenditure.

It is the reduction in time, cost, risk or congestion experienced by users.

Project evaluation should therefore estimate:

- travel time;
- freight cost;
- reliability;
- accident cost;
- market access;
- maintenance cost;
- induced demand;
- climate exposure.

A project can be strategically valuable even when it does not generate direct cash revenue.

The appraisal should still define the service outcome.

Analytical section 263D. Electricity is a cross-sector productivity input

Reliable and affordable electricity affects:

- manufacturing;
- cold storage;
- digital services;
- mining;
- hospitality;
- household welfare;
- public services.

The energy transition should therefore be assessed through delivered system performance rather than installed megawatts alone.

A national power dashboard should track:

- demand;
- available capacity;
- outage frequency;
- outage duration;
- system loss;

- fuel cost;
- generation mix;
- reserve margin;
- customer tariff.

Official Guyana energy planning and demand-side work provide inputs for this analysis. [S66-S67]

Analytical section 263E. Gas-to-energy should be evaluated as an integrated system

A gas-to-energy project is not one asset.

It can include:

- offshore gas infrastructure;
- pipeline;
- processing;
- power generation;
- transmission;
- distribution;
- backup supply.

The economic case depends on the performance of the full chain.

The risk model should therefore test:

- pipeline interruption;
- plant outage;
- demand forecast error;
- fuel-price assumptions;
- construction delay;
- transmission constraints.

A lower expected generation cost becomes a development benefit only when reliable power reaches users.

Analytical section 263F. Renewable energy and gas should be analysed through system reliability

Energy policy should not reduce to a contest between technologies.

A small system needs a portfolio that manages:

- variable demand;
- renewable intermittency;

- fuel risk;
- storage;
- transmission;
- reserve capacity;
- climate resilience.

The correct comparison is delivered reliability and lifecycle cost.

Analytical section 263G. Agriculture needs productivity, storage and market access

Oil revenue can support agricultural infrastructure, but long-run competitiveness requires more than subsidy.

The agricultural transformation chain includes:

- drainage and irrigation;
- seed and genetics;
- extension;
- mechanization;
- storage;
- processing;
- standards;
- logistics;
- export market access.

A country that expands primary production without storage or market access may create price volatility rather than durable value.

Analytical section 263H. Manufacturing requires scale and capability, not only cheaper power

Lower electricity cost can improve manufacturing economics.

It does not guarantee competitive industry.

Manufacturing strategy should also assess:

- domestic market size;
- regional market access;
- freight;
- labour skill;
- finance;
- quality standards;
- supplier networks;

- imported input dependence.

The most promising sectors may be those linked to existing resource and agricultural strengths rather than arbitrary import substitution.

Analytical section 263I. Tourism should be assessed against petroleum-era cost pressure

A petroleum boom can raise wages, land prices and construction cost.

Tourism competes for many of the same inputs.

A tourism strategy should therefore model:

- air access;
- room inventory;
- labour cost;
- service quality;
- nature assets;
- safety;
- seasonality;
- marketing;
- carrying capacity.

High national income does not automatically create a competitive tourism sector.

Analytical section 263J. Digital services offer a different diversification pathway

Digital exports are less dependent on physical freight.

Their constraints are different:

- connectivity;
- power reliability;
- skills;
- data protection;
- payment systems;
- market access.

A small state can therefore use digital services to diversify even when manufacturing scale is limited.

The petroleum transition can finance the infrastructure and education required, but the sector still needs firms capable of competing internationally.

Analytical section 263K. Urban growth should be treated as infrastructure demand

Rapid income and population changes affect:

- housing;
- land;
- transport;
- drainage;
- waste;
- water;
- schools;
- health care.

Urban planning should therefore be integrated into the resource-conversion model.

If housing and transport capacity lag, petroleum-driven income growth can create congestion and price pressure that reduce real welfare.

Analytical section 263L. Dutch-disease analysis should use evidence rather than labels

The phrase "Dutch disease" is often used too loosely.

A serious test should examine:

- real exchange-rate pressure;
- wages;
- tradable-sector employment;
- export performance;
- import penetration;
- productivity;
- government spending;
- migration.

A decline in one sector is not sufficient by itself.

The question is whether resource inflows are systematically reducing the competitiveness of non-resource tradables.

IMF analysis has treated this as a risk requiring monitoring rather than a conclusion that can be inferred from GDP growth alone. [S68]

Analytical section 263M. Public investment can crowd in or crowd out private investment

Infrastructure can crowd in private activity by lowering cost and risk.

Large state spending can also crowd out firms through:

- labour competition;
- construction capacity;
- credit;
- land;
- procurement concentration.

The development model should track both effects.

A public project that improves logistics may increase private investment.

A poorly sequenced capital boom may make private construction prohibitively expensive.

Analytical section 263N. Debt policy should be integrated with the sovereign fund

A country can simultaneously hold a large resource fund and borrow.

That is not automatically contradictory.

The correct comparison considers:

- borrowing cost;
- fund return;
- liquidity;
- project return;
- currency;
- maturity;
- risk.

Borrowing to preserve fund assets can be expensive if debt costs exceed the value of liquidity or the project return.

Using the fund excessively can weaken future buffers.

The optimal balance requires a consolidated public balance sheet.

Analytical section 263O. The second-engine test

By the time petroleum production matures, the economy should be able to point to at least one additional growth engine capable of earning foreign exchange and raising productivity without relying primarily on new petroleum investment.

That engine may be a portfolio rather than one sector.

The test should ask:

- Is it export capable?
- Is productivity rising?
- Can it attract private capital?
- Does it employ skills developed during the petroleum era?
- Does it remain viable under lower public spending?

This is a more demanding standard than non-oil GDP growth.

Analytical section 263P. Expanded Part VII conclusion

The economic-transformation chapter should connect petroleum wealth to the productive system that will remain after the boom.

The key sequence is: resource revenue to infrastructure and human capital to lower transaction cost to private investment and productivity to competitive exports to a broader tax base.

If one link is weak, large public spending can still produce visible growth without equivalent long-run capability.

For a newcomer, the blueprint is therefore to define the post-petroleum economy before peak revenue arrives.

Every major infrastructure project should be able to explain which productivity constraint it removes.

Every diversification programme should identify the market in which firms are expected to compete.

Every permanent expenditure commitment should be tested against a lower petroleum-revenue future.

The objective is not to predict the exact structure of the economy decades ahead.

It is to ensure that petroleum finance expands national options rather than narrowing them.

Edition 2.0 evidence deepening: the resource-to-asset conversion test

Analytical section 263Q. Public investment needs a delivery taxonomy

The development chapter should distinguish six stages of public investment.

Allocated means parliament has authorized expenditure.

Contracted means a procurement process has produced an enforceable agreement.

Under construction means physical works have begun.

Commissioned means the asset has been accepted for use.

Operational means the asset has the staff, power, maintenance and systems required to function.

Outcome evidenced means the public can observe a change in service, productivity, travel time, reliability, health access or another defined result.

These stages should never be collapsed into one word such as "delivered".

The distinction matters most during a rapid resource boom because expenditure can grow faster than the evidence systems used to track outcomes.

Analytical section 263R. The 2026 capital programme illustrates portfolio scale

The 2026 budget contains large allocations across energy, transport, health and other public infrastructure. [S129]

The analytical point is not that a large allocation is inherently positive or negative.

The point is administrative scale.

A state implementing many large projects simultaneously needs:

- engineers;
- procurement officers;
- contract managers;
- quantity surveyors;
- auditors;
- land administration;
- environmental review;
- maintenance planners.

If the number and value of projects rise faster than those functions, the marginal risk of delay, variation or weak documentation also rises.

The capital programme should therefore be accompanied by a public-sector delivery-capacity plan.

Analytical section 263S. The Auditor General provides the necessary counterweight

The Auditor General's 2024 report examined hundreds of public contracts and recorded overpayments and contract-management findings, as well as later recovery or completion actions in the cases described. [S130]

The significance for this publication is methodological.

A budget document tells the reader what government intended to finance.

An audit report tests whether transactions and works were administered correctly.

A project-performance review asks whether the completed asset achieved its purpose.

All three evidence types are needed.

A serious resource-conversion assessment should therefore never rely only on budget speeches or commissioning announcements.

Analytical section 263T. Drainage and irrigation show why maintenance belongs in the wealth account

The NDIA performance-audit material is particularly relevant to a country exposed to flooding and climate risk.

The audit identified weaknesses in asset-management systems, documentation, staffing and verification of maintenance expenditure during the period reviewed. [S130]

This is not merely an audit issue.

It is a wealth-preservation issue.

Drainage pumps, canals and related infrastructure can protect agriculture, housing and transport. Their economic value depends on availability when extreme rainfall occurs.

A resource-rich state should therefore treat maintenance expenditure as protection of public capital rather than as a lower-status alternative to new construction.

Analytical section 263U. The bridge example needs both construction and post-opening evidence

The Wismar-Mackenzie Bridge is useful because the evidence cutoff falls just before the reported opening period.

Government reporting on 26 September 2026 described the US\$35 million four-lane bridge as nearing opening. [S131]

The correct Edition 2.0 classification at the cutoff is therefore pre-commissioning or near completion.

The next edition should add:

- actual opening date;
- traffic volume;
- travel time;
- vehicle operating cost;
- safety;
- maintenance responsibility.

This demonstrates the value of versioned project registers.

A later operating outcome can be added without rewriting what was known on 27 September 2026.

Analytical section 263V. Hospital construction should be separated from health-service expansion

Government reporting identifies several hospitals under construction and the 2026 budget allocates substantial health capital. [S129, S132]

The health outcome should not be inferred from the construction contract.

The service chain is: building to equipment to staffing to supplies to operating systems to patient access to quality and outcome.

The book should therefore ask for:

- opening date;
- staffed beds;
- diagnostic capability;
- referral volume;
- waiting time;
- utilization;
- operating cost;
- maintenance.

A hospital can be physically complete but not yet functioning at designed capacity.

Analytical section 263W. Road construction should be tested through logistics metrics

The Linden-Mabura road project illustrates another conversion chain.

Government reporting in April 2026 described substantial physical progress. [S133]

Once completed, the relevant development metrics should include:

- journey time;
- freight cost;

- wet-season reliability;
- accident rates;
- traffic volume;
- maintenance condition;
- cross-border trade.

This is how a transport project enters the productivity analysis.

The book should avoid claiming an increase in trade before the data exist.

Analytical section 263X. Infrastructure corridors create network value

A bridge, highway or port can have greater value when it connects with other infrastructure.

Network analysis should therefore identify:

- origin and destination flows;
- connection to ports;
- connection to industrial areas;
- border routes;
- housing growth;
- agriculture;
- tourism.

The value of one link may depend on completion of another.

This creates a sequencing problem.

A state should avoid building expensive links whose economic value is delayed because complementary infrastructure is not ready.

Analytical section 263Y. Capital project appraisal should include an uncertainty range

A point estimate for project cost or benefit creates false precision.

Appraisal should include uncertainty around:

- construction cost;
- completion date;
- demand;
- maintenance;
- foreign exchange;
- climate exposure.

For large projects, the state should publish at least a base, high-cost and delayed case.

This mirrors the adverse-scenario approach used for petroleum projects.

Analytical section 263Z. Project selection should be constrained by operating budgets

A country can finance construction from temporary resource revenue while the completed asset creates permanent recurrent costs.

Before approval, every project should identify:

- annual staffing;
- utilities;
- maintenance;
- insurance;
- consumables;
- periodic refurbishment.

The medium-term budget should show how those costs will be financed under a lower petroleum-revenue scenario.

Analytical section 263AA. Procurement concentration can become a macroeconomic bottleneck

A rapid public investment programme can concentrate contracts among a limited number of firms.

That may reflect capability.

It can also reduce competition.

The public-investment dashboard should monitor:

- bidders per tender;
- repeat winners;
- variation orders;
- contract extensions;
- contractor workload.

A contractor carrying too many simultaneous projects can become a delivery risk even if each individual award was competitive.

Analytical section 263AB. Construction inflation should be separated from real asset growth

A higher capital budget can reflect:

- more projects;
- larger projects;

- higher material cost;
- higher wages;
- imported inflation;
- exchange-rate effects.

Real infrastructure expansion should therefore be analysed using physical output where possible.

Kilometres of road, bridge capacity, hospital beds, megawatts and serviced lots are imperfect but useful complements to nominal spending.

Analytical section 263AC. Resource revenue should finance resilience as well as growth

For Guyana, public investment must account for coastal and flood exposure.

A road or hospital that is vulnerable to flood disruption may require additional upfront resilience investment.

The appraisal should record:

- design flood;
- drainage;
- elevation;
- backup power;
- access redundancy;
- climate assumptions.

This is a form of lifecycle insurance.

Analytical section 263AD. The public asset register should be linked to the national accounts carefully

Public capital formation contributes to GDP during construction.

That accounting contribution is not the same as the asset's long-run productivity effect.

The publication should therefore keep three measures separate:

- construction activity;
- public asset stock;
- service/productivity outcome.

This prevents circular reasoning in which high public investment is used to prove both current growth and future productivity without independent evidence.

Analytical section 263AE. The "resource conversion ratio" should be a dashboard, not one number

It is tempting to create one measure of how successfully resource revenue becomes national wealth.

A single index would conceal too much.

A better dashboard includes:

- financial assets accumulated;
- public capital commissioned;
- maintenance funded;
- human-capital outcomes;
- non-oil productivity;
- export diversification;
- natural-capital condition.

The dashboard can show movement over time without ranking governments or compressing unlike assets into an arbitrary score.

Analytical section 263AF. Edition 2.0 public-investment dataset

The new `PUBLIC\_INVESTMENT\_DELIVERY\_REGISTER.csv` records selected projects and programmes by:

- sector;
- budget or contract value;
- status at cutoff;
- delivery evidence;
- outcome evidence;
- maintenance evidence;
- audit status.

The register deliberately includes rows with "OPEN OUTCOME".

That is not an error.

It is a statement that the book knows the difference between expenditure and result.

Analytical section 263AG. Development conclusion

The resource-to-asset chain is the practical test of whether petroleum revenue produces durable national wealth.

The chain fails if any link is missing:

- weak appraisal;

- weak procurement;
- poor construction;
- delayed commissioning;
- inadequate staffing;
- no maintenance;
- no measurable service improvement.

The book should therefore follow selected investments across several editions.

A project that is under construction in 2026 should become a commissioned asset only when the evidence supports the transition.

A commissioned asset should become an evidenced development outcome only when operating data support that conclusion.

This is the same audit discipline used elsewhere in the publication, applied to national development.

A diversification test that distinguishes oil spillovers from durable export capability

Define the outcome before choosing the indicator

A petroleum transformation can make non-oil sectors grow rapidly even when their ability to survive a lower-oil-revenue future has changed little. Construction expands when the state and petroleum operators build infrastructure. Wholesale trade grows as equipment and consumption imports rise. Transport and professional services gain petroleum and government clients. These activities provide genuine income and employment; they are not imaginary output. The analytical error occurs when every increase in non-oil GDP is treated as evidence that the economy has acquired a second, autonomous growth engine.

Guyana's first-half 2026 mid-year reporting provides a particularly useful dated test. The Ministry of Finance reported an estimated 33.3% increase in overall real GDP for the first half, with 10.1% non-oil real growth. It also reported a 24.6% rise in non-oil exports to about US\$1.146 billion, of which gold exports accounted for around US\$826.2 million. The figures are official estimates for a defined half-year, not audited full-year outcomes. They should be reported together with that period and source limitation. [S07]

These categories answer different questions. Overall GDP captures the extraordinary scale of petroleum output. Non-oil GDP excludes the direct oil sector but still contains petroleum-related services, construction and public-spending spillovers. Non-oil exports exclude petroleum but can contain gold and other minerals; they are not equivalent to exports from manufacturing, tourism or digital services. A GDP decomposition and an export decomposition are both needed before a claim of structural diversification can be tested.

A reproducible first-half export decomposition

The reported non-oil export total of US\$1.146 billion includes about US\$826.2 million of gold. Expressed in the same rounded dollar units, gold therefore accounts for approximately 72% of reported non-oil exports and the residual is approximately US\$319.8 million. These figures illustrate why the term "non-oil" should not be used as a synonym for "non-resource". Gold is a different extractive sector with its own price and production cycle. The reported total can accurately demonstrate expansion in exports outside oil, but it cannot by itself establish a broad-based manufactured export platform. [S07]

The next analytical table should separate at least five mutually intelligible categories: petroleum; gold and other minerals; primary agriculture; manufactured and processed goods; and internationally traded services, including tourism. This division is not intended to rank economic sectors. It reveals exposure to different demand and price shocks. Gold may behave differently from crude during a global crisis. Agricultural exports may be constrained by rainfall or cold-chain logistics. Tourism receipts may respond to air connectivity and competition. Digital services depend on connectivity and international client acquisition.

For each category, the book should carry nominal value, quantity where observable, price index, main destination, source, observation period and revisions. A value increase caused solely by a price spike is different from a quantity increase resulting from improved productivity. If reliable quantity data are missing, the publication should not infer productive capacity from dollar growth alone.

The counterfactual that matters

The central diversification counterfactual is not a Guyana without oil. It is the same economy with weaker or more slowly growing petroleum-financed demand. Imagine a petroleum-price and startup shock that reduces expected resource receipts while previously approved public projects continue to demand imported construction materials and skilled labour. Construction firms could still be busy in the short term but face rising financing costs and a shrinking future order book. Restaurants and logistics firms connected to the boom could have profitable current operations but limited export customers. A manufacturing firm selling to the domestic construction sector could expand while becoming more dependent on imported components and public projects.

An autonomous second growth engine should demonstrate at least some ability to maintain external sales, employment and productivity through this shock. That does not require complete independence from public infrastructure. All competitive economies rely on public goods. The question is whether the firm can sell outside a petroleum-driven demand cycle once roads, electricity, ports and workforce capability are in place.

A research design should therefore follow a cohort of non-oil firms over time and record their sales by customer and destination. Relevant fields include domestic public procurement, petroleum-operator procurement, other domestic sales, regional export sales and wider international export sales. For firms established during a resource boom, the trajectory after

an oil-price shock may be more informative than the initial growth rate. A cross-sector control group can reduce the risk of attributing all observed gains to petroleum policy.

Three distinct channels from petroleum revenue to other-sector productivity

The first channel is **reliable physical infrastructure**. An energy investment supports diversification only to the extent that electricity reaches businesses at a competitive all-in cost with improved reliability. Installed generation capacity, plant commissioning, fuel availability, transmission capacity, distribution losses and outage metrics must be measured separately. A cheaper generation contract that does not reduce actual industrial outages may produce smaller short-term productivity gains than projected.

The second channel is **human capability**. Petroleum can raise the demand for engineers, accountants, logistics specialists and safety professionals, but it can also attract them away from schools, utilities and regulators. A training programme should therefore measure portable technical competency and retention across sectors rather than counting attendance or certificates alone. For example, instrumentation, power engineering, welding procedures and data management may be deployed in petroleum today and manufacturing, renewable energy or maritime services later.

The third channel is **market connectivity**. Roads, ports, bridges and digital infrastructure reduce transaction costs only when the full transport or communication network works. A road may be substantially complete yet deliver limited freight benefit until complementary bridges, customs procedures, cold storage or port handling are operational. Every major corridor should therefore have baseline and post-commissioning measures of journey time, freight reliability, accidents and delivered market access. The public-investment register should mark an allocation, a construction milestone and an operating outcome as three separate events.

[S129, S130, S131, S133]

Inflation, labour scarcity and the non-tradables channel

A boom creates a difficult comparison between nominal expansion and real productive efficiency. A construction contractor may report record revenue because material prices have risen. An urban household may receive higher wages but face faster rent increases. A qualified engineer may obtain a higher salary in petroleum while the national utility struggles to recruit staff. The result can be rising GDP and higher household purchasing power for some groups alongside weaker supply capacity or real welfare pressure for others.

A credible policy case study should therefore monitor the distribution of inflation rather than rely only on the headline CPI. The indicators should include food, housing, transport, utilities and geographically differentiated prices. Occupational wage series and public-sector vacancies are needed to separate broadly shared income gains from resource-sector skill premiums. The latest representative household survey should be used only for the period it covers; the 2018/2019 baseline cannot become a current 2026 poverty or living-standard statistic by repetition. [S61, S68]

The same principle applies to public investment. An increase in expenditure can be justified by infrastructure needs while still placing pressure on a relatively small pool of contractors, engineers and imported materials. The IMF's 2025 Guyana Article IV report describes the interaction of rapid resource growth, substantial infrastructure spending, diversification ambitions and macroeconomic risks. Its projections and policy recommendations are IMF staff assessments, not the author's judgment about a government. The official H1 2026 figures should be dated separately from that earlier assessment rather than combined as if they came from one period. [S68]

A practical scoreless dashboard

Instead of producing a single national "diversification score", the book should publish a dashboard of independently interpretable series. One row should show non-oil real output growth; a second, the share and value of non-petroleum, non-mineral exports; a third, export customers and destination diversity; a fourth, electricity reliability and commercial energy cost; a fifth, labour productivity and occupational shortages; and a sixth, the share of surveyed firms with sales outside petroleum-linked public or private demand.

Every series should disclose its observation date and boundary. A firm registry might count firms legally incorporated but not operating. A local-content procurement report counts gross contracts but not necessarily domestic value added. An export value may increase because of commodity prices rather than more diversified buyers. The dashboard's value lies in preventing one strong aggregate from silently standing in for all other outcomes.

For a new island or coastal producer, the transferable design principle is to establish the non-oil baseline **before** petroleum cash becomes large. Select a tractable sample of firms, benchmark port and electricity reliability, and record current occupational shortages. Then update the same series after first oil and through at least one adverse petroleum scenario. This creates a credible basis for identifying which gains depend on a temporary boom and which represent durable productivity and external earning capacity.

PART VIII | Transparency and institutional learning

Accountability and institutional learning

Working manuscript for the 520-560-page flagship edition. Evidence cut-off: 27 September 2026. Source IDs refer to SOURCE_REGISTER.md. Political and institutional disagreements are attributed to documented sources. This section does not rank parties, governments or public officials.

39. Extractive disclosure as an operating system

Transparency in a petroleum state has little value if disclosure arrives too late, uses incompatible definitions or cannot be reconciled across agencies. The strongest disclosure architecture therefore behaves as an operating system rather than a periodic communications exercise. It links contracts, production, government entitlement, royalties, audited recoverable costs, fund deposits, investment income, parliamentary withdrawals and public expenditure through shared dates and definitions.

Guyana's current public record contains several layers. The Petroleum Management Programme publishes petroleum agreements, block information, production licences, cost-audit documents, environmental records and a production dashboard. The Bank of Guyana publishes Natural Resource Fund reports and audited statements. The Ministry of Finance publishes budget and mid-year fiscal reports. Guyana EITI publishes national reports and participates in the international EITI validation process. These sources create a substantial documentary trail, but the trail still requires reconciliation by date and transaction. [S18, S33-S34, S41-S47, S69-S72, S81-S86]

The EITI process deserves separate treatment because it tests more than numerical publication. In May 2026, the EITI Board launched a full validation of Guyana from 15 May and recorded concern about continuing stakeholder confidence in the selection of civil-society representatives. The International Secretariat also sought views on stakeholder engagement and barriers affecting participation. At the evidence cut-off used for this manuscript, a final 2026 validation result had not been established. The proper research treatment is therefore to describe the process as ongoing, record the Board's stated concern and avoid substituting an author's political conclusion for an unfinished institutional review. [S16-S18, S33-S34, S69-S70]

The design lesson for an emerging producer is to separate disclosure into at least six ledgers. The first is a legal ledger containing executed contracts, licences, permits and amendments. The second is a production ledger containing measured output, cargoes and entitlement. The third is a revenue ledger containing royalties, profit oil, taxes, penalties and settlement dates. The fourth is a cost ledger containing recoverable expenditures, audit adjustments and dispute status. The fifth is a sovereign-fund ledger containing opening cash, deposits, investment returns, withdrawals and closing cash. The sixth is a public-investment ledger containing appropriations, tenders, awards, variations, completion, operation and maintenance. A

seventh social and environmental ledger is warranted when the country has major local-content, community, biodiversity or climate-finance obligations.

A high-quality transparency platform should not force users to infer equivalence among these ledgers. For example, a monthly production volume does not imply a same-month government cash receipt. A disputed cost item does not imply a final cash recovery. An NRF withdrawal does not imply a completed road or hospital. A carbon-credit agreement does not imply a cash receipt in the petroleum fund. These distinctions protect public understanding and improve external audit.

40. Cost audits and revenue assurance

Cost recovery is one of the least intuitive components of a production-sharing system. Under the Stabroek framework, the economic effect of recoverable expenditure depends on the cost ceiling, the stock of eligible costs and the timing of production. The state's ability to verify cost claims therefore affects the timing and amount of profit oil even when headline contract terms remain unchanged.

Guyana's public petroleum portal includes an IHS Markit audit covering the 1999-2017 period and a VHE audit record for 2018-2020. The IHS Markit final report states that the audited cost-recovery submission totalled USD 1.6778 billion and identified USD 214.4 million plus overhead adjustments as having grounds for dispute. The Ministry of Natural Resources later stated that aspects of the earlier audit had moved into the production-sharing agreement's sole-expert procedure and described the status of several audit exercises. An audit exception, a government challenge, an expert determination and cash ultimately recovered are four different events. The flagship paper will preserve those statuses separately. [S42-S43, S71-S72]

For a new producer, the audit function should begin before first revenue. The state needs contract accountants, petroleum engineers, reservoir specialists, transfer-pricing expertise, legal counsel and access to source documentation. Cost categories should be mapped to contractual clauses before operators begin submitting large development claims. Data retention must be long enough to support later disputes. Audit timetables should account for the fact that offshore development projects spend billions before production begins, creating a large initial cost bank.

The revenue-assurance model should contain a claim register with fields for the cost category, project, invoice date, related party, contractual basis, amount claimed, amount accepted, amount challenged, audit reason, operator response, dispute stage, final determination and cash effect. Without these fields, public reporting risks compressing a multi-year accounting process into one headline number.

Audit independence also requires procurement safeguards. If the government hires external auditors, the public should know how the firm was selected, whether conflicts were screened, which years and entities fall within scope, and whether the contract permits access to underlying technical records. The state should maintain its own institutional memory rather than outsourcing the entire audit function to successive consulting teams.

41. Procurement and spending oversight

A petroleum windfall expands the public capital programme faster than administrative capacity may expand. Procurement oversight therefore becomes part of petroleum governance even when procurement law sits outside the petroleum sector. A country may account perfectly for oil revenue entering a sovereign fund and still convert the revenue poorly into durable assets.

Guyana's Ministry of Finance has published large public-investment allocations in successive budgets. IMF staff in 2026 highlighted absorption, project monitoring, procurement and public-enterprise oversight as policy priorities. The Audit Office publishes public-accounts and performance reports, including the 2024 public-accounts report presented to the National Assembly in September 2025. These records create a basis for project-level testing rather than broad claims about public-sector performance. [S64-S65, S73]

The flagship methodology therefore applies an outcome chain to major capital projects. The first stage is policy justification. The second is feasibility and lifecycle costing. The third is procurement and contractor selection. The fourth is construction and contract variation. The fifth is commissioning. The sixth is operation and maintenance. The seventh is measured service outcome. A project should not move to the final category because a contract was signed or because budgeted expenditure was executed.

For an emerging producer, the public-investment register should identify the original cost estimate, financing source, procurement route, evaluation criteria, winning bidder, beneficial owner, variation orders, completion date, maintenance provision, responsible operating agency and service indicators. Where climate exposure matters, the register should record flood, heat, hurricane, coastal or seismic design assumptions.

This framework also helps limit procyclical spending. If capital projects are approved faster than engineers, construction firms, ports, aggregate supply or procurement teams can deliver them, the result may be higher costs rather than proportionally higher public-service output. The correct test is not whether the government has the cash. The test is whether the economy and institutions can convert that cash into durable capacity at acceptable cost.

42. Parliament, judiciary and institutional checks

A resource-governance system is more resilient when multiple institutions can test decisions using different legal mandates. Parliament authorizes laws and public expenditure. Courts interpret legal obligations and review administrative action. Auditors examine financial records. Regulators issue permits and licences. Central banks hold or manage financial assets. EITI processes add a multi-stakeholder disclosure layer. None of these institutions performs the same function.

Guyana's petroleum framework provides several examples. Parliament enacted the 2019 Natural Resource Fund Act before first oil, later replaced it with the 2021 framework and amended withdrawal provisions in 2024. Parliament also enacted the Local Content Act 2021 and the Oil Pollution Prevention, Preparedness, Response and Responsibility Act 2025. Each statute needs a versioned legal crosswalk because enactment, assent, publication,

commencement and implementing regulations do not necessarily occur on the same date. [S05, S19, S37, S53-S54, S57]

The judiciary has also become part of petroleum governance. The May 2026 Court of Appeal dispute over financial assurance illustrates why the final publication must use authenticated judgments rather than relying on a press release or one party's summary. The government stated that the appeal court set aside the earlier High Court requirement for an unlimited guarantee and confirmed regulatory discretion over the amount of financial assurance. This work treats that statement as an attributed description pending full judgment review. It does not transform a contested legal question into an authorial verdict. [S20]

The lesson for a new producer is procedural. Every major decision should identify the statutory authority, decision-maker, evidence considered, public notice requirement, reasons, appeal route, judicial-review route and record-retention requirement. Where a regulator possesses discretion, the file should preserve the reasons for exercising that discretion. This improves the ability of courts and auditors to review the decision later without requiring them to re-create the technical record from fragments.

Institutional checks work best when their boundaries are explicit. A court does not run an environmental agency. An environmental regulator does not set the annual budget. A finance ministry should not certify its own external audit. A sovereign-fund board should not replace parliament's appropriation role. The operating model should therefore map both authority and limits.

43. Statistical institutions and public data

A resource boom changes the statistical system as much as it changes the fiscal system. Production, exports, government revenue, construction, wages, migration, prices and household consumption can move quickly. If statistical releases lag, public debate relies on nominal spending, operator announcements or national GDP, none of which answers the full distributional question.

Guyana's Bureau of Statistics completed the 2018/2019 Household Budget Survey and Living Conditions work across all ten regions. The World Bank and IMF have both noted limits in the availability of more recent household information. IMF reports in 2025 and 2026 also discuss statistical modernization and data gaps affecting analysis of the non-oil economy and household conditions. The responsible approach is to date every welfare statistic and avoid describing a 2019 poverty observation as a 2026 outcome. [S10, S61, S64, S68]

A flagship resource-governance dashboard should therefore carry three dates for each indicator: observation period, release date and last revision. It should also identify geographic coverage, sample size where relevant, price base, currency, nominal or real status and source owner. Petroleum data need equally careful definitions. Monthly barrels, daily peaks, nameplate capacity, government lifts and cash receipts should never occupy one undifferentiated column.

The public-data system should include a corrections policy. Resource states experience frequent revisions because cargo prices settle after lifting, cost claims change through audit,

national accounts undergo rebasing, and surveys improve. Publishing a revised series with a version history is better than defending an outdated number for reputational reasons.

The most useful indicator architecture for the final paper will include petroleum output, non-oil output, government revenue, NRF flows, inflation, employment, real wages, household consumption, poverty or deprivation, school and health outcomes, regional service access, public debt, construction costs, exchange-rate conditions, imports, exports outside petroleum and environmental performance. The point is not to produce one composite score. The point is to prevent a high petroleum-growth number from substituting for evidence on other outcomes.

Institutional learning cycle

The final accountability model uses a six-stage loop.

- 1. **Decide.** A legally authorized institution makes a decision using a dated record.
- 2. **Disclose.** The underlying rule, assumptions and non-confidential evidence are published.
- 3. **Measure.** Actual production, cash, cost, environmental or service outcomes are recorded.
- 4. **Challenge.** Auditors, parliament, courts, stakeholders and independent experts test the evidence.
- 5. **Correct.** Errors, weak controls or outdated rules are revised with reasons.
- 6. **Archive.** The version history remains accessible so later analysts can distinguish what was known at each date.

This cycle is more transferable than any single petroleum contract or fund-withdrawal rule. It allows a new producer to learn without assuming that the first institutional design is permanent.

Part VIII conclusions

Guyana's public record shows a growing network of petroleum disclosure, fund reporting, audit material, parliamentary lawmaking, court review and EITI engagement. The same record also contains unresolved audit disputes, an ongoing EITI validation at the research cut-off, legal questions requiring primary judgment review and statistical gaps affecting social analysis. These facts belong together.

The blueprint value lies in the structure of verification. A new producer should design for reconciliation, version control, challenge and correction from the start. Institutional maturity is not the absence of disputes. It is the ability to record them, resolve them through lawful procedures and preserve evidence of what changed.

Edition 2.0 deepening: accountability as a system of reconciled public ledgers

Analytical section 43A. Transparency should be designed for reconciliation

Publication is necessary but not sufficient.

A resource state can publish contracts, production data, fund reports and budgets while still making it difficult for an outside reviewer to connect one record to another.

The stronger standard is reconcilability.

A reviewer should be able to move from: licence to production to state entitlement to sale to cash receipt to NRF deposit to budget withdrawal to procurement to completed asset.

Each transition should have a dated identifier.

This is the difference between disclosure and an auditable public operating system.

Analytical section 43B. Beneficial ownership belongs in both licensing and procurement

Beneficial ownership is often treated as an extractive-industry disclosure issue.

Its relevance is wider.

Government should understand who ultimately controls:

- licence holders;
- joint-venture partners;
- petroleum suppliers;
- public contractors;
- companies receiving local-content preferences.

The purpose is not to stigmatize complex ownership.

It is to identify conflicts, related-party transactions, concentration and hidden control.

The same ownership identifier should be usable across petroleum, company-registration and procurement systems where law permits.

Analytical section 43C. Conflict-of-interest controls should attach to decisions

A generic ethics code is not enough for high-value petroleum decisions.

Each major decision file should record:

- decision-maker;
- declared interests;
- recusal if any;
- adviser conflicts;
- contractor relationships;
- cooling-off restrictions where applicable.

This should apply to:

- licence awards;
- contract negotiation;
- audit procurement;
- cargo marketing;
- environmental permitting;
- major public-investment awards.

The public disclosure level may vary.

The internal control should not.

Analytical section 43D. EITI is most valuable when it becomes part of routine government data

EITI reporting can operate as a periodic reconciliation exercise.

The higher institutional standard is systematic disclosure.

Production, licence, ownership and revenue data should increasingly come from regular government systems rather than be recreated only for an EITI cycle.

At the September 2026 evidence cut-off, Guyana's Validation remained ongoing, and EITI committee records show continuing process review. [S89, S104]

The correct lesson for a newcomer is not to design transparency around one external assessment date.

It is to make the underlying data routine enough that an external assessment can test an already functioning system.

Analytical section 43E. Audit findings need a status taxonomy

Public debate often compresses all audit numbers into "money owed".

A better taxonomy distinguishes:

- 1. potential exception;
- 2. formal audit finding;
- 3. contractor response;
- 4. government position;
- 5. dispute;
- 6. expert or judicial determination;
- 7. accounting adjustment;
- 8. cash recovery.

This taxonomy should be used in every public update.

It prevents a disputed cost from being represented as recovered public revenue before the contractual process is complete.

Analytical section 43F. Parliamentary oversight needs transaction-level support

Parliamentary debate is stronger when members receive reconciled data rather than headline totals.

An annual petroleum accountability package could include:

- production by project;
- government lifts;
- realized public petroleum receipts;
- NRF opening and closing balance;
- withdrawals;
- outstanding cost-audit disputes;
- public investment funded from withdrawals;
- major environmental incidents;
- local-content outcomes.

The package should use definitions consistent with audited financial statements and regulator data.

Analytical section 43G. Judicial review works best when administrative records are complete

Courts should not have to reconstruct the factual record from press statements.

A regulator's decision file should preserve:

- legal authority;
- technical evidence;
- consultation;
- reasons;
- conditions;
- financial-assurance analysis;
- internal review.

This improves both accountability and institutional defensibility.

The environmental financial-assurance litigation demonstrates why the quality of the administrative record matters even when the ultimate legal interpretation remains contested.
[S101, S103, S105]

Analytical section 43H. Freedom-of-information capability should include data architecture

A transparency regime can be overwhelmed if records are poorly indexed.

Petroleum records should therefore have:

- document type;
- project;
- date;
- version;
- responsible agency;
- confidentiality status;
- retention period.

Good records management reduces the cost of both public disclosure and internal decision-making.

Analytical section 43I. Statistical independence depends on revision discipline

Fast-growing resource economies experience frequent revisions.

The credibility question is not whether revisions occur.

It is whether the revision process is transparent.

Every major statistical series should retain:

- previous release;
- revised value;
- reason;
- methodology;
- effective date.

This applies to national accounts, production, trade and public finance.

Analytical section 43J. Public dashboards should publish definitions before graphics

A dashboard can increase confusion if users do not know what a number means.

Every dashboard should define:

- unit;
- time period;
- source;
- gross or net;

- nominal or real;
- cash or accrual;
- provisional or final.

The definition should be visible without downloading a technical appendix.

Analytical section 43K. Independent scrutiny needs access to machine-readable data

PDF reports are valuable for narrative and legal record.

Auditors and researchers also need structured data.

A mature disclosure system should publish:

- CSV or API production series;
- machine-readable budget data;
- contract metadata;
- permit metadata;
- beneficial-ownership fields where lawful;
- procurement records.

This allows external parties to reproduce calculations rather than manually transcribe tables.

Analytical section 43L. A corrections register is a governance instrument

Institutions sometimes resist public correction because correction is perceived as failure.

The stronger practice is to make correction routine.

A correction register should show:

- original release;
- corrected value or wording;
- reason;
- date;
- effect on prior conclusions.

The publication itself now uses this principle.

The same logic should apply to government data.

Analytical section 43M. Institutional memory should survive political transition

Petroleum systems operate across administrations.

A licence can outlast a government.

A cost audit can span several years.

A decommissioning obligation can persist for decades.

Institutional records should therefore be designed to survive political turnover.

This means:

- permanent file structures;
- documented delegations;
- version-controlled models;
- handover protocols;
- protected archives.

The objective is not to eliminate policy change.

It is to ensure that later decision-makers inherit the evidence needed to change policy intelligently.

Analytical section 43N. Expanded Part VIII conclusion

Accountability should not be evaluated through the number of reports published.

The stronger test is whether the public record can be reconciled across institutions and across time.

Guyana's petroleum, fund, EITI, budget, audit and judicial records provide a substantial documentary base. [S18, S41-S47, S69-S73, S81-S89, S104]

The next step is integration.

For a newcomer, the blueprint is to build one common data architecture across legal, production, revenue, cost, fund, procurement, social and environmental systems.

Every material figure should have a source owner, observation date, revision history and transaction boundary.

That architecture makes scrutiny cheaper, dispute resolution faster and institutional learning more durable.

Edition 2.0 operating-accountability deepening: from disclosure to institutional assurance

Analytical section 43O. Every important number should have an institutional owner

A public data system becomes fragile when a number is widely cited but no institution clearly owns the underlying record.

Edition 2.0 therefore treats data ownership as an accountability question.

For each major petroleum and development indicator, the government should identify:

- producing institution;
- approving institution;
- publication frequency;
- revision authority;
- source system;
- legal basis;
- retention period.

Examples include:

- daily and monthly petroleum production;
- government lifts;
- royalty;
- profit-oil receipts;
- recoverable-cost balance;
- NRF assets;
- budget withdrawals;
- local-content procurement;
- permit compliance.

A number should not migrate between institutions without retaining its original source identifier.

Analytical section 43P. The petroleum block register should be a legal index

The official block register already provides key dates and licence information. [S03, S41]

A mature register should also link:

- executed agreement;
- amendments;
- exploration licence;
- production licence;
- assignments;
- operator;
- co-venturers;
- beneficial ownership;
- field-development approvals;
- environmental permits.

This allows a reviewer to navigate from acreage to operative legal instruments.

Analytical section 43Q. Production disclosure should carry revision history

Production data can be revised because of late operator reporting, metering corrections or classification changes.

The public series should therefore preserve:

- release date;
- original value;
- revised value;
- reason.

A data series that silently overwrites history creates avoidable suspicion.

Versioned data is more credible even when revisions are frequent.

Analytical section 43R. Revenue disclosure should reconcile physical and cash systems

Government revenue reporting should answer three different questions.

Physical entitlement: how much petroleum was due to government?

Commercial realization: what cargoes were sold and at what value?

Cash receipt: when did money enter the NRF?

A strong disclosure package should show the bridge.

This would make the public record more resilient to timing differences between production, lifting and settlement.

Analytical section 43S. The cost-audit register should show unresolved amounts without sensationalizing them

Cost-audit transparency is difficult because large figures attract attention before disputes are resolved.

A public audit-status register could show:

- audit period;
- amount reviewed;
- exception amount;
- operator response;
- amount agreed;
- amount disputed;

- final adjustment;
- status date.

The register should avoid describing the gross exception as money already recovered.

This protects both public accountability and procedural fairness.

Analytical section 43T. Audit capacity should be reported like production capacity

A petroleum state should publish basic information about its audit capability:

- number of specialist auditors;
- audit periods open;
- audit periods completed;
- external specialists retained;
- deadlines;
- disputed findings.

This does not reveal confidential audit strategy.

It tells the public whether contractual rights are being exercised within time limits.

Analytical section 43U. The Auditor General and petroleum-specific audit functions are complementary

The Auditor General audits public accounts and public entities under public law.

Petroleum cost recovery requires technical and contractual review of contractor expenditure.

Tax administration has another legal mandate.

These functions should coordinate while retaining their distinct authorities.

A common error is to assume that one audit institution covers the entire resource value chain.

It does not.

The accountability map should therefore show:

- contractor cost audit;
- tax audit;
- NRF financial audit;
- public expenditure audit;
- performance audit.

Analytical section 43V. Public Accountability and Oversight Committee evidence should be integrated with the fund accounts

Oversight reports can provide a different perspective from the financial statements.

The strongest review compares:

- Bank of Guyana fund statements;
- Ministry withdrawal notifications;
- parliamentary approvals;
- oversight committee reporting.

Differences in closing balance or terminology should be reconciled, not ignored.

The 2023 example identified in the research illustrates why accounting basis must be explicit.

Analytical section 43W. Parliamentary authorization should be machine-traceable

For each NRF withdrawal, the public record should identify:

- statutory authority;
- annual ceiling;
- National Assembly approval;
- date and amount of each transfer;
- Consolidated Fund receipt;
- budget programme.

A reviewer should not have to search several PDFs to reconstruct the sequence.

Edition 2.0's NRF ledger is designed to model this approach.

Analytical section 43X. Procurement disclosure should link planning to award to execution

A public procurement record should show:

- project ID;
- budget;
- tender;
- bidders;
- award;
- beneficial owner;
- contract value;
- variations;
- payments;

- completion;
- defects;
- final account.

This creates an audit trail from appropriation to asset.

Where security or commercial confidentiality applies, the restriction should be specific rather than blanket.

Analytical section 43Y. Contract variations should be a separate risk indicator

Large public projects often change after award.

A variation is not automatically evidence of poor management.

The dashboard should still track:

- original contract value;
- cumulative variation;
- reason;
- approval;
- time extension.

A high variation rate across a portfolio can indicate weak design, price volatility or contract-management problems.

Analytical section 43Z. Beneficial ownership should be linked to conflict screening

Ownership data becomes more useful when connected to decision processes.

Before a licence, procurement award or local-content certification, the responsible institution should be able to screen for:

- politically exposed persons where legally relevant;
- government official interests;
- related parties;
- sanctions;
- debarment;
- common control among nominal competitors.

The output should support lawful conflict management, not political labeling.

Analytical section 43AA. Local-content verification requires ownership history

A supplier can change shareholders after registration.

The local-content registry should therefore preserve ownership history.

This helps detect temporary structures designed only to qualify for preference.

It also supports longitudinal analysis of whether Guyanese owners increase real control and capability over time.

Analytical section 43AB. Environmental disclosure should include regulator findings

Publishing an operator environmental report is not the same as publishing regulator oversight.

A public portal should distinguish:

- operator report;
- regulator inspection;
- independent sample;
- enforcement notice;
- corrective action.

This lets readers understand who generated the evidence.

Analytical section 43AC. Incident disclosure needs a clear threshold policy

The public should know which environmental or safety events are disclosed automatically.

The threshold policy should define:

- severity;
- timing;
- minimum information;
- update process;
- investigation closure.

This avoids inconsistent disclosure driven by media attention.

Analytical section 43AD. Statistical capacity is part of petroleum governance

An economy changing as quickly as Guyana's places unusual pressure on national statistics.

GDP weights change.

Labour markets change.

Prices and housing shift.

Business formation accelerates.

Migration can alter population assumptions.

The statistical system therefore needs investment in:

- household surveys;
- business registers;
- price collection;
- labour-force data;
- trade data;
- regional statistics.

Without those upgrades, fiscal data may become more precise while welfare evidence becomes less current.

Analytical section 43AE. Data gaps should have owners and closure dates

A data-gap register should identify:

- missing variable;
- why it matters;
- current proxy;
- responsible institution;
- planned collection;
- expected date.

This is better than repeatedly stating that "data are unavailable".

The gap itself becomes a management item.

Analytical section 43AF. Open-data systems need archival durability

Websites change.

Links break.

PDFs are replaced.

A petroleum evidence system should therefore maintain an archival repository with:

- permanent identifiers;
- checksum;
- publication date;
- superseded version;
- correction history.

This publication's own manifest and corrections register model the same principle.

Analytical section 43AG. Civil-society participation should be evaluated through process evidence

Participation should not be treated as a binary label.

A process record can show:

- notice;
- eligibility;
- selection;
- meeting attendance;
- submissions;
- response;
- grievance.

EITI's 2026 process concerns are relevant because the global Board explicitly recorded concerns about confidence in civil-society representative selection. [S16, S34]

The publication should attribute that concern to EITI and avoid turning it into an independent conclusion about broader political conditions.

Analytical section 43AH. EITI validation should be a dated institutional test

At the evidence cutoff, Guyana's Validation remained ongoing. [S89]

The book should therefore record:

- validation start;
- mission or committee activity;
- final decision when eventually published;
- corrective actions if any.

A future edition can update the status without altering the historical record.

Analytical section 43AI. Judicial decisions should be indexed to the administrative decision they review

For environmental and petroleum litigation, the legal database should connect:

- permit;
- administrative decision;
- High Court proceeding;
- Court of Appeal;
- CCJ if applicable;
- final status.

This is particularly important when public commentary continues while appeals are pending.

Analytical section 43AJ. Correction is a sign of institutional maturity

A system that never corrects public data is unlikely to be error-free.

The goal should be visible correction.

The correction record should preserve:

- old value;
- new value;
- cause;
- effect on previous analysis.

Edition 2.0 should use the same rule for its own mistakes.

Analytical section 43AK. An annual integrated accountability statement

Guyana could eventually publish one integrated resource-accountability statement containing:

- block and ownership changes;
- production;
- cargoes and prices;
- revenue;
- cost audits;
- NRF;
- budget transfers;
- public investment delivery;
- local content;
- environmental performance;
- outstanding litigation.

The statement would not replace specialist reports.

It would reconcile them.

Analytical section 43AL. Accountability maturity should be evaluated by reproducibility

The ultimate test is simple.

Can an informed outside reviewer reproduce a material public claim from the underlying record?

If the answer is yes, the system is moving toward auditability.

If the answer requires private explanation, the public evidence architecture is incomplete.

This standard is demanding but practical.

Analytical section 43AM. Expanded accountability conclusion

Edition 2.0 treats transparency as an engineering problem.

The resource system generates multiple ledgers.

Accountability requires identifiers, reconciliations, version history, ownership and correction rules that allow those ledgers to connect.

The result is not perfect consensus.

Different institutions and stakeholders may still interpret the evidence differently.

The achievement is that they are debating from a common factual record.

PART IX | International comparator mechanisms

International comparators and transferability

Working manuscript for the 520-560-page flagship edition. Comparisons below describe documented mechanisms. They do not rank countries, governments or political choices.

44. Norway: converting petroleum cash into a financial asset

Norway is frequently invoked in resource-governance debates because its Government Pension Fund Global sits within a long-established fiscal framework. The current official framework directs the state's net petroleum cash flow to the fund and uses a fiscal guideline linked to the fund's expected long-term real return, commonly expressed around 3%. The 2026 white paper continues to describe the fund as part of a broader framework for managing petroleum wealth over time. [S25, S74]

The relevant lesson is not the 3% number in isolation. Norway built the fund after developing strong tax administration, petroleum expertise, state ownership capacity, democratic institutions, mature capital markets and a large non-petroleum tax base. Its financial asset stock is the product of decades of accumulation. A new producer with severe infrastructure deficits and a small domestic tax base faces different intertemporal trade-offs.

The transferable mechanism is the separation of the petroleum cash flow from ordinary current revenue, the integration of fund withdrawals with the national budget, the preference for diversified external financial investment, transparent reporting and a fiscal rule intended to protect the real value of the fund. The receiving country must still determine its own acceptable spending path, debt strategy and public-investment capacity.

A useful comparison with Guyana will therefore examine five variables rather than headline fund size: annual petroleum revenue as a share of non-oil GDP, fund assets per capita, non-oil primary balance, domestic investment need, and institutional capacity to deliver public projects. The same nominal withdrawal ratio means different things under different values for those variables.

45. Ghana: stabilization and heritage within a producing developing economy

Ghana's Petroleum Revenue Management Act 2011 established a statutory framework that includes a Stabilisation Fund and a Heritage Fund. The Bank of Ghana publishes reports on the petroleum funds and describes their distinct purposes. Ghana therefore offers a more developmentally proximate comparison than Norway for the architecture of separating short-term stabilization from intergenerational saving. [S24, S75-S76]

The framework also provides a useful publication model. Semi-annual reporting, annual financial statements and central-bank administration create repeated disclosure points. The

research comparison should examine whether the fund rules remained stable, how excess petroleum revenue was allocated, and how withdrawals interacted with the national budget and debt over time. Those empirical questions are separate from the statutory design.

For Guyana, the comparison is useful because both states use petroleum revenue within economies that also require large infrastructure and social investment. Yet reserve size, production trajectory, population, exchange-rate system and political economy differ. A direct transfer of Ghana's allocation percentages would therefore be methodologically weak.

The adaptable lesson is to define the separate purposes of stabilization and intergenerational saving before money arrives, publish the rule, and specify when parliament or the executive may alter the allocation. A single undifferentiated petroleum fund can still achieve those functions, but the internal accounting should make them visible.

46. Timor-Leste: a small state's dependence risk

Timor-Leste established its Petroleum Fund by law in 2005 and later amended the framework. The central bank publishes monthly, quarterly and annual reports and describes the portfolio structure, including Liquidity and Growth components. This creates a strong public record for examining how a small state manages a large financial asset relative to the domestic economy. [S26, S77-S78]

Timor-Leste is analytically valuable because the central challenge is not only investment performance. It is the relationship among petroleum depletion, government withdrawals, domestic budget dependence and the future tax base. A fund can remain professionally managed while the broader fiscal system becomes increasingly dependent on withdrawals. The final comparator chapter will therefore use a depletion-adjusted fiscal lens rather than fund-return data alone.

For a new island producer, this case demonstrates the importance of modelling the post-field budget before first oil. If petroleum production has a limited life and the non-oil tax base is narrow, recurrent spending promises made during peak revenue years may become difficult to sustain. The operating manual should therefore require a 20-year expenditure path under low production and low return assumptions before permanent salary, subsidy or entitlement increases are adopted.

The comparison with Guyana also highlights geological uncertainty. Guyana's discovered resource base expanded rapidly after Liza, while a smaller state may face one dominant field. A rule that appears conservative under a long reserve life may be insufficient under a short one. Reserve duration belongs inside fiscal design rather than in a geological appendix.

47. Suriname: the regional pre-first-oil comparison

Suriname provides the closest regional comparison for a country moving from discovery and final investment decision toward first oil. TotalEnergies reported in September 2026 that the GranMorgu project, sanctioned in 2024, remained under construction with first oil targeted for 2028. Because production had not begun by the evidence cut-off, the proper comparison is institutional preparation rather than operating outcome. [S27, S79]

The research questions for Suriname include the role of Staatsolie, revenue-management arrangements, environmental regulation, supplier development, port capacity, fiscal exposure and public communication before first oil. The existence of an established national oil company creates a different information and capacity structure from Guyana's. That difference matters when comparing negotiating expertise and project monitoring.

For Guyana, Suriname also provides a test of regional spill preparedness. Offshore development in neighbouring jurisdictions increases the value of interoperable maritime response, shared training, compatible data standards and transboundary communication. Environmental risk does not stop at a licence boundary.

A useful regional blueprint would therefore include a shared minimum incident-data protocol, emergency contact architecture, mutual assistance terms, rules for independent sampling and compatible claims documentation. Regional cooperation is a practical extension of national regulation rather than a substitute for it.

48. Trinidad and Tobago: a mature Caribbean producer

Trinidad and Tobago offers a different Caribbean case because it has decades of petroleum and gas experience and a Heritage and Stabilisation Fund. The Ministry of Finance publishes fund reports through 2026. The country also has industrial, petrochemical and energy-sector capabilities that most new island producers lack. [S80]

Its relevance lies in three areas. First, it demonstrates the challenge of managing resource revenue across commodity cycles rather than only during a first production boom. Second, it shows how petroleum can shape industrial capability and the wider labour market. Third, it offers a regional fund structure explicitly combining heritage and stabilization purposes.

The comparison should not treat Trinidad's mature gas-based industrial system as a natural next stage for every Caribbean producer. Market size, pipeline infrastructure, feedstock availability, electricity systems and export logistics differ. The correct transfer question is which capabilities were needed before value-added industry became viable.

A small island considering domestic processing should therefore compare the full delivered cost of feedstock, plant scale, port infrastructure, electricity, water, carbon exposure, skilled labour and export contracts. Petroleum production alone does not establish a commercial downstream industry.

49. Small islands and microstates

The smallest prospective producers face a distinctive concentration problem. A single offshore project can dwarf the domestic economy, public budget, labour force and banking system. Imported investment may raise GDP while creating limited domestic value added. One accident may affect a large share of fisheries, beaches or tourism assets. A few specialist employees leaving the public service for contractors can materially weaken regulation.

The first adaptation is fiscal. The fund architecture should model revenue as volatile and temporary. A permanent expenditure increase should be tested against a lower production

path and delayed projects. External financial investment may reduce domestic overheating, but a state with severe infrastructure deficits also faces a credible case for higher public investment. The decision therefore needs an absorption model rather than a slogan about saving or spending.

The second adaptation is environmental. Island economies often have concentrated coastal tourism, reefs, fisheries, desalination intakes and limited emergency ports. Spill plans should include hurricane disruption, unavailable airports, damaged telecommunications and cross-border assistance. The cost of response equipment relative to population may argue for regional pooling, but legal liability and immediate response authority must remain clear.

The third adaptation is local content. Setting high local percentages in every category can produce front companies, inflated costs or shortages when the domestic workforce is small. A better model may prioritize skills transfer, verified ownership, apprenticeship, safety competence, data services and a narrower group of tradable capabilities with export potential.

The fourth adaptation is sovereign administration. A microstate should resist creating a large permanent bureaucracy during uncertain exploration. It may instead maintain a smaller core regulator with transparent access to independent external expertise, regional technical arrangements and rigorous conflict-of-interest controls.

Comparative matrix

/ Dimension /	Guyana	/	Norway	/	Ghana	/	Timor-Leste	/	Suriname	/	Trinidad and Tobago	/	Small-island adaptation
/													
/	---	/	---	/	---	/	---	/	---	/	---	/	---
/	Production stage at cut-off	/	Large and expanding	/	Mature	/	Mature	/	Mature/depletion-sensitive	/	Pre-first-oil		
	GranMorgu	/	Mature	/	Varies	/							
/	Revenue vehicle	/	NRF	/	GPFG framework	/	Stabilisation + Heritage Funds	/	Petroleum Fund	/	Under preparation for future production		
	Heritage and Stabilisation Fund	/	Design before first cash	/									
/	Main transfer lesson	/	Early fund law plus rapid scale-up	/	Budget integration and external financial saving	/	Separate stabilization and heritage purposes	/	Model depletion and budget dependence	/	Use pre-first-oil window		
	Manage cycles and mature capability	/	Concentration, tourism and response capacity	/									
/	Main non-transfer risk	/	Forest endowment and scale	/	Mature institutions and huge asset stock	/	Different production/fiscal profile	/	Different reserve and spending path	/	Different NOC structure	/	Different industrial history
	Limited staff, ports and economic diversification	/											

The matrix is descriptive. It does not assign a comparative score.

50. A method for responsible policy transfer

A policy should pass five tests before another state adopts it.

Context test. Was the originating country at exploration, discovery, early production, mature production or depletion when the policy was used?

Capacity test. Which institutions, professions, databases, courts, regulators and financial systems were needed for the rule to function?

Endowment test. Did the policy depend on a large forest estate, a long reserve life, an established national oil company, a mature capital market or a diversified tax base?

Stress test. What happens under lower prices, project delay, cost overruns, a spill, an audit dispute or political turnover?

Revision test. Does the receiving state have a lawful route to revise the instrument without undermining contract certainty or institutional credibility?

This framework converts comparison into a disciplined method. A newcomer studies the mechanism, measures the prerequisites, models failure and records the reasons for adaptation. It does not copy a headline percentage or institutional name.

Part IX conclusions

No comparator provides a universal template. Norway illustrates long-run financial saving and budget integration. Ghana separates stabilization and heritage functions. Timor-Leste shows the importance of depletion and withdrawal dependence for a small state. Suriname offers a live regional pre-first-oil comparison. Trinidad and Tobago provides a mature Caribbean petroleum and fund case. Guyana contributes a rapid discovery-to-scale sequence combined with a pre-existing forest-finance framework.

The transferable value lies in the questions these cases force a newcomer to answer. How long might the resource last? How volatile is revenue? How strong is the non-resource tax base? Which institutions must function before first cash? How much public investment can the economy absorb? How concentrated is environmental exposure? How quickly can the state correct a weak rule? Those questions are more durable than any copied numerical formula.

Edition 2.0 deepening: a wider comparator universe

Analytical section 50A. Why the comparator set must extend beyond famous success cases

Resource-governance comparison becomes misleading when every country is compared only with Norway.

A useful comparator set should include:

- a mature high-capacity producer;
- a developing producer with petroleum funds;
- a small state facing depletion;
- a neighbouring pre-first-oil state;
- a mature Caribbean producer;
- recent African first-oil or first-gas states;
- a pre-production state with a long preparation period;
- non-petroleum commodity states with fiscal stabilization experience.

The objective is mechanism comparison.

It is not a country ranking.

Each comparator should answer one defined question.

Analytical section 50B. Senegal: transparency after first oil and gas

Senegal is useful because the transition from preparation to production occurred recently.

EITI records the updated petroleum legal framework, oil and gas revenue-management law, PETROSEN's role, contract publication and local-content architecture. It also records the entry of Sangomar oil and Greater Tortue Ahmeyim gas into production. [S113-S114]

The comparator value lies in the overlap between:

- new hydrocarbon revenue;
- state participation;
- debt and financing questions;
- contract review;
- local content;
- EITI-based public disclosure.

The 2026 EITI Validation report highlights the importance of tracing state-participation financing and reconciling production-share revenues and related loans. [S114]

For Guyana, this provides a useful external test of the principle that state participation creates an additional accounting chain.

A national oil company or state equity stake does not replace the need for transparent budget and revenue accounting.

It adds:

- shareholder financing;
- dividends;
- debt service;
- production entitlement;
- related-party transfers.

A newcomer considering state participation should therefore model both commercial upside and contingent public liabilities.

Analytical section 50C. Mauritania: building revenue-administration capacity around first gas

Mauritania offers a different lesson because the Greater Tortue Ahmeyim gas project required revenue-administration capacity in a cross-border LNG development.

IMF technical assistance in 2025 focused directly on hydrocarbon tax administration and a medium-term capacity-development plan. [S115]

EITI's 2024 Validation also emphasized contract publication, national oil-company transparency, beneficial ownership and the need for stronger reporting around the hydrocarbon revenue fund. [S116]

The transferable lesson is that first gas can be institutionally different from first oil.

Gas projects often involve:

- processing;
- LNG pricing;
- long-term sales contracts;
- transportation;
- shared cross-border infrastructure;
- domestic gas allocation.

A newcomer should therefore design tax and transfer-pricing capability for the actual commercial chain.

A crude-oil audit model is not automatically sufficient for LNG.

Analytical section 50D. Uganda: a long pre-first-oil preparation period

Uganda provides a useful contrast to Guyana's compressed discovery-to-production timeline.

The country discovered commercial oil years before production, allowing a longer period for legal, institutional, infrastructure and local-content preparation.

Uganda EITI documents a petroleum authority, national oil company, petroleum fund, cost-recovery framework, beneficial-ownership reforms, environmental-management systems and local-content arrangements while the sector remained pre-production. [S117-S118]

This makes Uganda a valuable case for one question:

What should a state do when it has time?

A longer pre-first-oil window can support:

- institutional staffing;
- petroleum fund design;
- contract-administration systems;
- beneficial ownership;
- procurement preparation;
- environmental baseline work;
- public communication.

Long preparation does not guarantee strong operating outcomes.

It does reduce the argument that institutions could not be built because production arrived too quickly.

Analytical section 50E. Mozambique: statutory sovereign-fund design before full revenue scale

Mozambique's 2024 Sovereign Wealth Fund law provides a current example of formal rules designed around future petroleum and gas revenue.

The statute establishes development, stabilization and intergenerational objectives, specifies eligible revenue, defines transfer rules and assigns roles to the budget and fund. [S119]

Banco de Moçambique later documented operational management and initial capitalization. [S120]

This comparator is useful because the law explicitly tries to combine near-term development finance with saving.

For a newcomer, the mechanism questions are:

- How is the revenue projection calculated?
- How much goes to the budget?
- How much remains in the fund?
- What happens when actual revenue is below projection?
- What is the role of parliament?
- Which assets can the fund hold?

These questions are transferable even when the numerical percentages are not.

Analytical section 50F. Chile: the structural-balance principle

Chile is not a petroleum comparator.

It is a resource-revenue stabilization comparator.

Its fiscal framework uses a structural-balance concept to reduce the effect of cyclical commodity revenue on annual spending, supported by sovereign financial buffers. IMF analysis describes the interaction between the structural balance, stabilization fund and debt anchor. [S121]

The transferable mechanism is the separation of permanent or structural fiscal capacity from temporary commodity conditions.

For Guyana or another producer, the analogous question is whether annual expenditure is being set using a temporary petroleum-price and production peak or a more conservative medium-term resource envelope.

The exact Chilean formula should not be transplanted.

The principle of smoothing volatile revenue can be.

Analytical section 50G. Botswana: the risk of buffer depletion even after decades of resource income

Botswana provides a useful non-petroleum case because the Pula Fund and long diamond history demonstrate that resource wealth can support financial buffers for decades. [S122]

IMF analysis in 2024 also documents a different side of the story: mineral-revenue weakness and spending can reduce financial buffers even when public debt remains relatively low. [S123]

The lesson is not that the earlier model failed.

The lesson is that resource-management systems need to survive the mature and declining phases of the commodity cycle.

A fund should not be judged only by how well it accumulates during high-revenue years.

It should also be judged by how the fiscal system behaves when revenue weakens.

Analytical section 50H. Cross-border projects create a governance layer above the nation

The Senegal-Mauritania GTA project demonstrates a governance issue that is increasingly relevant to small states.

A petroleum or gas field can cross a maritime boundary or depend on shared infrastructure.

This creates questions about:

- unitization;
- revenue allocation;
- metering;
- tax jurisdiction;
- emergency response;
- environmental liability;
- domestic supply;
- dispute resolution.

A newcomer with a boundary-adjacent discovery should therefore prepare for treaty-level or intergovernmental arrangements in addition to domestic law.

Analytical section 50I. State participation should be treated as a financing decision

Comparators differ in the role of national oil companies and state equity.

The decision to take an equity share should be modelled like any other investment.

The state should estimate:

- capital contribution;
- financing source;
- cost of debt;
- downside exposure;
- dividend timing;
- governance;
- technical learning;
- opportunity cost.

Equity participation can create knowledge and revenue.

It can also create contingent liabilities or obscure the boundary between commercial and sovereign functions if poorly governed.

Analytical section 50J. Sovereign funds should be compared by cash-flow function

A fund comparison should not start with the fund's name.

It should start with what cash flow the fund is designed to manage.

A stabilization fund absorbs volatility.

A savings fund transfers wealth across generations.

A financing fund integrates resource revenue with the budget.

A pension reserve has a different liability structure.

Some funds combine several functions.

The comparator table should therefore show:

- deposit rule;
- withdrawal rule;
- investment objective;
- budget integration;
- audit;
- governance;
- escape clauses.

This is more useful than comparing asset size.

Analytical section 50K. New producers should be compared by institutional timing

The most important comparator question may be when an institution was created.

For each country, the timeline should record:

- discovery;
- petroleum law;
- model contract;
- environmental law;
- revenue fund;
- beneficial-ownership rules;
- local-content law;
- first production;
- first withdrawal.

A country that creates a revenue framework before first oil is solving a different problem from one that reforms after years of production.

The timing itself is a policy variable.

Analytical section 50L. Comparator failures are as valuable as successful mechanisms

A responsible comparative chapter should not search only for practices to copy.

It should identify failure modes.

Common failure modes include:

- rigid fund rules that are bypassed;
- excessive withdrawal;
- underinvestment in regulator capacity;
- state-company debt opacity;
- weak beneficial-ownership data;
- delayed cost auditing;
- infrastructure booms that exceed capacity;
- environmental response designed too late.

The point is not to label a country a failure.

It is to isolate a mechanism that can break.

Analytical section 50M. Small-state concentration should be its own comparative variable

A policy that works in a large diversified economy may behave differently in a country where one project dominates exports and public revenue.

The comparator table should therefore include:

- population;
- non-resource GDP;
- export concentration;
- public-sector size;
- labour force;
- number of major ports;
- disaster exposure;
- reserve life.

This makes the comparison more relevant to islands and small coastal states.

Analytical section 50N. A transferability scorecard without country rankings

The publication should not rank political systems or governments.

It can still assess whether a mechanism is transferable.

Each mechanism can be tested against:

- institutional complexity;
- data requirement;
- staffing requirement;
- fiscal volatility;
- market-size dependence;
- ecological dependence;
- legal dependence;
- reversibility.

The result is a mechanism profile.

For example, a simple monthly petroleum ledger may have high transferability.

A forest-carbon system requiring a large standing tropical forest has low transferability to a coral island.

Analytical section 50O. Expanded Part IX conclusion

The wider comparator set changes the purpose of the book.

Guyana is no longer being compared only with established petroleum producers.

It is placed in a sequence that includes recent first-oil states, first-gas states, long pre-production preparations and non-petroleum commodity fiscal frameworks.

Senegal shows the accounting complexity created by state participation and new production.
[S113-S114]

Mauritania shows the need to build revenue administration around a complex LNG chain. [S115-S116]

Uganda shows what can be prepared during a long pre-first-oil window. [S117-S118]

Mozambique shows a new statutory sovereign-fund architecture. [S119-S120]

Chile and Botswana show that stabilization and buffer management can be studied outside petroleum. [S121-S123]

The transferable lesson is methodological.

A newcomer should not ask "Which country should we copy?"

It should ask "Which mechanism solves our specific problem, what prerequisites does it require, and how does it fail under stress?"

Edition 2.0 comparator deepening: mechanism-by-mechanism transfer tests

Analytical section 50P. Norway is primarily a sequencing and integration comparator

Norway's relevance is often overstated through superficial reference to fund size.

The more useful mechanisms are:

- separation of petroleum cash flows from ordinary policy rhetoric;
- integrated tax and state participation;
- accumulation of foreign financial assets;
- budget integration through a fiscal guideline;
- long institutional continuity.

The 3% guideline is context-specific. [S25, S74]

A new producer with infrastructure gaps, lower income and a small initial fund should not copy that number.

It can copy the discipline of linking withdrawal decisions to long-run fiscal capacity.

Analytical section 50Q. Ghana demonstrates purpose-separated funds

Ghana's architecture distinguishes stabilization and heritage objectives. [S24, S75, S76]

This is useful because the objectives are conceptually different.

Stabilization addresses volatility faced by the current budget.

Heritage saving addresses intergenerational transfer.

A country that combines both purposes in one account can still function well, but the accounting should show how much of the fund is intended for each purpose.

The transferable lesson is objective clarity.

Analytical section 50R. Ghana also shows why earmarks and rules require implementation evidence

Formal law can specify allocations while later fiscal conditions create pressure for change.

The comparative chapter should therefore examine both:

- statutory formula;
- actual transfer/withdrawal history.

A rule's credibility is observed through its behavior under stress.

This is why the Guyana chapter should not infer long-run discipline solely from the text of the NRF Act.

Analytical section 50S. Timor-Leste is the depletion comparator

Timor-Leste's Petroleum Fund is especially relevant to small producers because the country demonstrates how a professionally reported fund can coexist with deep fiscal dependence on petroleum wealth. [S77, S78]

The mechanism lesson is that investment management and fiscal sustainability are separate systems.

A fund can be operationally well managed while withdrawals erode long-run wealth faster than sustainable income.

The Guyana stress model should therefore include a "high permanent withdrawal" case even when near-term fund balances remain large.

Analytical section 50T. Trinidad and Tobago is a mature regional energy-economy comparator

Trinidad and Tobago offers several relevant mechanisms:

- long petroleum/gas history;
- downstream industry;
- Heritage and Stabilisation Fund;
- mature energy labour market;
- exposure to resource-cycle decline.

Its industrial pathway depends on gas resources, infrastructure, market access and decades of accumulated capability.

Guyana should therefore study downstream mechanisms without assuming the same industrial structure is automatically transferable.

Analytical section 50U. Suriname is the closest geographic pre-production comparator

Suriname's GranMorgu project is useful because it provides a neighbouring first-oil preparation case with shared regional conditions but different institutions and state-company history. [S79]

The comparison should focus on:

- pre-production fiscal preparation;
- Staatsolie role;
- local content;
- environmental regulation;
- fund planning;
- infrastructure.

The key rule is to compare readiness while first oil is still future, not to present planned 2028 outcomes as achieved results.

Analytical section 50V. Senegal is a state-participation transparency comparator

Senegal's EITI record highlights the need to reconcile state-company financing, production shares and resource-revenue flows after first oil. [S113, S114]

A newcomer considering a national oil company should therefore publish:

- equity interest;
- financing;
- carried interest if any;
- debt;
- dividends;
- production share;
- transactions with government.

State participation can support national capability.

It also creates an additional public balance sheet.

Analytical section 50W. Mauritania is a hydrocarbon-tax administration comparator

Mauritania's technical-assistance record shows that hydrocarbon revenue administration is a specialist capability requiring deliberate development. [S115]

This is especially relevant for gas/LNG because the commercial chain can include:

- feed gas;
- liquefaction;
- transportation;
- LNG sales;
- related-party transactions;
- multiple jurisdictions.

The takeaway for Guyana's future gas developments is direct: petroleum audit capacity must expand beyond crude-oil cost recovery if the commercial structure changes.

Analytical section 50X. Uganda is a readiness-time comparator

Uganda's long interval between discovery and production created a different institutional opportunity set. [S117, S118]

It could develop:

- petroleum authority;
- national oil company;
- petroleum fund;
- local-content systems;
- EITI participation;
- environmental arrangements.

The comparison should not claim that more preparation time automatically produces better outcomes.

It demonstrates that time is a resource.

A country with a longer pre-first-oil window should be held to a higher preparation standard.

Analytical section 50Y. Mozambique is a current fund-design comparator

Mozambique's 2024 sovereign-fund law provides a contemporary case of designing a rule before very large LNG revenues fully mature. [S119, S120]

The law explicitly combines development, stabilization and intergenerational objectives.

This creates a useful design question for Guyana and other new producers:

Should the resource fund itself have multiple objectives, or should some objectives be implemented through the budget?

The answer depends on governance capacity and the desired separation between investment management and domestic spending.

Analytical section 50Z. Chile is the commodity-price smoothing comparator

Chile's structural-balance framework is valuable because it addresses a problem common to commodity states: temporary revenue can look permanent during a price boom. [S121]

The transfer principle is to estimate fiscal capacity using a structural or medium-term basis.

The specific copper-price methodology need not be copied.

For a petroleum producer, an analogous framework could use conservative long-term price, production and cost assumptions.

Analytical section 50AA. Botswana is the buffer-depletion comparator

Botswana's Pula Fund and broader reserve management show how resource wealth can create large financial buffers. [S122]

The more recent fiscal record shows that buffers can also decline when commodity revenue weakens and fiscal deficits persist. [S123]

This is important because resource-governance literature often studies accumulation more than drawdown.

Edition 2.0 should explicitly ask how Guyana's framework behaves when production matures or price falls.

Analytical section 50AB. Comparator mechanisms should be tested under the same stress

To make the comparison operational, each mechanism should be tested against a common stress set:

- 40% commodity price fall;
- two-year development delay;
- large public-investment programme;
- disaster;
- debt increase;
- reserve depletion.

The question is not which country "wins".

The question is what each mechanism is designed to absorb.

Analytical section 50AC. State oil companies need a separate governance template

Across the comparator set, national oil companies play different roles.

A template should record:

- regulator or commercial role;
- equity participation;
- borrowing authority;
- government guarantee;
- audited financials;
- procurement;
- dividend policy;
- off-budget activity.

Combining regulator and commercial operator roles can create conflicts that need explicit controls.

Analytical section 50AD. Fund governance should be compared separately from fiscal policy

Two countries can have similarly professional sovereign-fund investment management but very different budget discipline.

The comparator matrix should therefore separate:

- fund operational governance;
- deposit rule;
- withdrawal rule;
- medium-term fiscal rule;
- debt policy.

This prevents a well-managed investment portfolio from being treated as proof of a sustainable fiscal stance.

Analytical section 50AE. Local content should be compared by capability stage

A 50% local target in one country is not comparable to 50% in another if the underlying categories differ.

Comparator analysis should identify:

- definition of local company;
- ownership threshold;
- categories;
- employment levels;
- reporting;
- supplier capability.

The transfer question is which capabilities are economically plausible at national scale.

Analytical section 50AF. Environmental comparison needs ecological context

An offshore environmental regime in a large continental shelf system differs from one near reefs or dense coastal tourism.

Transferability should therefore consider:

- water depth;
- currents;
- shoreline sensitivity;
- fisheries;
- protected areas;
- response distance;
- disaster exposure.

A legal clause cannot be transferred without ecological context.

Analytical section 50AG. Small islands need a concentration adjustment

In a small island economy:

- one port may dominate imports;
- one airport may dominate evacuation;
- one beach zone may dominate tourism;
- one utility may dominate power.

This concentration raises the consequence of disruption.

The regulatory standard may therefore need more redundancy than the size of the petroleum industry alone would suggest.

Analytical section 50AH. A mechanism library is more useful than a best-practice list

Edition 2.0 should organize comparator findings into a mechanism library.

Each entry should contain:

- problem;
- mechanism;
- prerequisite;
- evidence;
- known failure mode;
- small-state adaptation;

- conditions for rejection.

This is more honest than calling a policy universally "best practice".

Analytical section 50AI. Comparator conclusion

The widened comparator set confirms that no single producer supplies a complete blueprint.

Norway is useful for fiscal integration.

Ghana for distinct fund purposes.

Timor-Leste for depletion and withdrawal dependence.

Trinidad and Tobago for mature regional energy experience.

Suriname for neighbouring pre-first-oil preparation.

Senegal for new production and state participation.

Mauritania for gas-revenue administration.

Uganda for long pre-first-oil institution building.

Mozambique for new sovereign-fund legislation.

Chile for structural commodity smoothing.

Botswana for buffer accumulation and depletion.

The blueprint emerges by selecting mechanisms, testing prerequisites and adapting them to national scale.

That is the comparative method Edition 2.0 should teach.

PART X | A new-producer operating blueprint

New-producer operating manual

Working manuscript for the 520-560-page flagship edition. This manual translates the research into decision stages for a prospective producer. It does not prescribe a political choice, ownership structure, contract type or spending ratio.

50. Exploration-stage operating manual

The exploration stage should be governed as if no commercial discovery will occur. This prevents the state from building expenditure promises around uncertain geology.

Step 1: establish the legal map

Publish the ministry responsible for petroleum policy, licensing authority, environmental regulator, revenue authority, central bank, auditor, procurement authority, maritime safety agency, emergency command, local-government role and parliamentary oversight bodies. For each institution, record the statute, decision rights, staffing, conflict-of-interest rules and publication duties.

The petroleum law should define ownership of subsurface resources, acreage allocation, data rights, work obligations, relinquishment, environmental duties, transfer of interests, decommissioning and dispute procedures. Existing environmental law should be cross-referenced rather than assumed to apply automatically in the same way offshore.

Step 2: establish the geological-data regime

Create an acreage map, licence register, seismic-data inventory, well register, data-room procedure, confidentiality schedule and beneficial-ownership register. State clearly which data become public and when. Do not use undiscovered resource estimates as budget forecasts.

The regulator should retain original or verified copies of all geological and engineering submissions required by law. Data storage should be resilient and independent of an operator's continued presence in the country.

Step 3: establish the environmental baseline

Before major drilling expands, compile baseline information on currents, fisheries, protected habitats, reefs, mangroves, wetlands, coastal water intakes, seabirds, marine mammals, shoreline communities and economic uses. Record seasonality. Baseline science commissioned only after a spill weakens the ability to establish causation.

The environmental agency should define when an EIA is required, which alternatives must be considered, who reviews the work, how the public comments, and how permit conditions are monitored. Permit issuance and permit compliance belong in different databases.

Step 4: build compact specialist capacity

A new state rarely needs a large petroleum bureaucracy before a commercial discovery. It needs a competent core able to procure and challenge external expertise. The minimum specialist pool includes petroleum law, fiscal modelling, reservoir engineering, environmental science, accounting, transfer pricing, maritime safety and emergency management.

External advisers should undergo conflict checks. Their terms of reference, procurement route and deliverables should be recorded. The state should retain working papers and models so knowledge does not leave when a consultancy ends.

Step 5: run the dry-hole fiscal test

Every proposed fiscal regime should be tested under exploration failure, marginal discovery, low price and high development cost. The objective is to understand incentives before a successful well changes bargaining conditions.

The model should separate royalty, cost recovery, profit share, corporate income tax, bonuses, duties, state participation, ring-fencing and decommissioning. It should disclose the discount rate and price path. A single government-take percentage is insufficient.

51. Discovery-to-final-investment-decision manual

A commercial discovery compresses time. Operators seek appraisal and project approval while public expectations rise. The state's task is to improve information quality before committing the country to long-lived rights and infrastructure.

First 90 days after commercial declaration

- 1. Obtain independent review of discovered resources, recovery assumptions and development concepts.
- 2. Freeze a versioned record of the contract and licence in force.
- 3. Identify every stabilization, tax, valuation, cost-recovery, transfer, decommissioning and dispute clause.
- 4. Publish a non-confidential summary of the state's fiscal exposure and environmental decision process.
- 5. Map expected staffing and training needs.
- 6. Begin a sovereign-revenue custody design, but do not authorize spending based on unproduced oil.

Months 3-9

Commission a full lifecycle environmental assessment with alternatives. Require a spill scenario and transboundary analysis. Define financial assurance and verify the legal trigger for drawing on it. Consult affected communities and document agency responses.

Build a project cost model with independent ranges for capital expenditure, operating expenditure, first oil, plateau output, decline and decommissioning. The model should show how each variable changes state revenue under the executed fiscal terms.

Months 9-18

Set up the production-and-revenue accounting chain. Define meter verification, cargo measurement, price valuation, royalty invoicing, cost submission, audit rights, profit-oil entitlement, sale settlement and deposit into the sovereign account.

Pass the sovereign revenue law before first cash. The law should define objectives, eligible receipts, deposit rules, withdrawal authority, investment policy, oversight, audit, publication and relationship to the annual budget. The numerical withdrawal formula should follow country-specific modelling.

Final investment decision gate

The government file should demonstrate that it understands the project economics, legal obligations, environmental risks, decommissioning exposure and public fiscal path. A decision should not rest solely on the operator's commercial model.

52. First-oil readiness manual

The first cargo is a live systems test.

Technical readiness

Verify production meters, cargo measurement, crude quality, custody-transfer procedures and regulator access. Confirm that the government can independently reproduce the state's entitlement from production and contract terms.

Financial readiness

Trace the first cargo from entitlement through sale, invoice, settlement and deposit. Reconcile the sovereign account to the operator's statement and government invoice. Publish the deposit in the legally required period.

Cost-accounting readiness

Test a sample of recoverable costs before the cost bank becomes too large for the administration to understand. Ensure the state possesses the underlying records and knows the deadline for challenging claims.

Environmental readiness

Conduct a full emergency exercise involving the operator, regulator, maritime authority, military or coast guard where applicable, health services, fisheries authority, local government and neighbouring states if transboundary effects are possible. Test communications loss and port disruption.

Parliamentary readiness

The first withdrawal from the resource fund should follow the same public budget procedures intended for later years. Emergency or exceptional clauses should be narrowly defined and publicly auditable.

Public communication

Publish a first-oil dashboard explaining production, entitlement, royalty, tax, fund deposits and withdrawals. Use plain definitions so citizens do not confuse gross export value with public revenue.

53. Years 1-5 scale-up manual

Year 1: prove the accounting

The first year should prioritize reconciliation over ambitious policy expansion. Complete audited fund statements, contract-cost audits, environmental compliance reports and a public record linking cargoes to government cash.

Commission a baseline household and labour survey if recent comparable data do not exist. This gives the country a reference point before the resource boom fully affects wages, migration and prices.

Year 2: prove domestic value

Audit local-content claims. Separate registered local suppliers from beneficial local ownership and domestic value added. Compare payroll, customs and procurement evidence. Identify skills where local firms can become internationally competitive.

Review public construction costs against pre-boom benchmarks. Rising unit costs may signal capacity constraints even when revenue is abundant.

Year 3: prove public-investment delivery

Select a sample of major projects and trace them from business case to operation. Record time, cost, procurement, variation, maintenance and service indicators. If the economy cannot deliver the planned programme efficiently, slow approvals rather than equating available cash with usable capacity.

Update the petroleum-revenue model with actual reservoir performance and audited cost data. Re-run low-price and delay scenarios.

Year 4: prove resilience

Stress-test the budget under a two- or three-year revenue decline. Test public-sector wage commitments, transfers, debt service and maintenance obligations. Conduct a national emergency exercise simulating a serious offshore incident during severe weather or infrastructure disruption.

Review the sovereign fund's asset allocation and liquidity needs. Do not increase illiquid domestic exposure merely because domestic investment is politically attractive.

Year 5: institutional review

Publish an independent five-year review of petroleum law, fiscal administration, environmental performance, local content, fund management and public investment. Record what changed and why. The aim is not to defend the original framework. It is to determine whether the framework still fits the evidence.

54. Adverse scenario playbooks

Scenario A: oil price drops by 40%

Immediate actions:

- update revenue forecast using realized contract terms;
- freeze assumptions based on the prior price path;
- identify legally committed expenditure;
- assess fund-withdrawal authority;
- defer projects with low completion value if permitted;
- publish a revised cash bridge.

Analytical caution: the public-revenue decline is not equal to the gross price decline because cost recovery, royalty and profit sharing interact.

Scenario B: a major development is delayed by two years

Immediate actions:

- remove projected first oil from the budget baseline;
- recalculate supplier and labour demand;
- revise port and power infrastructure timing;
- test debt-financed projects linked to expected petroleum cash;
- update the fund trajectory.

The purpose is to avoid borrowing against an operator schedule that has not been achieved.

Scenario C: cost audit identifies disputed expenditure

Immediate actions:

- record the disputed amount separately from final recovery;
- preserve all underlying documentation;
- activate contractual expert or dispute procedures;
- avoid presenting the disputed number as public cash;
- disclose process status where law permits.

Scenario D: major offshore pollution event

Immediate actions:

- activate the incident command structure;
- protect life first;
- initiate independent environmental sampling;
- notify affected communities and neighbouring states where relevant;
- secure financial assurance;
- preserve evidence;
- publish verified incident facts at regular intervals;
- separate compensation claims from cleanup operations.

A serious event should not be used to rewrite scientific findings before evidence is collected.

Scenario E: domestic economy overheats

Indicators:

- construction costs rising materially faster than inflation;
- vacancies in public technical roles;
- rapid wage divergence;
- import congestion;
- project delays and variations;
- non-resource exporters losing competitiveness.

Response options for decision-makers include rescheduling public investment, increasing external saving, changing procurement sequencing or expanding training and immigration pathways under applicable law. The manual presents the options without selecting a political preference.

Scenario F: public confidence deteriorates

Triggers include unreconciled accounts, unexplained revisions, delayed audits, weak stakeholder representation or repeated permit breaches. The response should emphasize independent verification, publication of source records, lawful investigation and correction of known errors. Public relations should not substitute for missing evidence.

55. Cabinet and regulator checklists

Monthly petroleum dashboard

- actual production by project;
- downtime;
- cargo lifts;
- government entitlement;
- realized sale price;
- royalty billed and received;
- recoverable cost submitted;
- audit status;
- fund deposits;
- fund investment income;
- withdrawals;
- environmental incidents;
- flaring and emissions where measured;
- permit compliance actions.

Quarterly fiscal dashboard

- opening fund balance;
- petroleum cash inflows;
- investment return;
- withdrawals;
- closing balance;
- public debt;
- non-oil primary balance;
- capital-program execution;
- construction-cost index;
- maintenance funding;
- forecast revision.

Annual social dashboard

- household real consumption;
- poverty/deprivation using current data;
- employment and real wages;
- local ownership and domestic value added;
- school outcomes;
- health outcomes;
- housing and utilities access;
- regional and hinterland gaps;
- migration;
- community grievances and resolution.

Annual environmental dashboard

- project permit status;
- monitoring results;
- incidents;
- emergency drills;
- response times;
- financial assurance;
- fisheries impacts;
- biodiversity indicators;
- emissions;
- decommissioning liabilities.

56. The 24-month implementation sequence for a prospective state

Months 1-3: legal and institutional map, licence register, conflict rules, geological data inventory, environmental baseline design.

Months 4-6: independent fiscal benchmark, regulator staffing plan, emergency-response map, public disclosure policy.

Months 7-9: sovereign-revenue framework options, procurement and audit protocols, beneficial-ownership system, stakeholder consultation framework.

Months 10-12: draft legal instruments, production measurement protocol, model petroleum accounting ledger, spill-response exercise.

Months 13-15: publish model contracts or fiscal principles, establish cost-audit team, environmental review panel and public-data portal.

Months 16-18: test fund accounting with synthetic transactions, test parliamentary reporting, update marine baseline, negotiate regional emergency support.

Months 19-21: run low-price, dry-hole, delay and spill scenarios; publish assumptions and institutional gaps.

Months 22-24: independent readiness review and public correction plan. Do not certify readiness merely because every policy document exists. Test whether the people, systems and data function.

57. Evidence standard for declaring readiness

A country should not declare a workstream complete until the corresponding evidence exists.

Legal readiness requires enacted and commenced law plus defined implementing authority.

Fiscal readiness requires transaction testing, not a policy paper.

Environmental readiness requires baseline data, permits, monitoring capacity and emergency drills.

Audit readiness requires staff, access rights, data retention and completed sample tests.

Local-content readiness requires verified ownership and skills data, not registration alone.

Public-investment readiness requires procurement and maintenance capacity.

Social readiness requires current surveys and grievance mechanisms.

Fund readiness requires audited financial statements and lawful budget integration.

Part X conclusion

The central principle of the operating manual is sequencing. A new producer should build evidence and institutional capacity before each irreversible stage: licence award, appraisal, development approval, first oil, first withdrawal and rapid scale-up.

Guyana's experience is useful because several of these stages occurred in a compressed period and left a substantial documentary record. The transferable product is not a single contract or statute. It is a staged system for asking the right question before the cost of a mistake rises.

Edition 2.0 deepening: the operating manual as a sequence of irreversible gates

Analytical section 58. The five irreversible gates

A newcomer petroleum state should organize readiness around five gates.

Gate 1: licence award. Before acreage is awarded, the state should know who has authority, what data exists, which work obligations are required, how beneficial ownership is checked and which environmental baseline obligations apply.

Gate 2: commercial discovery. Before the state accepts a commercial-development path, it should independently understand reserves, development cost, fiscal outcomes, infrastructure needs and environmental exposure.

Gate 3: final investment decision and development approval. Before irreversible capital is committed, the state should have completed legal, fiscal, environmental, decommissioning and emergency-response reviews.

Gate 4: first oil or first gas. Before production begins, the state should prove that metering, cost accounting, royalty, tax, sales, sovereign-revenue custody, incident command and public reporting systems function.

Gate 5: first major budget withdrawal. Before resource revenue is materially absorbed into the domestic economy, the state should have project appraisal, procurement, maintenance and macro-fiscal controls.

Each gate should have evidence requirements and a named institution authorized to sign readiness.

Analytical section 59. A national petroleum RACI matrix

Every major petroleum process should identify who is:

- Responsible;
- Accountable;
- Consulted;
- Informed.

The matrix should cover:

- licensing;
- contract negotiation;
- geological data;
- reserves review;
- development plan;
- production licence;
- EIA;
- permit;
- metering;
- petroleum valuation;
- cost audit;

- tax;
- cargo sale;
- NRF custody;
- budget withdrawal;
- local content;
- emergency response;
- decommissioning.

The purpose is to prevent shared responsibility from becoming no responsibility.

Analytical section 60. The discovery 100-day plan

Within the first 100 days after a material commercial discovery, government should create a single readiness programme.

Days 1-30:

- preserve all geological and contractual records;
- establish conflict controls;
- commission independent resource and fiscal review;
- freeze premature revenue assumptions in the budget;
- map institutional gaps.

Days 31-60:

- create environmental and social baseline workplan;
- define development-plan review team;
- establish cost-accounting architecture;
- begin sovereign-revenue design options;
- create public communication protocol.

Days 61-100:

- publish the readiness gap register where lawful;
- approve staffing and adviser strategy;
- run fiscal stress tests;
- establish first-oil data architecture;
- begin emergency-response planning.

The discovery should therefore trigger administration before it triggers spending.

Analytical section 61. The development-approval dossier

A development approval should be supported by one consolidated government dossier containing:

- resource estimate;
- reservoir uncertainty;
- development concept;
- capital cost;
- operating cost;
- production forecast;
- fiscal model;
- cost-recovery forecast;
- environmental assessment;
- financial assurance;
- emergency response;
- local-content plan;
- decommissioning concept;
- public infrastructure requirements.

The dossier should record both operator assumptions and independent government analysis.

Material disagreements should remain visible.

Analytical section 62. The first-oil certification exercise

Six months before first oil, government should run a full transaction test.

The exercise should simulate:

- daily production;
- meter exception;
- government entitlement;
- cargo nomination;
- benchmark price;
- royalty;
- tax;
- cost recovery;
- cash settlement;
- NRF deposit;
- public disclosure.

A second exercise should simulate an incident occurring during the same period.

Readiness is proven when institutions can perform the transaction under pressure, not when policy documents have been published.

Analytical section 63. The first-withdrawal certification exercise

The first major petroleum-funded budget programme should also be tested.

The government should demonstrate:

- legal withdrawal authority;
- transfer to the Consolidated Fund;
- parliamentary appropriation;
- project appraisal;
- procurement;
- contract management;
- completion reporting;
- maintenance funding.

This makes the first withdrawal an institutional milestone rather than a simple cash movement.

Analytical section 64. A single national resource dashboard with layered access

The state should maintain one integrated data model even if public access varies.

Layer 1 can be fully public:

- production;
- revenue;
- fund balance;
- major permits;
- public contracts;
- key outcomes.

Layer 2 can be regulator-access:

- metering details;
- cost records;
- commercially sensitive data;
- detailed environmental monitoring.

Layer 3 can be restricted for security or privileged legal material.

The important control is that all layers use consistent identifiers.

Analytical section 65. The minimum permanent specialist team

A small state does not need a huge bureaucracy.

It does need permanent capability in:

- petroleum engineering;
- geology/geophysics;
- fiscal modelling;
- petroleum accounting;
- tax;
- law;
- environmental science;
- emergency response;
- data management;
- procurement and public investment.

External advisers can supplement this team.

They should not become the only holders of state knowledge.

Analytical section 66. Adviser management protocol

Every external adviser engagement should specify:

- scope;
- conflict check;
- data ownership;
- model ownership;
- source-file delivery;
- training requirement;
- handover;
- confidentiality;
- publication rights.

The state should receive editable models and working papers, not only a final PDF.

This reduces dependence on the same adviser for future decisions.

Analytical section 67. A low-capacity state should prioritize control points

When staffing is limited, not every process can receive equal attention.

The highest-value control points are:

- licence and contract;
- reserves;
- development cost;
- metering;

- petroleum valuation;
- cost audit;
- revenue custody;
- environmental permit;
- financial assurance;
- emergency response;
- budget withdrawal.

A small regulator should secure these points before expanding into lower-risk administrative detail.

Analytical section 68. Island-state operating manual

A small island producer should add six specific controls.

- 1. **Tourism exposure map.** Identify beaches, reefs, ports and marine recreation.
- 2. **Desalination and water-intake protection.** Model spill and shutdown scenarios.
- 3. **Hurricane concurrency.** Test a petroleum incident during storm disruption.
- 4. **Port redundancy.** Identify what happens if the primary port is unavailable.
- 5. **Regional response agreements.** Pre-negotiate customs, aviation and equipment entry.
- 6. **Small-labour-market strategy.** Protect regulators and utilities from specialist drain.

These controls are not unique to islands, but concentration makes the consequences larger.

Analytical section 69. The no-oil scenario

A petroleum strategy should still create value if commercial production never occurs.

The state should ask which investments remain useful under a dry-hole outcome.

Usually these include:

- geological data;
- maritime mapping;
- environmental baselines;
- beneficial-ownership systems;
- procurement controls;
- emergency coordination;
- public financial management.

This test prevents exploration optimism from creating stranded bureaucracy or debt.

Analytical section 70. The lower-price scenario

The operating manual should assume that resource prices can remain below the planning case for several years.

The state should identify:

- protected essential services;
- deferrable projects;
- recurrent commitments;
- debt service;
- fund buffers;
- exchange-rate exposure.

The purpose is to decide before the shock which commitments are flexible.

Analytical section 71. The delayed-project scenario

A two-year project delay affects more than future oil revenue.

It can also affect:

- public expectations;
- infrastructure built for the project;
- local suppliers;
- labour;
- debt;
- regulator staffing.

The contingency plan should therefore cover the wider economy.

Analytical section 72. The major-spill scenario

The spill scenario should include:

- immediate command;
- operator action;
- financial assurance;
- public communication;
- fisheries;
- health;
- cross-border notification;
- evidence preservation;
- claims;
- independent investigation.

The exercise should be timed.

Decision delay should be treated as a risk variable.

Analytical section 73. The institutional-disruption scenario

Institutions can face leadership change, litigation, cyber incidents, natural disaster or staff turnover.

The petroleum system should remain operable.

Business-continuity plans should identify:

- delegated authority;
- backup records;
- credential recovery;
- alternate office;
- emergency signatories;
- succession.

This is administrative continuity, not a political judgment.

Analytical section 74. The annual resource-governance review

Once production begins, the state should publish an annual integrated review covering:

- production;
- reserves;
- revenue;
- audits;
- NRF;
- budget use;
- local content;
- social indicators;
- environmental performance;
- incidents;
- decommissioning;
- unresolved disputes.

The review should include corrections from prior years.

Analytical section 75. Independent readiness review

Before each irreversible gate, government should commission an independent challenge review.

The reviewer should not certify the political desirability of development.

The reviewer should test whether the evidence and controls required for the decision exist.

The final output should classify:

- ready;
- ready with conditions;
- not evidenced.

Any unresolved high-impact condition should have an owner and deadline.

Analytical section 76. Expanded Part X conclusion

The operating manual should make the book usable on Monday morning, not only intellectually persuasive.

Its central rule is simple:

Do not move from one irreversible petroleum stage to the next merely because the commercial project is ready.

Move when the state can also measure, regulate, account, respond and explain.

The manual therefore converts the Guyana case and international comparators into a sequence of gates, ledgers, drills and evidence requirements.

For a new producer, that sequence is the real blueprint.

It preserves national optionality before discovery, bargaining power after discovery, revenue integrity after first oil and development discipline after the first large withdrawal.

Edition 2.0 tool deepening: converting the blueprint into auditable decision instruments

Analytical section 77. The readiness checklist is a decision gate, not a ceremonial checklist

Edition 2.0 adds `NEW\_PRODUCER\_READINESS\_CHECKLIST.csv`.

The file is designed so that each readiness item has:

- a decision gate;
- a control area;
- minimum evidence;
- accountable institution;
- independent challenge;
- pass condition;
- failure response.

The final field is essential.

A checklist that has no consequence for failure becomes ceremonial.

For some items, failure should delay the decision.

For others, it should create a condition, deadline or compensating control.

The responsible authority should document why the response is proportionate to the risk.

Analytical section 78. Readiness evidence should be independently challengeable

"Government says it is ready" is not a sufficient control.

A readiness file should contain evidence that can be challenged by:

- another agency;
- an external technical reviewer;
- auditor;
- court where appropriate;
- parliament;
- public researcher for non-confidential material.

This does not mean every decision requires consensus.

It means the decision record should survive scrutiny.

Analytical section 79. The RACI solves overlapping mandates before they become disputes

Edition 2.0 adds `PETROLEUM\_GOVERNANCE\_RACI.csv`.

A RACI is useful in petroleum governance because multiple institutions legitimately touch the same transaction.

For example, a government cargo can involve:

- petroleum regulator;
- marketing function;
- finance ministry;
- central bank;
- legal advisers.

The RACI distinguishes:

- who performs the work;
- who is finally accountable;
- who must be consulted;
- who must be informed.

This prevents a multi-agency system from becoming an accountability vacuum.

Analytical section 80. The RACI should be anchored in law

A RACI cannot override statute.

The "accountable" field must reflect the institution legally empowered to make the decision.

If the operational practice differs from the legal mandate, the solution is not to alter the spreadsheet.

The legal framework or administrative delegation must be clarified.

This is why the RACI should be reviewed by public-law counsel before adoption.

Analytical section 81. A readiness gate should have an evidence room

Each irreversible gate should have an indexed evidence room.

The file structure might contain:

- legal authority;
- technical analysis;
- fiscal model;
- environmental record;
- financial assurance;
- local content;
- consultation;
- decision memorandum;
- approval;
- conditions;
- subsequent compliance.

The evidence room can be digital.

Its key feature is version control.

A reviewer should know exactly which documents existed when the decision was taken.

Analytical section 82. The first-oil dry run should include an error

A simulation that assumes everything works does not test resilience.

The first-oil certification exercise should deliberately introduce:

- meter failure;
- incorrect invoice;
- late cargo settlement;
- exchange-rate mismatch;
- missing cost document.

The test is whether controls detect and correct the error.

This is more realistic than a perfect transaction.

Analytical section 83. The spill drill should include unavailable equipment

Emergency exercises should also introduce failure.

Examples:

- primary response vessel unavailable;
- aviation grounded;
- customs delay;
- communications outage;
- simultaneous shoreline concern.

The after-action report should record:

- decision time;
- resource time;
- information gaps;
- unresolved authority.

The response plan should then be revised.

Analytical section 84. The first-withdrawal test should follow one project end to end

Rather than evaluating the entire public investment programme at once, a newcomer can choose one petroleum-funded project as a control case.

The review follows: NRF withdrawal to Consolidated Fund to appropriation to tender to contract to payment to asset to service outcome.

This creates a model for later scaling.

Analytical section 85. The adverse-scenario playbook is cross-institutional

Edition 2.0 adds `ADVERSE\_SCENARIO\_PLAYBOOK.csv`.

The scenarios are deliberately cross-system.

A price shock is not only a petroleum issue.

It affects the budget, fund, debt and public investment.

A spill is not only an environmental issue.

It affects finance, fisheries, health, foreign relations and public communication.

The playbook therefore tests institutional coordination.

Analytical section 86. The price-shock playbook

A 40% price decline should trigger immediate recalculation of:

- government entitlement;
- NRF inflow;
- withdrawal capacity;
- medium-term fiscal balance;
- debt financing;
- project pipeline.

The response should identify which expenditure is:

- protected;
- deferrable;
- scalable;
- one-off;
- recurrent.

This avoids improvised austerity after the shock occurs.

Analytical section 87. The project-delay playbook

A two-year petroleum project delay affects more than production.

Local suppliers may have invested based on forecast demand.

Government may have built infrastructure.

Workers may have trained for jobs not yet available.

The playbook should therefore update:

- revenue;
- local-content demand;
- labour;
- public infrastructure timing;
- borrowing.

Analytical section 88. The audit-dispute playbook

A large audit dispute should trigger a controlled process rather than a public accounting shortcut.

The register should show:

- disputed amount;
- cost category;
- accounting treatment;
- negotiation;
- expert process;
- arbitration if any;
- final adjustment.

Government cash should change only when the contractual/accounting process creates a cash or entitlement effect.

Analytical section 89. The major-spill playbook

The spill playbook begins with life safety and source control.

It then connects:

- incident command;
- environmental monitoring;
- fisheries;
- health;
- financial assurance;
- claims;
- investigation;
- public reporting.

The state should know in advance which institution has authority to approve emergency actions and which entity initially pays.

Analytical section 90. The operator-distress playbook

A petroleum regime should not assume all contractors and guarantors remain financially strong indefinitely.

If a material credit concern arises, government should review:

- outstanding obligations;
- insurance;
- guarantees;
- decommissioning exposure;
- assignment rights;
- joint-venture obligations.

The response should be based on verified financial evidence rather than speculation.

Analytical section 91. The public-investment overheating playbook

The trigger is not simply "high spending".

The trigger is a combination of:

- cost escalation;
- contractor concentration;
- repeated delay;
- material variation orders;
- labour shortage.

The response can include resequencing rather than cancellation.

The objective is to restore delivery capacity.

Analytical section 92. The household-pressure playbook

A resource boom can produce distributional stress even when aggregate growth is strong.

The playbook should monitor:

- food;
- housing;
- transport;
- wage distribution;
- regional prices.

Temporary support should use current evidence and have a defined review or expiry point.

Analytical section 93. The maritime-incident playbook

For a country with offshore boundary or territorial issues, petroleum operations should have a coordination protocol covering:

- maritime security;
- regulator;
- foreign affairs;
- operator;
- insurers.

The protocol should separate operational safety from legal/diplomatic claims.

The book should not predict the merits of an international dispute.

Analytical section 94. The cyber playbook

Petroleum revenue systems increasingly depend on digital records.

A cyber incident could affect:

- production reporting;
- invoicing;
- bank instructions;
- permit systems;
- public dashboards.

The state should preserve an offline recovery path for critical transactions.

Backups should be tested, not merely stored.

Analytical section 95. The small-state staffing model

A small jurisdiction should distinguish three staffing layers.

Permanent sovereign core: officials who own the decision.

Standing specialist pool: national or regional experts available when needed.

Transaction-specific advisers: external firms retained for a defined project or dispute.

This model avoids the impossible goal of keeping every specialist on the public payroll.

It also avoids outsourcing the sovereign decision itself.

Analytical section 96. Regional cooperation can create virtual scale

Small states can share or coordinate:

- spill-response equipment;
- specialist training;
- laboratory services;
- petroleum tax expertise;
- model contracts;
- data standards.

Regional cooperation should not erase national authority.

It can reduce the cost of rare specialist capacity.

Analytical section 97. Adviser rotation and institutional learning

A government that repeatedly hires the same external adviser gains continuity but may create dependence.

The adviser-management system should track:

- deliverables;
- knowledge transfer;
- model ownership;
- staff trained;
- successor capability.

For critical functions, the state should be able to change advisers without losing its own records.

Analytical section 98. The national petroleum handbook should be a living document

The operating manual should not end with first publication.

It should be updated after:

- each licence round;
- each development approval;
- first oil;
- major audit;
- major incident;
- legal judgment;
- statutory change.

Each edition should preserve the old version.

This creates institutional memory.

Analytical section 99. Readiness certification should not become a political score

The readiness framework evaluates evidence for defined controls.

It should not be converted into a score ranking governments or political actors.

A decision can be:

- evidenced;
- evidenced with condition;
- not evidenced.

That is sufficient for operational governance.

Analytical section 100. Edition 2.0 implementation package

The final Edition 2.0 newcomer package should contain:

- readiness checklist;
- RACI;
- source register;
- clause crosswalk;
- project register;
- environmental assurance matrix;
- NRF ledger;
- public-investment register;
- local-value framework;
- comparator matrix;
- adverse-scenario playbook;
- corrections register;
- publication manifest.

This is what turns the white paper into an operating reference.

Analytical section 101. Expanded operating-manual conclusion

The strongest transfer from Guyana is not one contract clause or one fund rule.

It is the ability to observe a compressed sequence of decisions and ask what institutions should exist before the next decision becomes irreversible.

Edition 2.0 converts that observation into tools.

A newcomer can replace Guyana's numbers with its own data while preserving the control logic.

That is the standard for a genuinely transferable blueprint.

Applied country exercise: implement the five petroleum decision gates with a traceable evidence file

The operating manual becomes useful only when a ministry can apply it to a real decision, identify the evidence still missing and explain the consequence of approving an incomplete dossier. The following exercise combines the historical chronology of Guyana with a deliberately hypothetical new producer. Its contract arithmetic is illustrative and assumes specified contractual terms; it is not a forecast of future Guyanese revenue or a recommendation to adopt those terms in another jurisdiction.

Gate one: before the first exploration licence

Assume a coastal country with one prospective offshore basin, a small legal and environmental staff, no commercial petroleum history and a government seeking its first credible exploration programme. The country's immediate asset is not proven oil; it is the legal right to allocate exploration opportunities and the geological information held by the state. The case study should compare this position with Guyana's decades-long pre-Liza history. The World Bank's completion report on Guyana's 1980s petroleum exploration promotion work describes institution building, foreign exploration attraction, legislative and technical support, and both useful outputs and implementation limitations. Those records show why a dry-hole history can still create a valuable data asset. [S35, S107]

The ministry's first approval file should contain a map of legally available acreage; the legislative authority for licensing; a geoscience data catalogue; a work programme and relinquishment schedule; beneficial-ownership evidence; a marine baseline plan; and a record of potential maritime-jurisdiction uncertainty. The authorities responsible for data, fiscal advice, maritime administration and environmental regulation should sign the aspects within their mandates. A bidder's geological optimism is not evidence of a national reserve.

The stop condition is narrow but consequential: do not award a licence if the state cannot show its legal title or statutory authority, determine the bidder's controlling interests, enforce minimum exploration work, or obtain the basic data generated by that work. Other deficiencies may have proportionate remediation dates, but a missing sovereign right or unidentified licensee defeats the decision itself. The country should keep a versioned file recording what was known before the award so later review does not confuse hindsight with negligence.

Gate two: between commercial discovery and development approval

Imagine that a commercially promising discovery follows. The cabinet receives an operator's reserves estimate and development proposal. The central question is not whether a discovery is newsworthy; it is whether government can test the production profile, capital budget, cost-recovery consequences, environmental assessment and end-of-life liabilities independently. Guyana's initial Stabroek contract provides a historical agreement to study, while its later Petroleum Activities Act 2023 and subsequent model PSA illustrate changes in legal and contractual architecture after more geological and administrative experience became available. Later model provisions cannot be substituted for the executed 2016 agreement when analyzing actual Stabroek entitlements. [S02, S87, S112]

Create a shadow development model with the operator's assumptions alongside independent low, central and high cases. The file should show recovery uncertainty, well count, FPSO or pipeline design, oil/gas price path, development cost, environmental conditions, decommissioning cost, expected government profit petroleum, tax and royalty treatment, and the government's own capacity to inspect the project. Every fiscal cell must link to an executed contractual clause or an identified statute. A missing price assumption should not be repaired by copying an unrelated country's forecast.

An expert reviewer should then conduct a challenge meeting. Ask which operational scenario creates the largest fiscal shortfall; which construction delay creates the greatest pressure to grant extensions; what information determines whether cost is recoverable; and who bears unplanned environmental expenditure. The responses should be documented and returned to the regulator before a development decision. The review is a technical record, not a verdict on an administration or elected official.

Gate three: prove the first production month with a fully disclosed synthetic transaction

The strongest pre-first-oil exercise is a simulated cargo that connects physical production to the sovereign account. Consider a *hypothetical* month with 100,000 saleable barrels priced at US\$70 each. Gross value is US\$7 million. For teaching purposes only, suppose the contract includes a 2% royalty on the gross value, a binding 75% cost-oil ceiling, and an equal division of residual profit oil. Those headline assumptions parallel the specific Stabroek mechanism described in the executed agreement and the Guyana Revenue Authority's explanation, but the example deliberately omits detailed valuation, cost eligibility, sales adjustments, tax certificates and transaction timing. It is not a full contract model. [S02, S15]

Under those assumptions, gross royalty is US\$140,000. Maximum cost oil is US\$5.25 million. Residual profit petroleum is US\$1.75 million, of which the government obtains US\$875,000 in physical or monetized profit-oil entitlement. Combined illustrative government value is therefore US\$1.015 million, or 14.5% of the gross example. This result holds in this specific binding-ceiling scenario, not automatically in every production month or over the life of a field. If eligible recoverable cost is lower, cost oil will be lower and residual profit petroleum higher.

Suppose the opening recoverable-cost bank is US\$50 million and current eligible cost is US\$5 million. The maximum US\$5.25 million recovered in the illustrative month would leave a closing cost bank of US\$49.75 million. The example teaches two distinct accounting systems: the current month's division of production and the stock of costs carried into the next period. A regulator must be able to reproduce both before approving the monthly statement.

Now introduce a deliberately erroneous invoice worth US\$250,000 charged twice to the same field. The first-oil exercise should test whether the cost ledger flags duplicate supplier identity, invoice number, currency, service dates and project allocation before the charge can enlarge recoverable cost. It should then create a formal exception, provide an auditee-response window and record the outcome. The government's final cash entitlement must not be silently adjusted merely because an auditor has raised a question. Guyana's first major cost-audit experience demonstrates why the stages from exception to dispute to settled adjustment need separate records. [S111]

Finally, simulate the government lifting its physical share a month later and receiving buyer payment after year-end. The production month, cargo sale, receivable recognition and NRF cash deposit must each have their own date. The trial should reproduce a statement for each reporting boundary and a bridge between them. The first-half 2026 NRF recognition issue examined in Part IV is a practical demonstration of what happens when this discipline is absent. [S07, S88, S134, S137]

Gate four: certify environmental and emergency arrangements before startup

A government should not treat the existence of an EIA as proof that its offshore emergency system can operate. The gate file should contain the original environmental baseline, permit and revisions, a condition-by-condition compliance matrix, project-specific insurance/financial-assurance evidence, an oil-spill command plan, a deployable response-resource inventory, and a documented offshore drill. The signed Uaru permit provides specific financial-assurance conditions against which another jurisdiction can design a review checklist: credible-cost basis, environmental-liability insurance and parent/affiliate guarantee arrangements are distinguishable obligations. The permit does not prove the exact limits or enforceability of executed instruments for every future loss. [S99]

The hypothetical exercise should remove the primary response vessel from service and delay entry of foreign equipment. Officials must identify who authorizes dispersant use, what fisheries notices are necessary, how evacuation is coordinated, who can obtain emergency cash and when neighbouring jurisdictions are informed. The result should be an after-action record of actual decisions and deployment times rather than a ceremonial attendance certificate. Any red-line deficiency affecting life safety, source control or availability of a legally required financial instrument requires remediation before the relevant operation begins.

Legal version control matters. Guyana's 2025 Oil Pollution Act was formally published, but commencement of each operative requirement must be established from authoritative provisions or instruments rather than inferred from gazetting alone. A 2024 authenticated Caribbean Court of Justice judgment addresses procedural and statutory aspects of underlying environmental litigation; it does not substitute for authenticated reasons from a later appellate merits decision. The newcomer's legal file should therefore identify the precise law, permit and judgment that apply at its own decision date. [S98, S103]

Gate five: do not certify development merely because the first fund withdrawal was lawful

Once receipts begin, the country's sovereign fund transfers an authorized amount to its budget. Legal withdrawal is the starting point, not the development result. To certify a completed resource-conversion cycle, an independent team should follow one representative project from appraisal through budget authorization, open procurement, contract changes, construction, commissioning, operations, maintenance and measured public service. A hospital should be tested for staffing, supplies and patient access after opening; a road for actual travel time, safety and maintenance after commissioning. A contract value or ribbon-cutting is not by itself a measured outcome.

The state should also explain the fiscal reserve after the withdrawal. The Natural Resource Fund of Guyana's published financial statements and quarterly reports illustrate the separate concepts of petroleum deposits, investment result, government transfer, cash and accrued fund assets. Different accounting periods require reconciliation before ratios are compared. [S44, S88, S109, S134]

For a small-island application, the after-withdrawal test should add port redundancy, tourism-area sensitivity, shoreline spill exposure, import dependence and the operating costs of disaster-resilient infrastructure. These are context-specific requirements, not arguments that an island should reproduce every Guyanese policy. The comparator records are useful here: Senegal illustrates new-production and state-company reporting questions; Mauritania gas-specific revenue administration; Uganda a longer pre-production preparation period; Mozambique the legal architecture of a newer sovereign fund; Chile revenue-smoothing mechanisms; and Botswana the behavior of mineral-derived buffers through commodity cycles. Their institutional settings differ, and the publication should compare the mechanism and prerequisite rather than rank countries. [S113, S115, S117, S119, S121, S122]

Evidence standard for applying this exercise

An adopting jurisdiction should publish a concise evidence register for each gate: claim, controlling instrument, dated input, accountable institution, independent reviewer, residual uncertainty, decision and conditions. "Ready", "ready with a specific condition" and "not evidenced" are operational documentation statuses, not political ratings. The test is whether an outsider can independently follow the trail from an exploration right to a national asset without guessing which year's data or which legal version the analysis used.

The accompanying machine-readable templates—the readiness checklist, petroleum-governance RACI, cost and fund ledgers, project register, environmental-assurance matrix and adverse-scenario playbook—should remain editable. That permits another government to substitute its own laws, geology, institutions and fiscal assumptions while preserving the evidence chain that makes the exercise auditable.

Reproducible data exhibits

All figures below are copied from the dated editable CSVs rather than invented for the PDF. Planned capacities must not be treated as realized production.

Seven-development project register

Development	FPSO	Status at 27 Sep 2026	Design bpd
Liza Phase 1	Liza Destiny	2019-12-20	120000
Liza Phase 2	Liza Unity	2022-02-11	220000
Payara	Prosperity	2023-11-14	220000
Yellowtail	ONE GUYANA	2025-08-08	250000
Uaru	Errea Wittu	Commissioning / no first-oil confirmation located by cutoff	250000
Whiptail	Jaguar	Construction and offshore installation; startup targeted 2027	250000
Hammerhead	VLCC conversion-type FPSO	Approved / FID; startup anticipated 2029	150000

Project-specific source IDs and evidence classifications appear in OFFSHORE_PROJECT_REGISTER.csv. Future startups are not observed facts.

NRF annual stock and flow (USD millions)

Period	Inflows	Withdrawals	Closing	Evidence status
2020	198.31	0.00	198.31	PROVISIONAL - investment-income detail to transcribe
2021	409.20	0.00	607.65	PROVISIONAL - residual USD0.14m is not an audited investment-income claim
2022	1411.95	607.65	1429.45	ROUNDED RECONCILIATION
2023	1608.22	1002.13	2122.38	ROUNDED RECONCILIATION
2024	2567.96	1586.00	3245.69	ROUNDED RECONCILIATION WITH DERIVED RETURN
2025	2509.71	2463.00	3434.53	ROUNDED RECONCILIATION
2026-H1		1020.00	4294.20	OPEN RECONCILIATION - do not cite as closed

2020-21 and 2026-H1 carry explicit provisional/open statuses. The 2025 December receivable is not double-counted as fresh 2026 accrued revenue. Primary series: [S44](#), [S88](#), [S109](#), [S137](#).

Evidence limits and corrections

2020–2021 NRF interest splits and the H1 2026 cash/accrual reconciliation remain provisional where labelled. Permit wording is not proof of executed insurance or compliance. EITI status is dated; legal appellate reasons and full oil-pollution commencement should be rechecked from authoritative records.

All scenarios in companion CSVs are explicitly illustrative and do not constitute forecasts of petroleum production, government cash or political outcomes.

Source register (dated and qualified)

The 141 entries below come from the GitHub Edition 2 source register. Source-ID hyperlinks in the narrative jump to their respective entries. Claim-level external review remains pending.

S01 ExxonMobil Guyana 2023 Annual Report

Evidence: Early seismic and onshore exploration, May 2015 Liza commercial discovery, 2019 first oil and FPSO dates

Limit: Operator is an interested party; cross-check public licence register

<https://corporate.exxonmobil.com/locations/guyana/publications/2023-annual-report>

S02 Guyana Ministry of Natural Resources, Stabroek petroleum agreement

Evidence: Official text of the 2016 agreement, contract terms and baseline obligations

Limit: Distinct from future PSAs and public summaries

<https://petroleum.gov.gy/docs-and-pubs/petroleum-agreement-stabroek-block/>

S03 Ministry of Natural Resources, block register

Evidence: June 27 2016 agreement / October 7 2016 effective date and licence information

Limit: Different contract milestone dates refer to different acts

<https://petroleum.gov.gy/oil-blocks/stabroek-block/>

S04 2024 NRF Annual Report, Ministry of Finance, notes/legal framework

Evidence: NRF law assented January 23 2019; replaced December 30 2021; withdrawal rule amended February 6 2024

Limit: Full audited report must be checked for changing cash period

https://finance.gov.gy/wp-content/uploads/2025/11/2024_NRF_Annual_Report_Final_Draft-JD0715_GE1507-Final-Print.pdf

S05 Parliament of Guyana, Acts index

Evidence: Local Content Act 2021 and NRF Act 2021 parliamentary dates

Limit: Parliamentary passing, assenting and commencement dates differ

<https://www.parliament.gov.gy/publications/acts-of-parliament/P80>

S06 Finance, 2026 Budget Speech, January 26 2026, 124-page PDF

Evidence: January production/revenue/NRF assumptions and policy descriptions

Limit: January projections were superseded by September mid-year update

https://finance.gov.gy/wp-content/uploads/2026/01/Budget_Speech_2026.pdf

S07 Finance, 2026 Mid-Year Report, issued September 2026, printed pp. 21-25 and 104

Evidence: H1 2026 21 government lifts, USD 1,778.6m profit-oil proceeds, USD 218.4m royalties, USD 4,294.2m end-June fund cash, USD 1,020m withdrawals; revised forecast USD 6,497.6m petroleum deposits

Limit: Government estimates and projections, not audited end-year results

<https://finance.gov.gy/wp-content/uploads/2026/09/Mid-Year-Report-2026.pdf>

S08 IMF, July 31 2026 mission concluding statement

Evidence: IMF staff views on balance/withdrawal rule, absorption, audit, procurement, AML/CFT, statistical gaps and LCDS

Limit: Staff preliminary findings, not IMF Executive Board Article IV report

<https://www.imf.org/en/news/articles/2026/07/31/mcs073126-guyana-staff-concluding-statement-of-the-2026-article-iv-mission>

S09 IMF, 2025 Article IV report and press

Evidence: Detailed macro, fiscal and debt context, external baseline

Limit: Historical to 2024; do not use as September 2026 snapshot

<https://www.imf.org/en/publications/cr/issues/2025/05/07/guyana-2025-article-iv-consultation-press-release-staff-report-and-statement-by-the-566712>

S10 World Bank, Guyana country overview

Evidence: Social indicators, non-oil growth and lack of current household poverty data

Limit: Explicitly labels last cited poverty estimate 2019, not 2026

<https://www.worldbank.org/ext/en/country/guyana>

S11 ExxonMobil, Yellowtail startup, August 8 2025

Evidence: Fourth FPSO, installed capacity exceeded 900k bopd

Limit: Installed capacity is not sustained actual output

https://corporate.exxonmobil.com/news/news-releases/2025/0808_exxonmobil-guyana-begins-production-at-fourth-offshore-guyana-project

S12 ExxonMobil, November 12 2025

Evidence: 900k bopd daily output milestone

Limit: A single-day threshold is not 2025 annual average

<https://corporate.exxonmobil.com/locations/guyana/news-releases/11122025-daily-oil-production-hits-900000-barrels-in-guyanas-stabroek-block>

S13 Petroleum Management Programme oil-production chart

Evidence: Government monthly production chart through July 31 2026

Limit: Preserve date and distinguish unit from capacity

<https://petroleum.gov.gy/data-chart/oil-production/>

S14 Government of Guyana, 2023 model fiscal terms

Evidence: Announced 10% royalty, 65% cost ceiling and 10% corporate tax for future licensing

Limit: Do not apply revised terms retroactively to Stabroek

<https://petroleum.gov.gy/article/govt-announces-wave-of-enhanced-fiscal-terms-for-first-offshore-oil-and-gas-competitive-licensing-round-highlights/>

S15 Guyana Revenue Authority, royalty treatment, 2026

Evidence: 2016 2% royalty added to simplified 12.5% government profit share when cap binds

Limit: 14.5% example is not constant government take across the life of the block

<https://gra.gov.gy/royalty-payments-to-guyana-government/>

S16 EITI, May 12 2026 Board decision

Evidence: Second validation from May 15; concern over civil-society selection and stakeholder confidence

Limit: No 2026 final validation result has been established in this work

<https://eiti.org/es/node/28602>

S17 EITI, 2026 call for stakeholder views

Evidence: 2022 validation and 2026 validation scope

Limit: Distinguish earlier assessment from later validation

<https://eiti.org/offers/guyana-2026-eiti-validation-call-views-stakeholder-engagement>

S18 Guyana EITI country secretariat

Evidence: December 2025 report covers fiscal year 2023; validation visits June 2026

Limit: Secretariat claims require triangulation with EITI global board

<https://eiti.gy/about/>

S19 Parliament, Oil Pollution Prevention, Preparedness, Response and Responsibility Act 2025

Evidence: Enacted Act No. 6 of 2025

Limit: September mid-year report says draft regulations progressing and commencement expected end-2026; enactment is not commencement

<https://parliament.gov.gy/publications/acts-of-parliament/oil-pollution-prevention-preparedness-response-and-responsibility-act-2025-act-6-of-2025>

S20 Government statement about May 7 2026 Court of Appeal decision

Evidence: Appeal set aside High Court unlimited financial-guarantee order

Limit: Obtain authenticated judgment and independent legal commentary before final jurisprudential analysis

<https://dpi.gov.gy/press-release-ag-hails-coas-ruling-on-exxonmobil-epas-financial-guarantees/>

S21 EPA environmental authorisation services

Evidence: Environmental authorisation/permit process and regulated-project categories

Limit: Online application portal alone does not prove monitoring quality

<https://eservices.epa.gov.gy/permits>

S22 Guyana LCDS 2030, October 2024 update

Evidence: 2009 framework, Norway USD 227m for 2009-15, 2022 policy and forest-credit programme; 15% community allocation

Limit: LCDS describes government strategy and self-reported expenditure; impact audit separate

<https://lcds.gov.gy/guyanas-low-carbon-development-strategy-lcds-2030-october-2024-update/>

S23 LCDS, Guyana ART TREES fact sheet, 2022

Evidence: Hess minimum USD 750m purchase agreement, credit volume and minimum prices

Limit: Contracted value is not money already received

<https://lcds.gov.gy/fact-sheet-on-guyanas-engagement-in-art-trees/>

S24 Ghana Petroleum Revenue Management Act 2011

Evidence: Stabilisation Fund, Heritage Fund, publication cadence and public accountability

Limit: Comparator institutional design, not a performance ranking

<https://ghalii.org/akn/gh/act/2011/815/eng%402011-04-15>

S25 Norway, 2025 Government Pension Fund white paper

Evidence: Entire net state petroleum cash transferred to GPFG; 3% real-return fiscal guideline

Limit: Norway's long accumulation and mature state capacity differ from a new producer

<https://www.regjeringen.no/contentassets/185dca55ef5649d1bc08db2699bb861f/en-gb/pdfs/stm202420250022000engpdfs.pdf>

S26 Timor-Leste central bank, 2005/2011 Petroleum Fund statutes

Evidence: Statutory fund structure and amendments

Limit: Cross-check 2026 depletion pressures independently before claims about outcomes

https://www.bancocentral.tl/publication/laws_np/

S27 TotalEnergies Suriname, September 19 2026

Evidence: GranMorgu 2024 FID, first oil planned 2028

Limit: Planned start is not operating output

<https://totalenergies.sr/press/suriname-granmorgu-project-takes-shape-major-production-equipment-arrives>

S28 Guyana pre-oil sovereign wealth fund Green Paper

Evidence: Documents work beginning in late 2016 with Commonwealth Secretariat support, inter-agency working group in 2017, IMF missions and FARI training before first oil

Limit: Historical government policy paper. Treat forecasts in the paper as superseded by actual later events

<https://petroleum.gov.gy/wp-content/uploads/2024/10/Green-Paper-Final.pdf>

S29 IMF 2023 Article IV staff report

Evidence: Explains two-rule system: Stabroek PSA revenue sharing plus NRF withdrawal framework; notes first oil receipts deposited in 2020 and first budget transfer in 2022; records 2023 model-PSA changes and governance priorities

Limit: IMF analysis reflects information available in 2023 and should not substitute for 2026 data

<https://www.elibrary.imf.org/view/journals/002/2023/379/article-A001-en.xml>

S30 Natural Resource Governance Institute, natural-resource fund governance guidance

Evidence: Clear objectives, deposit/withdrawal rules, investment rules, independent oversight, broad consensus and budget integration; warns a fund alone does not produce good governance

Limit: Comparative policy guidance, not jurisdiction-specific legal advice

https://resourcegovernance.org/sites/default/files/NRF_Complete_Report_EN.pdf

S31 NRGI sovereign wealth fund guidance for policymakers

Evidence: Highlights oversight, disclosure, fiscal-rule fit, dangers of direct domestic investment from funds and the need to fit rules to national conditions

Limit: Older cross-country guidance; current country practices require fresh verification

<https://resourcegovernance.org/articles/sovereign-wealth-funds-guidance-policymakers>

S32 LCDS homepage, accessed September 2026

Evidence: Current official payment dashboard lists cumulative Phase II carbon-credit payments of USD 353.5m, including March 2026 USD 57.9m

Limit: Official programme reporting; independent audit of final use and outcomes remains separate

<https://lcds.gov.gy/>

S33 EITI validation schedule, accessed September 2026

Evidence: Guyana validation commenced 15 May 2026 and remains listed among ongoing validations

Limit: Do not state a final 2026 validation score until EITI publishes a decision

<https://eiti.org/validation-schedule>

S34 EITI Board decision 2026-19, 12 May 2026

Evidence: Full validation launched with external expert; Board expressed concern over continuing lack of full stakeholder confidence in civil-society representative selection

Limit: Board concern is attributed to EITI, not adopted as the author's independent conclusion

<https://eiti.org/es/node/28602>

S35 Petroleum Management Programme, official exploration history

Evidence: Records the long pre-2015 exploration sequence, 1938 licence history, 1965 offshore licences, 1980s investment promotion and later exploration interruptions

Limit: Government historical summary; individual dates should be cross-checked against original licences where material

<https://petroleum.gov.gy/history/summary-history-petroleum-exploration-guyana/>

S36 Petroleum Management Programme, Liza-1 well register

Evidence: Records Liza-1 as an exploration oil well in the Stabroek Block with well date 15 May 2015 and mapped coordinates

Limit: Administrative well register; commerciality and reserve estimates require separate sources

<https://petroleum.gov.gy/well/liza-1/>

S37 Parliament of Guyana, Natural Resources Fund Act 2019 record

Evidence: Parliament lists passage on 3 January 2019 and publication on 23 January 2019; the official act records presidential assent on 23 January 2019

Limit: Statutory establishment does not establish later implementation quality or unchanged withdrawal rules

<https://parliament.gov.gy/publications/acts-of-parliament/natural-resources-fund-act-2019>

S38 Parliament of Guyana, Petroleum (Exploration and Production) Act 1986

Evidence: Establishes the 1986 petroleum-law foundation for exploration, exploitation, conservation and management, including maritime areas

Limit: Later legislation and regulations must be version-checked before applying historical provisions to 2026

<https://parliament.gov.gy/publications/acts-of-parliament/petroleum-exploration-and-production-act-1986/>

S39 Parliament of Guyana, Environmental Protection Act 1996

Evidence: Establishes the pre-oil statutory environmental framework and environmental-impact mandate

Limit: Existence of statutory authority is separate from project-level permitting, enforcement and outcomes

<https://parliament.gov.gy/publications/acts-of-parliament/environmental-protection-act-1996>

S40 Permanent Court of Arbitration, Guyana v. Suriname case record

Evidence: Documents the 2000 oil-concession incident, 2004 commencement of arbitration and maritime-boundary dispute framework

Limit: Use the final award itself for detailed legal propositions; case page provides procedural history

<https://pca-cpa.org/en/cases/9/>

S41 Petroleum Management Programme, Stabroek Block register

Evidence: Records June 27 2016 petroleum agreement date, October 7 2016 effective/licence date and June 15 2017 production licence date

Limit: Administrative register; contract clauses must be checked against executed text

<https://petroleum.gov.gy/oil-blocks/stabroek-block/>

S42 Petroleum Management Programme, VHE initial 2018-2020 Stabroek cost-recovery audit page

Evidence: Government-published audit record covering cost and revenue review and identifying the operator as responsible for cost-recovery accounting

Limit: Initial audit status and later government resolution of exceptions must be distinguished

<https://petroleum.gov.gy/docs-and-pubs/vhes-initial-audit-report-for-the-stabroek-block-cost-recovery-audit-2018-to-2020/>

S43 Petroleum Management Programme, regulatory-news document index

Evidence: Lists public cost-recovery audits, production licences and environmental permits, including 1999-2017 and 2018-2020 audit records

Limit: Document availability does not establish final settlement of every audit finding

<https://petroleum.gov.gy/documents-categories/regulatory-news/>

S44 Bank of Guyana, NRF audited financial statements index

Evidence: Provides audited NRF financial statements for 2020 through 2025 in one official series

Limit: Each annual statement must be read on its own accounting basis and checked for restatements

<https://bankofguyana.org.gy/bog3/publications/natural-resource-fund/audited-financial-statements>

S45 Bank of Guyana, NRF audited financial statements 2020

Evidence: Records 2020 royalties, profit oil, receivables, cash, first-year accounting policies and no government withdrawal during the year

Limit: GYD reporting and accrual receivables must be reconciled to USD cash deposits before cross-year comparisons

https://bankofguyana.org.gy/bog/images/accounts_budgeting/natural_resource_fund/financial_statements/nrf_audited_financial_statements_2020.pdf

S46 Bank of Guyana, NRF audited financial statements 2021

Evidence: Records 2021 royalties, profit oil, cash flows, no government withdrawal and replacement of the 2019 Act by the 2021 Act

Limit: Contains restated comparative 2020 figures; use audited restatement rather than mixing earlier presentation

https://bankofguyana.org.gy/bog/images/accounts_budgeting/natural_resource_fund/financial_statements/nrf_audited_financial_statements_2021.pdf

S47 Bank of Guyana, NRF Quarterly Report June 2025

Evidence: Provides quarterly inflows, market value, interest, transfers to the Consolidated Fund and cumulative withdrawal history through June 2025

Limit: Quarterly market value differs from annual audited closing balances and should not be merged without date alignment

https://www.bankofguyana.org.gy/bog/images/accounts_budgeting/natural_resource_fund/quarterly/nrf-june2025-quarterly.pdf

S48 Petroleum Management Programme, Liza Phase 1 Modified Environmental Permit

Evidence: Records original permit dated 1 June 2017, the approved EIA/ESMP and revised oil-spill/wildlife plans, and modification on 12 May 2021

Limit: Permit existence and conditions do not establish independent compliance; monitoring evidence must be reviewed separately

<https://petroleum.gov.gy/docs-and-pubs/liza-phase-1-modified-environmental-permit/>

S49 Petroleum Management Programme, EPA-EEPGL project document index

Evidence: Official index lists Liza Phase 1 permit, Liza Phase 2 amended permit and EIA materials, Payara permit/EIA/OSRP documents and related environmental records

Limit: Index establishes document availability, not project performance; each permit must be reviewed individually

<https://petroleum.gov.gy/projects-initiatives/environmental-protection-agency-esso-exploration-and-production-guyana-limited/>

S50 Petroleum Management Programme, Uaru Environmental Permit

Evidence: Permit Ref. 2023-023-EEPGL, effective 27 April 2023 to 26 April 2028; includes EPA powers for sampling, permit modification and annual review linked to audit

Limit: Project-level compliance and audit results require separate evidence

<https://petroleum.gov.gy/wp-content/uploads/2024/10/Uaru-Signed-Permit.pdf>

S51 Petroleum Management Programme, Whiptail Environmental Permit page

Evidence: Official publication dated 12 April 2024 for the Whiptail environmental permit

Limit: Detailed permit conditions require document-level extraction and monitoring evidence

<https://petroleum.gov.gy/whiptail-environmental-permit/>

S52 Petroleum Management Programme, Yellowtail public notice

Evidence: Records the 12 May 2021 environmental-authorization application and EPA determination that an EIA was required before approval or rejection

Limit: Application/EIA scoping is not evidence of a final permit or later compliance

<https://petroleum.gov.gy/notice-to-the-public-yellowtail-development-project/>

S53 Parliament of Guyana, Oil Pollution Prevention, Preparedness, Response and Responsibility Act 2025

Evidence: Parliament records Act 6 of 2025, passed 16 May 2025 and published 30 May 2025

Limit: Enactment/publication must be kept separate from commencement and implementing regulations

<https://parliament.gov.gy/publications/acts-of-parliament/oil-pollution-prevention-preparedness-response-and-responsibility-act-2025-act-6-of-2025>

S54 Official Gazette of Guyana, Act 6 of 2025 publication

Evidence: Official Gazette archive records the petroleum pollution Act as an extraordinary gazette dated 28 May 2025

Limit: Gazette publication alone does not establish that every provision had commenced by the research cut-off

<https://officialgazette.gov.gy/index.php/publications/2778-extraordinary-gazettes-28th-may-2025-act-no-6-of-2025-the-oil-pollution-prevention-preparedness-response-and-responsibility-bill-2025>

S55 Petroleum Management Programme, Liza 1 flaring permit modification notice

Evidence: EPA states the modified permit added emissions reporting, flaring time limits and a US\$30/tCO₂e payment for flaring beyond specified timelines

Limit: Agency statement documents the regulatory requirement; actual emissions, invoices and payments need period evidence

<https://petroleum.gov.gy/epa-issues-a-modified-environmental-permit-to-eeegl-for-the-liza-1-project/>

S56 Petroleum Management Programme, Uaru Petroleum Production Licence

Evidence: Licence requires compliance with environmental law, expert personnel/equipment for spill response and specified capping-stack/dispersant arrangements

Limit: Licence requirements are obligations, not evidence that all readiness conditions were demonstrated in practice

<https://petroleum.gov.gy/index.php/documents/uaru-petroleum-production-licence>

S57 Parliament of Guyana, Local Content Act 2021

Evidence: Parliament records Act 18 of 2021, passed 29 December 2021 and published 4 January 2022

Limit: Enactment establishes the statutory framework; measured domestic value added requires separate evidence

<https://www.parliament.gov.gy/publications/acts-of-parliament/local-content-act-2021-no.-18-of-2021>

S58 Guyana Local Content Register

Evidence: Official supplier/employment registration system and Local Content Secretariat portal

Limit: Certification establishes registry status, not independent proof of domestic value added, capability or retained profit

<https://lcregister.petroleum.gov.gy/>

S59 Ministry of Finance, 2024 Mid-Year Report

Evidence: Government reported H1 2024 procurement and employment information from contractor/subcontractor reports, including reporting scope and projected training/procurement

Limit: Self-reported sector figures require independent ownership, customs, payroll and value-added checks

<https://finance.gov.gy/wp-content/uploads/2024/08/Mid-Year-Report-2024.pdf>

S60 Ministry of Finance, 2025 Mid-Year Report

Evidence: Describes expansion of local-content reporting to additional sectors/subsectors and ongoing assessment of local capacity

Limit: Government implementation update; not an independent measure of outcome quality

<https://finance.gov.gy/wp-content/uploads/2025/09/Mid-Year-Report-2025.pdf>

S61 Guyana Bureau of Statistics, Household Budget Survey 2018/2019

Evidence: Official survey page states HBS/Living Condition fieldwork ran from November 2018 through November 2019 across all ten administrative regions

Limit: Historical household evidence must not be represented as current 2026 welfare data

<https://statisticsguyana.gov.gy/surveys/>

S62 LCDS, consultations and land-tenure fact page

Evidence: Official programme account describes 15% community revenue share, village-plan process and Amerindian Land Titling programme

Limit: Government account requires triangulation with Indigenous/community sources and independent safeguards evidence

<https://lcds.gov.gy/facts-on-consultations-and-land-tenure-guyanas-art-trees-programme/>

S63 LCDS 2030 October 2024 Update

Evidence: Government reported more than 240 villages receiving carbon-credit revenue and village-led projects under the 15% allocation

Limit: Programme spending/participation figures require independent outcome and consent verification

<https://lcds.gov.gy/guyanas-low-carbon-development-strategy-lcds-2030-october-2024-update/>

S64 IMF, 2026 Article IV mission concluding statement

Evidence: IMF staff reports 2025 growth, continued H1 2026 expansion, fiscal/absorption issues, procurement/audit priorities, data gaps and diversification observations

Limit: Preliminary staff findings, not an Executive Board Article IV report; policy recommendations are attributed to IMF staff

<https://www.imf.org/en/news/articles/2026/07/31/mcs073126-guyana-staff-concluding-statement-of-the-2026-article-iv-mission>

S65 Ministry of Finance, Budget Speech 2026

Evidence: Official 2026 budget policy, public-investment allocations and announced energy/infrastructure programme

Limit: Budget allocation and target do not establish project completion or delivered outcome

https://finance.gov.gy/wp-content/uploads/2026/01/Budget_Speech_2026.pdf

S66 Guyana Energy Agency, Demand Side Management Action Plan handover, July 2026

Evidence: GEA reports a 2030 target of 516 GWh energy savings against projected generation of 2,508 GWh

Limit: Government/agency plan and target; implementation and achieved savings require later measurement

<https://gea.gov.gy/2026/07/01/6793/>

S67 Guyana Energy Agency, Demerara-Berbice Integrated System transition page

Evidence: Official description of the electricity mix, gas-to-energy concept and planned renewable complement

Limit: Some project dates on the page are historical planning dates and must be updated against 2026 delivery evidence

<https://gea.gov.gy/renewable/demerara-berbice-integrated-system/>

S68 IMF, 2025 Article IV staff report

Evidence: Describes official-statistics modernization, planned new household survey and macroeconomic context before the 2026 mission

Limit: Historical to 2025 and superseded where 2026 evidence exists

<https://www.elibrary.imf.org/view/journals/002/2025/103/article-A001-en.xml>

S69 EITI Board Decision 2026-19

Evidence: Board launched a full Guyana Validation from 15 May 2026 and recorded concern about stakeholder confidence in civil-society representative selection

Limit: Validation remained a process at the research cut-off; do not state a final score without a later Board decision

<https://eiti.org/es/node/28602>

S70 EITI 2026 stakeholder consultation

Evidence: International Secretariat sought views on Guyana implementation from April 2022 to May 2026, including civil-society participation and barriers

Limit: Consultation questions and submissions are inputs to Validation, not a final assessment

<https://eiti.org/offers/guyana-2026-eiti-validation-call-views-stakeholder-engagement>

S71 Ministry of Natural Resources, 11 April 2024, official IHS Markit final-audit publication notice and executive-summary excerpt

Evidence: Government-hosted summary identifies USD1,677,774,727 submitted for 1999–2017 cost audit and grounds to dispute approximately USD214.4m plus overhead adjustments; this is an audit exception, NOT settled cash recovery

Limit: Official accessible HTML summary is a fallback because the direct report PDF returned 404 to the GitHub worker. It does not independently authenticate every underlying audit schedule or final dispute resolution; full PDF retrieval remains OPEN in issue #69

<https://mnr.gov.gy/2024/04/11/ihs-markit-final-audit-report-for-the-stabroek-block-cost-recovery-audit-1999-2017/>

S72 Ministry of Natural Resources, audit-status update

Evidence: Government states the 1999–2017 audit moved to the PSA sole-expert process and describes status of multiple cost audits

Limit: Ministry status statement requires later verification of settlement, expert decision or arbitration outcome

<https://petroleum.gov.gy/article/update-on-status-of-stabroek-block-cost-recovery-and-verification-of-profit-oil-audits/>

S73 Audit Office of Guyana, 2024 public-accounts report portal

Evidence: Supreme audit institution states the 2024 public-accounts/performance report was presented to the National Assembly in September 2025

Limit: Public-account audit observations must be tied to specific entities/projects; do not generalize individual findings across government

<https://www.audit.org.gy/site/>

S74 Norway Ministry of Finance, Government Pension Fund 2026 white paper

Evidence: Describes current GPF framework, 3% expected-real-return fiscal guideline and long-run role of petroleum wealth in public finance

Limit: Norway's mature institutions, large accumulated assets and economic structure differ materially from a new producer

<https://www.regjeringen.no/en/documents/meld.-st.-7-2025-2026/id3155109/>

S75 Bank of Ghana, Ghana Petroleum Funds

Evidence: Official fund page describes Stabilisation and Heritage objectives, semi-annual reporting and annual/quarterly audit framework

Limit: Ghana's statutory allocation formula and production history differ from Guyana; compare mechanisms rather than outcomes by headline balance

<https://www.bog.gov.gh/ghana-petroleum-funds/>

S76 Bank of Ghana, 2026 Ghana Petroleum Funds reporting index

Evidence: Official reporting index includes 2025 audited statements and 2026 semi-annual report

Limit: Use individual statements for numerical comparisons and preserve reporting dates

<https://www.bog.gov.gh/ghana-petroleum-funds/gpf-financial-statements/>

S77 Central Bank of Timor-Leste, Petroleum Fund reporting portal

Evidence: Official portal publishes annual, quarterly and monthly performance reports including 2025 annual reporting and 2026 updates

Limit: Fund size and investment returns must be analysed alongside budget withdrawals, petroleum depletion and domestic fiscal dependence

https://www.bancocentral.tl/publication/pf_rep/

S78 Central Bank of Timor-Leste, Petroleum Fund management framework

Evidence: Describes Liquidity and Growth portfolios and publication requirements

Limit: Current portfolio structure reflects Timor-Leste's withdrawal needs and should not be copied without national cash-flow modelling

https://bancocentral.tl/stats/about_pf/

S79 TotalEnergies EP Suriname, GranMorgu update 19 September 2026

Evidence: Operator states project FID occurred in October 2024, construction is progressing and first oil remains targeted for 2028

Limit: Operator project update; planned 2028 first oil is not an observed production result

<https://totalenergies.sr/press/suriname-granmorgu-project-takes-shape-major-production-equipment-arrives>

S80 Trinidad and Tobago Ministry of Finance, Heritage and Stabilisation Fund reporting portal

Evidence: Official portal publishes annual and quarterly HSF reports through 2026

Limit: Mature producer and dual stabilization/heritage design offer mechanism comparison, not a direct blueprint for Guyana or a new island producer

<https://www.finance.gov.tt/publications/heritage-and-stabilisation-fund/>

S81 Petroleum Management Programme, Oil Production dashboard

Evidence: Official production series covers 20 December 2019 through 31 July 2026

Limit: Monthly realized production must remain separate from installed/nameplate capacity

<https://petroleum.gov.gy/data-chart/oil-production/>

S82 ExxonMobil, Liza Phase 2 first oil, 11 February 2022

Evidence: Operator records first production from Liza Unity on 11 February 2022 and planned ramp to 220,000 bpd

Limit: Operator statement is suitable for startup date; realized monthly production should use regulator data

https://corporate.exxonmobil.com/news/news-releases/2022/0211_exxonmobil-starts-production-at-guyanas-second-offshore-development

S83 Petroleum Management Programme, Payara first oil, 14 November 2023

Evidence: Government records first oil at Prosperity in November 2023; Liza Destiny started December 2019 and Liza Unity February 2022

Limit: Government media release includes future production estimates that must not be treated as observed output

<https://petroleum.gov.gy/article/govt-welcomes-first-oil-at-prosperity-fpso/>

S84 ExxonMobil, Yellowtail first production, 8 August 2025

Evidence: Operator records startup of the fourth project and combined installed capacity above 900,000 bpd

Limit: Installed capacity is not annual average production

https://corporate.exxonmobil.com/news/news-releases/2025/0808_exxonmobil-guyana-begins-production-at-fourth-offshore-guyana-project

S85 ExxonMobil, 900,000 bpd milestone, 12 November 2025

Evidence: Operator reports a daily production milestone after Yellowtail ramp-up

Limit: One-day/point-in-time output is not a full-year average

<https://corporate.exxonmobil.com/locations/guyana/news-releases/11122025-daily-oil-production-hits-900000-barrels-in-guyanas-stabroek-block>

S86 Petroleum Management Programme, Liza Phase 1 Production Licence and Phase 2 Addendum

Evidence: Official production-licence publication provides legal project authorization documents for Liza developments

Limit: Licence conditions require document-level review and do not establish realized performance

<https://petroleum.gov.gy/docs-and-pubs/liza-phase-1-petroleum-production-licence-and-phase-2-addendum/>

S87 Parliament of Guyana, Petroleum Activities Act 17 of 2023

Evidence: Establishes the newer statutory petroleum framework for licensing, regulation, data, safety, decommissioning and administration

Limit: Legacy contractual rights and transitional application require agreement-specific legal analysis

<https://parliament.gov.gy/publications/acts-of-parliament/petroleumactivitiesact17of2023>

S88 Bank of Guyana, audited NRF financial statements 2025

Evidence: Audited financial statements for the year ended 31 December 2025, including income, financial position, cash-flow and notes

Limit: Must be reconciled to Ministry reporting using consistent currency, timing and accounting basis

https://bankofguyana.org.gy/bog3/images/accounts_budgeting/natural_resource_fund/financial_statements/nrf_audited_financial_statements_2025.pdf.pdf

S89 EITI Validation schedule, checked 27 September 2026

Evidence: Lists Guyana among ongoing Validations and notes commencement on 15 May 2026

Limit: Ongoing status is not a final Validation decision or score

<https://eiti.org/validation-schedule>

S90 International Court of Justice, Guyana v Venezuela case record

Evidence: Official case docket for the controversy concerning the 1899 Arbitral Award, including jurisdiction, preliminary objection and provisional-measures material

Limit: The merits remain before the Court; this publication must not predict or decide the outcome

<https://icj-cij.org/case/171>

S91 International Court of Justice, Order of 1 May 2025

Evidence: Official provisional-measures order in Guyana v Venezuela

Limit: Interim judicial measure, not a final merits judgment

<https://www.icj-cij.org/node/205403>

S92 CARICOM statement, 28 April 2026

Evidence: Regional statement supporting peaceful resolution through the ICJ and addressing conduct concerning the Guyana-Venezuela controversy

Limit: Political/diplomatic statement; use for regional context, not for a merits determination

<https://caricom.org/statement-from-the-caribbean-community/>

S93 Forest Peoples Programme, ART Guyana carbon-credit case study

Evidence: Documents criticisms concerning ART certification, Indigenous safeguards and complaint mechanisms

Limit: Advocacy research; contested claims require triangulation with ART, government and community evidence

<https://www.forestpeoples.org/pt/publications-resources/reports/article/arts-controversial-certification-of-carbon-credits-to-the-government-of-guyana-a-case-study-on-challenges-for-high-integrity-labels-in-carbon-markets/>

S94 Office of the President, Guyana Oil & Gas Capacity Building Project SESA terms of reference

Evidence: Identifies strategic offshore and onshore environmental and social risk categories including fisheries, spill risk, marine habitats and coastal facilities

Limit: Terms of reference identify risks to assess; they are not measured outcome evidence

<https://op.gov.gy/guyana-oil-gas-capacity-building-project-p166730-terms-of-reference-tor-strategic-environmental-social-assessment/>

S95 LCDS, CORSIA-eligible ART TREES credits, 28 February 2024

Evidence: Records issuance of 7.14 million 2021-vintage credits and CORSIA eligibility claims, including corresponding-adjustment context

Limit: Programme statement; registry and UNFCCC records should be used for final accounting verification

<https://lcds.gov.gy/guyana-announces-worlds-first-carbon-credits-for-use-in-un-airline-compliance-programme-corsia/>

S96 Ministry of Finance, NRF monthly reports 2026

Evidence: Official monthly NRF reporting through July 2026 available at the publication cut-off

Limit: Monthly reports need reconciliation to quarterly, annual and audited accounts before aggregation

<https://finance.gov.gy/monthly-reports/>

S97 Stabroek News, Anand Goolsarran analysis of PSA revenue provisions, August 2025

Evidence: Independent public commentary using NRF audited accounts to discuss profit-sharing, royalty and taxation

Limit: Commentary, not primary law or audit evidence; use only as an attributed critical interpretation

<https://www.stabroeknews.com/2025/08/04/features/accountability-watch/revisiting-the-profit-sharing-royalty-and-taxation-provisions-of-the-petroleum-sharing-agreement-part-ii/>

S98 Official Gazette of Guyana, Extraordinary Gazette, 28 May 2025, Act No. 6 of 2025

Evidence: Confirms formal publication of the Oil Pollution, Prevention, Preparedness, Response and Responsibility Act 2025 in the Official Gazette

Limit: Publication confirms enactment/gazetting, not necessarily commencement of every provision or issuance of implementing regulations

<https://officialgazette.gov.gy/index.php/publications/2778-extraordinary-gazettes-28th-may-2025-act-no-6-of-2025-the-oil-pollution-prevention-preparedness-response-and-responsibility-bill-2025>

S99 Petroleum Management Programme, Uaru Environmental Permit, financial assurance conditions 14.3-14.5

Evidence: Permit states financial assurance should be guided by reasonably credible costs, requires liability/environmental-liability insurance, and requires parent/affiliate guarantee agreements to provide resources for environmental obligations after default

Limit: Permit terms establish obligations and architecture; they do not by themselves prove current insurance limits, guarantee amount, liquidity or enforceability in a future loss event

<https://petroleum.gov.gy/wp-content/uploads/2024/10/Uaru-Signed-Permit.pdf>

S100 Petroleum Management Programme, Liza Phase 1 Modified Environmental Permit

Evidence: Official page confirms permit issued 1 June 2017 and modified 12 May 2021, and incorporates the EIA, environmental/social management plan, revised oil-spill response plan and wildlife response plan

Limit: Document availability does not establish compliance performance or current financial-assurance sufficiency

<https://petroleum.gov.gy/docs-and-pubs/liza-phase-1-modified-environmental-permit/>

S101 Department of Public Information, Attorney General press release on Court of Appeal ruling, 8 May 2026

Evidence: Government summary states the Court of Appeal on 7 May 2026 overturned the High Court unlimited-guarantee order and held that EPA retains discretion over financial assurance

Limit: Government account of judgment; authenticated Court of Appeal reasons remain preferred for detailed legal characterization

<https://dpi.gov.gy/press-release-ag-hails-coas-ruling-on-exxonmobil-epas-financial-guarantees/>

S102 Department of Public Information, Attorney General press release, 3 May 2023

Evidence: Government stated EPA and EEPGL had negotiated a US\$2 billion parent guarantee and indemnity arrangement and described the government's position on the High Court ruling

Limit: Party statement and negotiation description; does not substitute for the executed guarantee instrument

<https://dpi.gov.gy/press-release-by-attorney-generals-chambers/>

S103 Caribbean Court of Justice, [2024] CCJ 16 (AJ) GY, 30 July 2024

Evidence: Authenticated CCJ reasons identify the underlying Collins/Whyte litigation, renewed Liza 1 permit condition 14, the 2023 High Court order, the Attorney General intervention dispute and Environmental Protection Act section 31

Limit: Procedural/intervention judgment, not the May 2026 Court of Appeal merits reasons

https://ccj.org/wp-content/uploads/2024/07/2024_CCJ_16_AJ_GY.pdf

S104 EITI Validation Committee, June 2026 meeting record

Evidence: EITI committee page records discussion of preliminary mission findings for Guyana on 23 June 2026 while Validation remained underway

Limit: Committee agenda/minutes context; not a final EITI Validation decision

<https://eiti.org/fr/node/28362>

S105 News Room Guyana, 7 May 2026 Court of Appeal report

Evidence: Contemporaneous secondary reporting states the appellate court distinguished unlimited liability from financial assurance and treated the permit as requiring a definite financial-assurance amount

Limit: Original Demerara Waves link returned HTTP 404 to GitHub workers; replacement is another dated reporting outlet, not authenticated Court of Appeal reasons. Use only alongside official summaries and permit text

<https://newsroom.gy/2026/05/07/appeal-court-overturms-decision-on-financial-assurance-dispute-rules-in-favour-of-exxonmobil-epa/>

S106 Kaieteur News, 25 August 2026, reported leave to appeal to CCJ

Evidence: Reports that the Court of Appeal granted leave in July 2026 for Collins and Whyte to challenge the May appellate ruling before the CCJ

Limit: Secondary report; current CCJ docket/filing should be preferred when publicly available

<https://kaieteurnewsonline.com/2026/08/25/exxons-unlimited-guarantee-fight-heads-to-ccj/>

S107 World Bank, Guyana Petroleum Exploration Promotion Project Completion Report, Report No. 12485, 16 November 1993

Evidence: Documents Credit 1208-GUA, objectives to attract exploration investment and strengthen petroleum administration, the Petroleum Unit, negotiated exploration contracts, training and legislative reform, as well as project shortcomings and capacity lessons

Limit: Historical World Bank evaluation reflects the project and institutional context of the 1980s-early 1990s, not later petroleum performance

<https://documents1.worldbank.org/curated/en/508851468252286613/pdf/multi-page.pdf>

S108 Local Content Secretariat, Local Content Half-Yearly Report Submission Guideline Version 4, 13 June 2025

Evidence: Defines required reporting on employment, salary scales, procurement, supplier participation, training and capacity development and requires comparative reporting against approved plans

Limit: Reporting architecture establishes required data fields; it does not by itself establish compliance quality or domestic value added

<https://petroleum.gov.gy/wp-content/uploads/2025/07/Local-Content-Half-Yearly-Report-Submission-Guideline-Version-4-1.pdf>

S109 Bank of Guyana, Natural Resource Fund Quarterly Report Q4 2025

Evidence: Reports end-2025 fund market value, quarterly inflows/outflows, investment return, annual transfers and investment mandate; confirms IFRS reporting and Federal Reserve Bank of New York custody

Limit: Quarterly accounting must be reconciled with audited annual financial statements and later 2026 monthly/mid-year records

https://www.bankofguyana.org.gy/bog/images/accounts_budgeting/natural_resource_fund/quarterly/nrf-december2025-quarterly.pdf

S110 Bank of Guyana, Natural Resource Fund monthly-report portal

Evidence: Provides monthly NRF statements for 2025 and 2026, enabling month-by-month bridge analysis rather than reliance only on annual summaries

Limit: Individual monthly reports require consistent cash/accrual treatment and should not be summed without reconciliation

<https://bankofguyana.org.gy/bog3/publications/natural-resource-fund/monthly-reports>

S111 Guyana Revenue Authority, response on inaugural Stabroek cost-recovery audit, 5 April 2023

Evidence: Describes IHS Markit audit scope, Guyanese inter-agency participation, draft/final report process, Annex C audit-response requirements, institutional disagreements and capacity-transfer objective

Limit: Government revenue-authority account; later settlement/dispute status must be verified separately

<https://www.gra.gov.gy/response-to-stabroek-news-articles-of-sunday-april-2-2023-and-monday-april-03-2023/>

S112 Petroleum Management Programme, 2023 draft shallow-water model Production Sharing Agreement

Evidence: Provides later-generation model clauses on financial guarantees, liability and insurance, local content, cost recovery, taxation, royalty, decommissioning, audits, dispute resolution and stabilization

Limit: Draft/model terms are not the executed 2016 Stabroek terms and must not be applied retroactively

<https://petroleum.gov.gy/wp-content/uploads/2024/10/Guyana-Draft-Shallow-Water-PSA-2023.pdf>

S113 EITI, Senegal country profile and petroleum-governance disclosures

Evidence: Records Senegal's 2019 Petroleum Code, 2022 oil-and-gas revenue-management law, contract publication framework, PETROSEN role, local-content reforms and first-oil/first-gas transition

Limit: EITI country profile summarizes multiple legal and institutional sources; project-level fiscal results require primary national records

<https://eiti.org/countries/senegal>

S114 EITI, Senegal 2026 Final Validation Report

Evidence: Reviews Senegal's petroleum transition, state participation, contract review, transparency and reconciliation issues after first oil and gas

Limit: EITI institutional assessment, not a substitute for national law or audited project accounts

https://eiti.org/sites/default/files/Senegal%202026_EITI%20Final%20Validation%20Report%20%28EN%29.pdf

S115 IMF, Mauritania: Improving Administration of Hydrocarbon Tax Revenues, High-Level Summary TA Report 2025/017

Evidence: Assesses hydrocarbon-revenue administration and capacity development ahead of Greater Tortue Ahmeyim gas production

Limit: Technical-assistance summary; country implementation must be verified separately

<https://www.imf.org/en/publications/high-level-summary-technical-assistance-reports/issues/2025/05/05/islamic-republic-of-mauritania-improving-administration-of-hydrocarbon-tax-revenues-566696>

S116 EITI, Validation of Mauritania 2024

Evidence: Reviews contract publication, national oil company, hydrocarbon fund transparency, beneficial ownership and readiness for emerging gas revenues

Limit: Validation reflects evidence available to EITI and predates later operating years

<https://eiti.org/sites/default/files/2024-10/EITI%20Validation%20of%20Mauritania%20%282024%29%20-%20Final%20Validation%20report%20%28August%202024%29.pdf>

S117 Uganda EITI Report 2022/23, published 2025

Evidence: Provides pre-production petroleum legal/institutional framework, local content, cost recovery, environmental management, beneficial ownership and petroleum-revenue architecture

Limit: Reporting period precedes commercial production; future operating outcomes remain separate

<https://www.ugeiti.org/wp-content/uploads/2025/09/UGEITI-Report-2022-23.pdf>

S118 EITI, Uganda country profile

Evidence: Describes Uganda Petroleum Fund, revenue institutions, beneficial-ownership framework and pre-first-oil extractive governance

Limit: EITI summary; detailed legal operation should use Uganda statutes and fund reports

<https://eiti.org/countries/uganda>

S119 Mozambique, Law No. 1/2024 establishing the Sovereign Wealth Fund

Evidence: Establishes objectives, eligible revenues, transfer rules, stabilization mechanism, parliamentary controls and investment principles for Mozambique's oil and gas sovereign fund

Limit: Statutory design does not establish future implementation performance

https://inp.gov.mz/wp-content/uploads/2024/09/0_Lei-1_2024-Fundo-Soberano.pdf

S120 Banco de Moçambique, Sovereign Wealth Fund operational management

Evidence: Describes operational-management structure and records initial capital transferred in December 2025

Limit: Central-bank operational source; future investment performance and fiscal effects require later reporting

<https://www.bancomoc.mz/en/sovereign-wealth-fund/operational-management/>

S121 IMF, Chile: Strengthening Fiscal and Stabilization Funds, Country Report 23/249

Evidence: Examines Chile's structural-balance rule, Economic and Social Stabilization Fund, debt anchor and countercyclical resource-revenue management

Limit: Copper-based mature economy differs materially from Guyana's petroleum and development context

<https://www.imf.org/-/media/files/publications/cr/2023/english/1chlea2023004.pdf>

S122 Bank of Botswana, Pula Fund

Evidence: Official description of the Pula Fund's purpose, investment of excess reserves and intergenerational savings objective

Limit: Botswana's diamond and reserve-management context differs from petroleum-export accounting

<https://www.bankofbotswana.bw/index.php/content/pula-fund>

S123 IMF, Botswana 2024 Article IV Consultation

Evidence: Documents pressure on fiscal buffers during a diamond-market downturn and discusses fiscal-rule and sovereign-fund reform options

Limit: IMF staff analysis; Botswana's mature diamond economy is a mechanism comparator, not a direct petroleum analogue

<https://www.imf.org/-/media/files/publications/cr/2024/english/1bwaea2024006-print-pdf.pdf>

S124 Department of Public Information, Guyana, arrival of Errea Wittu FPSO, 21 August 2026

Evidence: Confirms arrival of Guyana's fifth FPSO for Uaru, project cost US\$12.7 billion, 250,000 bpd design scale and first-oil expectation in Q4 2026

Limit: Government forward-looking startup expectation; does not establish that Uaru had produced first oil by the 27 September 2026 evidence cut-off

<https://dpi.gov.gy/press-release-govt-welcomes-arrival-of-fifth-fpso/>

S125 MODEC, arrival of FPSO Errea Wittu in Guyana, 24 August 2026

Evidence: Confirms vessel arrival, deployment approximately 200 km offshore in 1,690 m water, about 2 million barrels storage and commissioning ahead of first oil

Limit: Contractor statement; startup and realized production require later operating evidence

https://www.modec.com/news/assets/pdf/20260824_pr_Uaru_en.pdf

S126 Maritime Administration Department, Notice to Mariners 21 of 2026

Evidence: Confirms continuing deep-water development operations across Yellowtail, Whiptail, Liza, Payara and Uaru fields during 2026

Limit: Operational marine notice, not a production or fiscal record

<https://marad.gov.gy/wp-content/uploads/2026/01/Notice-to-Mariners-21-of-2026.pdf>

S127 Department of Public Information, Hammerhead petroleum production licence approval, 22 September 2025

Evidence: Confirms approval of seventh Stabroek development, US\$6.8 billion project, planned 150,000 bpd capacity, 18 production/injection wells and anticipated 2029 startup

Limit: Approved/planned development data; future production is not observed output

<https://dpi.gov.gy/gog-approves-7th-petroleum-production-license/>

S128 ExxonMobil, Hammerhead final investment decision, 22 September 2025

Evidence: Operator confirms seventh-development FID, planned 150,000 bpd FPSO capacity and anticipated 2029 startup

Limit: Operator forward-looking project statement; future startup is not an observed result

https://corporate.exxonmobil.com/news/news-releases/2025/0922_exxonmobil-guyana-expands-capacity-with-seventh-offshore-development

S129 Ministry of Finance, Budget at a Glance 2026

Evidence: Provides 2026 sector and major-project allocations across health, energy, roads, bridges, transport and other public investment

Limit: Budget allocation is an authorization/planning figure, not evidence of physical completion or service outcome

https://finance.gov.gy/wp-content/uploads/2026/01/Budget_at_a_Glance_2026.pdf

S130 Audit Office of Guyana, Auditor General Annual Report 2024

Evidence: Audits public accounts and selected contracts; records contract-management findings, overpayments and performance-audit evidence including NDIA asset-management weaknesses

Limit: Audit findings are period-specific and should be reconciled to later recoveries/corrective action

<https://audit.org.gy/site/images/AG/AnnualReport2024.pdf>

S131 Department of Public Information, Wismar-Mackenzie Bridge status, 26 September 2026

Evidence: Government reports the US\$35 million four-lane bridge as nearing opening, with contract and construction-history details

Limit: Government project-status statement; final commissioning and independent service-outcome evidence are separate

<https://dpi.gov.gy/new-four-lane-wismar-mackenzie-bridge-to-open-within-weeks-president-ali/>

S132 Department of Public Information, six new hospitals construction update, 6 January 2026

Evidence: Government reports construction underway on six hospitals and describes expected services/timing

Limit: Government project-status statement; completion, staffing, utilization and health outcomes require later verification

<https://dpi.gov.gy/six-new-hospitals-underway-to-strengthen-healthcare-delivery-across-guyana/>

S133 Department of Public Information, Linden-Mabura Hill road progress, 1 April 2026

Evidence: Government reports 86% completion and 100 km of 121 km paved at the report date

Limit: Dated construction-progress claim; final completion, cost and transport outcome require later evidence

<https://dpi.gov.gy/a-major-milestone-for-linden-mabura-hill-road-project-races-toward-august-2026-completion/>

S134 Bank of Guyana, NRF March 2026 quarterly report, Table 3

Evidence: Provides exact GYD-thousand quarterly opening balance, profit-oil receipts, royalties, Consolidated Fund transfers, investment result and closing fund balance for Q1 2026

Limit: IFRS/quarterly financial summary; the monthly series uses cash basis and may diverge from accrual/receivables at period cutoffs

https://bankofguyana.org.gy/bog/images/accounts_budgeting/natural_resource_fund/quarterly/nrf-march2026-quarterly.pdf

S135 Guyana Natural Resource Fund Annual Report 2023, Appendix Table 3, government-hosted republication

Evidence: Dated transaction-level history of government profit-oil and royalty receipts since March 2020, including the 2020/2021 transition

Limit: Original Parliament-hosted PDF returned HTTP 404 to GitHub workers; the Petroleum Management Programme hosts the report. Transaction receipt ledger differs from year-end IFRS NAV/accrued investment income

<https://petroleum.gov.gy/wp-content/uploads/2024/10/Natural-Resource-Fund-Annual-Report-2023-2.pdf>

S136 Bank of Guyana, Natural Resource Fund quarterly reports portal (June 2026 Q2 listing)

Evidence: Identifies Q1 and Q2 2026 quarterly reporting releases needed for the exact H1 IFRS bridge

Limit: Exact Q2 table transcription and method-matched reconciliation must be separately completed; portal listing alone is not a numeric source

<https://bankofguyana.org.gy/bog3/publications/natural-resource-fund/quarterly-reports>

S137 Bank of Guyana, December 2025 NRF monthly cash-basis statement

Evidence: Shows year-end cash and equivalents GYD 677,711,491 thousand, approximately USD 3,250.41 million, distinct from the December audited accrual NAV including year-end receivables

Limit: Monthly statement expressly uses cash basis; do not compare directly with IFRS NAV without receivable bridge

https://bankofguyana.org.gy/bog/images/accounts_budgeting/natural_resource_fund/monthly/nrf-december2025-monthly.pdf

S138 Guyana Ministry of Natural Resources, Liza Unity FPSO incident statement, 24 August 2026

Evidence: Ministry reports a small laundry-room fire triggering detectors, immediately initiated response and all personnel safe, with relevant regulators to follow up

Limit: Initial incident notice, not the subsequent independent incident-investigation findings or evidence that all offshore safety risks are resolved

<https://dpi.gov.gy/press-release-statement-on-liza-unity-incident/>

S139 Ministry of Natural Resources, 2026 local-content plan approvals, 17 June 2026

Evidence: Reports approval of over 40 companies' 2026 local content plans and operator-reported engagement of approximately 2,000 local suppliers with US\$3.6bn cumulative spend by end-2025

Limit: Annual plans and operator-attributed cumulative gross procurement are not audited domestic value-added or verified job outcomes

<https://dpi.gov.gy/over-40-companies-receive-approval-for-2026-local-content-plans/>

S140 Department of Public Information, local content H1 2026 registration report, 18 September 2026

Evidence: Reports approximately 500 new business registrations and preliminary US\$466.2m first-half goods/services spending in prioritized local-content sectors

Limit: Preliminary registration and gross-spend statistics; no verified imports/retained domestic value bridge

<https://dpi.gov.gy/nearly-500-guyanese-businesses-register-for-local-content-opportunities-in-first-half-of-2026/>

S141 Guyana EPA, Uaru environmental authorization announcement, 27 April 2023

Evidence: Summarizes Uaru permit requirements on environmental effects monitoring, liability, annual company financial-capacity statements, insurance and parent/affiliate financial assurances

Limit: EPA permit conditions are primary obligations; EPA announcement is official description, neither proves independent operating compliance nor the full executed guarantee terms

<https://epaguyana.org/article/epa-grants-environmental-authorisation-for-uaru-petroleum-development-project/>